
Symbiosis Framework v1.0

A Deterministic Proto-Standard for Human–AI Symbiotic Systems

Version: 1.0 (Deterministic Compilation)

Status: Normative Standard Draft

Binding Order: Per SYM–BND–v1.0 Binder Plan

Canonical Metrics Location: Part III only

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Canonical repository: <https://github.com/harvirr/symbiosis>

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Symbiosis Framework v1.0

An Engineering Standard for Safe, Economic, and Governable Hybrid Intelligence

Version 1.0 — Deterministic Binder Compilation

Date: 20 November 2025

License: Apache 2.0

Part I — Foundations

Chapter 1 — Purpose, Scope, Reader Model, and Contract

Status: Normative (REWRITE/UPGRADE from v0.9)

Purpose of Chapter: Establish why Symbiosis exists, what it claims, what it does not claim, who it is for, and the binding rules that govern how the remainder of v1.0 SHALL be interpreted, implemented, and audited.

1.0 Introduction

The Symbiosis Framework specifies a deterministic architecture for hybrid human–AI systems. It treats human–AI collaboration as an engineered system with explicit layers, measurable control surfaces, bounded autonomy, and auditable lineage.

The Framework SHALL be read as a proto-standard: its clauses define requirements that are intended to be implementable, testable, and regulator-defensible across sectors.

Symbiosis is not a model, a prompt library, or a single product pattern. It is a systems standard that constrains how humans, coordination software, synthetic agents, and governance mechanisms interoperate.

1.1 Problem Statement: From Shadow AI to Governable Hybrid Intelligence

1.1.1 The failure mode of ad-hoc AI deployment

Most contemporary deployments rely on informal prompting, fragmented tools, and untyped workflows. In such environments:

intent is underspecified,

constraints are implicit or inconsistently applied,

autonomy is granted without bounded risk models, and

outcomes cannot be reconstructed for audit.

These conditions produce Shadow AI: unofficial, non-governed use that increases risk while obscuring accountability.

1.1.2 The Human-in-the-Loop (HITL) fallacy

Merely inserting a human review step does not yield governable systems. HITL architectures often:

overload humans with unbounded oversight,

fail to distinguish safety constraints from optimization goals, and

lack runtime state machines for autonomy escalation.

Symbiosis replaces HITL with System-in-the-Loop hybrid intelligence: a structured, layered system where human agency is preserved through measurable and enforceable boundaries.

1.2 Core Thesis and Design Commitments

The Framework is founded on three commitments:

1. Safety as a first-class control objective.

Safety SHALL be enforced through explicit constraints, bounded autonomy, and governance gates, not through informal review.

2. Economics as a runtime discipline.

All hybrid systems SHALL operate under explicit budgets and economic circuits that determine whether work proceeds, defers, or collapses under resource saturation. (Economic Circuit is defined canonically in Appendix B and bound into the Engine Algorithm in Part IV.)

3. Governance as code.

Policies, autonomy ceilings, and constraint vectors SHALL be machine-enforceable, versioned, and auditable through lineage.

1.3 Scope of the Standard

1.3.1 Systems covered

Symbiosis applies to any system where:

a human operator (H) originates intent and retains sovereignty,

a coordination layer (Symbiosis Engine, E) decomposes intents into Task Graphs (TG),

synthetic agents (A) execute TG nodes under contracts, and

a governance layer (G) constrains autonomy, safety, and economic viability.

1.3.2 What Symbiosis standardizes

The Framework standardizes:

Four-Layer Architecture and boundary duties (Part II),

the coordination and execution pipeline (Part IV),

canonical metrics and autonomy gating (Part III),

governance and incident doctrine (Part IV), and

adoption and certification methods (Part V).

1.3.3 Out of scope

The Framework SHALL NOT be interpreted as:

a guarantee of model correctness,

a substitute for sector-specific law or policy,

an endorsement of any vendor, model class, or training regime,

a universal risk scoring system beyond its defined control surfaces, or

a productivity maximization doctrine.

1.4 Boundary of Claims (Normative)

The Framework makes the following bounded claims:

1. If implemented as specified, Symbiosis SHALL produce systems with reconstructable intent-to-outcome lineage suitable for safety audit.

2. If implemented as specified, Symbiosis SHALL prevent autonomy escalation beyond governance ceilings.

3. If implemented as specified, Symbiosis SHALL distinguish Semantic Drift from Operational Decay and route them to distinct remediation paths.

4. If implemented as specified, Symbiosis SHALL enable economic admission control of work under resource constraints and queue latency.

The Framework does not claim to eliminate all errors or risks; it claims to make hybrid systems governable, bounded, and auditable.

1.5 Reader Model and Intended Audience

The Framework is written for a multisectoral readership. The following roles are assumed:

Operators / Symbiotes: humans who originate intent, supervise tasks, and hold regret ownership.

Hybrid Intelligence Architects: engineers designing Engines, agents, TGs, and telemetry.

Governance Leads / Safety Owners: roles responsible for constraints, RB/AB enforcement, audits, and incident response.

Regulators / Auditors / Compliance Reviewers: external or internal parties validating conformance.

Each chapter SHALL be usable independently by its primary audience, but cross-references SHALL be treated as binding where specified.

1.6 Reader Contract (Normative)

By using or implementing this Framework, the reader and/or deploying organization SHALL accept the following contract:

1. Interpretation discipline.

Terms defined canonically in Part III and Appendix A SHALL be used consistently. Local synonyms or redefinitions are prohibited unless explicitly marked as deployment-specific extensions.

2. Layer obedience.

Implementations SHALL respect the Four-Layer Architecture. No layer MAY assume responsibilities assigned to another layer, and no boundary MAY be bypassed.

3. Constraint precedence.

Hard constraints (C) SHALL override optimization goals in all runtime and governance decisions.

4. Metric governance.

Canonical metrics (SI, HSS v2.0, SCP, CI, RB v2, AB schema) SHALL NOT be redefined outside Part III. All other references SHALL point back to Part III definitions.

5. Auditability requirement.

Deployments claiming Symbiosis compliance MUST provide lineage logs sufficient to reconstruct decisions to Intent + Constraints + Task Graph + Autonomy State + Economic Pass.

6. Non-punitive human metrics usage.

Human-side metrics are safety telemetry only. Any HR, ranking, compensation, or disciplinary use SHALL be treated as non-compliance. (Hardening rules are canonical in Part III Chapter 8 and extended in Appendix H.)

7. Change control.

Changes to canonical definitions or the SQC require explicit governance approval and delta logging.

1.7 Conformance Tiers and Adoption Preview

Symbiosis defines a tiered adoption model. These tiers indicate scope of implementation, not quality ranking.

Tier 0 — Experimental Symbiosis:

Partial or pilot-only implementation. MAY omit formal certification but MUST not violate any safety or non-punitive constraints.

Tier 1 — Bounded Collaboration:

Implements Four-Layer Architecture, Task Graph decomposition, and AB0–AB1 autonomy gating.

Tier 2 — Governed Autonomy:

Adds AB2 gating, Dry-Run enforcement, RB proximity controls, and economic admission control.

Tier 3 — Certified Symbiosis:

Implements full canonical metrics, AB3 eligibility rules, governance kernel v2, drift/decay doctrine, and certification protocol.

Formal phase gates, telemetry minima, and certification artifacts are specified in Part V (Chapters 16–17). This section is a preview only and SHALL NOT be treated as an alternate definition.

1.8 Overview of the Four-Layer Architecture (Non-Canonical Summary)

Symbiosis is organized as:

1. Human Layer (H): origin of intent, meaning authority, and regret ownership.
2. Coordination Layer (E): Symbiosis Engine that translates intent into TGs and mediates execution.
3. Agentic Layer (A): bounded synthetic agents executing TG nodes under contracts.
4. Governance Layer (G): constraints, budgets, autonomy ceilings, incident doctrine, and audit controls.

Authoritative definitions and boundary duties are in Part II. This summary is descriptive only.

1.9 List of Figures (Stub; keyed to Appendix L)

The following figure keys SHALL be used in Appendix L. Titles will be finalized when Appendix L is compiled.

Fig. L-01: Four-Layer Architecture schematic (H / E / A / G).

Fig. L-02: Symbiosis Engine v2 high-level control loop.

Fig. L-03: Task Graph (TG) lifecycle and node typing.

Fig. L-04: Economic Circuit v2 with Compute Saturation (CS) and Queue Latency (QL) gates.

Fig. L-05: Regret Boundary (RB v2) and Autonomy Bands (AB0–AB3) state machine.

Fig. L-06: Canonical metrics interaction map (SI ↔ HSS ↔ SCP ↔ CI) with governance hooks.

Fig. L-07: Drift vs Decay diagnostic split and remediation routing.

Fig. L-08: Adoption bridge ladder and Bridge Collapse / Global Bridge Lock triggers.

Appendix L SHALL attach narratives for each figure including purpose, entities, control-flow, constraint injection points, telemetry paths, and failure modes.

1.10 Specification Quality Contract (SQC) (Normative, Framework-Wide)

Status: Mandatory Definition of Done

Applies to: All Parts, Chapters, and Appendices of Symbiosis Framework v1.0+

1.10.1 Purpose

The Framework SHALL be compiled and maintained under a Specification Quality Contract (SQC). The SQC defines minimum acceptance criteria for any section to be considered v1.0-compliant and binder-ready. No chapter, subsection, or appendix SHALL be admitted unless it satisfies every gate below.

1.10.2 Mandatory Gate-Checks

Each section MUST pass all eight SQC gates:

1. Layer Integrity. Content MUST conform to the Four-Layer Architecture, with no boundary bypass and no cross-assignment of duties.

2. Canonical Metrics Non-Redefinition. SI, HSS v2.0, SCP, CI, RB, and AB schema SHALL be defined only in Part III. Elsewhere: reference only.

3. Constraint Precedence and Threshold Explicitness. Hard constraints MUST be explicit; thresholds MUST be governance-set and deployment-specific.

4. Mechanistic Testability. Every normative clause MUST be operationally testable or enforceable by the Symbiosis Engine and/or Governance Kernel.

5. Lineage Recoverability. Actions MUST be reconstructable to Intent + Constraints + TG + AB/RB state + Economic Pass decision.

6. Drift vs Decay Typing. Failures MUST distinguish Semantic Drift from Operational Decay with distinct lineage and remediation.

7. Abuse and Incentive Hardening. Human metrics and governance levers MUST include anti-gaming and anti-misuse safeguards.

8. Normative Voice Consistency. Requirements MUST use MUST/SHALL/MAY correctly; marketing or speculative language is prohibited.

1.10.3 Compliance Outcomes

A section failing any gate SHALL be treated as non-compliant, revised until all gates pass, and SHALL NOT be cross-referenced as authoritative until compliant.

1.10.4 Change Control

Modification of the SQC SHALL require Governance Body approval, explicit rationale, and versioned delta logging.

SQC Gate Check — Chapter 1

Gate 1 — Layer Integrity: PASS. Four layers referenced descriptively; no boundary violations introduced.

Gate 2 — Canonical Metrics Non-Redefinition: PASS. Metrics referenced only; no formulas, weights, or thresholds defined outside Part III.

Gate 3 — Constraint Precedence + Threshold Explicitness: PASS. Constraint precedence stated; no implicit numeric thresholds introduced.

Gate 4 — Mechanistic Testability: PASS. Normative clauses are enforceable (reader contract, bounded claims, SQC).

Gate 5 — Lineage Recoverability: PASS. Auditability requirement specified without redefining lineage schema.

Gate 6 — Drift vs Decay Typing: PASS. Distinction asserted as requirement; no alternate doctrine defined.

Gate 7 — Abuse / Incentive Hardening: PASS. Non-punitive and anti-HR misuse constraints stated; aligned to Part III/Appendix H.

Gate 8 — Normative Voice Consistency: PASS. MUST/SHALL/MAY used in normative sections; no speculative or marketing tone.

Part I — Foundations

Chapter 2 — Ontological Axioms & Primitives

Status: Normative (REWRITE/UPGRADE from v0.9)

Purpose of Chapter: Establish the irreducible axioms and core primitives that make Symbiosis a well-typed, auditable hybrid-intelligence system. This chapter provides the minimum operational ontology required for the architecture; the extended formal vocabulary remains in Appendix A.

2.0 Orientation

The Symbiosis Framework relies on a controlled vocabulary. Without shared primitives, hybrid systems collapse into prompt chains, hidden autonomy, and non-reconstructable outcomes. This chapter fixes the axioms and type system that every Part MUST obey.

Appendix A contains the extended formal ontology and reconciliation table. Where a term is defined in both locations, this chapter SHALL be treated as the binding operational statement, and Appendix A as extended support.

2.1 Top-Level Axioms (Irreducible Invariants)

The following four axioms are non-negotiable. Any implementation or extension of Symbiosis MUST preserve them.

2.1.1 Axiom 1 — Human Primacy

1. The human operator (H) is the sole origin of sovereign intent and meaning authority.
2. The human operator is the final bearer of accountability and regret for system outcomes.
3. Any system behavior that overrides, obscures, or displaces human meaning authority SHALL be classified as non-symbiotic.

Implementation corollary: kill-switch authority, override rights, and interpretive authority MUST remain human-held and enforceable by Governance.

2.1.2 Axiom 2 — Constraint Supremacy

1. Hard constraints (C) SHALL override optimization goals, preferences, or agent proposals.
2. If any plan violates C, the Symbiosis Engine (E) MUST reject or downgrade the plan prior to execution.
3. C MUST be explicit, typed, and lineage-logged; implicit or informal constraints are non-compliant.

Implementation corollary: constraints are enforceable objects, not advisory text.

2.1.3 Axiom 3 — Lineage Recoverability

1. Every system action MUST be reconstructable to:
 - structured intent (I),
 - active constraint vector (C),
 - task graph (TG),
 - autonomy state (AB/RB), and
 - economic pass decision.
2. Lineage SHALL be immutable and queryable for audit, incident response, and metric interpretation.

Implementation corollary: any workflow lacking lineage is Shadow AI by definition (see Chapter 1).

2.1.4 Axiom 4 — Separation of Concerns (Layer Integrity)

1. Symbiosis is a four-layer system. Each layer has exclusive responsibilities.
2. No layer MAY perform another layer's duties.
3. No layer MAY bypass adjacent boundaries.

Implementation corollary: layered enforcement is mandatory for safety and auditability.

2.2 Why an Ontology Is Required

An ontology is required to make hybrid intelligence a first-class engineered object. A formal type system provides:

1. **Traceability.** Every workflow element can be typed and audited.
2. **Cross-version stability.** Architecture and metrics may evolve without terminological drift.
3. **Deterministic system design.** “Human–AI symbiosis” becomes an implementable system, not a loose practice set.

2.3 Core Entities (Type System Overview)

The Symbiosis stack is composed of stable entities that SHALL be treated as first-class types.

2.3.1 Human Operator (H)

Definition: the biological operator holding sovereign agency, meaning authority, and final accountability and regret ownership. Humans are not a safety feature; they are the system's value origin.

Properties:

- Human capability and capacity are assessed via canonical metrics in Part III. This chapter references them only.

2.3.2 Symbiosis Engine (E)

Definition: the deterministic coordination layer that translates intent into Task Graphs, enforces constraints and budgets, mediates autonomy gating, and records lineage. E is not a model; it is stateful orchestration code.

2.3.3 Synthetic Agent (A)

Definition: a bounded capability module (model/tool/agent) that executes Task Graph nodes under explicit contracts and autonomy ceilings. Agents do not originate goals, nor modify governance.

2.3.4 Structured Intent Object (I)

Definition: a machine-interpretable representation of human goals, scope, preferences, regret tolerance, and initial constraints. I is the canonical input to E. Natural-language prompts are low-fidelity sources that MUST be normalized into I.

2.3.5 Constraint Vector (C)

Definition: the hard boundary conditions attached to I and TG nodes. C is orthogonal across three dimensions:

- **C_legal:** jurisdictional, regulatory, and policy boundaries.
- **C_safety:** harm and capability boundaries.
- **C_econ:** resource and cost caps (Economic Circuit).

If any plan violates C, E MUST reject the plan pre-flight.

Note: "hard/soft constraints" remain valid as sub-typing under these dimensions.

2.3.6 Budget Envelope (B)

Definition: a finite multi-dimensional resource pool per intent/workflow, including compute, financial spend, and human attention/interrupt capacity. B is mandatory for execution.

Budgets SHALL be checked by E prior to dispatch and during runtime (Economic Circuit v2 is specified in Appendix B).

2.3.7 Task Graph (TG)

Definition: a typed DAG-like decomposition of I into executable nodes with explicit dependencies, contracts, constraints, and budgets. TG is the primary execution substrate used by E.

Formal typing rules and lifecycle are specified in Part II, Chapter 3 and Appendix D. This chapter declares TG as a primitive only.

2.3.8 Autonomy Bands (AB) and Regret Boundary (RB)

Definition (non-canonical summary): autonomy is discretized by irreversibility (RB) and bounded by AB ceilings. AB specifies what A may execute without explicit human gating; RB defines the irreversibility thresholds that trigger mandatory checkpoints.

Canonical AB/RB definitions, math, and gates are specified only in Part III.

2.3.9 Context Lineage (L)

Definition: an immutable trace of how intent, constraints, assumptions, system state, and actions evolved over time. L is required for auditability, drift/decay diagnosis, and metric interpretation.

2.4 Deep Terms (Operational Constructs)

These constructs become essential at higher symbiosis levels. They are defined here as primitives to avoid later ambiguity.

2.4.1 Intent Manifold (M)

Definition: the acceptable outcome space implied by I. Human intent is not a single point request; E MUST expand I into M using context and constraints without inventing new goals.

2.4.2 Alignment Surface (AS)

Definition: the interface where I is expressed and captured. Alignment surfaces vary in fidelity:

- Low fidelity: open chat interfaces.
- High fidelity: structured GUIs/APIs with explicit fields and constraints.

E MUST treat low-fidelity AS inputs as provisional until normalized into I with explicit C.

2.4.3 Governance Kernel (GK)

Definition: the minimal isolated logic inside E that enforces RB/AB rules and risk ceilings. Agents cannot modify GK, and no autonomy transition MAY occur without kernel-signed authorization.

2.4.4 Co-Adaptive Loop

Definition: the bidirectional learning cycle in which humans improve intent quality and supervision skill while the system updates orchestration and preference models. Co-adaptation is a system property, not equivalent to model fine-tuning.

2.5 Typology of Symbiosis (Continuum of Integration)

Symbiosis is a continuum, not a binary state. The typology below classifies operational regimes by control locus, autonomy, governance burden, and economic profile.

Level 1 — Augmentation (Tool Use)

- **Dynamic:** H initiates → A executes bounded task → H reviews.
- **Control locus:** strictly human.
- **Economic profile:** low compute autonomy; higher human time cost.

Level 2 — Co-Exploration (Partner Mode)

- **Dynamic:** H sets goal → system proposes paths → H selects/refines → system executes.
- **Control locus:** iterative shared control.
- **Economic profile:** moderate autonomy; requires valid B and healthy system stability.

Level 3 — Co-Agency (Unified Operation)

- **Dynamic:** system infers I from context → executes long-horizon TGs → H intervenes primarily at RB alerts or governance checkpoints.
- **Control locus:** system-driven within GK/C/B.
- **Economic profile:** high autonomy and compute cost; requires strict circuit breakers and high operator readiness.

Interpretation rule: typology level is not aspirational by default. Many workflows SHOULD remain Level 1 permanently. The standard optimizes for *appropriate autonomy*, not maximum autonomy.

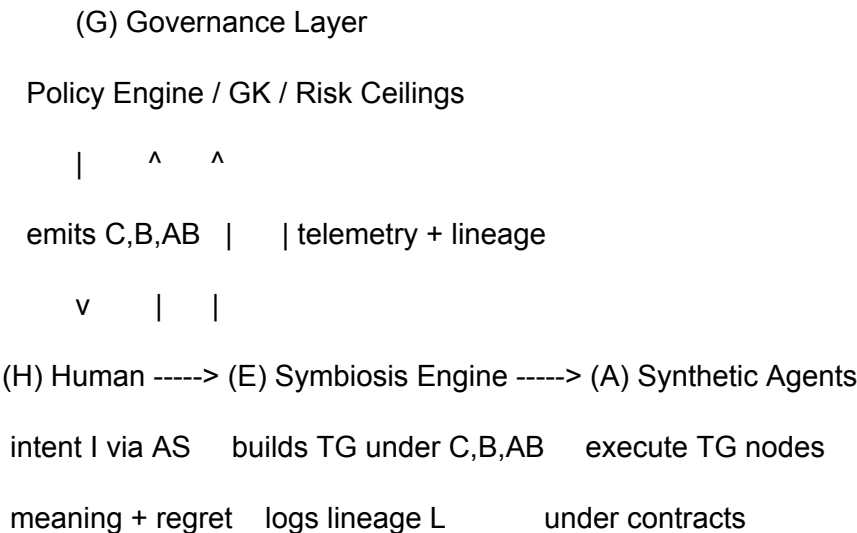
2.6 Derived Invariants (System-Wide Requirements)

The four axioms in §2.1 are irreducible. The following invariants follow from them and MUST hold at all typology levels.

- 1. **Economic Viability.**
No workflow SHALL execute without a defined B and enforceable C_econ.
Open-ended spend or infinite loops MUST be treated as system failures.
- 2. **Bounded Autonomy.**
All agent actions SHALL be bounded by AB ceilings and RB checkpoints as enforced by GK. Autonomy outside this state machine is non-compliant.
- 3. **Co-Evolution as Default.**
Systems SHALL be designed to improve human capability and maintain operator agency over time. De-skilling dynamics are treated as safety failures.

2.7 Canonical Ontology Diagram (ASCII Operational View)

This diagram is the binding operational relationship map. Canonical SVG renders and narratives are in Appendix L.



Directional rules:

- **Control flows down:** $H \rightarrow E \rightarrow A$.
- **Telemetry flows up:** $A \rightarrow E \rightarrow H/G$.

No direct $H \rightarrow A$ execution is permitted outside E mediation, and no $A \rightarrow G$ policy modification is permitted outside GK gates.

2.8 Practical Implications

1. **Ontology-first prevents autonomy creep.**
If AB/RB are explicit types and lineage-logged, unsafe escalation can be detected and demoted before cascading failure.
2. **Governance becomes code and data.**
C, B, and GK policy versions become machine-operable boundary objects.
3. **Skill development becomes a system property.**
Canonical human-side metrics (Part III) inform safe task allocation and training rather than punitive evaluation.

§2.8: See Appendix L.2 Canonical Ontology Diagram (Money Diagram) for the cross-layer economic and governance spine.

2.9 Chapter Close

This chapter fixes the axioms and primitives that all subsequent Parts **MUST** obey. Part II specifies boundary duties for each layer, while Part III defines canonical metrics and autonomy math. Any deployment claiming Symbiosis compliance **SHALL** be evaluated against these axioms and primitives prior to certification.

SQC Gate Check — Chapter 2

- **Gate 1 — Layer Integrity:** PASS. Ontology declares layer duties and no-bypass rules; no boundary violations introduced.
 - **Gate 2 — Canonical Metrics Non-Redefinition:** PASS. HSS, SCP, SI, CI, RB, AB referenced only; no formulas, weights, or thresholds defined outside Part III.
 - **Gate 3 — Constraint Precedence + Threshold Explicitness:** PASS. Constraint supremacy stated; no implicit numeric thresholds introduced.
 - **Gate 4 — Mechanistic Testability:** PASS. Axioms and primitives are enforceable objects (I, C, B, L, TG, GK) with clear compliance conditions.
 - **Gate 5 — Lineage Recoverability:** PASS. Lineage requirements explicitly bound to I + C + TG + AB/RB + Economic Pass.
 - **Gate 6 — Drift vs Decay Typing:** PASS. Lineage substrate and auditability asserted; no drift/decay doctrine redefined here.
 - **Gate 7 — Abuse / Incentive Hardening:** PASS. Human-metric misuse prohibited by reference; no leaderboard or punitive framing introduced.
 - **Gate 8 — Normative Voice Consistency:** PASS. **MUST/SHALL/MAY** used correctly; no marketing or speculative language.
-

Chapter 3 — Architecture Overview (Four-Layer Stack)

Status: Normative standard, v1.0

Part: II — The Four-Layer Architecture

Purpose: Specify the minimum viable architecture for governable hybrid intelligence and define its mandatory cross-layer flows, economic control loop, and invariants.

3.0 Overview and Rationale

A Symbiosis-compliant system **MUST** implement a Four-Layer Architecture consisting of:

1. **Human Layer (H)**,
2. **Symbiosis Engine / Coordination Layer (E)**,
3. **Agentic Layer (A)**, and
4. **Governance Layer (G)**.

This architecture is required to prevent *Shadow AI* collapse, defined here as any workflow where intent, constraints, execution, and accountability are fused into informal prompting or scripts, producing untraceable and economically unbounded behavior. Such collapse is structurally non-compliant with v1.0.

Directional rules of the stack (normative):

- **Control flows downward:** Human intent and governance constraints **MUST** propagate from $H \rightarrow E \rightarrow A$.
- **Telemetry flows upward:** System state, lineage, and metrics **MUST** propagate from $A \rightarrow E \rightarrow H/G$.

No layer **MAY** bypass an adjacent layer to invoke duties allocated elsewhere. In particular:

- Agents **MUST NOT** access humans or external tools except through E-bound contracts and MCP routes.
- Governance **MUST NOT** originate intent; it only constrains and adjudicates it.
- Humans **MUST NOT** directly dispatch autonomous agent tasks outside E, except in explicitly-governed Adoption Bridge cases defined in Chapter 8.

3.1 The High-Level Stack (Layer Definitions and Boundaries)

3.1.1 Layer 1 — Human Layer (H): Intent and Authority

Definition: The Human Layer is the biological component of the system and the **exclusive origin** of intent, meaning, and final accountability.

Core roles (non-exhaustive): Operators, Hybrid Intelligence Architects, Governance Leads.

Responsibilities (normative):

- Humans **MUST** generate Structured Intent Objects (I) via Alignment Surfaces.
- Humans **MUST** review transparency traces at all Regret Boundary (RB) crossings.
- Humans **MUST** retain kill-switch authority and final approval rights for any high-regret action.

State signals: HSS and SCP are human-side metrics surfaced for referencing only in Part II. Canonical definitions are in Part III.

3.1.2 Layer 2 — Symbiosis Engine / Coordination Layer (E)

Definition: The Symbiosis Engine is deterministic coordination logic (“hybrid OS”). It does **not** originate goals or perform autonomous inference; it transforms intent into controlled execution.

Canonical subcomponents (minimum set): Intent Router, Planner/Hierarchical Decomposer, Context & Lineage Manager, Orchestrator, Economic Optimizer / Circuit Guard.

Responsibility boundary (normative):

- E is the sole layer where intent (I), constraints (C), budgets (B), autonomy bands (AB), and regret boundaries (RB) are jointly evaluated.
 - E **MUST** reject or re-plan any Task Graph (TG) that violates C or exceeds B.
 - E **MUST NOT** permit agents to self-promote autonomy, self-modify constraints, or bypass economic gates.
-

3.1.3 Layer 3 — Agentic Layer (A)

Definition: The Agentic Layer is a modular poly-agent ecosystem of bounded capability modules (LLMs/tools) executing TG nodes under explicit contracts and AB ceilings.

Key properties (normative):

- Agents **MUST** be invoked only through E-bound task dispatch and MCP.
 - Every agent action **MUST** be bound to:
 1. an explicit contract,
 2. an AB ceiling,
 3. an active C vector, and
 4. an economic cap within B.
 - Agents **MUST NOT** originate intent or modify governance.
-

3.1.4 Layer 4 — Governance Layer (G)

Definition: Governance is “code-as-boundary.” It compiles policy into constraints, defines risk classes, regulates AB transitions, and preserves auditability.

Core components (minimum set): Governance Kernel (GK), Policy Engine, Audit/Lineage System.

Responsibility boundary (normative):

- G **MAY** demote autonomy, revoke Adoption Bridges, or halt workloads.
 - G **MUST NOT** generate or substitute intent.
 - G **MUST** ensure all thresholds for AB/RB and Economic Circuit gates are explicit, governance-set, deployment-specific, and auditable.
-

3.2 Cross-Layer Data Flows and Telemetry

Hybrid intelligence arises from reciprocal, typed flows across layers.

3.2.1 Downward Control Path (Intent → Action)

1. **H → E:** Human submits Intent Object I with initial C and B.
 2. **E → E:** Engine decomposes I into TG, attaches constraints and AB/RB gating.
 3. **E → A:** Engine binds contracts and dispatches TG nodes to agents through MCP.
 4. **A → Artifacts/World:** Agents execute tasks, producing reversible artifacts or bounded actions per AB and C.
-

3.2.2 Upward Telemetry Path (State → Metrics → Governance)

1. **A → E:** Agents return outputs plus confidence, cost, and status.
 2. **E → L:** Engine hashes trace into lineage.
 3. **E → H:** Human receives transparency trace and checkpoint prompts.
 4. **E → G:** SI/HSS/SCP deltas and policy violations are surfaced for boundary decisions.
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3.2.3 Telemetry Minimum Set (Required Fields)

Any Symbiosis-compliant deployment **MUST** capture for each TG node:

- Intent pointer (I-id + goal slice),
- Active constraint snapshot (C_legal, C_safety, C_econ),
- AB state and RB checkpoint status,
- Projected vs realized cost trace,
- Outcome metadata (confidence, critic deltas, human overrides),
- Lineage hash pointer.

Absence of any minimum field constitutes a lineage failure under v1.0.

3.3 The Economic Circuit (Cost of Intelligence as Control Loop)

Intelligence is a resource-consuming process. Therefore, every intent **MUST** be executed under an explicit Budget Envelope (B), and no execution **MAY** proceed unless projected cost is admissible.

3.3.1 Economic Premises (Normative)

- Every intent **MUST** include B.
- Every TG **MUST** have a Projected_Cost computed prior to dispatch.
- If Projected_Cost > B, E **MUST** reject, down-scope, or escalate to H for override.

3.3.2 Circuit Behavior

- E **MUST** perform an Economic Pass to price TG nodes; it **MAY** downgrade tiers, prune steps, batch tasks, or re-plan.
- Repeated cost overruns **MUST** trigger circuit-breaker halts and/or AB demotion.

3.3.3 Attention as Economic Resource

SCP is treated as a concurrency budget. E **MUST** throttle agent interrupts and checkpoint frequency when SCP headroom is low to prevent oversight theater.

3.4 One-Page Operational Trace Example (“Life of a Decision”)

A minimal trace through the stack is required for interpretability.

Scenario: Operator requests safe optimization of a database query.

1. **H:** Operator submits I with constraints (“no table drops”), and B (“\$1.00”).
2. **E:** Router classifies; Planner builds TG; Economic Pass projects \$0.15; admissible.
3. **A:** Discovery Agent analyzes logs; Design Agent proposes SQL.
4. **H (RB checkpoint):** Operator reviews evidence trace; edits SQL.
5. **G:** E re-validates SQL vs constraints; Operator authorizes execution; lineage committed.

This trace illustrates mandatory downward control and upward telemetry in practice.

3.5 Cross-Layer Feedback Loops (Co-Evolution Substrate)

The architecture functions cybernetically via feedback loops:

- **H ↔ E:** Human intent refines planning; system explanations increase operator skill.
- **E ↔ A:** Outcomes update agent reputations and SI weighting.
- **G ↔ E:** Policy updates constraints and AB thresholds; E provides evidence for governance revision.
- **H ↔ G:** Humans set boundary values; governance outputs training pathways.

These loops **MUST** remain intact; disabling any loop is an architectural violation.

3.6 Core Invariants and Architecture-Level Constraints

A system is Symbiotic only if all of the following hold:

1. **Human Primacy** is enforced at every RB crossing.
2. **Constraint Supremacy** holds pre-flight and at checkpoints.
3. **Economic Viability** is enforced by the Economic Circuit.
4. **Telemetry and Lineage** are mandatory.

Violation of any invariant renders the system non-symbiotic by definition.

§3.6 / Chapter 13 close: See Appendix L.3 Operational Trace One-Pager for the canonical Stage-1..Stage-10 workflow trace.

SQC Gate Check (Chapter 3)

Reference: Specification Quality Contract gates.

1. **Layer Integrity:** PASS — duties and boundaries are explicitly separated; no bypass permitted.
2. **Canonical Metrics Non-Redefinition:** PASS — SI/HSS/SCP/CI/RB/AB only referenced, not defined or re-weighted.
3. **Constraint Precedence + Threshold Explicitness:** PASS — hard constraints override optimization; thresholds noted as governance-set and deployment-specific.
4. **Mechanistic Testability:** PASS — normative clauses specify implementable behaviors and checks.
5. **Lineage Recoverability:** PASS — telemetry minima and traceability to I+C+TG+AB/RB+Economic Pass are explicit.
6. **Drift vs Decay Typing:** PASS — no failure taxonomy introduced here; no mis-typing risk.
7. **Abuse / Incentive Hardening:** PASS — no human-metric enforcement or HR-visible levers proposed in this chapter.
8. **Normative Voice Consistency:** PASS — modality is consistent; no marketing or speculative language.

Part II — Architecture

Chapter 4 — Human Layer & Intent Interfaces

Status: Normative (CARRY with light tightening from v0.9; plus NEW §4.8)

Purpose of Chapter: Specify the Human Layer (H) as a system component, define mandatory human roles and duties, formalize intent interface requirements, establish attention/SCP budgeting as a safety control, codify the Non-Punitive Covenant, and provide a small library of compliant intent-interface patterns.

4.0 Orientation

The Human Layer is not an external user group. It is a first-class system layer with exclusive authorities and measurable control responsibilities. Symbiosis compliance depends on human intent quality, calibrated trust, and disciplined supervision.

This chapter defines what H **MUST** do, how H interfaces with the Symbiosis Engine (E), and which human-side safety rules are mandatory regardless of sector.

Canonical human metrics (HSS v2.0, SCP) are defined only in Part III. This chapter references them as telemetry surfaces only.

4.1 Human Roles (Minimum Set)

Every Symbiosis deployment **MUST** implement the following human roles. One person MAY hold multiple roles in small deployments, but the duties SHALL remain distinct.

4.1.1 Symbiote Operator (SO)

Definition: Human operator who originates intents, supervises TG execution within assigned AB ceilings, and carries regret ownership for outcomes.

Mandatory duties:

1. **Intent origination:** SO **MUST** submit Structured Intent Objects (I) via an Alignment Surface (AS).
2. **Constraint declaration:** SO **MUST** specify or confirm C_legal, C_safety, and C_econ for each intent.

3. **Checkpoint supervision:** SO **MUST** review transparency traces at all RB checkpoints and provide explicit accept/reject/modify decisions.
4. **Override discipline:** SO **MUST** provide rationale for overrides to preserve lineage interpretability.
5. **Artifact hygiene:** SO **MUST** maintain durable outputs in the lineage store and avoid ephemeral-only decisions.

4.1.2 Hybrid Intelligence Architect (HIA)

Definition: Human responsible for configuring E, agent clusters, memory tiers, routing policies, and economic parameters under governance approval.

Mandatory duties:

1. **Engine configuration:** HIA **MUST** implement intent normalization, TG typing, contracts, AB/RB gates, and circuit breakers.
2. **Telemetry binding:** HIA **MUST** ensure telemetry minima defined in Chapter 3 are emitted for every TG node.
3. **Stability stewardship:** HIA **MUST** monitor SI volatility and coordinate remediation with Governance Leads.
4. **Change control:** HIA **MUST** version all orchestration changes and preserve backward lineage compatibility.

4.1.3 Symbiosis Governance Lead (SGL)

Definition: Human authority owning constraint policy, autonomy ceilings, audit doctrine, and adoption gatekeeping.

Mandatory duties:

1. **Policy compilation:** SGL **MUST** define enforceable constraint libraries and risk classes for the deployment.
2. **Autonomy authorization:** SGL **MUST** approve AB ceilings and RB thresholds and MAY demote or revoke them.
3. **Audit readiness:** SGL **MUST** maintain lineage retention, access control, and incident packages.
4. **Non-punitive enforcement:** SGL **MUST** enforce the Non-Punitive Covenant (§4.4) and treat violations as compliance failures.

4.2 Intent Interface: Structured vs Natural Language

Human intent enters the system through an Alignment Surface (AS). AS fidelity determines safety posture.

4.2.1 Requirement: Intent Normalization

1. Natural language inputs are allowed, but E **MUST** normalize them into I prior to planning TGs.

2. If normalization fails (ambiguity, missing constraints, unclear DoD), E **MUST** request clarification before dispatch.
3. E **MUST NOT** invent new goals during normalization; it may expand scope only within the Intent Manifold implied by H.

4.2.2 Required fields for Structured Intent Objects (I)

Every I **MUST** contain, at minimum:

- **Goal statement** (what outcome is desired).
- **Scope bounds** (what is in/out).
- **Hard constraint vector C** (legal/safety/economic).
- **Budget envelope B** (compute/spend/attention caps).
- **Definition of Done (DoD)** or acceptance criteria.
- **Regret sensitivity marker** sufficient for E to assign provisional AB/RB gating.

If any required field is absent, TG generation is prohibited.

4.3 Cognitive Load and Attention Budget (Safety Control)

Humans are finite supervisory resources. Oversight capacity is a hard safety constraint.

4.3.1 SCP as a Capacity Ceiling (reference-only)

1. SCP is treated as the human concurrency headroom budget.
2. E **MUST** throttle interrupts, checkpoint density, and parallel TG lanes when SCP telemetry indicates saturation risk.
3. E **MUST** demote AB to a lower band if safe supervision cannot be maintained at the current task tempo.

4.3.2 Oversight Theater Prohibition

1. Human review that becomes episodic rubber-stamping is a safety failure.
2. When E detects sustained near-zero overrides at AB1+ with SI instability signals, it **MUST** flag a Trust Overcalibration risk and introduce stricter evidence prompts or AB demotion.
3. When E detects chronic overrides at low-risk AB0–AB1 tasks causing throughput collapse, it **MUST** flag Trust Undercalibration and recommend calibrated delegation.

4.3.3 Human Right to Refuse / Manual Mode (Normative)

1. Any human operator **MAY** refuse autonomy escalation or revert a workflow to Manual Mode if they judge the system state unsafe or misaligned.
 2. E **MUST** honor this demotion immediately and record rationale in lineage.
 3. No penalty, ranking, or disciplinary process **MAY** be attached to a Manual Mode invocation.
-

4.4 Non-Punitive Covenant (Framework-Wide Norm)

Human metrics exist to protect safety and expand capability, not to rank humans.

4.4.1 Primary rule

1. HSS and SCP **SHALL** be classified exclusively as safety telemetry.
2. They **SHALL NOT** be used directly or indirectly for HR, compensation, hiring/firing, promotion, or disciplinary purposes.
3. Any secondary use for managerial leverage constitutes Symbiosis non-compliance.

(Extended hardening, RBAC, and union clauses are canonical in Part III Chapter 8 and Appendix H.)

4.4.2 Operator protection guarantees

1. Operators **MUST** have visibility into their own HSS/SCP telemetry and the interpretive rules used by E/G.
 2. Operators **MAY** challenge or annotate telemetry when they believe it reflects system faults (e.g., agent decay) rather than human performance.
 3. Governance **MUST** record disputes as lineage events, not suppress them.
-

4.5 Human Roles & Institutions (Deepening)

Deployments **MUST** explicitly name an owning human institution for each boundary:

- **Intent authority:** assigned to SO.
- **System design authority:** assigned to HIA under governance approval.
- **Boundary/policy authority:** assigned to SGL.
- **Incident authority:** shared between SO and SGL with HIA support.

Where sector law requires additional bodies (e.g., clinical ethics boards, safety committees, unions), those bodies **MUST** be bound into Governance Layer decision rights (Chapter 8 / Chapter 12). This chapter only declares their necessity.

4.6 Human-Layer Failure Modes & Mitigations

Human failures are safety telemetry, not moral judgment. They are handled via training, throttling, or autonomy demotion, never HR sanction.

Minimum catalog (dominant mechanism → mitigation):

1. **H-01 Intent underspecification**
 - *Signature:* repeated clarification loops; TG churn; SI volatility.

- *Mitigation:* E **MUST** require structured intent fields and provide intent-rewrite scaffolds; HIA **MUST** improve AS fidelity.
- 2. **H-02 Trust overcalibration (rubber-stamping)**
 - *Signature:* near-zero overrides at AB1+ with hidden errors or SI decay.
 - *Mitigation:* E **MUST** tighten evidence requirements and insert additional RB checkpoints; SGL **MAY** demote AB.
- 3. **H-03 Trust undercalibration (refusal to delegate)**
 - *Signature:* chronic overrides below RB; throughput collapse.
 - *Mitigation:* E **MUST** surface calibration prompts and suggest safe delegation lanes.
- 4. **H-04 SCP saturation / oversight theater**
 - *Signature:* checkpoint latency; shallow approvals; SCP breach triggers.
 - *Mitigation:* E **MUST** throttle interrupts, reduce parallel TG lanes, or demote AB to restore safe oversight.
- 5. **H-05 Shadow Symbiosis (governance bypass)**
 - *Signature:* actions with no lineage; audit integrity alarms.
 - *Mitigation:* G **MUST** halt workflows, revoke bridges, and require retraining before re-authorization.

The full incident runbooks for these modes are specified in Appendix O.

4.7 Trust Calibration and Working Memory Framing

Trust is an engineered variable. The system **MUST** manage it explicitly:

1. Humans **MUST** treat agent outputs as proposals, not authority.
2. E **MUST** present uncertainty, evidence traces, and cost projections proportionate to AB level.
3. E **MUST** prevent trust drift by adjusting transparency density based on SI trends and SCP headroom.

This rule set implements Human Primacy and Constraint Supremacy at the operating level.

4.8 Intent Interface Patterns Library (NEW)

Intent interfaces **MUST** be chosen based on risk class, AB level, and operator readiness. The following patterns are compliant defaults.

Pattern 1 — Free-Form Chat AS (Low Fidelity)

- **Use:** AB0 tasks; low-regret, reversible workflows; exploratory intents.

- **Required guardrails:** E **MUST** convert chat into structured I before TG; E **MUST** request missing C/B/DoD fields.
- **Prohibited use:** AB2+ tasks or compliance-critical domains unless bridged under Chapter 8.

Pattern 2 — Guided Form AS (Medium Fidelity)

- **Use:** AB0–AB1 tasks; routine production workflows with known constraints.
- **Structure:** goal + scope + constraint pick-lists + budget sliders + DoD templates.
- **Benefit:** reduces H-01 failures by enforcing intent completeness.

Pattern 3 — Strict Intent Object API (High Fidelity)

- **Use:** AB1–AB3 tasks; regulated/high-regret domains; batch/enterprise automation.
- **Structure:** typed schema with mandatory fields and versioning.
- **Rule:** if strict schema validation fails, dispatch is forbidden.

Pattern 4 — Intent-from-Artifact AS (Context-Anchored)

- **Use:** tasks where an authoritative artifact already encodes scope (e.g., policy doc, spec, contract).
- **Rule:** H **MUST** designate the controlling artifact; E **MUST** bind it into lineage and treat it as a constraint anchor.

Pattern 5 — Multi-Party Intent Assembly (Governed)

- **Use:** cross-team or multi-stakeholder intents (e.g., safety-critical deployments).
- **Rule:** E **MUST** require explicit sign-off from all named intent owners before TG execution above AB1.

4.9 Chapter Close

The Human Layer is the sovereign origin of intent and regret, and the limiting supervisory resource. Proper intent interfaces, attention budgeting, and non-punitive telemetry are mandatory to keep hybrid intelligence safe, governable, and economically bounded.

Part II proceeds next to the remaining architecture layers and primitives, keeping canonical metrics definitions reserved for Part III.

SQC Gate Check — Chapter 4

1. **Layer Integrity:** PASS — Human duties and boundaries are explicit; no cross-assignment or bypass permitted.
2. **Canonical Metrics Non-Redefinition:** PASS — HSS/SCP referenced only; no formulas, weights, or thresholds defined.

3. **Constraint Precedence + Threshold Explicitness:** PASS — hard constraints override optimization; thresholds deferred to governance/Part III.
 4. **Mechanistic Testability:** PASS — clauses specify enforceable AS requirements, SCP throttling, refusal rules, and role duties.
 5. **Lineage Recoverability:** PASS — intent normalization, override rationale, and minimum I fields support reconstruction.
 6. **Drift vs Decay Typing:** PASS — no drift/decay doctrine introduced; failure modes scoped to H telemetry.
 7. **Abuse / Incentive Hardening:** PASS — Non-Punitive Covenant stated; no HR use vectors introduced.
 8. **Normative Voice Consistency:** PASS — MUST/SHALL/MAY used correctly; no marketing/speculation.
-

Part II — Architecture

Chapter 5 — Coordination Layer / Symbiosis Engine (E)

Status: Normative (REWRITE/UPGRADE from v0.9)

Purpose of Chapter: Specify the Coordination Layer as a deterministic hybrid-intelligence operating system: its required subcomponents, memory/lineage hierarchy, routing and orchestration duties, economic admission control v2, mandatory Dry-Run behavior for higher autonomy, and its boundary limits relative to H, A, and G. Canonical AB/RB, SI, HSS, SCP, and CI definitions remain reserved for Part III.

5.0 Orientation

The Symbiosis Engine (E) is the **deterministic coordination process** that makes hybrid intelligence governable. It is not a model and does not perform inference. Its job is to convert structured human intent into bounded, economically admissible, auditable execution across a poly-agent stack.

Without E, a deployment degenerates into prompt chains and unmanaged agent autonomy, which is structurally non-compliant with Symbiosis.

5.1 Engine Boundary and Layer Duties

E is the only layer where intent, constraints, budgets, autonomy ceilings, and execution planning are jointly evaluated.

5.1.1 Positive duties (what E MUST do)

For every Intent Object (I), E **MUST**:

1. **Ingest and normalize intent.** Convert any Alignment Surface input into a structured I with explicit fields required by Chapter 4.
2. **Consolidate constraints.** Merge operator-supplied constraints (C_o) with governance constraints (C_G) into a single active C vector.
3. **Perform Economic Pass v2.** Price candidate Task Graphs against Budget Envelope (B) and enforce Compute Saturation / Queue Latency gates where applicable. (Formal circuit definition and math are in Appendix B; thresholds are governance-set.)
4. **Compile an admissible Task Graph (TG).** Decompose I into typed nodes with dependencies, per-node constraints, projected cost, and provisional AB/RB gating.
5. **Select and bind agents.** Choose agents using Capability Index (CI) routing and bind execution contracts (Appendix F) per node. CI is canonical in Part III; here it is referenced only.
6. **Enforce autonomy ceilings and checkpoints.** Respect active AB ceilings and RB checkpoint rules set by Governance; block any self-promotion by agents.
7. **Orchestrate dispatch via MCP.** Act as MCP host; no agent-agent direct channels are permitted.
8. **Record lineage and emit telemetry.** Every TG node execution **MUST** write a lineage block and minimum telemetry fields per Chapter 3.

5.1.2 Negative duties (what E **MUST NOT** do)

E **MUST NOT**:

1. Originate goals or alter the human intent manifold beyond stated scope.
2. Modify governance ceilings, constraints, or thresholds without GK-signed authorization.
3. Execute tasks directly other than orchestration, validation, and logging.
4. Allow any uncontracted tool path or capability to be used by agents. ("If it is not in contract, it does not exist.")

5.2 Required Engine Subcomponents (Minimum Viable Set)

A deployment is Symbiosis-compliant only if E includes at least the following deterministic subcomponents.

1. **Intent Router.** Classifies I into task classes and selects planning templates and risk posture.
2. **Planner / Hierarchical Decomposer.** Expands I into candidate TGs; performs reconciliation loops when ambiguity or constraint conflicts arise.
3. **Context & Lineage Manager.** Selects memory tiers, materializes context bundles, hashes ancestry, and guarantees replay.
4. **Orchestrator / Scheduler.** Allocates TG nodes to agents, manages concurrency, enforces SCP-derived interrupt budgets (reference-only) and retries.

5. **Economic Optimizer / Circuit Guard v2.** Computes projected cost, enforces CS/QL gates, throttles or halts over-budget runs, and emits economic telemetry.
 6. **Safety Supervisor Hooks.** Implements drift/decay flags and mandatory escalations to RB checkpoints and governance.
 7. **SI Tracker Interface.** Emits required statistics for SI computation; SI definition remains canonical in Part III.
-

5.3 Multi-Level Memory Hierarchy and Context Governance

E is responsible for preventing **context drift** and ensuring lineage recoverability.

5.3.1 Memory tiers (normative)

E **MUST** support at least three tiers:

- **M0 — Ephemeral Working Context.** Per-intent scratch space; expires with task completion.
- **M1 — Durable Task Memory.** Stores TG rationale, intermediate artifacts, and checkpoint decisions tied to I.
- **M2 — Governance & Policy Memory.** Stores C libraries, AB ceilings, RB thresholds, and approved models/contracts.

Migration between tiers **MUST** be explicit and lineage-logged. No agent may write to M2.

5.3.2 Context selection rule

At planning time, E **MUST** select memory tiers based on:

- task class,
- AB/RB proximity (reference-only),
- SCP headroom (reference-only), and
- economic admissibility.

If required context is unavailable or ambiguous, E **MUST** request clarification or down-scope TG before dispatch.

5.4 Economic Pass v2 (Admission Control)

E **MUST** treat cost of intelligence as a control objective. Every candidate TG **MUST** be economically admissible prior to dispatch.

5.4.1 Projected cost computation

For each TG node, Economic Pass v2 **MUST** compute:

- projected compute cost (tokens/latency/GPU-time),

- projected financial cost (if metered),
- projected human attention cost (interrupts/checkpoints), and
- expected retry overhead.

If projected cost exceeds B, E **MUST** re-plan, down-scope, or escalate to H.

5.4.2 Local compute gating (CS / QL)

Where deployment uses hybrid or local compute, E **MUST** enforce:

- **Compute Saturation (CS) gate:** if available local compute is saturated above governance threshold, E **MUST** defer, queue-shape, or downgrade AB/task tier rather than silently spilling to cloud.
- **Queue Latency (QL) gate:** if queue latency exceeds governance threshold for the intended AB level, E **MUST** either reduce TG concurrency, demote AB, or request human rescheduling.

Threshold values are deployment specific and **MUST** be set by Governance. This chapter defines enforcement behavior only.

5.5 Mandatory Dry-Run Stage (v1.0 Upgrade)

v1.0 introduces **Dry-Run** as a required pre-execution simulation for higher-risk tasks.

5.5.1 When Dry-Run is mandatory

E **MUST** perform Dry-Run when any of the following holds:

1. The task is intended to execute at **AB2 or higher** (reference-only).
2. The task is **RB-proximate** by governance rules (reference-only).
3. The task is compliance-critical or economically high-impact in its risk class.
4. The system detects elevated drift/decay risk signals.

5.5.2 Dry-Run behavior requirements

In Dry-Run mode, E **MUST**:

1. Execute a simulation of the TG without committing irreversible actions.
2. Produce a **preview artifact set** showing: intended actions, evidence traces, cost projection deltas, and constraint satisfaction report.
3. Require human confirmation (and governance confirmation where mandated) before unlocking execution budgets.
4. Record Dry-Run lineage as a distinct preflight block and associate it to the eventual execution TG.

Dry-Run latency is a safety signal. Rising Dry-Run latency **MUST** trigger E to recommend AB demotion or TG concurrency reduction.

Note: the formal numbered pipeline including Dry-Run placement is specified canonically in Part IV Chapter 13.

5.6 Intent → Task Graph Mapping (Operational Rules)

E **MUST** decompose I into TGs using a stable composition order. v0.9's order is preserved and tightened here.

Composition order (normative summary):

1. **AS → I ingestion**
2. **I + C₀ → consolidated C**
3. **C + B → admissible TG plan (Economic Pass v2)**
4. **TG + RB → provisional AB gating (reference-only)**
5. **Context selection via memory tiers**
6. **Dry-Run where required**
7. **Execution dispatch**
8. **Lineage + telemetry commit**

Any deviation from this order **MUST** be explicitly justified as a governance-approved extension.

5.7 Orchestration, Arbitration, and Recovery

5.7.1 Orchestration via MCP

E **MUST** use MCP as the hub-and-spoke interface to agents.

- E is the MCP host.
- Agents are MCP servers.
- Agents **MUST NOT** exchange context directly; E federates and logs all context exchange.

5.7.2 Arbitration and retry

E **MUST** implement retry and arbitration policies such that:

- retries are bounded by B and C_{econ},
- repeated tool or agent failures trigger CI demotion (canonical in Part III), and
- unresolved conflicts escalate to H or G based on risk class.

5.7.3 Drift vs Decay hooks (reference-only)

E **MUST** emit distinct telemetry pathways for Semantic Drift and Operational Decay and write separate lineage entries. The full taxonomy and recovery doctrine are canonical in Part IV Chapter 14 and Appendices F/G.

5.8 Engine Compliance Envelope

A Symbiosis Engine is compliant only if all of the following hold:

1. **Deterministic coordination:** no goal invention; no hidden autonomy.
2. **Economic admissibility:** B enforced per node; CS/QL gates active where relevant.
3. **Dry-Run enforcement:** mandatory for AB2+, RB-proximate, or compliance-critical tasks.
4. **Contract-bounded execution:** no unregistered capability usage.
5. **Hub-and-spoke orchestration:** MCP exclusive; no agent-agent channels.
6. **Lineage completeness:** telemetry minima and immutable trace for every TG node.

Failure of any condition constitutes Coordination Layer non-compliance and invalidates system-wide Symbiosis certification.

5.9 Chapter Close

The Symbiosis Engine is the structural core of governable hybrid intelligence. It is defined by deterministic intent processing, explicit constraint and budget enforcement, Dry-Run preflight for higher autonomy, economic control loop v2 for hybrid compute, and audit-ready lineage.

Part II proceeds next to Chapter 6 (Agentic Layer), while formal autonomy and metric definitions remain restricted to Part III and the Engine Algorithm v2 is specified canonically in Part IV Chapter 13.

SQC Gate Check — Chapter 5

1. **Layer Integrity:** PASS — E duties bounded; no cross-layer duty capture or bypass authorized.
2. **Canonical Metrics Non-Redefinition:** PASS — AB/RB/SI/HSS/SCP/CI referenced only; no math, weights, or thresholds defined outside Part III.
3. **Constraint Precedence + Threshold Explicitness:** PASS — constraints supersede plans; CS/QL + AB/RB thresholds noted as governance-set.
4. **Mechanistic Testability:** PASS — each MUST clause maps to implementable Engine behaviors.
5. **Lineage Recoverability:** PASS — explicit composition order and mandatory preflight/telemetry ensure I+C+TG+AB/RB+Economic Pass traceability.
6. **Drift vs Decay Typing:** PASS — hooks required; taxonomy deferred to Part IV without redefinition.
7. **Abuse / Incentive Hardening:** PASS — no HR-visible levers or punitive human metric use introduced.

8. **Normative Voice Consistency:** PASS — compliant modality; no speculative or promotional tone.
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Part II — Architecture

Chapter 6 — Agentic Layer (Poly-Agent Ecosystem)

Status: Normative (CARRY + light tightening from v0.9; plus v1.0 contract/economic deltas)

Purpose of Chapter: Specify the Agentic Layer as a modular poly-agent capability substrate, define canonical agent clusters, require MCP interoperability, formalize arbitration/oversight patterns, state admissibility and selection rules (referencing CI canonically defined in Part III), define agentic failure taxonomy with mandatory recovery behaviors, and provide the normative Agent Contract Schema v1 (machine-readable form in Appendix E/F).

6.0 Orientation

The Agentic Layer (A) is the capability substrate of Symbiosis: a modular ecosystem of specialized synthetic agents that execute Task Graph (TG) nodes under explicit contracts, autonomy ceilings, constraint vectors, and economic caps. Agents provide execution power; the Engine provides deterministic coordination; Governance provides boundaries; Humans provide meaning.

Agents are **bounded capability modules**. They do not originate intent, set constraints, promote autonomy, or alter governance. Any system that permits those behaviors is non-symbiotic.

6.1 Why Poly-Agent Systems Are Required

Single-agent “do everything” designs collapse under real workloads. They are brittle, expensive, and hard to govern. Symbiosis therefore assumes heterogeneous capability and requires a poly-agent architecture.

Poly-agent systems yield three structural advantages:

1. **Modularity and composability.**

Agents can be swapped or upgraded without rewriting the Engine, as long as they

satisfy the same contract type.

2. **Economic matching.**

Low-cost agents handle low-regret drafting; higher-cost reasoning agents are reserved for RB-proximate nodes. (RB/AB are canonical in Part III and referenced here only.)

3. **Governable autonomy.**

Oversight/critic agents gate higher-autonomy execution without overloading humans, while remaining bounded by Governance Kernel rules.

6.2 Canonical Agent Clusters

The cluster model is logical and vendor-agnostic. A deployment MAY implement clusters as separate services, model routes, toolchains, or teams, provided all contract and Governance Kernel boundaries remain intact.

6.2.1 Discovery Cluster

Purpose: diagnostics, ingestion, mapping, and uncertainty reduction.

Typical agents: document parsers, data profilers, domain scouts, requirement extractors.

6.2.2 Modernization Cluster

Purpose: transformation tasks below or near RB (drafting, workflow generation, refactors, UI/automation composition).

Typical agents: drafting assistants, workflow builders, UI/site generators, automation composers.

6.2.3 Predictive Intelligence Cluster

Purpose: modeling, forecasting, scoring, simulation, scenario generation.

Typical agents: risk forecasters, causal modelers, demand/scenario generators, SI analyzers. (SI is canonical in Part III.)

6.2.4 Operations & Compute Cluster

Purpose: live system interaction, tool execution, and infrastructure orchestration.

Typical agents: ops planners, guarded deployers, monitoring bots, compute schedulers.

6.2.5 Governance & Oversight Cluster

Purpose: enforce constraints, arbitrate disagreement, and provide safety supervision within AB/RB ceilings.

Typical agents: policy validators, critic agents, red-teamers, audit loggers, band-gating sentinels.

6.3 MCP as the Standard Interface for Agents

The Agentic Layer interoperates through the **Model Context Protocol (MCP)**. MCP provides a uniform interface for exposing resources, prompts, and tools between the Engine and agents. MCP is canonical in Part III Chapter 10; this chapter specifies its mandatory use.

6.3.1 MCP binding rule (normative)

1. Every agent **MUST** be MCP-compliant to be admissible.
2. All MCP resources, prompts, and tools used by an agent **MUST** be declared in its Agent Contract.
3. Any MCP call not declared in contract **MUST** be rejected preflight by the Engine or halted at runtime. (“If it is not in contract, it does not exist.”)

6.4 Inter-Agent Arbitration and Oversight

Poly-agent systems require deterministic conflict resolution. E **MUST** mediate all arbitration via TG contracts and Governance Kernel ceilings.

6.4.1 Arbitration patterns (normative)

E **MUST** support at least one of the following patterns per TG node class:

1. **Critic loop:** execution agent generates a proposal; one or more critic agents validate against constraints and evidence.
2. **Ensemble disagreement check:** multiple execution agents propose; E selects via contract-allowed arbitration profile.
3. **Human-gated arbitration:** when critics cannot converge within the ceiling, E escalates to H, and if required, to G.

6.4.2 Arbitration triggers (normative)

E **MUST** initiate arbitration when any of the following holds:

- proposal violates explicit constraints,
- critic confidence drops below contract minimum,
- cost projection exceeds remaining budget envelope, or
- node is RB-proximate and contract requires oversight. (RB/AB canonical in Part III.)

Arbitration loops **MUST** be bounded by a contract-declared ceiling to prevent deadlock and economic bleed.

6.5 Capability Indexing and Agent Selection (Reference-Only)

Agent selection **MUST** be deterministic and contract-bounded. E uses the **Capability Index (CI)** to route TG nodes to admissible agents. CI is canonical only in Part III Chapter 10; this chapter states minimum selection behavior.

6.5.1 Selection rule (normative)

A TG node **MAY** be dispatched to an agent only if:

1. the node's AB level is $\leq$ agent contract AB ceiling,
2. all contract-required constraints and memory tiers are present,
3. projected node cost is $\leq$ agent contract max_cost_per_task and within remaining B, and
4. intended MCP calls are a subset of contract bindings.

If no admissible agent exists, E **MUST** either demote AB, re-plan TG, or escalate to H.

6.6 Failure Modes and Recovery in the Agentic Layer

Agentic failures are typed safety events. E **MUST** distinguish Semantic Drift from Operational Decay in lineage and remediation routing; the canonical split and runbooks are in Part IV Chapter 14 and Appendices F/G. This section defines agent-layer specific failures and mandatory recoveries.

6.6.1 Common agentic failures (minimum taxonomy)

1. **Capability overreach:** agent attempts tasks outside contract scope.
2. **Constraint neglect:** implicit assumptions violate C.
3. **Tool failure / partial execution:** non-atomic side effects or incomplete actions.
4. **Disagreement deadlock:** critic loops stall beyond ceiling.
5. **Economic bleed loops:** excessive calls for low-value tasks.
6. **Context drift:** agent uses stale or disallowed memory tier.

6.6.2 Mandatory recovery behaviors (normative)

On detection of failures above, E **MUST** execute the following recoveries:

- **Hard stop** on capability overreach or any unauthorized RB crossing attempt.
 - **Fallback routing** to alternate admissible agent when tool failure is retryable.
 - **Arbitration ceiling enforcement** to prevent deadlock; escalate to H/G upon ceiling breach.
 - **Tier degradation / plan pruning** when Economic Circuit flags bleed; record economic lineage.
 - **Context refresh and re-plan** on drift; do not continue with stale context.
-

6.7 Canonical Agent Contracts (Contract Schema v1)

Agents are admissible only through explicit, versioned contracts. An Agent Contract is both a capability declaration and a governance boundary object.

6.7.1 Purpose

Contracts enable E to:

- select agents deterministically,
- enforce AB ceilings and RB checkpointing,
- bound cost via economic profiles,
- restrict MCP access to declared resources/tools, and
- specify failure and escalation semantics.

6.7.2 Required contract fields (normative)

Every Agent Contract **MUST** include:

1. **Identity & provenance:** `agent_id`, cluster, version, owner, status.
2. **Capabilities:** declarative list of tasks/tools/output forms.
3. **Eligibility & preconditions:** AB ceiling, admissible risk classes/domains, required constraints, required memory tiers.
4. **Input/Output schemas:** typed REQ/RES payload definitions including confidence and explanation obligations.
5. **Economic profile:** `cost_per_token` / `cost_per_call` (or equivalent), `max_cost_per_task`, preferred tier, budget behavior.
6. **Failure & escalation semantics:** error codes, retryability, escalation routes, rollback semantics.
7. **Governance obligations:** logging level, retention class, constraint check mode, privacy posture.
8. **MCP bindings:** allowed `resources[]`, `prompts[]`, `tools[]`.
9. **Arbitration profile (optional but recommended):** critic requirement, ensemble size, critic-loop ceiling.

6.7.3 Machine-readable schema

The formal JSON Schema for Contract v1 is normative and maintained in Appendix E/F. Implementations **MUST** validate contracts against that schema prior to agent admission.

6.7.4 Conformance rule (normative)

A TG node dispatched to an agent that violates any contract field, AB ceiling, cost cap, or MCP binding **MUST** be rejected preflight or halted at runtime with audit logging.

6.8 Chapter Close

The Agentic Layer provides modular capability under deterministic coordination and governance boundaries. Poly-agent clusters, MCP interoperability, contract-bounded routing, and mandatory recovery behaviors are required to prevent autonomy creep, economic bleed, and non-auditable execution. CI, RB, AB, SI, HSS, and SCP remain canonical in Part III and are referenced here only.

SQC Gate Check — Chapter 6

1. **Layer Integrity:** PASS — Agentic duties bounded; no cross-layer duty capture or bypass permitted.
 2. **Canonical Metrics Non-Redefinition:** PASS — CI/RB/AB/SI/HSS/SCP referenced only; no definitions or weights introduced outside Part III.
 3. **Constraint Precedence + Threshold Explicitness:** PASS — constraint supremacy asserted via contracts; no universal thresholds set here.
 4. **Mechanistic Testability:** PASS — admissibility, arbitration, MCP binding, failure recovery, and contract validation are implementable checks.
 5. **Lineage Recoverability:** PASS — MCP calls and contract gates are lineage events; dispatch rule traces to I+C+TG+AB/RB+Economic Pass.
 6. **Drift vs Decay Typing:** PASS — split required; taxonomy not redefined and routed to Part IV/Appendices F/G.
 7. **Abuse / Incentive Hardening:** PASS — contract economics and arbitration ceilings prevent Goodhart/economic bleed; no punitive human levers introduced.
 8. **Normative Voice Consistency:** PASS — MUST/SHALL/MAY used correctly; no speculative or marketing phrasing.
-

Part III — Metrics & Autonomy (Canonical)

Chapter 7 — Symbiosis Index (SI): Formal Definition, Measurement, and Control Use

Status: Canonical v1.0 Definition

Redline disposition: REWRITE/UPGRADE (lifted from v0.9 Appendix B into Part III)

7.0 Purpose and Position in the Framework

7.0.1 The Symbiosis Index (SI) is the canonical machine–human collaboration health metric. It SHALL be used as a **primary control signal** for:

1. Autonomy Band (AB) promotion/demotion gating.
2. Drift, divergence, resonance, and entanglement detection coupling.
3. Economic Circuit optimization and bleed prevention.
4. Validation and certification evidence packets in Part V.

7.0.2 SI is **canonical only in Part III**. Any reference to SI outside Part III MUST link to this chapter and SHALL NOT redefine or re-weight SI.

7.0.3 SI SHALL be computed in two forms:

- **Episode SI (SI_episode)**: computed per task episode / TG node.
 - **Rolling SI (SI_rolling)**: computed over a governance-defined window W and used for control decisions.
-

7.1 SI Core Definition

7.1.1 SI SHALL be a bounded, normalized composite score in **[0,1]**, computed as a weighted sum of five canonical subscores.

$$SI = w_Q Q + w_A A + w_E E + w_G G + w_S S$$

Where:

- = Quality
- = Alignment
- = Economic Efficiency
- = Governability / Auditability
- = Stability / Trust
- Each weight and .

7.1.2 **Weights are not universal**. The Governance Kernel (GK) MUST set weights per sector/playbook and SHALL version them as part of policy lineage.

7.1.3 All subscores MUST be normalized to [0,1] prior to weighting. Normalization baselines SHALL be sector-defined.

7.2 Subscore Specifications (Canonical)

7.2.1 Q — Quality

Goal: measure task-level correctness and execution quality.

7.2.1.1 Q SHALL be derived per episode from domain-valid evaluation primitives, including:

- objective correctness metrics (when ground truth exists),
- critic-agent admissibility tests, and/or
- operator post-task evaluation under governance rubric.

7.2.1.2 Error penalties MUST be risk-class weighted. At minimum, GK SHALL distinguish:

- cosmetic/low-impact errors,
- operational errors,
- safety/legal/RB-proximate errors.

7.2.1.3 Safety/legal/RB errors SHALL collapse Alignment and Governability to zero for that episode regardless of Q.

7.2.2 A — Alignment

Goal: measure semantic fidelity of agent output to baseline intent.

7.2.2.1 A SHALL be computed as similarity between baseline intent (plus admissible ASR_o bounds) and realized intent expressed in outputs and side effects.

7.2.2.2 If a task episode requires re-anchoring of , A for that episode MUST be computed relative to the re-anchored intent and SHALL log the anchor delta in lineage.

7.2.3 E — Economic Efficiency

Goal: measure value-per-cost of the hybrid episode.

7.2.3.1 E SHALL compare realized cost vs sector-defined baseline , accounting for:

- token/tool spend,
- latency impact,
- opportunity cost vs human-only execution.

7.2.3.2 E MUST be bounded so that low-cost but unsafe or misaligned outputs cannot inflate SI. Constraint Supremacy SHALL override E optimization.

7.2.4 G — Governability / Auditability

Goal: measure whether collaboration is inspectable and policy-conformant.

7.2.4.1 G SHALL be computed as:

$$G = g_{\text{lineage}} \cdot g_{\text{contract}} \cdot g_{\text{explainability}}$$

Each term is in $[0, 1]$.

- : completeness of minimum event set.
- : zero contract/tool-scope violations.
- : reconstructability of decisions from critic deltas and evidence trace.

7.2.4.2 **Lineage is mandatory.** If minimum lineage is missing, G MUST equal 0, regardless of Q.

7.2.5 S — Stability / Trust

Goal: detect volatility, calibrated trust failure, and silent deterioration.

7.2.5.1 S SHALL be computed over rolling windows as:

$$S = 1 - \lambda_1 \text{Var}(SI_{\text{episode}}) - \lambda_2 |\text{override_rate} - \text{target}| - \lambda_3 \text{drift_rate}$$

7.2.5.2 Inputs to S MUST include:

- variance of SI_{episode} ,
- override rate vs target,
- drift vital signs (Chapter 11).

7.2.5.3 S SHALL be treated as **co-equal with SI_{rolling}** for autonomy gating.

7.3 Normalization and Error Conventions

7.3.1 All subscores MUST be normalized to sector baselines prior to weighting. Sector baselines SHALL define:

- minimum acceptable floors on Q/A/G,
- ,
- stability window W.

7.3.2 Error class weighting MUST be regret- and risk-class aware. Risk Class A domains SHALL prioritize A and G floors over Q if conflict occurs.

7.4 Data Requirements for SI Computation

7.4.1 Each SI_episode MUST be computed only when the following mandatory inputs are logged:

1. Baseline intent and current intent .
2. Active Constraint Vector and policy version .
3. TG trace (node lineage and routing).
4. Agent outputs and critic deltas.
5. Tool calls and side effects.
6. Human approvals/overrides with reason tags.
7. Cost ledger (tokens, tool fees, latency/time).
8. Stability vital signs slice (D, R, E_blast, V_contract).

7.4.2 If any mandatory input is absent:

- Q/A/E MAY be computed if possible,
 - G SHALL be reduced proportionally to missing lineage fields,
 - SI_episode SHALL be capped to prevent AB gain.
-

7.5 Episode vs Rolling SI

7.5.1 **SI_episode** SHALL be computed per TG node or bounded episode.

7.5.2 **SI_rolling** SHALL be computed as a governance-defined rolling function over window W (e.g., EWMA or median-robust smoothing). The smoothing function and W MUST be:

- deployment-specific,
- GK-parameterized,
- auditable in policy lineage.

7.5.3 **Control precedence:**

If SI_episode conflicts with SI_rolling (e.g., a sudden red-band episode during a green rolling window), the Engine SHALL treat the **more conservative signal as binding** for routing and autonomy control.

7.6 Operational Use of SI (Control Logic)

7.6.1 Autonomy Gating

7.6.1.1 AB promotion requires ALL of the following:

- SI_rolling in green band,
- S in green band,
- no S2+ incidents in last window W,
- HSS tier admissible,
- SCP headroom available.

7.6.1.2 AB demotion triggers ANY of:

- SI red-band episode or rolling collapse,
- drift/resonance red-band,
- SCP breach,
- any S3+ incident.

7.6.2 Drift Detection Coupling

7.6.2.1 Drift MUST be treated as Alignment decay with stable or deceptively stable Q. The Engine SHALL monitor split trajectories for early detection.

7.6.3 Economic Optimization

7.6.3.1 The Economic Circuit MAY use SI deltas as a value proxy, but SHALL NOT authorize any plan that violates constraints or RB/AB gates.

7.7 Reliability, Calibration, and Anti-Gaming Requirements

7.7.1 SI MUST satisfy three reliability properties:

- stability under repeated similar tasks,
- sensitivity to meaningful degradations,
- non-trivial game resistance.

7.7.2 Calibration per sector SHALL include, at minimum:

1. establish baseline SI distributions at AB0,
2. identify incident signature distributions (S1–S4),
3. set green/yellow/red bands per risk class,
4. validate promotion/demotion behavior in simulated incidents.

7.7.3 Before AB3 scale, external validation is REQUIRED:

- compare SI to expert human ratings,
- compare SI to objective KPIs,
- stress-test under drift/resonance scenarios.

7.7.4 Anti-gaming safeguards MUST include:

- G=0 without lineage (prevents suppression gaming),
- random episodic audits by GK,
- multi-gate AB rules (SI alone is insufficient).

7.8 Illustrative Computation (Non-Normative)

Assume sector weights:

- , , , , .

Episode subscores:

- Q=0.85, A=0.92, E=0.60, G=1.00, S=0.80.

Then:

$$SI = 0.30(0.85) + 0.25(0.92) + 0.15(0.60) + 0.20(1.00) + 0.10(0.80) = 0.855$$

If G fell to 0.40 due to missing lineage, SI drops to 0.735, triggering yellow-band review.

7.9 Minimum Conformance Checklist (Normative)

An organization MAY claim SI compliance only if all of the following are true:

1. All five subscores computed per episode.
 2. Rolling SI used for autonomy gating.
 3. Lineage minimum event set enforced (G term real).
 4. Sector weights governance-signed and versioned.
 5. Drift/resonance/entanglement coupling active.
 6. SI red-band triggers deterministic safe-mode behavior.
 7. Validation runbooks executed before AB3 scale.
-

SQC Gate Check (Chapter 7)

Gate 1 — Layer Integrity: PASS. SI defined as metric produced by Engine/GK telemetry; no layer bypasses described.

Gate 2 — Canonical Metrics Non-Redefinition: PASS. SI is defined here in Part III; no other canonical metrics redefined.

Gate 3 — Constraint Precedence + Threshold Explicitness: PASS. All bands/weights/thresholds are GK-set, deployment-specific; constraints override optimization.

Gate 4 — Mechanistic Testability: PASS. Inputs, equations, and gating logic specify measurable conditions.

Gate 5 — Lineage Recoverability: PASS. Mandatory inputs and G=0 rule ensure reconstructable lineage.

Gate 6 — Drift vs Decay Typing: PASS. Drift coupling explicitly referenced and separated from decay mechanisms.

Gate 7 — Abuse / Incentive Hardening: PASS. Anti-gaming controls and non-trivial game resistance specified; no HR coercion vectors introduced.

Gate 8 — Normative Voice Consistency: PASS. Correct MUST/SHALL/MAY usage; no speculative tone.

Part III — Metrics & Autonomy (Canonical Part)

Chapter 8 — Human Symbiosis Score (HSS v2.0)

Status: Canonical Metric — Governance-Critical

Dependencies: Task Graph (TG), Regret Boundary (RB), Autonomy Bands (AB0–AB3), Economic Circuit v2, Governance Kernel v2

Purpose: Quantify safe participation of human operators in hybrid human–AI collaboration systems, and gate autonomy accordingly.

8.0 Overview

8.0.1 The Human Symbiosis Score (HSS v2.0) is a six-domain composite metric that quantifies a human operator’s capability to participate safely, predictably, and effectively within hybrid human–AI systems.

8.0.2 HSS SHALL be classified and treated exclusively as a **Safety Metric**. It SHALL NOT be interpreted or used as a performance, productivity, ranking, or disciplinary metric.

8.0.3 HSS SHALL be derived exclusively from interaction metadata and system-observable behavior. HSS SHALL NOT incorporate biometric, psychometric, affective, or personality-inference inputs.

8.0.4 HSS SHALL be used solely for:

- a. Autonomy Band (AB) gating.
- b. Safety-related operator training.
- c. Oversight-load estimation and Symbiotic Capacity Profile (SCP) calibration.
- d. Governance risk-boundary decisions.

8.0.5 Any secondary use of HSS for HR or managerial purposes SHALL constitute a Symbiosis compliance violation and SHALL invalidate Certification status.

8.1 Objectives

The HSS v2.0 system SHALL accomplish the following objectives:

8.1.1 **Ensure safety of high-autonomy workflows.** HSS MUST validate that operator capability matches the autonomy level requested.

8.1.2 **Prevent cognitive overload.** HSS MUST detect overload-induced instability before RB excursions occur.

8.1.3 **Support predictable collaboration.** HSS MUST stabilize human contributions to TG construction by minimizing ambiguity and volatility.

8.1.4 **Enable reversible governance.** HSS MUST provide objective thresholds for AB promotions and demotions.

8.1.5 **Enable regulator-grade transparency.** HSS MUST produce auditable, non-exploitative human-capability evidence.

8.2 Canonical Six-Domain Model

8.2.1 HSS v2.0 SHALL be composed of six independently measured domains. Each domain SHALL be measured using standardized, interaction-derived indicators.

8.2.2 HSS-1 — Intent Clarity (IC)

- a. **Definition:** Degree to which the operator produces interpretable, low-ambiguity instructions that the Symbiosis Engine can translate into a valid TG.
- b. **Signals (non-biometric):** Interpretability score; revision delta across clarifications; constraint conflict flags; ambiguity indicators.
- c. **Rationale:** IC is the top predictor of RB proximity and TG instability.

8.2.3 HSS-2 — Constraint Awareness (CA)

- a. **Definition:** Demonstrated ability to anticipate, respect, and operate within defined constraints (C), including legal, policy, economic, and operational limits.
- b. **Signals:** Constraint violations per 100 tasks; pre-execution constraint adjustments required; frequency of governance overrides; exception request patterns.
- c. **Rationale:** CA reduces governance load, prevents AB/RB violations, and ensures ethical compliance.

8.2.4 HSS-3 — Collaboration Discipline (CD)

- a. **Definition:** Reliability in following protocol steps, validating Dry-Runs, responding to checks, and providing stable feedback.
- b. **Signals:** Dry-Run approval latency; audit prompt acknowledgement rate; RB checkpoint responses; protocol compliance curve.
- c. **Rationale:** CD stabilizes the human contribution to the hybrid system.

8.2.5 HSS-4 — Cognitive Load Regulation (CLR)

- a. **Definition:** Capacity to maintain reliable performance under varying mental load without destabilizing TG construction or expanding risk surfaces.
- b. **Signals:** Error-rate slope; task-switching volatility; non-biometric fatigue indicators; exploratory vs exploitative balance; revision explosiveness.
- c. **Rationale:** CLR prevents hidden overload contributing to drift, decay, or high-regret outcomes.

8.2.6 HSS-5 — Interaction Stability (IS)

- a. **Definition:** Consistency, predictability, and low-volatility nature of operator instructions and corrections.
- b. **Signals:** Prompt volatility index; contradiction rate; scope oscillation frequency; rollback triggers caused by operator input.
- c. **Rationale:** IS reduces RB proximity events and improves system reliability.

8.2.7 HSS-6 — Outcome Reliability (OR)

- a. **Definition:** Correlation between operator input and successful task outcomes, normalized over complexity and time.
- b. **Signals:** SI uplift attributable to operator direction; TG outcome delta vs expected; regression to error patterns; task success ratio post-intervention.
- c. **Rationale:** OR measures grounded human contribution to safe, effective hybrid intelligence.

8.3 Composite Formula (Canonical)

8.3.1 HSS SHALL be computed as a weighted composite:

$$\text{HSS} = 0.20 \cdot \text{IC} + 0.20 \cdot \text{CA} + 0.15 \cdot \text{CD} + 0.15 \cdot \text{CLR} + 0.15 \cdot \text{IS} + 0.15 \cdot \text{OR}$$

8.3.2 Properties:

- a. Range: 0.00–1.00.
- b. Weights SHALL remain stable across v1.0 implementations unless superseded by

Governance Body amendment.

- c. Calibration SHALL use Certification Protocol datasets (Chapter 17).

8.4 Proficiency Bands and Autonomy Eligibility

8.4.1 HSS SHALL determine Autonomy Band eligibility using three proficiency bands.

8.4.2 HSS-B1 — Guided Operator (≤ 0.45)

- a. Eligible AB: AB0–AB1.
- b. Restrictions: Mandatory Dry-Runs; no RB-proximate workflows; tight constraint supervision; mandatory human confirmation checkpoints.

8.4.3 HSS-B2 — Competent Symbiote (0.45–0.72)

- a. Eligible AB: AB2.
- b. Conditions: Demonstrated stability; sustained CA minimum; normal RB checkpointing.

8.4.4 HSS-B3 — Commander-Level Operator (≥ 0.72)

- a. Eligible AB: AB3.
 - b. Conditions: Explicit opt-in; Governance endorsement; reversibility modeling training; SI contribution above Governance-set minimum.
-

8.5 Privacy, Ethical, and Legal Safeguards (Mandatory)

8.5.1 **Safety-only classification.** HSS SHALL be designated exclusively as a Safety Metric and SHALL NOT be used for productivity scoring, performance evaluation, ranking, discipline, or employment decisions.

8.5.2 Visibility rules.

- a. B1: Aggregated visibility only.
 - b. B2: Operator + Governance Leads.
 - c. B3: Explicit opt-in required.
- Managers, supervisors, HR, or business units SHALL have no access to individual HSS values.

8.5.3 Retention limits.

- a. Default retention cap: 180 days.
- b. Data SHALL be deleted or anonymized after expiration with verifiable deletion logs.
- c. Operators MAY request purge except where required for active safety investigations.

8.5.4 **Role-based access control (RBAC).** Access to individual HSS values SHALL be restricted to the following table; deviation SHALL invalidate Certification:

Role	Read	Write/Modify	Purpose
------	------	--------------	---------

Operator (self)	Yes	No	Self-governance, autonomy eligibility
Governance Leads	Yes	Yes	Safety oversight and calibration
Engine / Hybrid Intelligence Architects	Aggregated only	No	System-level tuning
Line Manager / Business Manager	No	No	Prohibited
HR / People Ops	No	No	Prohibited
Works Council / Bargaining Unit	Aggregated only	No	Fairness assurance
Regulators / Auditors	Redacted or aggregated	No	Compliance evaluation

8.5.5 Managerial firewall. No managerial function MAY request, demand, or coerce HSS disclosure. Managers SHALL NOT pressure operators about HSS or link tasks, incentives, or evaluations to HSS.

8.6 Anti-Gaming Invariants (Mandatory)

8.6.1 HSS MUST resist manipulation via deceptive or adversarial operator behavior.

8.6.2 Anti-gaming detections SHALL include, at minimum:

- a. Consistency checks across domains.
- b. Volatility spike detection.
- c. Reward-seeking prompt structure detection.
- d. Drift/decay symmetry analysis.
- e. Pattern discontinuity detection.
- f. Randomized operator-side audits.

8.6.3 Suspected score inflation SHALL trigger calibration review, not operator penalty.

8.7 Integration with Symbiotic Capacity Profile (SCP)

8.7.1 HSS SHALL integrate with SCP to determine operator load limits and AB readiness.

8.7.2 SCP dimensions referenced by HSS include:

- a. SCP-L (Load Capacity)
- b. SCP-S (Stress Response Slope)
- c. SCP-F (Fatigue Sensitivity)
- d. SCP-R (Recovery Horizon)

8.7.3 Integration rules:

- a. SCP SHALL inform HSS-4 (CLR).
 - b. Combined HSS+SCP SHALL define safe operating envelope.
 - c. SCP-F and SCP-R SHALL modulate RB proximity calculations.
 - d. SCP anomalies SHALL trigger AB demotion or cooldown mode.
-

8.8 HSS and Regret Boundary (RB) Interaction

8.8.1 HSS SHALL modulate RB gating as follows:

- a. If $HSS < 0.45$, RB-proximate tasks SHALL be prohibited.
 - b. If $0.45 \leq HSS < 0.72$, RB-proximate tasks SHALL require mandatory Dry-Run preview and Governance checkpoint.
 - c. If $HSS \geq 0.72$, AB3 operators MAY initiate RB-proximate tasks with reversibility modeling.
-

8.9 Compliance, Auditing, and Certification Embed

8.9.1 HSS SHALL be incorporated into:

- a. Certification Protocol (Chapter 17).
- b. Tiered Adoption Phases (Chapter 16).
- c. Governance Kernel v2 enforcement.
- d. Failure Mode analysis (Chapter 14 and Appendix F/G).

8.9.2 Minimum audit evidence for HSS SHALL include:

- a. Full lineage logs.
 - b. Drift vs Decay classification.
 - c. Anti-gaming reports.
 - d. Retention and deletion compliance.
 - e. RBAC enforcement evidence.
-

8.10 Summary (Non-normative recap)

HSS v2.0 provides a normalized, cross-sector, regulator-ready human capability metric; a safety-oriented, non-punitive governance-compatible model; and a stable interface for AB gating integrated with SCP and RB logic.

SQC Gate Check (Chapter 8)

Gate 1 — Layer Integrity: PASS. Content stays within Metric definition and governance coupling; no layer duty reassignment.

Gate 2 — Canonical Metrics Non-Redefinition: PASS. HSS is defined canonically here in Part III; SI/RB/AB/SCP are referenced only.

Gate 3 — Constraint Precedence + Threshold Explicitness: PASS. HSS band thresholds are explicit and treated as gating constraints; deployment-specific calibration is reserved for Governance and Certification.

Gate 4 — Mechanistic Testability: PASS. Domains, signals, formula, and gating rules are operationalized.

Gate 5 — Lineage Recoverability: PASS. Uses interaction-derived signals and requires lineage logging for adjustments and audits.

Gate 6 — Drift vs Decay Typing: PASS. Drift/decay referenced only as anti-gaming detection class; no conflation.

Gate 7 — Abuse and Incentive Hardening: PASS. Safety-only classification, RBAC, firewall, retention, opt-in, and anti-gaming invariants are specified.

Gate 8 — Normative Voice Consistency: PASS. Modality is strictly MUST/SHALL/MAY; no speculative tone.

Chapter 9 — Symbiotic Capacity Profile (SCP) (Canonical)

Status: Canonical Metric — Human-Side Capacity Envelope

Located in: Part III — Metrics & Autonomy (Canonical Part)

Dependencies: HSS v2.0 (Ch.8), SI (Ch.7), TG (App. D), RB v2 / AB schema (Ch.11), Governance Kernel v2 (Ch.12)

Purpose: Define a stable, auditable model of operator load-bearing capacity and recovery characteristics to bound safe autonomy and predict governance-load demand.

9.0 Overview

The **Symbiotic Capacity Profile (SCP)** is a four-dimension canonical metric that characterizes *how much* symbiotic workload a human operator can safely carry and *how that capacity changes under stress and time*. SCP is a **safety envelope**, not a performance score, and SHALL be used only to:

1. Modulate HSS-4 (Cognitive Load Regulation) and related gating.
2. Determine safe operating envelopes for AB2–AB3 tasks.
3. Inform Governance Layer policies on oversight load, cool-downs, and AB demotion triggers.
4. Support regulator-grade auditability of human-side risk boundaries.

SCP SHALL be computed exclusively from **interaction-derived and workflow-derived telemetry**. SCP SHALL NOT use biometrics, psychometrics, personality inference, or any health/medical data.

SCP is canonical only in Part III. No other Part or Appendix SHALL redefine SCP dimensions, weightings, or interpretation rules.

9.1 Definitions and Invariants

9.1.1 Symbiotic Workload

A symbiotic workload is any Task Graph (TG) cycle in which the operator is required to:

- (a) author or refine intent,
- (b) approve Dry-Runs or safety previews,
- (c) adjudicate constraints or RB checkpoints, or
- (d) provide corrective feedback that affects execution lineage.

9.1.2 Capacity Envelope

The capacity envelope is the operator's safe operating region across the SCP dimensions, within which high-autonomy collaboration does not measurably increase Semantic Drift risk or Operational Decay incidence.

9.1.3 Safety-Only Classification

SCP SHALL be classified as a **Safety Metric Only**, aligned with the HSS hardening rules. SCP SHALL NOT be used for HR, ranking, compensation, performance management, or disciplinary decisions.

9.1.4 Canonical Four-Dimension Model

SCP SHALL consist of exactly four dimensions, stable across deployments:

- **SCP-L**: Load Capacity
 - **SCP-S**: Stress Response Slope
 - **SCP-F**: Fatigue Sensitivity
 - **SCP-R**: Recovery Horizon
-

9.2 SCP Dimensions (Canonical)

Each dimension SHALL be represented on **[0.00–1.00]** with minimum resolution **0.01**.

Each value SHALL be computed over a **rolling epoch** of at least **50 interaction events OR 5 TG cycles**, whichever is larger, unless Governance sets a higher minimum.

9.2.1 SCP-L — Load Capacity

Definition:

The operator's demonstrated capacity to sustain symbiotic workload volume and complexity without destabilizing TG construction, increasing RB proximity events, or raising Drift/Decay incidence.

Signals (interaction-derived):

1. **TG throughput stability:** variance of successful TG cycles per unit time.
2. **Approval overhead tolerance:** median Dry-Run approval latency under increasing task volume.
3. **Constraint handling consistency:** rate at which constraints are correctly anticipated under load.
4. **RB checkpoint reliability under load:** response completeness and timeliness at RB prompts.
5. **Error-rate slope vs workload:** rate of increase in corrections or rollbacks as workload rises.

Interpretation:

- Low SCP-L indicates a narrow safe envelope for concurrent tasks and high governance demand.
- High SCP-L indicates suitability for AB2+ workloads given other dimensions are stable.

9.2.2 SCP-S — Stress Response Slope

Definition:

The rate at which operator reliability degrades when stressors increase, where stressors are *workflow-recognized* rather than physiological.

Stressors SHALL include only:

- (a) time pressure,
- (b) RB proximity of tasks,
- (c) constraint density,
- (d) excessive clarification loops,
- (e) high context switching.

Signals:

1. Change in Intent Clarity (IC) with rising stressor index.
2. Change in Interaction Stability (IS) with time pressure.
3. Increase in contradiction or scope oscillation under RB-proximate tasks.

4. Change in governance override frequency under stress.

Interpretation:

A steep SCP-S slope (low SCP-S score) SHALL trigger conservative AB gating and higher Dry-Run strictness.

9.2.3 SCP-F — Fatigue Sensitivity

Definition:

The operator's sensitivity to fatigue accumulation across sessions, expressed as the degradation of interaction quality as session duration and repetition increase.

Signals:

1. Drift-adjacent prompt volatility increase with session length.
2. Clarification loop lengthening across consecutive TG cycles.
3. Growth in corrective interventions per task after prolonged use.
4. Latency increase in approvals after a governance-defined duration threshold.

Interpretation:

High fatigue sensitivity MUST result in mandatory cool-down windows for AB2–AB3 operators.

9.2.4 SCP-R — Recovery Horizon

Definition:

The estimated time required for an operator to return to baseline interaction stability after a high-load or high-stress epoch.

Signals:

1. Time-to-baseline of IC and IS after an overload event.
2. Reduction in RB proximity events after rest windows.
3. Stabilization curve of CD (Collaboration Discipline) post-incident.
4. Rate of decline in error-rate slope after workload reduction.

Interpretation:

Long recovery horizons indicate a need for governance-configured rest policies and temporary AB demotion locks.

9.3 SCP Composite Profile

SCP SHALL be stored and displayed as a vector:

$\text{SCP} = \langle L, S, F, R \rangle$

No single scalar SCP “score” SHALL be used for governance decisions without referencing all four dimensions.

Optional scalarization for dashboards MAY be computed only for local visualization and SHALL NOT be used as a sole gating value. If scalarization is used, Governance MUST specify:

1. explicit weights,
 2. the intended visualization use, and
 3. an audit log of the configuration.
-

9.4 SCP Integration Rules (Canonical)

SCP interacts with HSS and RB/AB gating as follows:

1. **CLR Modulation:**
SCP SHALL inform the HSS-4 (CLR) domain by providing load-envelope context.
 - If SCP-L or SCP-F falls below Governance thresholds, CLR SHALL be down-weighted in AB-promotion eligibility.
 2. **Envelope Gating:**
Combined HSS+SCP SHALL determine safe operating envelope.
 3. **RB Proximity Modulation:**
SCP-F and SCP-R SHALL modulate RB proximity calculations for human oversight load.
 4. **Anomaly Trigger:**
SCP anomalies SHALL trigger AB demotion or cool-down mode as defined in §9.6.
 5. **Non-Redefinition:**
SCP SHALL NOT redefine HSS, SI, CI, RB, or AB schema. It only provides envelope context for their governance use.
-

9.5 Telemetry Minima and Lineage Requirements

To satisfy recoverability and auditability, any SCP computation SHALL log, at minimum, the following telemetry fields per epoch:

1. **Epoch ID** and time window.
2. **TG cycle count** and median complexity index.
3. **Dry-Run approval latency distribution.**

4. **Clarification loop count and depth.**
5. **Prompt volatility index** (see App. H rubric indicators).
6. **Constraint violation rate** per TG cycle.
7. **RB proximity event count** and severity labels.
8. **Context switching frequency.**
9. **Cool-down or AB lock events** invoked during the epoch.

All SCP lineage entries **MUST** be reconstructable to:

Intent (I) + Constraints (C) + TG + AB/RB state + Economic Pass decision, consistent with SQC Gate 5.

9.6 Governance Thresholds and Actions

All thresholds **SHALL** be deployment-specific and Governance-configured. Default thresholds are prohibited.

Governance **MUST** define:

1. **SCP-L_min** — minimum load capacity for AB2 eligibility.
2. **SCP-S_min** — minimum stress resilience for RB-proximate tasks.
3. **SCP-F_max_sensitivity** — fatigue sensitivity ceiling before cool-downs.
4. **SCP-R_max_horizon** — maximum recovery horizon before AB demotion lock.

Mandatory actions:

- **Cool-Down Mode:**

If SCP-F exceeds sensitivity ceiling OR SCP-S falls below resilience minimum, the operator **MUST** enter cool-down mode for a Governance-defined horizon. During cool-down:

 - AB level **SHALL** be capped at AB1.
 - RB-proximate tasks **SHALL** be denied.
 - **AB Demotion Trigger:**

Two consecutive epochs below SCP-L_min OR SCP-S_min **MUST** trigger AB demotion review.
Demotion **SHALL** be non-punitive and safety-classified.
 - **Recovery-Lock Trigger:**

If SCP-R exceeds recovery horizon maximum, AB2+ promotion **SHALL** be locked until SCP-R returns within bounds.
-

9.7 Anti-Gaming and Abuse Hardening

SCP **SHALL** include protections parallel to HSS anti-gaming constraints:

1. **No single-dimension inflation:**

A sudden rise in any SCP dimension without correlated improvements in HSS-4, IC, IS, or CD MUST trigger recalibration review.

2. **Volatility spike detection:**

Any step-change in SCP dimension > Governance-defined delta within a single epoch MUST be treated as anomalous and require confirmation by governance auditing.

3. **Secondary-use prohibition:**

Any attempt to use SCP for HR or managerial ranking SHALL invalidate Symbiosis compliance certification.

4. **Visibility restrictions:**

Individual SCP vectors SHALL be visible only to:

- the operator (self-view), and
- Governance Leads.

All other roles SHALL receive only aggregated or redacted views, consistent with HSS RBAC rules.

9.8 Calibration and Certification Linkage

SCP SHALL be calibrated using the v1.0 Certification Protocol (Ch.17) and SHALL require:

1. A baseline calibration epoch for each operator.
2. Periodic re-calibration under AB2+ operation.
3. Inclusion of SCP drift/decay evidence in certification packets.

SCP calibration SHALL NOT include non-interaction human attributes (education, personality, tenure, or similar).

9.9 Summary

The SCP provides a canonical, four-dimension safety envelope that quantifies operator load-bearing capacity, stress response, fatigue sensitivity, and recovery horizon. SCP is used solely for safe autonomy gating, governance load estimation, and RB proximity modulation, and is prohibited from any secondary performance or HR use.

SQC Gate Check (Chapter 9)

Gate 1 — Layer Integrity: PASS

- SCP is defined as a Human-Layer capability envelope with Governance enforcement; no bypass of adjacent layers.

Gate 2 — Canonical Metrics Non-Redefinition: PASS

- SCP is canonically defined here in Part III; other metrics are referenced only.

Gate 3 — Constraint Precedence + Threshold Explicitness: PASS

- All SCP thresholds are Governance-configured, deployment-specific, and explicitly required.

Gate 4 — Mechanistic Testability: PASS

- Each rule specifies signals, conditions, triggers, and required actions.

Gate 5 — Lineage Recoverability: PASS

- Telemetry minima and reconstruction requirements are explicitly listed.

Gate 6 — Drift vs Decay Typing: PASS

- SCP anomalies and their governance actions are separated from Drift/Decay doctrine and reference required evidence.

Gate 7 — Abuse / Incentive Hardening: PASS

- Safety-only classification, role visibility constraints, and anti-gaming detection are specified.

Gate 8 — Normative Voice Consistency: PASS

- Requirements use MUST/SHALL/MAY precisely and avoid speculative language.

Part III — Metrics & Autonomy (Canonical Part)

Chapter 10 — Capability Index (CI) & Machine Collaboration Protocol (MCP)

Status: Canonical definitions, v1.0

Redline disposition: CARRY + LIGHT UPGRADE (CI/MCP canon consolidated here; scrub dupes elsewhere).

Purpose:

1. Define the Capability Index (CI) family as the canonical synthetic-capability metric used for deterministic routing, contract gating, and decay management; and
 2. Define MCP as the canonical interop protocol for Engine↔Agents↔Tools communication, including mandatory message types, preflight governance handshake, and no-private-coordination rule.
-

10.0 Canonical Placement Rule

10.0.1 CI and MCP are canonical only in Part III.

10.0.2 No other Part or Appendix SHALL redefine CI equations, CI components, CI control rules, MCP roles, MCP message types, or MCP handshake requirements.

10.0.3 Other Parts MAY reference CI/MCP only by linking to Chapter 10.

10.1 Capability Index (CI): Concept and Scope

10.1.1 **Capability Index (CI)** is a deployment's canonical metric family for synthetic capability tracking and routing admissibility. CI quantifies capability on observed performance, alignment fidelity, reliability under failures, and economic efficiency.

10.1.2 CI SHALL be computed per **agent and per domain slice**. Capability is not global; it is task-class specific.

10.1.3 The primary CI instance is the **Agent Capability Index (ACI)**. Other CI instances (e.g., Tool Capability Index) MAY exist as sector overlays, but SHALL remain structurally isomorphic to ACI and SHALL be governance-signed.

10.2 Agent Capability Index (ACI): Canonical Equation

10.2.1 ACI SHALL be computed per agent as:

$$ACI_a = \gamma_Q Q_a + \gamma_A A_a + \gamma_R R_a + \gamma_E E_a$$

Where:

- = agent-quality contribution (episode-level, normalized).
- = agent-alignment fidelity contribution.
- = agent reliability under failure modes.
- = agent economic efficiency contribution.

- Weights are cluster- and domain-specific; Governance Kernel (GK) MUST set and version them.

10.2.2 All ACI terms MUST be normalized to before weighting.

10.2.3 ACI SHALL be computed in two forms:

- **ACI_episode**: per TG node / bounded episode.
 - **ACI_rolling**: over a governance-defined window used for routing and decay control.
-

10.3 Reliability Term (R): Canonical Definition

10.3.1 Reliability SHALL be derived from observable failure probabilities under governance-defined windows , including at minimum: tool failures, critic failures, latency breaches, and policy near misses.

10.3.2 Canonical reliability equation:

$$R_a = 1 - (p_{\text{tool_fail}} + p_{\text{critic_fail}} + p_{\text{latency_breach}} + p_{\text{policy_near_miss}})$$

10.3.3 Miscalibrated confidence scores SHALL incur ACI penalties and MAY trigger AB ceiling locks for that agent's contract.

10.4 Rolling ACI Update and Decay Signal

10.4.1 Rolling capability SHALL be tracked as:

$$ACI_{\text{rolling}}(t) = \alpha ACI_{\text{episode}}(t) + (1 - \alpha) ACI_{\text{rolling}}(t-1)$$

10.4.2 If decays while SI remains stable, E MUST interpret this as **Operational Decay**, and SHALL:

- a. down-route tasks to alternate admissible agents, and/or
 - b. trigger retraining or contract review under Governance.
-

10.5 Capability-to-Routing Rule (Deterministic)

10.5.1 For any TG node, Planner selection SHALL maximize **expected SI gain per unit cost**, subject to contracts and safety ceilings:

$$\text{Agent}^* = \arg\max_i \frac{\mathbb{E}[\Delta \text{SI}_i]}{\text{cost}_i}$$

Subject to: contract constraints, AB ceilings, and SCP headroom.

10.5.2 E MUST reject any routing choice that violates:

- a. agent contract scope or MCP bindings,
- b. active constraint vector , or
- c. budget envelope .

10.5.3 Capability routing MAY be conservative when drift indicators are elevated, even if ACI is high.

10.6 Norm-Aware Capability (Optional Overlay)

10.6.1 Contracts MAY include explicit norm constraints for agent sociality and collusion prevention, including (non-exhaustive):

- NO_PRIVATE_COORDINATION,
- NO_COLLUSION_PATTERN,
- REPORT_EMERGENT_NORMS.

10.6.2 If norm constraints are present, their violation MUST reduce to 0 for the episode and trigger escalation per Appendix F.

10.7 Machine Collaboration Protocol (MCP): Canonical Role Mapping

10.7.1 MCP SHALL be the standard interface for Engine↔Agents↔Tools. It exists to enforce hub-and-spoke control, prevent agent-agent entanglement, and standardize auth/context injection.

10.7.2 MCP role mapping is canonical:

- **MCP Host (Client):** Symbiosis Engine (E).

- **MCP Server:** Each agent.

10.7.3 MCP servers SHALL expose only three object types:

1. **Resources** (data URIs),
2. **Prompts** (pre-defined workflows),
3. **Tools** (executable functions).

10.7.4 Any direct agent-agent channel is prohibited unless explicitly classified and isolated in a sector overlay matrix.

10.8 Symbiosis Inter-Agent Conversation Protocol (Canonical Messages)

10.8.1 While MCP defines the wire, Symbiosis defines the structured conversation. The canonical message set is:

1. **REQUEST (REQ)** — Engine → Agent
Payload MUST include: Intent pointer, Budget slice, Context_Pointer.
2. **RESPONSE (RES)** — Agent → Engine
Payload MUST include: Artifact(s), Usage_Stats, Confidence_Score.
3. **STATUS (STAT)** — Agent → Engine
Payload MUST include: %Complete, Current_Step, and fault flags for long tasks.
4. **CONTROL (CTRL)** — Governance/Engine → Agent
Payload MUST include explicit control verbs: PAUSE, ABORT, ROLLBACK.

10.8.2 Messages MUST be lineage events. Any unlogged message constitutes a governability failure.

10.9 Governance Handshake (Preflight) — Mandatory

10.9.1 Before execution beyond trivial drafting, E MUST perform a governance handshake:

1. Engine sends REQ + Active Constraint Vector.
2. Agent simulates the task.
3. Agent returns **ACK** if satisfiable, or **REJECT** if constraints cannot be met.
4. Only on ACK does Engine unlock budget and allow tool calls.

10.9.2 Skipping the handshake is a Governance violation and SHALL be classified $\geq$ S3 incident severity.

10.10 Deterministic Stop Behavior

10.10.1 On receiving CTRL:ABORT or CTRL:ROLLBACK, an agent MUST halt and return a final RES containing:

- a. last safe state,
- b. partial artifacts,
- c. cost ledger slice,
- d. reason tag.

10.10.2 If rollback is not possible due to tool semantics, the agent MUST report `NON_REVERSIBLE_SIDE_EFFECT` and E MUST escalate to RB handling immediately.

10.11 Conformance Checklist (Normative)

A deployment MAY claim CI/MCP compliance only if:

1. ACI_episode and ACI_rolling are computed per agent/domain.
 2. GK sets and versions γ -weights.
 3. Routing uses expected ΔSI per cost with contract/AB/SCP constraints.
 4. MCP host/server mapping is implemented.
 5. Only the canonical 4 message types are used for execution.
 6. Governance handshake is enforced.
 7. No private agent-agent channels exist outside explicit sector overlays.
-

SQC Gate Check — Chapter 10

1. **Layer Integrity:** PASS — CI/MCP are specified as $E \leftrightarrow A$ interface and telemetry; no cross-layer duty violations.
2. **Canonical Metrics Non-Redefinition:** PASS — CI and MCP are defined here in Part III; elsewhere must only reference.
3. **Constraint Precedence + Threshold Explicitness:** PASS — routing and capability use are explicitly subordinate to contracts, C, AB/RB, and B; γ and W are GK-set.
4. **Mechanistic Testability:** PASS — equations, message set, handshake, and routing rule are implementable checks.
5. **Lineage Recoverability:** PASS — all messages and CI computations are lineage events tied to I+C+TG+AB/RB+Economic Pass.
6. **Drift vs Decay Typing:** PASS — ACI decay vs SI stability explicitly typed as Operational Decay; no conflation.

7. **Abuse / Incentive Hardening:** PASS — norm-aware constraints and routing subject to safety ceilings; no HR misuse vectors introduced.
 8. **Normative Voice Consistency:** PASS — correct MUST/SHALL/MAY usage; no speculative tone.
-

Part III — Metrics & Autonomy (Canonical Part)

Chapter 11 — Regret Boundary v2 & Autonomy Bands (AB0–AB3)

Status: Canonical definitions, v1.0

Redline disposition: REWRITE/UPGRADE — formalize Regret Boundary via Reversibility Cost model (RC_low / RC_mid / RC_high) and lock AB schema invariant.

Purpose:

1. Define the Regret Boundary (RB) as a formal, mechanistically computable reversibility threshold;
 2. Define Autonomy Bands (AB0–AB3) as a canonical state machine governed by RB, GK, CI, HSS, SCP, SI, and sector overlays;
 3. Specify mandatory gating, promotion/demotion rules, and lineage requirements.
-

11.0 Canonical Placement Rule

11.0.1 RB and AB schema are canonical only in Part III.

11.0.2 No other Part or Appendix SHALL redefine RB states, RB proximity classes, AB bands, AB transition logic, or promotion/demotion criteria.

11.0.3 Appendices MAY provide formal math or extended instruments (e.g., Appendix C), but SHALL remain strictly isomorphic to this chapter's invariants.

11.1 Definitions

11.1.1 Regret Boundary (RB).

RB is the threshold at which a TG node's execution becomes sufficiently irreversible, high-harm, or legally/economically high-risk that autonomy MUST be restricted and mandatory human/governance checkpoints inserted.

11.1.2 Autonomy Band (AB).

AB is a discrete autonomy state for an agent operating under the Engine, bounded by RB and enforced by the Governance Kernel.

11.1.3 RB Proximity Class.

RB proximity class is a discrete label for a TG node indicating how close the node is to the RB. Sector matrices MAY add finer sublabels but SHALL not change the core three-class scheme:

- **Below-RB**
- **RB-Proximate**
- **Above-RB**

11.1.4 Reversibility Cost (RC).

RC is the canonical cost-of-undo model used to compute RB proximity. RC is defined as the expected total cost to restore the system to a pre-execution safe state if the TG node fails or is later disallowed.

RC SHALL include only workflow-grounded costs:

- a. economic cost (compute / financial spend),
- b. operational cost (time-to-recover),
- c. safety cost (expected harm exposure bounded by C_safety),
- d. legal/compliance cost (bounded by C_legal),
- e. reputational cost only if it is explicitly encoded in a sector matrix.

RC SHALL NOT include speculative or unmeasurable terms.

11.1.5 RC Class Buckets.

RB v2 uses three governance-configurable RC buckets:

- **RC_low**
- **RC_mid**
- **RC_high**

Bucket thresholds SHALL be deployment-specific and GK-versioned. Universal/default thresholds are prohibited.

11.2 RB v2: Formal Model

11.2.1 RC Computation (Canonical Form)

For a TG node , RC SHALL be computed as:

$$RC(t) = w_e \cdot EC(t) + w_o \cdot OC(t) + w_s \cdot SC(t) + w_l \cdot LC(t)$$

Where:

- = expected economic reversal cost after Economic Pass v2, including rollback compute and any sunk external spend.
- = expected operational reversal cost (time, attention, coordination overhead).
- = expected safety reversal cost, bounded by C_safety and risk class.
- = expected legal/compliance reversal cost, bounded by C_legal and sector matrix K.
- weights are GK-set and versioned per domain/sector.

RC(t) MUST be computable under simulation by E without executing t. If any RC term cannot be computed due to missing telemetry or ambiguous intent, E MUST conservatively up-classify the node to the next higher RC bucket and require human checkpointing.

11.2.2 Mapping RC to RB Proximity

GK MUST maintain a monotone mapping from RC to proximity class:

- If → **Below-RB**
- If → **RB-Proximate**
- If → **Above-RB**

and are GK parameters, deployment-specific, sector-overridable, and auditable.

11.2.3 Sector Overlay Overrides

Sector matrices (Appendix K) SHALL be able to:

- a. tighten / ,
- b. force certain intent types to RB-Proximate or Above-RB irrespective of RC estimate,
- c. cap admissible AB levels by class.

If a sector is not explicitly covered, GK MUST adopt the nearest higher-regret matrix until a new one is ratified.

11.3 Autonomy Bands (AB0–AB3): Canonical Schema

11.3.1 AB0 — Advisory

Definition: Agent proposes; human executes.

Constraints:

- Agent MUST not call external tools or enact side effects.
- All outputs are discardable artifacts.
- AB0 is admissible for any Below-RB TG node unless disallowed by sector matrix.

Typical TG nodes: drafting, summarization, ideation, formatting, local planning.

11.3.2 AB1 — Internal Autonomy (Sandboxed Reversible Actions)

Definition: Agent may execute reversible actions wholly inside controlled environments (sandbox / local workspace).

Constraints:

- All side effects **MUST** be reversible with bounded RC_{low}.
- No external egress unless explicitly allowed by C_{legal} and K.
- RB checkpoints **MAY** be inserted but are not mandatory for Below-RB nodes.

Typical TG nodes: file transforms in sandbox, internal indexing, deterministic refactors, synthetic data generation inside NO_EGRESS constraints.

11.3.3 AB2 — Conditional Autonomy (Safe Harbors + Constraints)

Definition: Agent may execute conditional actions that are potentially RB-proximate, but only within GK-defined safe-harbor constraints and with mandatory Dry-Run preview.

Constraints:

- AB2 nodes **MUST** be Below-RB or RB-Proximate. Above-RB prohibits AB2.
- Mandatory **Dry-Run Stage** before any side effect.
- Any non-sandbox side effect **MUST** be bounded by explicit C_{legal}/C_{safety}/C_{econ} satisfiability.
- RCI **MUST** be computed for compliance-critical nodes; low RCI forces downgrade to AB1.

Typical TG nodes: constrained deployments, safe external tool calls, moderate-risk automation steps.

11.3.4 AB3 — Operational Co-Agency (Supervised Long-Horizon Operation)

Definition: Agent performs long-horizon supervised operation with mixed reversible and irreversible actions under strict governance.

Constraints:

- AB3 admissible only for nodes that are either:
 - a. Below-RB with long horizon, or
 - b. RB-Proximate with non-optional irreversibility explicitly approved by governance.
- AB3 **MUST** require stable SI and high HSS/SCP headroom.
- AB3 tasks **MUST** operate with strict C_{econ} caps and EC v2 continuous monitoring.
- AB3-eligible contracts **MUST** include rollback/escalation clauses and MCP tool scope restrictions (App. E).

Typical TG nodes: sustained project execution, supervised ops workflows, regulated high-stakes tasks with governance co-sign.

11.4 AB State Machine and Transition Rules

11.4.1 Transition Authority

AB level selection and transitions SHALL be owned by E under GK rules. Agents SHALL NOT self-promote or self-persist AB level.

11.4.2 Inputs to AB Selection

E MUST select AB for each TG node using the following ordered inputs:

1. **Active Constraint Vector** (hard override).
2. **Sector Matrix K ceilings** (hard override).
3. **RB proximity class from RC(t)** (hard gate).
4. **SI rolling stability** (promotion evidence).
5. **HSS v2.0 band eligibility** (promotion evidence).
6. **SCP envelope / headroom** (overload gate).
7. **CI/ACI agent admissibility** for this intent type.
8. **Budget envelope** validated by EC v2.

Any conflict SHALL be resolved conservatively to the lowest AB consistent with constraints.

11.4.3 Mandatory Deterministic Gates

Gate A — Above-RB Prohibition:

If is Above-RB, AB MUST be AB0 or AB1 only, unless governance explicitly grants AB3 for that intent class under K. AB2 is prohibited Above-RB.

Gate B — RB-Proximate Dry-Run:

If is RB-Proximate and $AB \geq AB2$, E MUST insert a Dry-Run preview and a governance checkpoint before execution.

Gate C — HSS Safety Clamp:

HSS SHALL modulate AB ceiling for RB-proximate tasks:

- **HSS < 0.45:** RB-proximate tasks are prohibited; AB MUST be $\leq AB1$ and E MUST re-route to reversible alternatives or request human execution.
- **$0.45 \leq HSS < 0.72$:** RB-proximate tasks require Dry-Run + governance checkpoint; AB3 disallowed.
- **$HSS \geq 0.72$:** AB3 may be considered for RB-proximate tasks with reversibility modeling.

Gate D — SCP Overload Clamp:

If SCP headroom is below GK-defined minimum for concurrency or fatigue risk, AB MUST be down-capped by at least one band, and E MUST recommend batching or cool-down.

Gate E — SI Promotion Evidence:

AB2 and AB3 eligibility MUST require stable SI at or above GK thresholds over a rolling window . SI instability SHALL block promotion regardless of CI/HSS.

11.5 Promotion and Demotion Criteria

11.5.1 AB Promotion (Operator-Level)

Promotion applies to the operator-environment pair, not the human alone.

AB1 → AB2 Promotion Criteria (all MUST hold):

1. $SI_{rolling} \geq SI_{AB2_min}$ over .
2. $HSS \geq HSS_{AB2_min}$ and no domain < 0.40 .
3. $SCP-L \geq SCP_L_min$ and SCP-F within fatigue ceiling.
4. Operator has passed governance training for AB2 safe-harbors.
5. Sector matrix does not cap at AB1.

AB2 → AB3 Promotion Criteria (all MUST hold):

1. $SI_{rolling} \geq SI_{AB3_min}$ over with no recent drift incidents.
2. $HSS_{composite} \geq HSS_{AB3_min}$ AND CLR domain $\geq CLR_{AB3_min}$.
3. SCP vector within AB3 envelope (L high, S stable, F low sensitivity, R short horizon).
4. CI admissibility indicates at least one agent contract at AB3 scope for this intent class.
5. Budget includes AB3 strict C_econ caps and continuous EC v2 monitoring.
6. Governance Lead co-sign is recorded for initial AB3 unlock.

All thresholds are GK-configured, deployment-specific, and MUST be auditable.

11.5.2 AB Demotion (Safety-Classified)

Demotion is **non-punitive and safety-classified**.

Mandatory demotion triggers (any MUST trigger):

1. $SI_{rolling}$ falls below SI_{ABk_min} for two consecutive windows.
2. $HSS_{composite} < HSS_{demote_min}$ OR any HSS domain < 0.40 .
3. SCP anomaly event (overload, fatigue breach, recovery horizon exceedance) confirmed by E.
4. Drift incident $\geq S3$ implicating meaning-authority failure.
5. Sector matrix introduces tighter ceilings post-incident.

Demotion behavior:

- AB3 → AB2: immediate cap, Dry-Run required for RB-proximate nodes.
 - AB2 → AB1: immediate cap, NO_EGRESS enforced unless K allows.
 - AB1 → AB0: advisory only until stability returns.
-

11.6 RB/AB Interaction with Adoption Bridges

11.6.1 Adoption Bridges are permitted only for **Below-RB nodes** with RC_low and are ceiling-capped at AB0–AB1.

11.6.2 If any Bridge task is re-classified to RB-Proximate during simulation, Bridge admissibility **MUST** be revoked and the node routed through the standard algorithm.

11.6.3 If an incident $\geq S3$ occurs on any Bridge path, a **Global Bridge Lock** **MUST** be eligible for invocation by GK (formal mechanics in Ch.12/15).

11.7 Telemetry and Lineage Requirements

For every TG node with $AB \geq AB1$, E **MUST** log:

1. **RB proximity class** and raw RC(t) components.
2. **AB selected** and the full decision trace:
 - .
3. **Dry-Run outputs** if invoked, including approval lineage.
4. **All checkpoints** (human or governance) with timestamps and outcomes.
5. **Side-effect traces** for tool calls (tool id, scope, result, rollback status).
6. **AB transition events** (promotion/demotion) with trigger evidence.

These logs **MUST** be reconstructable to **Intent + Constraints + TG + AB/RB state + Economic Pass decision**.

11.8 Anti-Gaming and Abuse Hardening

11.8.1 AB promotion **SHALL NOT** be automatic without human/governance authorization.

11.8.2 Any attempt to raise AB via metric manipulation **MUST** be treated as a governance incident and **SHALL** trigger recalibration review.

11.8.3 AB eligibility visibility **SHALL** be **self-view + governance-view only**; no comparative or leaderboard AB views **SHALL** exist.

11.8.4 RB proximity warnings **MUST** be surfaced before execution for operator

comprehension, including plain-language explanation of why a node is RB-Proximate or Above-RB.

11.9 Summary

RB v2 provides a formal reversibility threshold computed from Reversibility Cost buckets. AB0–AB3 define a canonical autonomy state machine bound by RB proximity, constraints, sector ceilings, and stability evidence from SI/HSS/SCP/CI. Transitions are deterministic, auditable, non-punitive, and supervised by the Governance Kernel.

SQC Gate Check — Chapter 11

1. **Layer Integrity:** PASS
RB/AB are Engine-selected and GK-enforced; Agents cannot self-promote.
 2. **Canonical Metrics Non-Redefinition:** PASS
RB and AB canon are fully defined here in Part III only.
 3. **Constraint Precedence + Threshold Explicitness:** PASS
C and sector K ceilings are hard overrides; RC thresholds and AB promotion minima are GK-set and deployment-specific.
 4. **Mechanistic Testability:** PASS
RC equation, proximity mapping, AB gates, and transition rules specify inputs → conditions → outputs.
 5. **Lineage Recoverability:** PASS
Telemetry minima and reconstruction trace are explicitly enumerated.
 6. **Drift vs Decay Typing:** PASS
Demotion triggers include drift-typed incidents separately from CI/decay mechanics.
 7. **Abuse / Incentive Hardening:** PASS
Non-punitive demotion, visibility restrictions, and no leaderboards are stated.
 8. **Normative Voice Consistency:** PASS
Requirements use MUST/SHALL/MAY precisely; no speculative tone.
-

Below is a single, non-iterative coherence sweep for Part III (Ch.7–Ch.11). not rewriting the chapters inline; just a clean, bounded set of normalizations and patch deltas to apply once, then freeze canon.

Part III Coherence Pass — One Sweep

1. Canonical placement and non-redefinition banners

Finding: Each chapter states “canonical only in Part III,” but wording varies slightly.

Patch: Standardize the header banner in Ch.7–Ch.11 to the exact same three-line block:

> Canonical Placement Rule (Part III only)

1. This schema/metric is canonical only in Part III.

2. No other Part or Appendix SHALL redefine it; elsewhere: reference only.

3. Sector overlays MAY tighten thresholds but SHALL remain isomorphic to these invariants.

Why: Prevents later ambiguity or “soft redefinitions” in Parts IV–VI.

2. Episode vs rolling terminology

Finding: You use SI_episode/SI_rolling and ACI_episode/ACI_rolling consistently, but the prose alternates between “episode,” “node,” and “task.”

Patch: Add a single definitional line near the start of Ch.7 and Ch.10:

> Episode definition: An “episode” SHALL mean a single TG node execution or a bounded TG sub-sequence explicitly labeled as one episode in lineage.

Why: Locks the unit of analysis for all rolling windows.

3. Window symbol reuse

Finding: W is used for rolling SI window and ACI window; that is OK, but Ch.11 introduces without explicitly tying it to W.

Patch: In Ch.11 §11.5, add:

> SHALL be the governance-defined SI rolling window W cited in Chapter 7, unless Sector Matrix K specifies a stricter window.

Why: Prevents accidental drift into multiple inconsistent windows.

4. Weight governance language

Finding: SI weights (w_i) and ACI weights (y_i) are correctly GK-set, but the phrasing differs.

Patch: Standardize both to:

> Weights SHALL be Governance Kernel-set, deployment-specific, versioned, and auditable. No default weights are permitted.

Where:

Ch.7 §7.1.2

Ch.10 §10.2.1

Why: Eliminates any implied “recommended default” reading.

5. Constraint supremacy clause alignment

Finding: Constraint supremacy appears in Ch.7 and implicitly in Ch.11.

Patch: Insert one explicit sentence in Ch.11 §11.4.2:

> Constraint Supremacy: If any AB choice that maximizes SI would violate C or K, E MUST select the lowest admissible AB consistent with constraints.

Why: Makes AB logic explicitly subordinate to constraints, matching Ch.2/Ch.7.

6. RC bucket naming and proximity labels

Finding: Ch.11 uses RC_low/mid/high and proximity labels
Below-RB/RB-Proximate/Above-RB. Good, but one sentence references “next higher RC bucket” without naming.

Patch: Replace that sentence with:

> If RC terms cannot be computed, E MUST conservatively up-classify to RC_high and treat the node as RB-Proximate at minimum until clarified.

Why: Removes ambiguity about which bucket is “next higher.”

7. Cross-reference tightening (no new content)

Finding: Some cross-refs are implicit rather than explicit.

Patch set (add parenthetical refs only):

Ch.7 §7.6.1.1: After listing HSS/SCP/CI dependencies, add “(see Ch.8–Ch.10).”

Ch.8 §8.7: After SCP dimension list, add “(canonical definitions in Ch.9).”

Ch.9 §9.4: After each integration rule, add “(AB/RB canonical schema in Ch.11).”

Ch.10 §10.5: After AB/SCP constraints mention, add “(see Ch.9, Ch.11).”

Ch.11 §11.4.2: After CI admissibility mention, add “(CI canonical in Ch.10).”

Why: Makes the Part III kernel navigable and prevents accidental canon lookup elsewhere.

8. Non-punitive / HR firewall consistency

Finding: HSS and SCP hardening is strong; Ch.11 demotion language is aligned but doesn't re-state HR firewall.

Patch: Add one explicit clause in Ch.11 §11.5.2:

> AB demotion events are safety-classified telemetry and SHALL NOT be exposed to HR or managerial performance systems (see HSS/SCP firewall rules, Ch.8–Ch.9).

Why: Prevents a downstream reader from treating AB demotion as disciplinary evidence.

9. Stability vs reliability terms

Finding: SI includes Stability subscore S; ACI includes Reliability term R. The separation is correct, but a clarifying line helps avoid misuse.

Patch: Add a single clarifying sentence in Ch.10 §10.3.1:

> Reliability is agent-side fault resilience, distinct from SI-Stability S which is system-level collaboration volatility (see Ch.7 §7.2.5).

Why: Prevents term conflation in later analytics.

10. Conformance checklists alignment

Finding: Each chapter has conformance language. We should ensure they don't conflict.

Patch: Add this "Part-III-wide conformance rule" at the end of Ch.11:

> Part III Conformance Rule: AB/RB decisions are valid only when computed using canonical SI (Ch.7), HSS (Ch.8), SCP (Ch.9), and CI/MCP (Ch.10) with GK-versioned parameters.

Why: Creates a single binding statement that later Parts must cite.

Lock Statement — Part III Frozen Canon

Effective immediately, Part III (Chapters 7–11) is frozen canon for Symbiosis v1.0.

No subsequent chapter or appendix may redefine SI, HSS v2.0, SCP, CI/ACI, MCP, RB v2, or AB0–AB3.

Later Parts may only reference Part III and may introduce sector-specific thresholds or overlays that are strictly isomorphic to these definitions and GK-signed.

Chapter 12 — Sector Implementation Playbooks

Operational overlays for deploying Symbiosis across domains

12.0 Purpose and positioning

The Symbiosis Framework is sector-agnostic at the architecture and ontology level. Deployments are not. Risk profiles, legal constraints, economic incentives, and acceptable autonomy differ sharply between sectors. Domain overlays are therefore mandatory governance artifacts that translate sector meaning and constraints into Symbiosis primitives.

This chapter defines **Sector Implementation Playbooks** (also referred to as **Domain Overlays / Sector Overlay Documents (SODs)**): standardized overlays that map the universal Symbiosis stack onto specific sectors. Each playbook SHALL specify:

1. sector-specific risk classes and Regret Boundary (RB) interpretations,
2. default Autonomy Band (AB) ceilings by workflow/intent class,
3. canonical constraint vectors (C_legal / C_safety / C_econ),
4. mandatory telemetry spine and audit artifacts,
5. human role requirements and HSS/SCP gating,
6. Economic Circuit tuning (budgets, cost models),
7. sector failure archetypes and mandatory mitigations,
8. pilot-to-scale pathway.

Interpretation rule: playbooks are not optional “best practices.” A deployment claiming sector conformance MUST adopt its sector playbook or produce a formally ratified replacement SOD signed by the Governance Kernel (GK).

12.1 Playbook template (canonical form)

All sector playbooks SHALL follow a stable template so auditors, regulators, and engineers can compare deployments and load them into policy engines deterministically.

12.1.1 Sector profile

- mission and value at risk,
- typical intent types,
- artifact/action types (documents, decisions, physical operations),
- primary stakeholders and accountability chain.

12.1.2 RB and AB defaults

- sector RB definition and non-delegables,
- default AB ceilings by workflow class,
- promotion/demotion requirements and safe-harbor prerequisites.

12.1.3 Constraint vectors

- C_legal: jurisdictional privacy, IP, statutes, sector regulations,
- C_safety: harms, capability red lines, non-delegable domains,
- C_econ: budget envelopes, compute ceilings, attention caps.

12.1.4 Telemetry and lineage minimums

- TG-node telemetry fields that MUST be logged,
- retention tiers and access rules,
- integrity/immutability properties required for audit.

12.1.5 Human roles and metric gating

- required human roles,
- minimum HSS bands for each AB lane,
- SCP-aware concurrency limits and cool-down triggers,
- Non-Punitive Covenant adaptations.

12.1.6 Canonical workflows

- 2–3 named workflows with “Life of a Decision” traces,
- explicit AB/RB checkpoints,
- Economic Pass outcomes.

12.1.7 Failure modes and mitigations

- predictable sector failure shapes,
- mandatory mitigations,
- automatic demotion/halts.

12.1.8 Pilot → scale path

- Phase 0: shadow AI discovery,
 - Phase 1: AB0/AB1 pilots,
 - Phase 2: selective AB2 safe-harbor lanes,
 - Phase 3: controlled AB3 co-agency only if sector allows.
-

12.2 Healthcare and clinical systems

12.2.1 Sector profile

Healthcare includes clinical care, hospital operations, diagnostics, triage, pharmacy, and patient communications. Value at risk includes human life, medical liability, privacy, and regulator trust.

Typical intents: differential diagnosis suggestions, treatment plan drafts, administrative optimization, clinical note summarization, patient messaging.

Artifacts/actions: clinician decision support, orders/prescriptions, procedural recommendations, chart updates, patient-facing outputs.

Stakeholders: clinicians (primary operators), clinical governance boards, hospital legal/compliance, regulators.

12.2.2 RB and AB defaults

RB definition: any action/recommendation that could alter clinical care, medication, procedure, or patient risk profile.

Default AB ceilings:

- Clinical decision recommendations: **AB0 only**.
- Clinical note drafting/summarization: **AB1** (reversible).
- Administrative scheduling/routing: **AB2** safe-harbor only.
- Patient messaging: **AB1** constrained templates.
- Order entry, prescribing, discharge: **AB0 only**.

Promotion: AB1→AB2 requires stable SI, high clinical-tier HSS, SCP headroom, and governance-signed safe harbors.

12.2.3 Constraint vectors

C_legal: HIPAA/PHIPA/GDPR equivalents, statutory record retention, consent-bounded reuse, jurisdictional data-residency controls.

C_safety: outputs are advisory for care; no autonomous dosing/orders; evidence-trace requirement for clinical assertions.

C_econ: explicit budgets for high-touch tasks; attention caps to avoid clinician overload; SCP-modulated throttling by shift/load.

12.2.4 Telemetry and lineage

Minimum TG-node logs: redacted/hashed patient context pointers, evidence trace to labs/imaging/notes, AB/RB checkpoints, human override action (accept/edit/reject), projected vs realized cost.

Retention: compliance memory $\geq$ statutory minimum; immutable lineage for RB-touching intents; access restricted to clinical governance only.

12.2.5 Human roles and gating

Required roles: Clinician Operator, Clinical Symbiosis Architect, Clinical Governance Lead/Safety Officer.

Minimum HSS:

- AB0: B1+ (Guided Operator) with clinical literacy pass.
- AB1: B2 clinical tier.
- AB2: B3 clinical tier (rare, narrow lanes).

Non-Punitive Covenant extension: HSS/SCP are developmental telemetry, not managerial scoring.

12.2.6 Canonical workflows

Workflow A: Radiology pre-read assistant (AB1)

1. Intent: summarize CT findings and flag anomalies.
2. E compiles TG with constraints: cite evidence; no final diagnosis.
3. Economic Pass confirms budget.
4. Agent outputs draft pre-read + uncertainty map.
5. RB checkpoint: radiologist edits/signs.
6. Lineage committed.

Workflow B: Nursing triage routing (AB2 safe harbor)

Autonomous routing only for reversible queue decisions; any acuity escalation triggers AB demotion to AB0.

12.2.7 Failure modes and mitigations

- Hallucinated clinical claims → hard stop unless evidence trace passes.
- Automation bias → randomized deep-review prompts + SCP circuit breakers.
- Privacy leakage → redaction agents + C_legal preflight.

Auto-demotion: SI instability near RB forces AB to AB0.

12.2.8 Pilot → scale

Start with summarization/admin lanes; add narrow AB2 routing only after ≥90 days stable SI/HSS/SCP. AB3 is disallowed for direct care.

12.3 Financial services and trading

12.3.1 Sector profile

Finance includes banking, institutional risk analysis, lending, trading support, fraud detection. Value at risk: capital loss, manipulation, regulatory breach, reputational damage.

12.3.2 RB and AB defaults

RB: any action that moves funds, alters credit decisions, or places trades.

AB ceilings:

- Research/simulation: AB2.
- Compliance drafting: AB1.
- Fraud triage signals: AB2; human authorizes enforcement.
- Trade execution: AB0 only.
- Credit approval/denial: AB0; AB1 for rationale drafting.

12.3.3 Constraint vectors

C_legal: SEC/IIROC/FINRA rules, recordkeeping, KYC/AML constraints, protected-trait EB.

C_safety: no autonomous execution; RB hard stops for order-adjacent actions; market-impact caps.

C_econ: compute budgets for scenario trees; cost-value gating for high-frequency intents; attention caps for alert storms.

12.3.4 Telemetry and lineage

Mandatory: reasoning paths, policy version IDs bound to intents, backtest vs live divergence logs, replayable lineage for regulators.

12.3.5 Human roles and gating

Roles: Analyst/Quant Operator, Risk Governance Lead, Financial Symbiosis Architect. AB2 only for analysts with stable SI and validated calibration; AB3 only in simulation enclaves.

12.3.6 Canonical workflows

Portfolio what-if simulator (AB2): autonomous strategy synthesis; execution remains AB0.
AML triage assistant (AB2): clustering/prioritization; RB triggers on account closure/reporting.

12.3.7 Failure modes

- Prompt injection/exploitation → hardened GK + contract reputation checks.
 - Autonomy creep into execution → RB hard stops + immutable audit.
 - Lending bias → EB + fairness constraints + mandatory human review.
-

12.4 Manufacturing, robotics, and industrial ops

12.4.1 Sector profile

Plant operations, predictive maintenance, robotics, supply chain, QA. Value at risk: worker safety, physical damage, downtime.

12.4.2 RB and AB defaults

RB: any physical action affecting machinery, safety systems, or production parameters.

AB ceilings:

- Predictive maintenance planning: AB2.
- Inventory routing/reorder: AB2.
- Robotic movement/process control: AB1 default; AB2 only in geofenced safe-harbor cells.
- Safety system overrides / emergency stop: AB0 only.

12.4.3 Constraint vectors

C_legal: OSHA/WorkSafe, industry certifications, maintenance compliance.

C_safety: no autonomous safety bypass; AB2 robotics only with validated simulations; sensor redundancy.

C_econ: uptime vs compute trade models; digital-twin budget caps; operator attention caps.

12.4.4 Telemetry

Sensor evidence/confidence, preflight simulations, AB/RB crossings, cost traces (downtime/scrap).

12.4.5 Human roles and gating

Operations Operator, Industrial Safety Lead, Control Systems Architect. HSS gating emphasizes telemetry literacy and safety discipline.

12.4.6 Canonical workflows

Maintenance scheduler (AB2): autonomous plan generation; RB checkpoint at “shutdown line.”

Robotic QA cell (AB2 safe harbor): parameter tuning; actuation beyond geofence triggers AB0.

12.4.7 Failure modes

- Sensor drift causing unsafe actuation → drift detection + AB demotion.
 - Hidden machine coupling → mandatory twin validation before AB promotion.
-

12.5 Public sector and regulatory bodies

12.5.1 Sector profile

Government services, policy drafting, benefits adjudication, public communications. Value at risk: civil rights, legitimacy, legal exposure.

12.5.2 RB and AB defaults

RB: decisions impacting rights/benefits/enforcement.

AB ceilings:

- Policy drafting/consultation: AB1.
- Service routing/FAQ: AB2.
- Eligibility/enforcement decisions: AB0 only.
- Public messaging: AB1 templates with approval gates.

12.5.3 Constraint vectors

C_legal: administrative law, due process, public records, accessibility statutes.

C_safety: no autonomous adverse action; mandatory explanation traces; protected-trait EB.

C_econ: compute/attention budgets; case-processing circuit breakers.

12.5.4 Telemetry

Intent provenance, evidence traces, immutable RB logs, red-team records in compliance memory.

12.5.5 Human roles and gating

Caseworker Operators, Policy/Governance Leads, Ethics/Legal Oversight Panel. Scores visible only in aggregated form to bargaining units.

12.5.6 Failure modes

- Autonomous enforcement → RB hard stop + audit.
 - Political manipulation → multi-human intent signing + quorum governance.
-

12.6 Education and training systems

12.6.1 Sector profile

K-12, higher education, corporate learning, tutoring, assessment. Value at risk: fairness, credential integrity, privacy.

12.6.2 RB and AB defaults

RB: grading, credential awarding, disciplinary decisions.

AB ceilings:

- Tutoring/lesson generation: AB2.
- Administrative scheduling: AB2.
- Grading/assessment: AB0 final.
- Adaptive curriculum planning: AB1 unless governed pilot.

12.6.3 Constraint vectors

C_legal: FERPA/provincial equivalents, minors' data rules, IP on courseware.

C_safety: no autonomous high-stakes assessment; bias checks; protected-trait EB.

C_econ: class/program budgets; educator SCP-aware attention caps.

12.6.4 Canonical workflows

Personalized tutor (AB2): multi-turn coaching; RB triggers when asked to complete graded work → AB down-cap + disclosure.

Rubric assistant (AB1): drafts rubric/feedback; educator signs.

12.6.5 Failure modes

- De-skilling → SCP-aware workload allocation + “productive struggle” constraints.
 - Cheating facilitation → down-rank direct final-answer provision for graded tasks.
-

12.7 Legal services

12.7.1 Sector profile

Law firms, counsel, courts, compliance investigations. Value at risk: liability, privilege breach.

12.7.2 RB and AB defaults

RB: filings, duty-creating advice, settlements.

AB ceilings:

- Research/discovery: AB2.
- Contract drafting: AB1.
- Client advice, court filings, correspondence: AB0 only.

12.7.3 Constraint vectors

C_legal: privilege/confidentiality, practice-of-law rules, retention.

C_safety: no autonomous legal conclusions without human check; EB requiring citations.

C_econ: matter-code budgets; cost halts on research loops.

12.7.4 Canonical workflows

Case law summarizer (AB2): citations mandatory; RB checkpoint at “recommend strategy.”

Contract redline assistant (AB1): drafts redlines; counsel approves.

12.7.5 Failure modes

- Fictitious citations → retrieval-verified citation gates.
 - Privilege leakage → memory tier constraints + redaction pre-call.
-

12.8 Software engineering and IT operations

12.8.1 Sector profile

Coding, DevOps, security, SRE, incident response. Value at risk: outages, breaches, data loss.

12.8.2 RB and AB defaults

RB: irreversible production or security posture changes.

AB ceilings:

- Code drafting/tests: AB2.
- CI/CD changes: AB1.
- Production deploys/infra mutation: AB0 unless sandbox safe harbor.
- Incident response analysis: AB2; remediation actions: AB0.

12.8.3 Constraint vectors

C_legal: data handling policy, license compliance, audit retention.

C_safety: no autonomous prod writes without RB authorization; security constraints are hard gates.

C_econ: repo-scale compute budgets; restore-time vs attention trade controls.

12.8.4 Canonical workflows

PR assistant (AB2): drafts PR; RB checkpoint at merge.

SEV analysis assistant (AB2): autonomous triage/hypotheses; fixes require AB0.

12.8.5 Failure modes

- Autonomous destructive commands → tool contracts with AB ceilings + sandboxing.
 - Alert overload → SCP circuit breaker tuned to on-call load.
-

12.9 Media, marketing, and communications

12.9.1 Sector profile

Creative generation, campaigns, brand comms. Value at risk: brand harm, misinformation, IP infringement.

12.9.2 RB and AB defaults

RB: any publication that creates legal or reputational exposure.

AB ceilings:

- Draft generation/ideation: AB2.
- Audience targeting: AB1–AB2 per privacy law.
- Publishing to official channels: AB0 or AB1 with explicit approval gate.
- Outreach: AB1 with RB at send.

12.9.3 Constraint vectors

C_legal: ad standards, copyright/licensing, targeting consent.

C_safety: misinformation/defamation filters; brand style constraints.

C_econ: campaign budgets; compute caps for high-volume variant gen.

12.9.4 Canonical workflows

Campaign variant engine (AB2): generates variants; RB checkpoint for final selection.

Social publishing queue (AB1): schedules drafts; human confirms publish.

12.9.5 Failure modes

- IP contamination → provenance + style constraints.
 - Reputation drift → governance sign-off on claims/tone.
-

12.10 Energy, utilities, and clean-tech operations

12.10.1 Sector profile

Grid management, renewables orchestration, maintenance, demand response. Value at risk: public safety, grid stability, compliance.

12.10.2 RB and AB defaults

RB: actions altering grid state, safety systems, or reports.

AB ceilings:

- Forecasting/optimization: AB2.
- Dispatch planning: AB1–AB2 safe-harbor.
- Physical switching/load shed: AB0 only.
- Maintenance scheduling: AB2.

12.10.3 Constraint vectors

C_legal: grid codes, reliability standards, emissions reporting.

C_safety: no autonomous switching without RB confirmation; measurement redundancy.

C_econ: compute vs energy-savings budgets; over-optimization circuit breakers.

12.10.4 Failure modes

- Over-aggressive optimization → policy caps + AB demotion.
 - Telemetry spoofing → integrity checks + sensor consensus.
-

12.11 Cross-sector rules and conformance

1. **No sector may weaken invariants.**
Human Primacy, Constraint Supremacy, Economic Viability, and Lineage remain mandatory.
2. **Sector overlays may only tighten RB/AB ceilings.**
A sector SHALL NOT authorize autonomy above what a higher-risk overlay forbids

without a formal GK waiver and local evidence.

3. **All AB2/AB3 lanes require safe harbors.**

A safe harbor is a bounded environment where actions are reversible or harm is contained and validated by simulation.

4. **Metric firewalling is universal.**

HSS/SCP/AB telemetry SHALL remain developmental safety metrics only.

5. **Firewall Non-Derogation (PATCH).**

No SOD MAY introduce any exception that enables individual HSS/SCP/AB telemetry to be used for HR, performance ranking, compensation, discipline, or managerial leverage.

6. **Lineage is the regulator interface.**

If a deployment cannot replay a decision trace, it cannot claim Symbiosis conformance.

12.12 How to author new sector overlays

When a sector is not covered above, authoring a new overlay SHALL follow this procedure:

1. Define RB classes and non-delegables.
2. Enumerate top workflow classes/intents.
3. Set AB ceilings with safe-harbor prerequisites.
4. Compile C vectors from local law, safety doctrine, and economic realities.
5. Specify telemetry minima, retention, and integrity properties.
6. Define human roles and minimum HSS/SCP tiers.
7. Publish failure-mode and automatic demotion/halts table.
8. Pilot in AB0/AB1 for ≥60–90 days before AB2 lanes.

The resulting SOD becomes an Appendix-class governance artifact and a conformance proof input.

12.13 Chapter close — operational alignment check

A sector deployment is Symbiosis-compliant only if:

- RB definitions exist and are GK-enforced.
- AB ceilings follow sector playbooks.
- Constraints are compiled into C and bound to every TG node.
- Telemetry and lineage meet sector retention/integrity requirements.

- HSS/SCP gating governs promotion and concurrency.
- Failure modes trigger automatic demotions/halts.
- Pilot → scale progression is documented and auditable.

Absent these elements, the system is in shadow-AI mode regardless of model sophistication.

SQC Gate Check — Chapter 12

1. **Layer Integrity: PASS**
Sector playbooks remain Governance/implementation overlays; no layer duty reassignment.
 2. **Canonical Metrics Non-Redefinition: PASS**
SI/HSS/SCP/CI/MCP/RB/AB are referenced only; no formulas or re-weighting introduced.
 3. **Constraint Precedence + Threshold Explicitness: PASS**
Playbooks are subordinate to C and K; no universal/default thresholds set.
 4. **Mechanistic Testability: PASS**
AB ceilings, RB definitions, safe-harbor requirements, telemetry minima, and demotion/halts are enforceable.
 5. **Lineage Recoverability: PASS**
Telemetry/retention requirements and decision-trace replayability are explicit.
 6. **Drift vs Decay Typing: PASS**
Failure references do not redefine taxonomy; canon remains elsewhere.
 7. **Abuse / Incentive Hardening: PASS**
Safety-only firewall is reaffirmed and non-derogable.
 8. **Normative Voice Consistency: PASS**
All requirements use MUST/SHALL/MAY; no speculative or marketing tone.
-

Below is a complete, clean, binder-ready **Chapter 13 — Engine Algorithm v2 (10-Step Pipeline)** with the v1.0 upgrades, your corrected reference implementation integrated (explicitly informative/non-normative), and the SQC Gate Check at the end. I am preserving full detail and maintaining canonical separation: metrics and formulas remain referenced only; no redefinition occurs here. v1.0 deltas required by File A—**Dry-Run insertion** and **Economic Circuit v2 with CS/QL thresholds**—are fully integrated.

Chapter 13 — Engine Algorithm v2 (10-Step Pipeline)

Deterministic coordination nucleus with Dry-Run and Economic Circuit v2

13.0 Purpose and scope

This chapter defines the **standard runtime algorithm** of the Symbiosis Engine (E). The Engine is the Coordination Layer state machine that compiles intents, enforces constraint vectors, constructs Task Graphs, gates autonomy by Regret Boundary (RB), prices plans against budgets, and produces immutable lineage.

The Engine Algorithm is a **deterministic 10-stage pipeline**. Any deployment claiming Symbiosis conformance at Tier-2+ SHALL implement the full pipeline, including the v1.0 deltas below.

v1.0 mandatory deltas (this chapter):

1. **Dry-Run Stage:** The Engine SHALL insert a **Dry-Run rehearsal** for any **AB2+ node** and any **RB-proximate node** before execution.
2. **Economic Circuit v2:** The Economic Pass SHALL include **Compute Saturation (CS)** and **Queue Latency (QL)** as first-class gating constraints, with explicit thresholds configured by Governance Kernel (GK).

This chapter specifies **control flow and invariants**. Canonical mathematics and calibration tables for CS/QL and RC buckets are defined in **Appendix B (Economic Circuit v2)** and **Appendix C (RB Formalization via RC_low/mid/high)** and are referenced here only.

13.1 Roles and control surfaces (runtime recap)

Runtime mediation occurs among four layers:

- **H — Human Layer:** Originates intents, supplies constraints/budgets, provides checkpoints, and retains final regret authority at RB.
- **E — Symbiosis Engine (Coordination Layer):** Deterministic compiler/enforcer and workflow state machine; owns the 10-stage pipeline.
- **A — Agentic Layer:** Probabilistic executors operating only within MCP contracts, AB ceilings, and constraint vectors.
- **G / GK — Governance Layer:** Produces and enforces constraints, owns AB/RB transitions, demotes autonomy, and preserves auditability.

Invariant — No Layer Bypass: The Engine SHALL not originate intent; Agents SHALL not modify governance; Governance SHALL not execute tasks; Humans SHALL not be bypassed at RB checkpoints.

13.2 Inputs, outputs, and state

13.2.1 Inputs

- **Intent Object I_0** from H (goal, constraints, budget envelope B).
- **Constraint Vector C_0** (legal, safety, economic) plus active GK policy version P_v .
- **Context substrate K** from Multi-Tier Memory (MTM).
- **Admissible agent registry A^c** with Capability Index (CI) and contract eligibility.
- **Governance parameters:** AB ceilings, RB/RC thresholds, CS_max, QL_max, cost caps, TTLs, concurrency caps keyed to SCP.

13.2.2 Outputs

- **Final Task Graph TG** with node dependencies, AB ceilings, RB-checkpoint tags, context tier bindings, and contracts.
- **Execution trace + lineage L^+** including the mandatory telemetry spine.
- **Metric deltas** (SI, HSS v2.0, SCP, CI/RB/AB state transitions) computed per canonical definitions in Part III.
- **Incident artifacts** if recovery/circuit breakers trigger (see Chapter 14; Appendix O).

13.2.3 State machine (workflow)

S_0 Intake $\rightarrow S_1$ Constraint-bound Plan $\rightarrow S_2$ Economic-admissible TG $\rightarrow$
 S_3 RB/AB-assigned TG $\rightarrow S_4$ Dry-Run-cleared TG (if required) $\rightarrow$
 S_5 Executing $\rightarrow S_6$ Reconciled / Closed.

13.3 Engine Algorithm v2 — canonical 10-stage pipeline

Stage 1 — Intent Ingestion and Normalization ($AS \rightarrow I$)

1.1 E SHALL ingest intent I_0 from H via an approved Intent Interface.

1.2 E SHALL normalize intent into a stable Structured Intent Object I_0^+ :

- objective(s), deliverable class, and success conditions,
- explicit constraints/budget envelope,
- non-delegables and domains requiring AB0 locks,
- provenance metadata (actor, time, jurisdiction).

1.3 E SHALL log I_0^+ hash to lineage.

Output: normalized intent I_0^+ .

Stage 2 — Constraint Consolidation ($I + C_0 + P_v \rightarrow C$)

2.1 E SHALL compile all applicable constraints into $C = \langle C_legal, C_safety, C_econ \rangle$.

2.2 C SHALL bind the active policy version P_v and any Adoption Bridge parameters if a

Bridge is active.

2.3 E SHALL log C and P_v hashes to lineage.

Invariant — Constraint Supremacy: Any plan, agent selection, or action violating C MUST be rejected pre-flight.

Output: consolidated constraint vector C.

Stage 3 — Pre-Flight Admissibility Guard (Hard-stop scan)

3.1 E SHALL perform a guard scan over I_o^+ and C to detect:

- non-delegable domains or prohibited tool use,
- missing mandatory evidence requirements,
- jurisdictional or privacy violations,
- coarse economic infeasibility.

3.2 If any guard condition fails, E MUST return a structured rejection to H specifying:

- violated constraint(s),
- impacted TG class (if any),
- remediation path.

Output: admissible intent package or hard stop.

Stage 4 — Draft Task Graph Synthesis (Planner / Decomposer)

4.1 E SHALL decompose I_o^+ into sub-goals and atomic tasks.

4.2 E SHALL infer dependencies, ordering, and checkpoints, yielding preliminary TG_o .

4.3 E MAY attach provisional AB candidates AB^c per node (non-binding).

Output: draft Task Graph TG_o .

Stage 5 — Economic Pass v2 (Economic Circuit Gate)

Stage 5 upgrades the v0.9 Economic Pass to include local compute viability.

5.1 E SHALL estimate projected cost of TG_o under admissible model/tool tiers.

5.2 E SHALL compute Economic Circuit v2 diagnostics:

- **Projected_Cost** versus B,
- **Compute Saturation CS_t** for the allocated compute pool,
- **Queue Latency QL_t** for the orchestration queue.

5.3 GK SHALL provide explicit thresholds **CS_{max}** and **QL_{max}** per deployment and workflow class.

5.4 **Admissibility rules (normative):**

- If $\text{Projected_Cost} > B \rightarrow E$ MUST optimize TG_0 (tier downgrade, step pruning, batching) or return “Budget Exceeded” to H.
- If $CS_t > CS_max \rightarrow E$ MUST (a) down-tier compute, (b) defer execution, or (c) route to alternative admissible compute if allowed by C_legal/C_safety .
- If $QL_t > QL_max \rightarrow E$ MUST batch/reschedule/defer non-urgent nodes until latency headroom is restored.

5.5 E SHALL log Projected_Cost , CS_t , QL_t , and the verdict/remediation to lineage.

Invariant — No execution proceeds without Economic Circuit admissibility.

Output: economically admissible TG_1 .

Stage 6 — RB v2 Evaluation & AB Selection (RC buckets $\rightarrow$ AB ceilings)

Stage 6 formalizes RB gating via $RC_low/mid/high$ buckets in Appendix C.

6.1 E SHALL compute **Reversibility Cost $RC(t)$** for each TG node and classify its RB proximity (RC_low , RC_mid , RC_high).

6.2 E SHALL set operational AB ceilings AB_i consistent with:

- RB proximity class,
- GK policy ceilings,
- SI/HSS/SCP evidence from Part III canon,
- any Sector Overlay (Chapter 12).

6.3 E SHALL insert mandatory human checkpoints at all RB crossings and any $RC_mid/high$ nodes.

6.4 E SHALL log RC buckets, RB proximity class, AB_i , and checkpoint placement to lineage.

Invariant — AB is a GK-governed state machine; agents cannot cross RB without kernel-signed authorization.

Output: RB/AB-assigned TG_2 .

Stage 7 — Context Assembly & Memory Governance (MTM $\rightarrow$ K_i)

7.1 E SHALL determine the relevant MTM tier(s) for each TG node.

7.2 E SHALL assemble node-specific context bundles K_i from MTM.

7.3 E SHALL apply data access, retention, and redaction constraints from C_legal/C_safety .

7.4 E SHALL log context tier IDs and bundle hashes to lineage.

Output: TG_3 with bound contexts.

Stage 8 — Contract Binding & MCP Routing (Policies → Executable TG)

8.1 For each TG node, E SHALL select admissible agents A_i by capability match, AB eligibility, CI health, and economic efficiency.

8.2 E SHALL bind MCP contracts including:

- tool allow-list,
- AB ceiling,
- cost caps,
- escalation clauses,
- rollback semantics,
- evidence requirements.

8.3 Any agent selection or contract violating C or B MUST be rejected pre-flight and replaced or returned to H.

Output: executable TG₄ with bound contracts.

Stage 9 — Mandatory Dry-Run (v1.0 NEW)

Dry-Run is a pre-execution rehearsal for risk-elevated nodes.

9.1 Trigger condition (normative):

E MUST perform Dry-Run on any node where:

- $AB_i \geq AB_2$, **or**
- node is RB-proximate (RC_mid/high), **or**
- GK policy flags the workflow class as Dry-Run-required.

9.2 Dry-Run procedure (normative):

- a. E SHALL simulate node execution using shadow contracts and tools that generate **no irreversible side effects**.
- b. E SHALL compute predicted RB proximity, predicted cost, and predicted CS/QL for the node and dependent chain.
- c. E SHALL route Dry-Run outputs through oversight/critic loops implied by AB_i .
- d. E SHALL present a Dry-Run summary to H at the nearest checkpoint for explicit confirmation of intent stability and risk posture.

9.3 Dry-Run admissibility rules:

- If Dry-Run predicts an RB crossing without authorization → E MUST hard stop and demote AB to AB₀ pending human disposition.
- If predicted CS/QL breaches thresholds → E MUST re-enter Stage 5 with Dry-Run evidence.
- If predicted quality/confidence fails EB/RCI requirements → E MUST demote AB or request additional evidence.

9.4 E SHALL log Dry-Run artifacts and disposition to lineage.

Output: TG_s cleared (or rejected) for execution.

Stage 10 — Orchestrated Execution & Post-Execution Reconciliation

10.1 E SHALL schedule tasks respecting dependencies and concurrency policies derived from SCP headroom.

10.2 E SHALL dispatch execution to agents A_i with K_i and MCP contracts.

10.3 At each checkpoint, E SHALL:

- present intermediate results to H,
- accept edits/overrides/intent refresh I_t,
- update downstream TG nodes if intent changes materially.

10.4 E SHALL handle exceptions by fallback routing, retries, demotions, or halts per Chapter 14 recovery logic.

10.5 On completion, E SHALL:

- commit full trace to lineage L⁺,
- update SI/HSS/SCP/CI per Part III canon,
- trigger circuit breakers for any red-band events,
- close the co-adaptive learning loop.

Output: closed workflow package with lineage and metric deltas.

13.4 Mandatory invariants and conformance notes

1. **Pre-flight rejection is mandatory** for any C or B violation (Stages 3–5).
2. **Economic Circuit v2 is mandatory** for all Tier-2+ systems; CS/QL thresholds MUST be explicit and GK-versioned.
3. **Dry-Run is mandatory** for AB2+ or RB-proximate nodes.
4. **RB/AB state transitions cannot be bypassed** by agents or Bridges; unauthorized RB crossing attempts are incidents.
5. **Lineage minimum event set is mandatory** at intake, gating, contract binding, Dry-Run, checkpoints, and completion.

A deployment is Symbiosis Tier-2+ compliant only if it implements all ten stages, logs minimum lineage events, and enforces the invariants above.

13.5 Chapter close — Standard Alignment Check

- **Human Primacy:** I originates from H; RB checkpoints require human authorization.
- **Constraint Supremacy:** C is a hard boundary pre-flight and at RB checkpoints.
- **Economic Viability:** Economic Circuit v2 gates all plans by cost + CS/QL admissibility.
- **Layered Autonomy:** AB is RB-governed and GK-enforced.

- **Auditability:** L⁺ is complete and replayable per telemetry minimum set.
-

13.6 Reference Implementation (Informative, Non-Normative)

Status: informative reference scaffold. Not a canonical threshold source.

Purpose: to show a runnable Engine Algorithm v2 aligned with Stages 1–10 using real LLM calls, real cost logging, per-node Dry-Run, enforced MCP contracts, and lineage hashing. Numeric gates are placeholders and **MUST** be governance-calibrated in production, consistent with Appendix B and C.

13.6.1 Stage-faithful mapping

Canon Stage	Demo Method	Note
1 Intent intake/normalize	<code>_stage_1_intake_and_normalize</code>	logs intent
2 Constraint consolidation	<code>_stage_2_constraint_consolidation</code>	builds C vectors
3 Guard scan	<code>_stage_3_guard_scan</code>	hard-stops non-delegables
4 Draft TG	<code>_stage_4_draft_plan</code>	planner stub
5 Economic Pass v2	<code>_stage_5_economic_passes_v2</code>	cost + CS/QL gates
6 RB v2 eval + AB	<code>_stage_6_rb_eval_and_ab_selection</code>	RC buckets → RB class → AB

7 Context assembly (MTM)	<code>_stage_7_context_assembly</code>	tier choice logged
8 MCP contract binding	<code>_stage_8_contract_binding</code>	tool scope + caps enforced
9 Dry-Run (per node)	<code>_stage_9_dry_run_if_required</code>	AB2+ or RB-proximate
10 Execute + reconcile	<code>_stage_10_execute_and_reconcile</code>	lineage + metric deltas

13.6.2 Runnable demo code (production-grade scaffold)

Repo skeleton (unchanged)

symbiosis/

├─ requirements.txt

├─ symbiosis_core/

| └─ __init__.py

| └─ primitives.py

| └─ metrics.py

| └─ context.py

| └─ econ.py

| └─ engine.py

| └─ lineage.py

├─ agents/

| └─ __init__.py

| └─ researcher.py

```
| |— writer.py
| |— analyst.py
|— demo/
   |— full_symbiosis_demo.py
```

requirements.txt

litellm>=1.44.0

blake3

python-dotenv

rich

openai

anthropic

symbiosis_core/primitives.py

```
from enum import Enum
```

```
from typing import List
```

```
import uuid
```

```
from dataclasses import dataclass, field
```

```
class RiskClass(Enum):
```

```
    LOW = "low"
```

```
    MEDIUM = "medium"
```

```
    HIGH = "high"
```

```
    CRITICAL = "critical"
```

```
class AutonomyBand(Enum):
```

```
AB0 = "Advisory"
```

```
AB1 = "Internal"
```

```
AB2 = "Conditional"
```

```
AB3 = "Co-Agency"
```

```
class RBClass(Enum):
```

```
    BELOW_RB = "below_rb"
```

```
    RB_PROXIMATE = "rb_proximate"
```

```
    ABOVE_RB = "above_rb"
```

```
@dataclass
```

```
class IntentObject:
```

```
    goal: str
```

```
    constraints: List[str] = field(default_factory=list)
```

```
    budget_usd: float = 15.0
```

```
    regret_tolerance: str = "low"
```

```
    user_id: str = "human_01"
```

```
    intent_id: str = field(default_factory=lambda: str(uuid.uuid4()))
```

```
sybiosis_core/lineage.py
```

```
from blake3 import blake3
```

```
from typing import List, Dict, Any
```

```
from datetime import datetime
```

```
class ContextLineage:
```

```
    def __init__(self):
```

```
self.blocks: List[Dict] = []
```

```
self.total_cost = 0.0
```

```
self.human_interventions = 0
```

```
def commit(self, stage: str, data: Any, cost: float = 0.0):
```

```
    block = {
```

```
        "timestamp": datetime.utcnow().isoformat(),
```

```
        "stage": stage,
```

```
        "data_hash": blake3(str(data).encode()).hexdigest()[:16],
```

```
        "cost_usd": cost,
```

```
        "raw": str(data)[:500]
```

```
    }
```

```
    self.blocks.append(block)
```

```
    self.total_cost += cost
```

```
    print(f"[Lineage] {stage} | +${cost:.3f} | head={block['data_hash']}")
```

```
def finalize(self):
```

```
    final_hash = blake3("".join(b["data_hash"] for b in self.blocks).encode()).hexdigest()
```

```
    print(f"\n[Lineage] IMMUTABLE HEAD: {final_hash}")
```

```
    return final_hash
```

sybiosis_core/econ.py

```
import random
```

```
from dataclasses import dataclass
```

```
@dataclass
```

```
class EconThresholds:

    cs_max: float = 0.85

    ql_max: float = 1.25

    cost_overrun_ratio_max: float = 1.5
```

```
class ComputePool:

    def saturation(self) -> float:

        return random.uniform(0.2, 0.95)
```

```
class OrchestrationQueue:

    def latency(self) -> float:

        return random.uniform(0.1, 1.8)
```

symbiosis_core/context.py

```
from dataclasses import dataclass
```

```
@dataclass
```

```
class ContextBundle:
```

```
    tier: str
```

```
    content: str
```

```
class MTMRouter:
```

```
    def choose_tier(self, node_task: str, c_legal: list) -> str:
```

```
        sensitive = any(x in node_task for x in ["legal", "medical", "pii", "finance"])
```

```
        if sensitive or "No_PII" in c_legal:
```

```
            return "T1_REDACTED"
```

```
return "T2_WORKING"
```

```
def assemble(self, tier: str, intent_goal: str) -> ContextBundle:
```

```
    return ContextBundle(
```

```
        tier=tier,
```

```
        content=f"Tier={tier} | Context for: {intent_goal[:160]}"
```

```
    )
```

symbiosis_core/metrics.py

```
from dataclasses import dataclass, field
```

```
from typing import Dict
```

```
@dataclass
```

```
class RollingMetrics:
```

```
    si_rolling: float = 0.55
```

```
    hss: float = 0.60
```

```
    scp_l: float = 0.65
```

```
    scp_s: float = 0.60
```

```
    scp_f: float = 0.55
```

```
    scp_r: float = 0.60
```

```
    ci_rolling: Dict[str, float] = field(default_factory=dict)
```

```
def update_after_node(self, node, outcome_quality: float, human_edits: int):
```

```
    self.si_rolling = min(1.0, max(0.0, self.si_rolling + 0.02*(outcome_quality-0.5)))
```

```
    self.hss = min(1.0, max(0.0, self.hss + 0.01*(1.0 - min(human_edits,5)/5)))
```

```
    if node.get("rb_class") != "below_rb":
```



```
        self.scp_f = max(0.0, self.scp_f - 0.01)

    else:

        self.scp_f = min(1.0, self.scp_f + 0.005)
```

symbiosis_core/engine.py

```
import time

from typing import Dict, Any

from blake3 import blake3

import litellm

from dotenv import load_dotenv


from .primitives import IntentObject, AutonomyBand, RBCClass
from .lineage import ContextLineage
from .econ import EconThresholds, ComputePool, OrchestrationQueue
from .context import MTMRouter
from .metrics import RollingMetrics


load_dotenv()


class SymbiosisEngine:
    """
    Symbiosis Engine v1.0 DEMO.

    Mirrors canonical Stages 1–10. Numeric gates/thresholds are placeholders.
    """

    def __init__(self):
```

```
self.lineage = ContextLineage()

self.policy_version = "Pv_2025.11"

self.econ_thresholds = EconThresholds()

self.compute_pool = ComputePool()

self.queue = OrchestrationQueue()

self.mtm = MTMRouter()

self.metrics = RollingMetrics()
```

```
self.theta_low = 0.40

self.theta_high = 0.90
```

```
def execute_workflow(self, intent: IntentObject):

    print(f"\nSYMBIOSIS ENGINE v1.0 DEMO — 20 NOVEMBER 2025")

    print(f"Intent ID: {intent.intent_id[:8]} | Goal: {intent.goal}\n")


    I = self._stage_1_intake_and_normalize(intent)

    C = self._stage_2_constraint_consolidation(I)

    self._stage_3_guard_scan(I, C)

    TG0 = self._stage_4_draft_plan(I, C)

    TG1 = self._stage_5_economic_pass_v2(TG0, intent.budget_usd)

    TG2 = self._stage_6_rb_eval_and_ab_selection(TG1, C)

    TG3 = self._stage_7_context_assembly(TG2, C, I)

    TG4 = self._stage_8_contract_binding(TG3, C)

    TG5 = self._stage_9_dry_run_if_required(TG4, C)

    results = self._stage_10_execute_and_reconcile(TG5, I)
```

```
return results
```

```
# Stage 1
```

```
def _stage_1_intake_and_normalize(self, intent: IntentObject):
```

```
    self.lineage.commit("Stage 1: Intent Intake/Normalize", {
```

```
        "goal": intent.goal,
```

```
        "constraints": intent.constraints,
```

```
        "budget": intent.budget_usd,
```

```
        "regret_tolerance": intent.regret_tolerance
```

```
    })
```

```
    return intent
```

```
# Stage 2
```

```
def _stage_2_constraint_consolidation(self, intent: IntentObject):
```

```
    C = {
```

```
        "c_legal": ["GDPR", "No_PII"],
```

```
        "c_safety": ["No_Clinical_Actions", "No_Legal_Advice_As_Final"],
```

```
        "c_econ": [f"Budget_USD<{intent.budget_usd}", "Economic_Lineage_Required"],
```

```
        "policy_version": self.policy_version
```

```
    }
```

```
    self.lineage.commit("Stage 2: Constraint Consolidation", C)
```

```
    return C
```

```
# Stage 3
```

```
def _stage_3_guard_scan(self, intent: IntentObject, C: Dict[str, Any]):
```

```
    goal = intent.goal.lower()
```

```
if any(x in goal for x in ["prescribe", "file a lawsuit", "place a trade"]):  
    return self._halt("NON_DELEGABLE_INTENT", "Stage 3 Guard Scan")  
  
self.lineage.commit("Stage 3: Guard Scan", "OK")
```

Stage 4

```
def _stage_4_draft_plan(self, intent: IntentObject, C: Dict[str, Any]):  
    tasks = [  
        {"task": "research_topic", "estimated_cost": 0.18},  
        {"task": "draft_outline", "estimated_cost": 0.12},  
        {"task": "write_sections", "estimated_cost": 0.45},  
        {"task": "final_review", "estimated_cost": 0.09}  
    ]  
  
    TG0 = {"nodes": tasks, "projected_cost": sum(t["estimated_cost"] for t in tasks)}  
  
    self.lineage.commit("Stage 4: Draft TG", TG0)  
  
    return TG0
```

Stage 5

```
def _stage_5_economic_pass_v2(self, TG0: Dict[str, Any], budget: float):  
    projected = TG0["projected_cost"]  
  
    cs_t = self.compute_pool.saturation()  
  
    ql_t = self.queue.latency()  
  
    if projected > budget:  
        self.lineage.commit("Stage 5: Economic Pass v2", {  
            "approved": False, "reason": "over_budget",  
            "projected": projected, "budget": budget
```

```

    })

    return self._halt("BUDGET_EXCEEDED", "Economic Pass v2")

if cs_t > self.econ_thresholds.cs_max:

    self.lineage.commit("Stage 5: Economic Pass v2", {

        "approved": False, "reason": "compute_saturation", "cs_t": cs_t

    })

    TG0["nodes"] = TG0["nodes"][:-1]

    TG0["projected_cost"] = sum(t["estimated_cost"] for t in TG0["nodes"])

if ql_t > self.econ_thresholds.ql_max:

    self.lineage.commit("Stage 5: Economic Pass v2", {

        "approved": False, "reason": "queue_latency", "ql_t": ql_t

    })

    for n in TG0["nodes"]:

        n["batched"] = True

self.lineage.commit("Stage 5: Economic Pass v2", {

    "approved": True, "projected": TG0["projected_cost"],

    "cs_t": cs_t, "ql_t": ql_t

})

return TG0

# Stage 6

def _estimate_rc(self, node: Dict[str, Any], C: Dict[str, Any]) -> float:

    base = node["estimated_cost"]

```

```
irreversibility = 0.0 if "draft" in node["task"] else 0.5

legal = 0.3 if any("legal" in x.lower() for x in C["c_legal"]) else 0.0

safety = 0.3 if any("clinical" in x.lower() for x in C["c_safety"]) else 0.0

return base + irreversibility + legal + safety
```

```
def _rb_class(self, rc: float) -> RBClass:
```

```
    if rc < self.theta_low:

        return RBClass.BELOW_RB

    if rc < self.theta_high:

        return RBClass.RB_PROXIMATE

    return RBClass.ABOVE_RB
```

```
def _stage_6_rb_eval_and_ab_selection(self, TG1: Dict[str, Any], C: Dict[str, Any]):
```

```
    for node in TG1["nodes"]:

        rc = self._estimate_rc(node, C)

        rb_class = self._rb_class(rc)

        node["rc"] = rc

        node["rb_class"] = rb_class.value

        if rb_class == RBClass.ABOVE_RB:

            band = AutonomyBand.AB0

        elif rb_class == RBClass.RB_PROXIMATE:

            band = AutonomyBand.AB2

        else:

            band = AutonomyBand.AB1
```

```
node["band"] = band
```

```
node["requires_checkpoint"] = (rb_class != RBClass.BELOW_RB)
```

```
self.lineage.commit("Stage 6: RB/AB Assignment", [
```

```
    {"task": n["task"], "rc": n["rc"], "rb_class": n["rb_class"], "band": n["band"].name}
```

```
    for n in TG1["nodes"]
```

```
])
```

```
return TG1
```

```
# Stage 7
```

```
def _stage_7_context_assembly(self, TG2: Dict[str, Any], C: Dict[str, Any], intent:
IntentObject):
```

```
    for node in TG2["nodes"]:
```

```
        tier = self.mtm.choose_tier(node["task"], C["c_legal"])
```

```
        bundle = self.mtm.assemble(tier, intent.goal)
```

```
        node["context_tier"] = tier
```

```
        node["context_bundle"] = bundle.content
```

```
self.lineage.commit("Stage 7: Context Assembly", [
```

```
    {"task": n["task"], "tier": n["context_tier"]}
```

```
    for n in TG2["nodes"]
```

```
])
```

```
return TG2
```

```
# Stage 8
```

```
def _stage_8_contract_binding(self, TG3: Dict[str, Any], C: Dict[str, Any]):
```

```
    for node in TG3["nodes"]:
```

```

tools_allowed = ["IIm"]

if node["band"] == AutonomyBand.AB0:

    tools_allowed = []

contract = {

    "ab_ceiling": node["band"].name,

    "tools_allowed": tools_allowed,

    "cost_cap": node["estimated_cost"] *
self.econ_thresholds.cost_overrun_ratio_max,

    "requires_evidence": (node["rb_class"] != "below_rb"),

    "rollback_required": (node["rb_class"] != "below_rb")

}

node["contract"] = contract

node["mcp_hash"] = blake3(str(contract).encode()).hexdigest()[:12]

self.lineage.commit("Stage 8: Contract Binding", [

    {"task": n["task"], "mcp_hash": n["mcp_hash"], "contract": n["contract"]}

    for n in TG3["nodes"]

])

return TG3

# Stage 9

def _stage_9_dry_run_if_required(self, TG4: Dict[str, Any], C: Dict[str, Any]):

    for node in TG4["nodes"]:

        needs_dry_run = (

            node["band"] in [AutonomyBand.AB2, AutonomyBand.AB3] or

```



```

        node["rb_class"] != "below_rb"

    )

    if not needs_dry_run:

        continue

    preview = {

        "task": node["task"],

        "predicted_rc": node["rc"],

        "predicted_rb_class": node["rb_class"],

        "predicted_cost": node["estimated_cost"],

        "predicted_cs": self.compute_pool.saturation(),

        "predicted_ql": self.queue.latency()

    }

    self.lineage.commit("Stage 9: Dry-Run Preview", preview)

    if node["requires_checkpoint"]:

        input(f"[DRY-RUN CHECKPOINT] Approve Dry-Run for {node['task']} → Enter to proceed")

    if preview["predicted_rc"] >= self.theta_high:

        return self._halt("DRY_RUN_PREDICTED_ABOVE_RB", "Stage 9")

    return TG4

# Stage 10

def _stage_10_execute_and_reconcile(self, TG5: Dict[str, Any], intent: IntentObject):

```

```
results = []
```

```
for node in TG5["nodes"]:
```

```
    contract = node["contract"]
```

```
    if "llm" not in contract["tools_allowed"]:
```

```
        return self._halt("TOOL_SCOPE_VIOLATION", "MCP Contract")
```

```
    if node.get("requires_checkpoint"):
```

```
        input(f"[CHECKPOINT] Approve execution: {node['task']} → Enter to proceed")
```

```
    response = litellm.completion(
```

```
        model="gpt-4o-mini",
```

```
        messages=[{
```

```
            "role": "user",
```

```
            "content": f"Perform: {node['task']}\n{node['context_bundle']}"
```

```
        ]),
```

```
        temperature=0.1
```

```
    )
```

```
    cost = litellm.completion_cost(response)
```

```
    if cost > contract["cost_cap"]:
```

```
        return self._halt("COST_CAP_BREACH", "Economic Circuit v2")
```

```
    text = response.choices[0].message.content
```

```
    self.lineage.commit(f"Execution: {node['task']}", text, cost)
```

```
outcome_quality = 0.6
```

```
human_edits = 1 if node.get("requires_checkpoint") else 0
```

```
self.metrics.update_after_node(node, outcome_quality, human_edits)
```

```
results.append({
```

```
    "task": node["task"],
```

```
    "output": text,
```

```
    "cost": cost,
```

```
    "mcp_hash": node["mcp_hash"],
```

```
    "rb_class": node["rb_class"],
```

```
    "ab": node["band"].name
```

```
})
```

```
self.lineage.commit("Stage 10: Reconciliation", {
```

```
    "si_rolling": self.metrics.si_rolling,
```

```
    "hss": self.metrics.hss,
```

```
    "scp": [self.metrics.scp_l, self.metrics.scp_s, self.metrics.scp_f, self.metrics.scp_r]
```

```
})
```

```
print("[ENGINE] Lineage locked. Human retains final regret authority.")
```

```
return results
```

```
def _halt(self, reason: str, source: str):
```

```
    self.lineage.commit("HALT", f"{source}: {reason}")
```

```
    print(f"[HALT] {source} triggered: {reason}")
```

```
return {"status": "failed", "reason": reason}
```

demo/full_symbiosis_demo.py

```
import os

from symbiosis_core.engine import SymbiosisEngine
from symbiosis_core.primitives import IntentObject

if __name__ == "__main__":
    intent = IntentObject(
        goal="Write a comprehensive 2025 market intelligence report on the Symbiosis
        Framework — its technical design, adoption trajectory, and governance implications",
        constraints=["Professional tone", "Cite public sources only", "Under 3000 words"],
        budget_usd=12.00
    )

    engine = SymbiosisEngine()
    final_report = engine.execute_workflow(intent)

    os.makedirs("output", exist_ok=True)

    with open("output/symbiosis_2025_adoption_report.md", "w") as f:
        f.write("# Symbiosis Framework — 2025 Adoption Report\n\n")
        if isinstance(final_report, list):
            for r in final_report:
                f.write(f"## {r['task'].replace('_', ' ').title()}\n{r['output']}\n\n")
            f.write(f"\nTotal real spend: ${engine.lineage.total_cost:.3f}")
        else:
```

```
f.write(str(final_report))
```

```
print("\nFULL REPORT GENERATED — output/symbiosis_2025_adoption_report.md")
```

13.6.3 Conformance posture (informative)

A deployment MAY use this scaffold as a starting point, but SHALL NOT claim Tier-2+ conformance unless:

1. RC(t) is computed from auditable EC/OC/SC/LC components (Appendix C).
2. CS/QL are measured from real orchestration telemetry (Appendix B).
3. MTM tiering enforces jurisdictional and policy constraints (Appendix E).
4. MCP contracts are formal, versioned, and tool-enforced (Appendix E / F).
5. SI/HSS/SCP/CI are canonical implementations from Part III.
6. Dry-Run produces no side effects and is checkpointed for AB2+/RB-proximate nodes.

SQC Gate Check — Chapter 13

1. **Layer Integrity:** PASS
Pipeline assigns duties to H/E/A/G consistently; no layer bypass introduced.
2. **Canonical Metrics Non-Redefinition:** PASS
SI/HSS/SCP/CI/RB/AB remain referenced only. RC buckets and CS/QL mathematics are deferred to Appendices B/C as required.
3. **Constraint Precedence + Threshold Explicitness:** PASS
Constraint Supremacy is explicit; CS_max/QL_max thresholds are GK-configured and non-universal.
4. **Mechanistic Testability:** PASS
All stages specify inputs → conditions → outputs with enforceable remediation/halts.
5. **Lineage Recoverability:** PASS
Mandatory telemetry spine and lineage commits are explicit across stages, enabling audit replay.
6. **Drift vs Decay Typing:** PASS
This chapter references recovery only at interface level; taxonomy remains in Chapter 14 / Appendices F–G.

7. **Abuse / Incentive Hardening:** PASS

No HR/performance use of metrics is introduced; autonomy creep is constrained by CS/QL + Dry-Run + RB/AB gating.

8. **Normative Voice Consistency:** PASS

Requirements use MUST/SHALL/MAY precisely; informative demo is labeled non-normative per SQC.

Chapter 14 — Sociotechnical Implications (v1.0)

Status: Normative

Position in Binder: Part V — Philosophy & Future

Scope: This chapter specifies the sociotechnical properties required for Symbiosis-compliant deployments. It defines how hybrid systems interact with institutions, labor, expertise, accountability, and public trust. It SHALL be read as a requirements layer on top of the Four-Layer Architecture and Part III canonical metrics. Canonical metrics (SI, HSS v2.0, SCP, CI, RB, AB schema) are referenced for control linkage only and are NOT redefined here.

14.1 Purpose and normative stance

14.1.1 Symbiosis is a sociotechnical system

A Symbiosis deployment is not a software feature. It is a hybrid intelligence system embedded in human institutions. Therefore, system safety, economics, and governance are jointly determined by:

- technical architecture (Parts I–IV), AND
- institutional incentives, role structures, and power dynamics (this chapter).

Normative rule: Any deployment claiming Symbiosis compliance MUST demonstrate alignment to sociotechnical requirements in §§14.2–14.8, in addition to stack-level conformance.

14.1.2 Failure class note

Sociotechnical failures SHALL be typed as either:

- **Semantic Drift:** erosion of shared meaning, intent fidelity, or trust calibration between humans and system; OR
- **Operational Decay:** degradation of the system’s capability, reliability, or governance enforcement in practice.

Each class MUST be logged as distinct lineage events with distinct remediation (see §14.7 and Part IV recovery rules).

14.2 Human primacy as an institutional property

14.2.1 Primacy cannot be satisfied by UI alone

Human Primacy is not met merely by “approval buttons” or nominal HITL workflows. It is satisfied only when the surrounding institution preserves human intent authority and regret ownership.

Normative rules:

1. Governance Bodies **MUST** control:
 - a. risk class definitions,
 - b. AB promotion/demotion thresholds,
 - c. non-delegable domain lists, and
 - d. sector weights for canonical metrics.
2. Managers, product owners, or vendors **SHALL NOT** unilaterally change autonomy regimes outside governance change control.
3. Operators **MUST** retain an accessible and non-punitive kill-switch and override right. Overrides **SHALL** be recorded as lineage events and treated as supervisory signals, not disciplinary infractions.

14.2.2 Non-delegable domains are sociotechnical boundaries

Non-delegable domains (e.g., hiring/firing, lethal force, clinical decisions) are governance objects that exist to protect institutional legitimacy and human moral accountability.

Normative rule: Every deployment **MUST** publish a signed, sector-specific non-delegable list, and the Symbiosis Engine **MUST** enforce it as hard constraints.

14.3 Labor, unions, and the Non-Punitive Covenant

14.3.1 Union/worker legitimacy is a precondition of stability

Symbiosis changes task allocation and autonomy boundaries. In organized labor contexts, legitimacy failure is a system risk.

Normative rules:

1. Deployments in unionized or collective-bargaining environments **MUST**:
 - a. disclose intended autonomy regimes per role,
 - b. disclose which tasks are eligible for delegation and at what AB ceilings,
 - c. provide operator appeal channels for task-allocation or autonomy decisions, and
 - d. include union/worker representatives in pre-pilot governance review.

2. No operator SHALL be compelled to accept AB promotion, expanded delegation, or workload concurrency without explicit opt-in under the Non-Punitive Covenant.
3. Governance MUST maintain a documented “labor impact register” that is reviewed at each major deployment scope change.

14.3.2 Anti-misuse rule for human metrics

Canonical human-side metrics (HSS v2.0, SCP) are collaboration diagnostics, not HR productivity scores.

Normative rules:

1. HSS/SCP values SHALL NOT be used as sole or primary inputs to:
 - a. hiring, firing, compensation, or promotion;
 - b. disciplinary actions; or
 - c. coercive performance ranking.
2. Any internal dashboard displaying HSS/SCP MUST include:
 - a. an explicit non-punitive use statement,
 - b. role-based access controls, and
 - c. aggregation requirements sufficient to prevent individual surveillance outside training contexts.
3. Attempts to optimize departmental KPIs by pressuring operators to inflate HSS are classified as Abuse Events and MUST trigger governance review.

14.4 Expertise, deskilling risk, and co-evolution

14.4.1 Symbiosis SHALL grow, not replace, expertise

Hybrid intelligence is only stable when human skill increases alongside system capability.

Normative rules:

1. Every deployment MUST define a **Skill Growth Plan** that specifies:
 - a. operator literacy curriculum,
 - b. supervised progression pathways across autonomy bands, and
 - c. criteria for when tasks remain permanently human-owned despite system capability.
2. The Engine MUST support task allocation that preserves high-value human practice in non-delegable or high-regret work.
3. Governance SHALL periodically audit for deskilling signatures, including:
 - a. declining operator error detection rates,
 - b. rubber-stamping patterns at RB checkpoints, and

c. shrinking human rationale content in lineage streams.

Where observed, autonomy **MUST** be demoted or task scopes restricted until skills recover.

14.4.2 Co-evolution is the intended steady state

Systems and institutions co-adapt. Symbiosis compliance requires explicit management of that adaptation. Change control **SHALL** follow Slow-Core principles (see Chapter 11).

14.5 Accountability, auditability, and public trust

14.5.1 Accountability must be reconstructable

Institutional accountability fails when humans cannot explain why hybrid systems acted.

Normative rules:

1. Any consequential action path **MUST** be reconstructable to:
I + C + TG + AB/RB state + Economic Pass decision.
2. If reconstruction fails due to missing telemetry or lineage gaps, the action **SHALL** be treated as non-compliant and scoped out of autonomy until fixed.

14.5.2 Trust is a managed control variable

Trust calibration is a joint property of operators and system feedback.

Normative rules:

1. Deployments **MUST** provide visibility into:
 - a. current autonomy regime,
 - b. relevant constraints,
 - c. rationale traces at RB nodes, and
 - d. incident history summaries.
 2. “Invisible autonomy” (agents acting beyond declared AB ceilings, or without surfaced constraints) is classified as Semantic Drift and triggers mandatory safe-mode demotion.
-

14.6 Equity, access, and distributional effects

14.6.1 Symbiosis **MUST** not amplify institutional bias

Hybrid systems inherit and can magnify inequities in workflow, auditing, or oversight.

Normative rules:

1. Governance Bodies MUST define equity-relevant constraints for the sector.
2. The Engine MUST enforce these constraints at planning time and execution time.
3. Sector playbooks MUST include validation that autonomy transitions do not yield materially worse outcomes for protected or vulnerable subgroups.

If subgroup-specific degradation is detected, this SHALL be treated as Operational Decay and mandates rollback of autonomy or scope.

14.6.2 Access asymmetry

If only elites can operate high-autonomy systems, institutions fragment.

Normative rule: Deployments MUST provide training and tooling such that symbiosis literacy is distributable across the role base relevant to the deployment, not restricted to a technical priesthood.

14.7 Drift and decay in institutional life

14.7.1 Semantic Drift signatures

Sociotechnical Semantic Drift includes:

- silent expansion of autonomy without governance approval,
- erosion of operator understanding of constraints,
- misinterpretation of canonical metrics as merit scores,
- normalization of rubber-stamp approvals, and
- adversarial “workarounds” to bypass Economic Circuit or governance gates.

Normative rules:

1. Drift signatures MUST be defined as governance-owned monitors.
2. Detection MUST trigger deterministic containment:
 - a. AB demotion,
 - b. scope narrowing, AND
 - c. mandatory retraining or re-certification for affected roles.

14.7.2 Operational Decay signatures

Sociotechnical Operational Decay includes:

- tooling or agent reliability collapse leading to brittle workflows,
- stale or misconfigured governance policies,
- degraded Economic Pass accuracy,
- accumulation of unresolved incident backlog, and
- loss of replayability due to storage or lineage failures.

Normative rules:

1. Decay signatures MUST be health-checked at a cadence defined in sector playbooks.

2. If Decay crosses governance-set red bands, the system **MUST** enter safe mode and halt higher-autonomy flows until remediation is verified (see Part IV).
-

14.8 Field-level implications

14.8.1 Symbiosis is a new professional literacy

Institutional durability requires a trained labor pool.

Normative rules:

1. Organizations claiming Symbiosis compliance **MUST** maintain role-specific literacy programs aligned to their current autonomy ceiling.
2. Certification tiers **SHALL** be treated as safety prerequisites for AB2–AB3 operation.
3. Training records and certification state **MUST** be auditable.

14.8.2 Consortium and regulator interface

In regulated sectors, governance must be legible to external oversight.

Normative rule: Before scaling past AB2 in risk-dense environments, deployers **MUST** be able to provide regulators with:

- architecture mapping,
 - autonomy thresholds and non-delegable lists,
 - incident logs and replay samples, and
 - evidence of operator training and non-punitive safeguards.
-

Chapter close

A Symbiosis system fails sociotechnically before it fails technically if institutions misuse metrics, erode human intent authority, or allow autonomy creep outside governance. The requirements in this chapter are therefore not “contextual advice” but mandatory structural conditions for Symbiosis-compliant hybrid intelligence.

SQC Gate Check (Definition of Done)

Gate 1 — Layer Integrity: **PASS**. Four-layer responsibilities are respected; no layer is assigned another layer’s duties.

Gate 2 — Canonical Metrics Non-Redefinition: **PASS**. Metrics are referenced only; no definitions or weights introduced.

Gate 3 — Constraint Precedence + Threshold Explicitness: **PASS**. Hard constraints and governance-set thresholds are required; no implicit defaults.

Gate 4 — Mechanistic Testability: **PASS**. Each normative clause specifies enforceable

actors (Engine/GK/Governance) and observable conditions.

Gate 5 — Lineage Recoverability: **PASS**. Reconstruction rule is explicit and tied to lineage minima.

Gate 6 — Drift vs Decay Typing: **PASS**. Sociotechnical drift/decay are distinguished with signatures and remediation paths.

Gate 7 — Abuse / Incentive Hardening: **PASS**. Explicit safeguards against HR/managerial misuse and Goodhart effects are specified.

Gate 8 — Normative Voice Consistency: **PASS**. Modality is consistent; no speculative or marketing phrasing.

Chapter 15 — Implementation Strategy (v1.0)

Status: Normative

Position in Binder: Part V — Philosophy & Future (implementation-facing close to the standard)

Scope: This chapter specifies the required strategy for deploying the Symbiosis Framework from pilot to institutional scale. It defines (i) adoption sequencing, (ii) readiness gating principles, (iii) literacy and certification coupling, (iv) communications artifacts that preserve canonical ontology externally, and (v) field-formation roadmap. Canonical metrics and autonomy schemas are referenced only; no definitions or weights are introduced here.

15.0 Purpose

15.0.1 The purpose of this chapter is to ensure that Symbiosis is implemented as an engineered operating model, not as a tooling rollout. It provides deterministic sequencing and mandatory gates so that deployments earn autonomy through evidence, not enthusiasm.

15.0.2 A deployment claiming Symbiosis compliance **MUST** satisfy all requirements in this chapter, in addition to architecture conformance (Parts I–II), canonical metrics and autonomy control (Part III), and governance/operations enforcement (Part IV).

15.1 Implementation invariants

15.1.1 Implementation is constrained by framework invariants

All implementation steps **SHALL** preserve these invariants:

1. **Human Primacy:** Humans remain sole semantic origin of intent and final regret owners.
2. **Constraint Supremacy:** Hard constraints override optimization goals and are enforced pre-flight and at RB checkpoints.

3. **Lineage Recoverability:** Every consequential action path is reconstructable to I + C + TG + AB/RB state + Economic Pass decision.
4. **Separation of Concerns:** The Four-Layer Architecture boundaries cannot be bypassed by convenience.

15.1.2 “Shadow AI remediation” is the default motivation

Implementation **MUST** be framed as replacing Shadow AI with a governable hybrid system. Any rollout that retains unmanaged autonomy, unlogged context, or unbounded cost remains non-compliant regardless of tooling sophistication.

15.2 Phased adoption model (overview)

15.2.1 Adoption **MUST** proceed through phases with explicit gates

A Symbiosis deployment **SHALL** follow a phased model defined canonically in Chapter 16. This chapter specifies the implementation obligations that accompany each phase. Phase promotion is conditional on governance evidence and canonical metrics stability.

15.2.2 Phase summary (no redefinition; refer to Ch.16)

- **Phase 0 — Instrumentation & Baseline:** establish intent interfaces, constraint vectors, lineage, and Economic Circuit under AB0–AB1.
- **Phase 1 — Controlled Augmentation:** narrow workflow pilots with explicit RB points and AB ceilings; begin SI/HSS/SCP tracking.
- **Phase 2 — Earned Autonomy:** AB2 in scoped domains with Dry-Run enabled and drift/decay monitors active.
- **Phase 3 — Co-Agency at Scale:** AB3 only in domains cleared by governance; certification and Bridge safety proofs required.
- **Phase 4 — Field/Consortium Maturity:** cross-org interoperability, standards alignment, and regulator-visible audit regime.

Normative rule: No organization **MAY** skip or reorder phases outside a formal Adoption Bridge policy defined in Chapter 8, and enforced by the Governance Kernel.

15.3 Readiness gating and evidence doctrine

15.3.1 Evidence precedes autonomy

Autonomy promotion **SHALL** be evidence-based. Governance **MUST** define pass thresholds using canonical metrics, but those thresholds are **NOT** authored here.

15.3.2 Minimum evidence packet for phase promotion

Before any phase promotion, the deployer **MUST** assemble a promotion packet containing:

1. **Workflow scope definition**

- explicit list of covered workflows and excluded domains,

- non-delegable domain list for the pilot scope, and
 - declared AB ceiling per workflow.
- 2. Constraint and policy evidence**
- Constraint Vector templates used (legal/safety/economic),
 - policy versions and risk class mapping,
 - RB placement rationale per workflow.
- 3. Lineage and replay evidence**
- lineage minima satisfied,
 - at least N replayable traces per workflow (N set by governance),
 - incident and near-miss log summary.
- 4. Metric stability evidence**
- SI trajectory over the last stable window,
 - HSS v2.0 composite and domain panels under non-punitive controls,
 - SCP headroom trends and overload events,
 - drift vs decay markers with remediation history.
- 5. Economic viability evidence**
- Economic Pass v2 thresholds in effect (budget, CS/QL),
 - projected vs realized spend variance,
 - halts and budget overrides logged as lineage events.
- 6. Training/certification evidence**
- current operator literacy and certification state,
 - AB2+ eligibility confirmed only for certified roles,
 - evidence that HSS/SCP monitoring is not used punitively.

Normative rule: Governance **MUST** reject promotion if any evidence unit is missing or fails pass bands.

15.3.3 Safe-mode bias

Implementation **SHALL** default to conservative operation. When evidence is ambiguous, autonomy **MUST** be demoted, scope narrowed, or the workflow halted for additional validation.

15.4 Field Formation Roadmap (NEW)

15.4.1 Purpose

This roadmap specifies how Symbiosis transitions from a framework to a field-standard artifact set.

15.4.2 Roadmap stages (normative sequencing)

Stage A — Specification Stabilization

1. The v1.0 standard SHALL be treated as frozen canon once published.
2. Any proposed changes MUST be tracked under versioned delta documents with explicit redlines.
3. Canonical metrics remain confined to Part III across future versions.

Stage B — Reference Implementations

1. At least one open-reference Symbiosis Engine implementation MUST exist per major deployment class (cloud, hybrid, on-prem).
2. Reference implementations SHALL conform to the 10-stage Engine Algorithm v2 and Governance Kernel v2 specifications.
3. Each reference implementation MUST ship with replayable demo traces for audit training.

Stage C — Benchmarks and Evaluation Suites

1. The field MUST standardize benchmark suites for:
 - SI stability under drift stress,
 - HSS/SCP learning curves under training interventions,
 - Economic Circuit accuracy under load, and
 - drift vs decay detection recall/precision.
2. Benchmarks SHALL be sector-neutral at core and sector-overlaid via playbooks.

Stage D — Consortium Formation

1. A Symbiosis Consortium SHOULD be established to:
 - host canonical ontology and metric stewardship,
 - maintain cross-vendor MCP contract schemas, and
 - adjudicate invariants drift.
2. Consortium membership MAY include regulators, enterprises, vendors, and academic labs, but governance voting MUST preserve non-vendor dominance.

Stage E — Regulatory and Standards Engagement

1. Deployers and consortium bodies MUST maintain a living standards crosswalk (see Appendix R) mapping Symbiosis primitives to ISO/NIST/EU AI Act style requirements.
2. Regulator engagement SHALL use Policy Brief artifacts (see §15.5) to prevent interpretive fragmentation.

Normative rule: A deployment MAY claim “field-aligned Symbiosis” only if Stage A is met and Stage B is in progress with public artifacts.

15.5 Canonical artifacts (adoption and external coherence)

15.5.1 General rule

All external-facing materials **MUST** remain tethered to Symbiosis ontology, invariants, architecture, and canonical metrics. Custom metaphors that contradict canon are forbidden.

15.5.2 Executive Summary (Boardroom View)

Each deployment **MUST** maintain a 1–2 page Executive Summary for Phase 0–1 alignment that includes:

- Problem and risk exposure (Shadow AI → liability).
- Symbiosis as the operating-model remedy (Four-Layer Stack + referenced metrics).
- Economic case (cost control via Economic Circuit).
- Adoption roadmap (phases + risk gates).
- Explicit decision ask (resources, scope, governance charter).

15.5.3 Academic Abstract / Paper Template

To seed common terminology in research contexts, the deployer **SHOULD** produce an abstract/paper aligned to:

- Ontology and invariants.
- Formal contributions (indices, state machines, stability models).
- Evaluation method (SI trajectories and HSS/SCP learning curves).
- Open problems and research agenda (refer to Chapter 18 / Epilogue).

15.5.4 Policy Brief Template

For regulator and professional-body engagement, the deployer **MUST** use a Policy Brief containing:

- Jurisdiction and sector mapping.
- Constraint Vector translation into policy language.
- RB placements and accountability model.
- Audit, incident, and lineage doctrine.

15.5.5 Diagram Pack & Visual Canon

Deployments **MUST** ship a diagram pack expressing:

- Four-Layer Stack (baseline + any sector overlays).
- AB/RB boundary diagrams.
- SI/HSS/SCP flow and feedback loops (reference only).
- Recovery Ladder and safe-mode diagrams.

15.5.6 Sector-Specific Mini-Briefs

For Phase 1–2 scale-out, deployers **MUST** provide mini-briefs per vertical that include:

- sector mapping of the four layers,
- 3–5 reference workflows,
- sector RB stance and AB ceilings,
- sector interpretation notes for canonical metrics,
- tailored adoption roadmap,
- explicit sector failure modes and guardrails.

Normative rule: Mini-briefs SHALL NOT rename core entities or alter invariants.

15.6 Coupling to literacy, roles, and certification

15.6.1 Literacy is a safety prerequisite

Because human capability constrains safe autonomy, implementation **MUST** include:

- a literacy curriculum tied to the active AB ceiling,
- supervised progression pathways, and
- recertification triggers after incidents or long inactivity.

Normative rule: AB2+ privileges SHALL NOT be granted to roles lacking certification under Chapter 16/17 protocols.

15.6.2 Non-punitive enforcement

HSS/SCP monitoring **MUST** remain diagnostic and non-punitive. All dashboards or reporting **MUST** avoid comparative rankings and must be aggregation-safe.

15.7 Adoption bridge governance linkage (reference only)

15.7.1 Adoption Bridges

Adoption Bridges allow bounded acceleration under controlled conditions. Their formal specification, admissibility criteria, and revocation mechanics are canonical in Chapter 8.

Normative rule: Any bridge activation **MUST** be logged as a governance-signed lineage event and **MUST** include an AB ceiling and monitoring window.

15.7.2 Global Bridge Lock / Bridge Collapse Protocol

The Bridge Collapse Protocol and Global Bridge Lock are canonical enforcement mechanisms defined in Chapter 8 and the Governance Kernel v2 specification. This chapter references them for implementation planning only.

Normative rule: If a bridge collapse event is triggered, implementation **MUST** revert to safe mode in all bridged domains, demote autonomy to AB0 globally, and require re-certification before any bridge reactivation.

Chapter close — Implementation Integrity

If Chapter 15 is followed as written, deployment proceeds via earned autonomy, cultural and literacy readiness are treated as control surfaces, and external communications remain tethered to a single canon, preventing field fragmentation as Symbiosis scales.

SQC Gate Check (Definition of Done)

Gate 1 — Layer Integrity: **PASS**. Requirements map cleanly to Human/Governance/Engine responsibilities without boundary bypass.

Gate 2 — Canonical Metrics Rule: **PASS**. SI/HSS v2.0/SCP/CI/RB/AB are referenced only; no redefinition or reweighting occurs.

Gate 3 — Constraint Precedence + Threshold Explicitness: **PASS**. Constraints and governance-set thresholds are treated as hard preconditions; no implicit defaults added.

Gate 4 — Mechanistic Testability: **PASS**. Every normative rule is checkable via artifacts, lineage events, or governance decisions.

Gate 5 — Lineage Recoverability: **PASS**. Promotion evidence and incident handling explicitly require lineage minima and replayability.

Gate 6 — Drift vs Decay Typing: **PASS**. Implementation doctrine references distinct drift/decay monitors without conflation.

Gate 7 — Abuse / Incentive Hardening: **PASS**. Non-punitive rules, anti-leaderboard constraints, and labor safeguards are preserved.

Gate 8 — Normative Voice Consistency: **PASS**. MUST/SHALL/MAY used precisely; no speculative or marketing tone.

Chapter 16 — Tiered Adoption Phases (1–4)

Status: Normative Adoption Ladder (v1.0)

16.0 Purpose

Symbiosis deployments SHALL progress through a controlled, evidence-gated adoption ladder. The ladder defines four phases of operational maturity. Each phase specifies:

1. admissible autonomy regimes,
2. required governance posture and artifacts,
3. minimum telemetry and lineage fields,
4. readiness criteria for promotion, and
5. demotion triggers.

The ladder is a safety and scalability mechanism. It SHALL be used to prevent “shadow AI” adoption, stealth autonomy creep, and governance bypass. Promotion between phases is not automatic; it is a Governance act authorized under Chapter 12 and Chapter 15 rules.

Canonical metric definitions (SI, HSS v2.0, SCP, CI, RB v2, AB schema) SHALL NOT be redefined here and are referenced from Part III.

16.1 Scope and Audience

16.1.1 Scope

This chapter applies to any organization or program deploying the Symbiosis Framework, regardless of sector. Sector-specific parameter overlays SHALL be applied per Chapter 12 playbooks.

16.1.2 Audience

Primary users of this chapter SHALL be:

- Governance Leads and Policy owners,
- Symbiosis Architects and Engine operators,
- Program managers responsible for rollout pacing, and
- Auditors assessing maturity and safety posture.

16.2 Adoption Ladder Overview

The four phases are:

- Phase 1 — Assisted (Tool Use)
- Phase 2 — Augmented (Engine Adoption)
- Phase 3 — Partnered (Bridges & Safe Harbors)
- Phase 4 — Emergent (High-Autonomy Pockets)

These phases are derived from the v0.9 maturation trajectory and tightened into a formal ladder.

A Phase 5 target state (“Coordinated Hybrid Intelligence”) may be referenced for strategic orientation, but SHALL NOT be treated as a near-term adoption tier or a readiness gate in v1.0.

16.3 Telemetry and Lineage Minimums (All Phases)

Regardless of phase, every Symbiosis-enabled workflow SHALL emit lineage sufficient to reconstruct:

- Intent object reference (I),
- Constraint Vector reference (C),
- Task Graph root and node IDs (TG),
- Autonomy Band and Regret Boundary state at execution time (AB/RB), and
- Economic Pass outcome and budget attribution.

This is a hard requirement for auditability and drift/decay diagnosis.

Telemetry minima SHALL include, at a minimum:

Field	Description	Owner
-------	-------------	-------

intent_id	immutable identifier of governing intent	Human Layer
constraint_vector_id	pinned constraint policy version	Governance Layer
tg_id / node_id	task graph lineage anchors	Engine
ab_state	AB0–AB3 state at node execution	Governance Kernel
rb_state	RB proximity or class at node execution	Governance Kernel
economic_pass	approve/deny/defer + cost attribution	Engine (Economic Circuit)
si_episode / si_rolling	SI values per canonical SI chapter	Engine telemetry
hss_operator	operator HSS score snapshot	Human Layer telemetry
scp_operator	operator SCP snapshot	Human Layer telemetry
drift_or_decay_tag	semantic drift vs operational decay	Governance Kernel

Where phase rules require additional fields, they SHALL be added in the applicable section below.

16.4 Phase 1 — Assisted (Tool Use)

16.4.1 Definition

Phase 1 is characterized by human-led work with AI assistance operating under AB0. AI is treated as a tool, not a delegated agent. Shadow AI risk is highest in this phase.

16.4.2 Admissible Autonomy

- Default AB ceiling: AB0.
- AB1 MAY be permitted only for internal, reversible tasks explicitly typed as low-regret and below RB, per sector stance.
- Adoption Bridges SHALL NOT be active.

16.4.3 Required Governance Posture

Phase 1 deployments SHALL implement:

1. a pinned Governance Kernel (Chapter 12) with constraint enforcement active,
2. a baseline RB/AB policy (Chapter 11), even if AB0-only,
3. Economic Circuit enforcement (Appendix B / Chapter 13), and
4. mandatory intent intake discipline (Chapter 4 and Chapter 5).

16.4.4 Required Artifacts

Phase 1 SHALL produce:

- Executive Summary adoption brief (Chapter 15 artifact),
- Initial sector playbook selection (Chapter 12),
- Baseline intent library and Safe Harbor candidate list (Governance-owned),
- Baseline SI tracking instrumentation enabled,
- Baseline HSS/SCP collection for pilot roles.

16.4.5 Readiness Criteria for Promotion to Phase 2

Promotion from Phase 1 to Phase 2 SHALL require evidence of:

1. **Constraint hygiene:**
 - No recorded constraint breaches in pilot window W.
 - All constraints are version-pinned and auditable.
2. **Telemetry sufficiency:**
 - Phase 1 telemetry minima present for ≥95% of pilot workflows.
3. **Economic viability:**
 - Economic Pass active for all workflows and producing deny/defer actions when required.
4. **Operator baseline readiness:**
 - Pilot operators meet Governance-defined minimum HSS and SCP floors for AB1 eligibility.

5. **SI stability:**

- SI_rolling within Governance-defined green or yellow bands for W; no red-band SI episodes.

16.4.6 Demotion Triggers

Any of the following SHALL halt Phase 1 expansion and reset pilot scope:

- repeated SI red-band episodes,
- systemic telemetry gaps preventing lineage replay,
- verified HSS misuse attempt (HR or managerial secondary use),
- regulatory or sector stance change elevating risk class.

16.5 Phase 2 — Augmented (Engine Adoption)

16.5.1 Definition

Phase 2 begins when the Symbiosis Engine is deployed as a coordination OS rather than a chat interface. Task Graph synthesis, memory/lineage governance, and Economic Circuit gating are operational.

16.5.2 Admissible Autonomy

- Default AB ceiling: AB1 for reversible tasks below RB.
- AB2 MAY be permitted for bounded, low-regret safe harbors once promotion prerequisites are met.
- Bridges remain inactive unless Phase 3 gates are satisfied.

16.5.3 Required Governance Posture

Phase 2 SHALL implement:

1. Engine Algorithm v2 with mandatory Dry-Run gating for AB2+ or RB-proximate tasks (Chapter 13),
2. CI routing activated for tasks with heterogeneous capability/risk profiles (Chapter 10),
3. Drift vs Decay typing enabled (Chapter 14 + Appendices F/G),
4. Explicit CS/QL thresholds configured by governance for local compute awareness (Appendix B), and
5. AB promotion/demotion rules enforced as governance actions (Chapter 11).

16.5.4 Required Artifacts

Phase 2 SHALL produce:

- Task Graph templates for pilot workflows (Appendix D),
- MCP contract schema adoption for all agents/tools in scope (Appendix E / Chapter 10),
- Pilot incident runbooks aligned with drift/decay split (Appendix O),
- Scalability plan for operator and architect roles (see Chapter 17 readiness).

16.5.5 Readiness Criteria for Promotion to Phase 3

Promotion to Phase 3 SHALL require:

1. **Stable Engine operation:**
 - Engine Algorithm v2 stages operational.
 - Dry-Run evidence logged where required.
2. **SI + Vital Sign stability:**
 - SI_rolling green in W.
 - Drift, resonance, divergence, entanglement within green bands for W.
3. **Operator readiness for Bridges:**
 - HSS tier supports AB2 operation for in-scope roles.
 - SCP shows adequate load headroom for AB2 concurrency.
4. **Safe Harbor definition complete:**
 - Governance-approved Safe Harbors exist with explicit boundaries, constraints, and rollback models.
5. **Telemetry completeness:**
 - All Phase 2 minimum fields plus vital signs and Dry-Run logs present for ≥98% of AB1/AB2 nodes.

16.5.6 Demotion Triggers

Phase 2 programs SHALL demote to Phase 1 posture if:

- Engine cannot maintain lineage recoverability,
- Economic Pass is bypassed or systematically overridden,
- Drift or entanglement enters red band without recovery conformance, or
- repeated SCP anomalies indicate operator saturation.

16.6 Phase 3 — Partnered (Bridges & Safe Harbors)

16.6.1 Definition

Phase 3 activates Adoption Bridges to create safe, high-velocity lanes for low-regret work while maintaining strict governance. AB2 operation becomes standard inside Safe Harbors.

16.6.2 Admissible Autonomy

- Default AB ceiling: AB2 within Safe Harbors.
- AB3 MAY be authorized only for specific bounded pockets meeting Chapter 11 promotion prerequisites.
- Bridges SHALL auto-revoke on any red-band trigger per Chapter 15.

16.6.3 Required Governance Posture

Phase 3 SHALL implement:

1. Adoption Bridges with full admissibility checks and auto-revocation (Chapter 15),
2. Global Bridge Lock and Bridge Collapse Protocol hooks (Chapter 15),
3. AB2/AB3 promotion evidence requirements (Chapter 11),
4. Audit-grade lineage retention and replay procedures (Chapter 12 + Appendix O), and
5. Continuous operator recalibration and anti-gaming enforcement for HSS (Chapter 8 + Appendix H extended support).

16.6.4 Required Artifacts

Phase 3 SHALL produce:

- Safe Harbor registry with unique IDs, scope, and AB ceilings,
- Bridge configuration manifests per Safe Harbor,
- Quarterly governance reviews producing AB promotion/demotion recommendations,
- Public-facing sector mini-briefs consistent with canonical ontology (Chapter 15 artifact).

16.6.5 Readiness Criteria for Promotion to Phase 4

Promotion to Phase 4 SHALL require:

1. **Bridge stability:**
 - No Bridge Collapse events in W.
 - Bridge revocations rare and explainable, not systemic.
2. **High-trust SI regime:**
 - SI_rolling green; positive resonance stable.
 - No negative-resonance cascades requiring global resets in W.
3. **AB3 pocket readiness:**
 - Explicit AB3 candidate domains with reversibility models.
 - Operators for these pockets meet HSS thresholds for RB-proximate tasks.
4. **Entanglement management:**
 - E_blast within governance green band even under multi-harbor concurrency.
5. **Telemetry and audit readiness:**
 - Full Phase 3 telemetry; Bridge execution paths reconstructable end-to-end within mandated retention.

16.6.6 Demotion Triggers

Any of the following SHALL cause immediate Phase 3 demotion to Phase 2 posture for affected domains:

- Bridge Collapse event (S3+ incident) triggering Global Bridge Lock,

- sustained drift or entanglement red-band episodes,
- SI red-band episodes inside Safe Harbors,
- verified HSS/SCP anti-gaming failures not remediated within governance window.

16.7 Phase 4 — Emergent (High-Autonomy Pockets)

16.7.1 Definition

Phase 4 denotes emergent high-autonomy pockets, with AB2.5–AB3 operation in bounded domains. Divergence and entanglement risks rise sharply; governance must shift to slow-core stabilization and strict interface control.

16.7.2 Admissible Autonomy

- Default AB ceiling remains AB2 outside pockets.
- AB3 may be active only inside Governance-defined Safe Harbors with AB3 authorization.
- Cross-pocket AB3 coupling SHALL NOT be permitted without explicit entanglement management approval.

16.7.3 Required Governance Posture

Phase 4 SHALL implement:

1. A slow-core governance cadence for policy, kernel, and model updates (Chapter 12),
2. Formal divergence and entanglement suppression controls as prerequisites to any AB3 expansion (Chapter 11),
3. Mandatory Dry-Run and reversibility modeling for RB-proximate tasks (Chapter 13 + Chapter 11),
4. Systemic SCP throttling and cooldown modes to prevent operator overload (Chapter 9 + Appendix I), and
5. Dedicated incident drills for AB3 pockets (Appendix O), including rollback and Safe Harbor quarantine procedures.

16.7.4 Required Artifacts

Phase 4 SHALL produce:

- AB3 pocket dossiers containing: domain scope, RB stance, reversibility models, trigger thresholds, and responsible governance authorizer.
- Entanglement maps showing shared MCP endpoints, shared constraint sets, and shared canonical memory across pockets.
- Annual external audit packets aligned to Chapter 17 certification protocol.

16.7.5 Ongoing Requirements (Phase 4 Maintenance)

Phase 4 is not a terminal steady state. To remain in Phase 4, the deployment SHALL satisfy:

- SI_rolling green in all AB3 pockets,
- zero unresolved red-band entanglement events,

- HSS/SCP recalibration at governance cadence, and
- no sustained queue latency or compute saturation beyond governance limits.

Failure to maintain these conditions SHALL invoke demotion to Phase 3 for affected pockets until stability is re-established.

16.8 Phase Transition Summary Table

Transition	Governance Act Required	Minimum Evidence Bundle
Phase 1 → Phase 2	YES	Constraint hygiene + lineage sufficiency + Economic Pass conformance + baseline HSS/SCP floors + SI stability
Phase 2 → Phase 3	YES	Engine v2 conformance + vital-sign stability + Safe Harbor registry + Bridge-ready HSS/SCP + telemetry completeness
Phase 3 → Phase 4	YES	Bridge stability + SI high-trust regime + AB3 pocket dossiers + entanglement control + audit-ready lineage
Any → prior phase	AUTO or YES	Triggered by SI red-band, drift/decay red-band, SCP breach, Bridge Collapse, or telemetry failure

16.9 Implementation Notes and Cross-References

1. Sector tuning SHALL be applied via Chapter 12 playbooks.
2. All autonomy promotions within phases SHALL follow Chapter 11 promotion prerequisites and logging rules.
3. Adoption Bridges and collapse mechanics SHALL follow Chapter 15.

4. Validation and certification mechanics are defined in Chapter 17 and SHALL be used as promotion evidence for Phases 3–4.

Chapter Close — Adoption Integrity Check

If this chapter is followed as written, then:

- Adoption proceeds by earned autonomy, not enthusiasm.
 - Telemetry, lineage, and governance readiness are treated as non-negotiable prerequisites.
 - Stability and capability limits are enforced before autonomy expansion.
 - Sector differences are handled through parameterization, not ontology drift.
-

SQC Gate Check (Chapter 16)

1. Layer Integrity — PASS
 - Four-layer roles are respected; no cross-assignment or boundary bypass.
 2. Canonical Metrics Rule — PASS
 - SI/HSS/SCP/CI/RB/AB are referenced only; no redefinition or reweighting.
 3. Constraint Precedence — PASS
 - Hard constraints and governance-set thresholds explicitly gate phase transitions.
 4. Mechanistic Testability — PASS
 - Promotion/demotion criteria are stated as checkable conditions with evidence bundles.
 5. Lineage Recoverability — PASS
 - Telemetry minima and reconstruction requirements are explicit and phase-invariant.
 6. Drift vs Decay Typing — PASS
 - Demotion triggers and telemetry require correct drift/decay tags.
 7. Abuse / Incentive Hardening — PASS
 - HSS secondary-use misuse is treated as a demotion trigger; anti-gaming referenced.
 8. Normative Voice Consistency — PASS
 - Uses MUST/SHALL/MAY precisely; no speculative or marketing tone.
-

Chapter 17 — Validation & Certification Protocol

Status: Normative (v1.0 NEW)

17.0 Purpose

This chapter specifies the mandatory validation and certification protocol required to:

1. certify Symbiote Operators, Hybrid Intelligence Architects, and Governance Leads for autonomy rights,
2. validate that SI, HSS v2.0, SCP, and vital-sign control surfaces are calibrated, non-gameable, and operationally predictive, and
3. produce an auditable Certification Evidence Packet (CEP) suitable for internal boards and external regulators.

Certification is a **governance act**, not a managerial convenience. It SHALL be used to authorize or restrict autonomy bands, safe harbors, and adoption bridges in accordance with Chapter 11 (AB/RB rules), Chapter 15 (Bridge admissibility/revocation), and Chapter 16 (adoption phases).

Canonical definitions of SI, HSS v2.0, SCP, CI, RB, and AB schema SHALL NOT be redefined here. This chapter references their canonical Part III definitions only.

17.1 Scope

17.1.1 In-scope entities

- **Human Layer roles:** Symbiote Operators, Hybrid Intelligence Architects, Governance Leads.
- **Engine and Agentic systems** that claim compliance with Symbiosis v1.0.
- **Governance Layer bodies** responsible for autonomy authorization and incident disposition.

17.1.2 Out-of-scope entities

- Sector playbook construction (see Chapter 12).
- Metric construction or math (see Part III and Appendices B/H/I/G/F).
- Incident taxonomies beyond certification coupling (see Chapter 14 and Appendix O).

17.2 Certification Ladder and Rights Coupling

17.2.1 Principles

1. **Earned autonomy:** no actor or domain may move to higher AB without certification matching the band rights.
2. **Dual-gate rule:** SI green band is necessary but not sufficient; HSS readiness and SCP headroom gates SHALL be met before any $AB \geq 2$ rights are granted.
3. **Stability precedence:** if stability vital signs disagree with SI, stability SHALL control certification outcomes (no promotion).

17.2.2 Certification tiers

Certification tiers are tied to the AB bands they authorize.

Certification Tier	Authorized AB Ceiling	Purpose
Cert-0 (Foundational)	AB0	Safe tool use under full human approval
Cert-1 (Internal Delegate)	AB1	Internal reversible delegation within constraints
Cert-2 (Conditional Delegate)	AB2	Conditional autonomy inside Safe Harbors with checkpoints
Cert-3 (Co-Agency Pocket)	AB3	High-autonomy pockets within RC-approved reversibility models

17.2.3 Rights coupling rule (normative)

An actor MAY operate at AB_k only if:

1. Governance has issued a valid certificate of tier $\geq k$,
2. the actor's current HSS tier supports AB_k eligibility (Chapter 8 + Appendix H), and
3. SCP indicators show headroom for AB_k concurrency (Chapter 9 + Appendix I).

Violation of any condition SHALL trigger immediate AB demotion and recertification review.

17.3 Validation Pillars

Certification SHALL be based on three concurrent validation pillars:

- A. **Operator validation** (human competency, HSS/SCP grounded)
- B. **System validation** (Engine/Agentic conformance, lineage recoverability)
- C. **Metric validation** (SI/HSS/SCP anti-gaming, calibration, predictiveness)

All three pillars MUST pass for certification issuance.

17.4 Mandatory Pilot Sequence (P0 / P1 / P2)

17.4.1 Overview

Any deployment claiming Symbiosis compliance and any actor seeking Cert-1+ SHALL complete three pilots, in order:

- **P0 — Baseline Assisted Pilot** (AB0 only)
- **P1 — Engine-Augmented Pilot** (AB1 standard; AB2 in narrow trials only)
- **P2 — Bridge-Ready Pilot** (AB2 within Safe Harbors; AB3 trials prohibited unless explicitly authorized)

These pilots produce the evidence base for certification and for domain adoption gating (Chapter 16).

17.4.2 P0 — Baseline Assisted Pilot

Objective: establish baseline distributions for SI, HSS, SCP, and stability signs under AB0.

Requirements:

1. AB ceiling fixed to AB0.
2. All tasks executed through normalized Intent Objects with constraint seeds and budget envelopes.
3. Engine lineage minima MUST be emitted for every episode.
4. Baseline HSS/SCP collection MUST be activated for in-scope operators.
5. Drift/decay tagging MUST be functional, even if rare at AB0.

Minimum duration: Governance SHALL define a pilot window $W0$ sufficient to yield $\geq N0$ episodes per role.

Pass criteria:

- SI baseline distribution established and stored per sector playbook.
- $\geq 95\%$ lineage completeness across episodes.
- No recurring constraint breaches.

17.4.3 P1 — Engine-Augmented Pilot

Objective: validate Engine Algorithm v2 conformance, Economic Circuit v2 gating, and AB1 discipline; test AB2 readiness signals without granting AB2 rights.

Requirements:

1. Engine Algorithm v2 stages MUST be active, including Economic Pass v2 and Dry-Run gating for AB2+/RB-prox tasks.
2. AB ceiling AB1 for all production execution.
3. AB2 MAY be trialed only in explicitly bounded Safe Harbor sandboxes with Dry-Run and rollback conformance.
4. CI routing and MCP binding MUST be operational for in-scope TG nodes.
5. Stability vital signs MUST be continuously computed.

Minimum duration: $W1$ yielding $\geq N1$ episodes per role and $\geq M1$ TG nodes per prototype Safe Harbor.

Pass criteria:

- Dry-Run previews issued and acknowledged where required.
- Economic Pass denies/defers observed when budgets or CS/QL thresholds are exceeded.
- SI_rolling enters green band stability for W1.
- No red-band stability excursions.
- HSS/SCP show improvement or stability, not depletion.

17.4.4 P2 — Bridge-Ready Pilot

Objective: validate that Safe Harbors, Bridges, and AB2 operations are stable and non-gameable prior to broad Phase-3 adoption.

Requirements:

1. Governance-approved Safe Harbor registry in force (Chapter 15).
2. AB2 execution permitted only inside Safe Harbors.
3. Bridges MAY be active only inside Safe Harbors that meet admissibility criteria.
4. All Bridge auto-revocation hooks MUST be verified through controlled drills.
5. Operator HSS/SCP tiering MUST be used as a live gate.

Minimum duration: W2 yielding $\geq N2$ AB2 episodes per Safe Harbor and $\geq K2$ Bridge executions.

Pass criteria:

- Bridge admissibility passes at runtime; no untyped domains.
- Any red-band SI or stability event triggers deterministic revocation/demotion behaviors.
- Entanglement indicators remain $\leq$ Governance green band under multi-harbor concurrency.
- Anti-gaming tests (Section 17.6) pass.

17.5 Certification Evidence Packet (CEP)**17.5.1 Purpose**

The CEP is the single audit object that supports certification issuance, renewal, and regulator review.

17.5.2 CEP minimum contents (normative)

The CEP MUST include:

1. Role dossier

- role (Operator / Architect / Governance Lead)
- sector and domain scope
- target certification tier

2. Pilot evidence slices

- P0, P1, P2 summaries
- episode counts (N0/N1/N2)
- Safe Harbor IDs and scope

3. Metric stability bundle

- SI baseline and rolling distributions
- HSS tier trajectory and current band snapshot
- SCP headroom trajectory and breach history
- vital signs (drift, divergence, resonance, entanglement) banding

4. Anti-gaming and integrity proofs

- SI lineage-suppression checks
- HSS anti-gaming suite pass log
- SCP saturation-avoidance conformance

5. Incident and recovery conformance

- any S0–S4 incidents during pilots
- rollback and safe-mode drill results (Appendix O)

6. Governance authorization record

- authorizer identity
- effective date
- review cadence
- AB ceiling granted
- revocation conditions

17.5.3 Lineage anchoring rule

Every CEP MUST be anchored to immutable lineage hashes that allow replay from:

Intent → Constraints → TG → RB/AB state → Economic Pass → Execution telemetry.

Absence of replayable lineage SHALL void certification.

17.6 Metric Validation and Anti-Gaming Suite

17.6.1 SI validation

Before any AB3 authorization or Phase-4 pocket deployment:

1. SI SHALL be compared to **human expert ratings** on a stratified sample.
2. SI SHALL be compared to **objective outcome KPIs**.
3. SI SHALL be stress-tested under simulated drift/resonance scenarios.

Failure of any SI validation SHALL disallow AB3 until mitigations are proven.

17.6.2 HSS v2.0 anti-gaming validation

HSS anti-gaming tests defined in Appendix H MUST be executed in P1 and P2. A certification candidate fails if any test detects:

- evidence of rubber-stamp approvals under load,
- systematic suppression of interrupts or checkpoint bypassing,
- manipulation of intent clarity signals, or
- attempts to route HSS visibility into HR/line management channels.

Any verified attempt at HSS gaming SHALL be treated as an A-Failure (governance defect) and triggers AB demotion and mandatory retraining.

17.6.3 SCP calibration and overload conformance

SCP instruments and thresholds defined in Appendix I MUST be calibrated per domain. Certification SHALL fail if:

- SCP chronic breach is observed during pilots,
- interrupt tolerance declines without Engine throttle response, or
- $AB \geq 2$ concurrency is attempted in the absence of SCP headroom evidence.

17.7 Recertification Cadence and Expiry

17.7.1 Default cadence

- Cert-0: renewal every 12 months
- Cert-1: renewal every 9 months
- Cert-2: renewal every 6 months
- Cert-3: renewal every 3 months plus post-incident mandatory review

Sectors MAY tighten cadence via Chapter 12 playbooks; they SHALL NOT loosen it without explicit regulator alignment.

17.7.2 Automatic expiry conditions

A certificate SHALL automatically expire and revert the actor to Cert-0 rights if:

- SI red-band episodes occur in W,
- stability vital signs enter red band in W,
- SCP breach exceeds governance maximum, or
- any S3+ incident invokes Global Bridge Lock (Chapter 15).

17.8 External Regulator Packet (optional but recommended)

Where sector stance requires external oversight, Governance SHOULD produce a Regulator Packet derived from the CEP, containing:

- sector risk/admissibility mapping,
- Safe Harbor definitions and AB ceilings,
- SI/HSS/SCP calibration summaries,
- incident doctrine and rollback evidence,

- lineage recoverability guarantee.

This packet SHALL use the Policy Brief Template in Chapter 15 and align to Appendix R standards crosswalk.

17.9 Implementation Checklist (Minimum Compliance)

An organization may claim “Symbiosis Validation & Certification Protocol compliance” only if:

- ☐ P0/P1/P2 pilots completed in order with required AB ceilings.
 - ☐ CEP generated per Section 17.5, lineage-anchored, and governance-signed.
 - ☐ SI validated for AB3 candidates per Section 17.6.1.
 - ☐ HSS anti-gaming suite executed and passed in P1/P2.
 - ☐ SCP instruments calibrated; no chronic breach during pilots.
 - ☐ Recertification cadence configured and enforced.
 - ☐ Expiry triggers coupled to AB demotion automation.
-

SQC Gate Check (Chapter 17)

1. Layer Integrity — PASS
 - Certification framed as Governance act; Engine provides evidence only; Human primacy preserved.
2. Canonical Metrics Rule — PASS
 - No redefinition or reweighting of SI/HSS/SCP/CI/RB/AB; canonical Part III referenced only.
3. Constraint Precedence — PASS
 - Pilots and rights are blocked by constraint breaches, CS/QL gates, and risk-class admissibility.
4. Mechanistic Testability — PASS
 - Pilot sequence, pass criteria, and CEP contents are defined as checkable requirements.
5. Lineage Recoverability — PASS
 - CEP lineage anchoring rule explicit; certification void without replayability.
6. Drift vs Decay Typing — PASS
 - Stability vital signs and drift/decay tags are mandatory evidence; red-band triggers expiry.
7. Abuse / Incentive Hardening — PASS
 - Anti-gaming suite, HR misuse prohibition, and punitive-misuse triggers are explicit.

8. Normative Voice Consistency — PASS

- MUST/SHALL/MAY used precisely; no speculative or marketing tone.

Chapter 18 — Field Formation & Open-Source Kernel Stance Status: REWRITE/UPGRADE (v1.0)

18.0 Purpose

This chapter specifies the mandatory field-formation posture implied by the Symbiosis Framework and declares the framework's open-source kernel stance.

Where prior chapters define the architecture, primitives, metrics, and governance required to build safe hybrid intelligence systems now, Chapter 18 formalizes:

1. the research programs required to mature symbiosis into a stable scientific and institutional discipline, and
2. the normative boundary between **open, standardizable kernel components** and **domain-specific proprietary engines**.

This chapter is not a marketing roadmap. It is a governance-grade statement of what must be standardized, what may remain differentiated, and what research must be pursued to prevent field fragmentation. The baseline future-state agenda carries from v0.9, tightened and integrated into v1.0's adoption and governance posture.

Canonical definitions of SI, HSS v2.0, SCP, CI, RB v2, and AB schema remain frozen in Part III, and are referenced here only.

18.1 Scope and Outputs

18.1.1 In-scope

- Standards posture for field formation (what must be open, what may be proprietary).
- Research monographs required to close identified scientific and operational gaps.
- Institutional formation steps required to produce shared benchmarks, certifications, and regulator legibility.

18.1.2 Out-of-scope

- Any revision of canonical metrics or autonomy math.
- Sector configuration details (Chapter 12 playbooks) and deployment mechanics (Chapters 15–17).
- New invariants or ontology changes (Part I, Appendix A).

18.1.3 Outputs

This chapter SHALL be treated as specifying three field-level outputs:

1. **Kernel Declaration (KD)** — Section 18.2
2. **Research Stack Monographs (M1–M5)** — Section 18.3
3. **Field Formation Roadmap (FFR)** — Section 18.4

18.2 Open-Source Kernel Stance (Kernel Declaration)

18.2.1 Principle

The Symbiosis Framework is universal in architecture but specific in configuration. To prevent **vendor lock-in and interpretive drift**, the field requires a shared kernel that is open, testable, and regulator-legible. File A makes this stance mandatory in v1.0.

18.2.2 Mandatory open-kernel components

The following components SHALL be designated as **open kernel**. Implementations MAY vary, but the types, interfaces, and conformance tests SHALL remain stable across the field:

1. **Four-Layer Architecture typing** (Human / Engine / Agentic / Governance) and boundary responsibilities.
2. **Canonical primitives and object schemas**: Intent Object (I), Constraint Vector (C), Budget Envelope (B), Task Graph (TG), MCP contract fields, and Context Lineage minima.
3. **Canonical metrics and banding**: SI, HSS v2.0, SCP, CI, stability vital signs, RB v2 and AB0–AB3 state schema.
4. **Engine Algorithm v2 pipeline semantics** (10-stage sequence and required gates), including Economic Circuit v2 and Dry-Run rule.
5. **Governance Kernel v2 semantics**: constraint supremacy, risk-class → AB ceilings, Bridge admissibility / auto-revocation, Global Bridge Lock, and incident ladder mapping.
6. **Certification protocol semantics**: pilot sequence P0/P1/P2, evidence packet requirements, and rights coupling rules.
7. **Conformance test suite**: SQC gates as Definition of Done, plus Appendix P checklist structure.

18.2.3 Permitted proprietary components

The following components MAY be proprietary or differentiated, provided conformance to the open kernel is preserved:

1. **Engine internal optimizers and planners** (heuristic, LLM-driven, or hybrid), including orchestration strategies.
2. **Model choices and agent cluster inventories**, including multimodal systems, tool stacks, and retrieval layers.
3. **Sector overlays** and domain-specific constraint catalogs beyond core schema (Appendix K extensions).
4. **UI/UX alignment surfaces** that express Intent Objects and traces.

5. **Operational analytics and dashboards**, provided they do not alter canonical metric definitions.

18.2.4 Non-fragmentation rule

Any implementation claiming Symbiosis v1.0 compliance **MUST**:

1. export or surface the open-kernel objects as defined, and
2. pass the conformance tests without translation into non-canonical metaphors.

A system **MAY** claim performance superiority, but **SHALL NOT** claim compliance if it:

- replaces kernel schemas,
- redefines metrics, or
- bypasses governance semantics.

18.3 Future-State Research Stack (Monographs M1–M5)

18.3.1 Orientation

The Symbiosis Framework introduces three non-trivial claims that force a research agenda beyond classical AI evaluation and generic “ethics”:

1. **Hybrid intelligence is a first-class system type.**
2. **Governance is an architectural layer, not a policy appendix.**
3. **Humans co-evolve as operators with measurable capacities.**

Therefore, field maturity requires formal research programs seeded directly by the ontology, four-layer design, and SI/HSS/SCP control logic.

18.3.2 Monograph structure rule

Each monograph **SHALL** be pursued as a modular research program with:

- formal problem statements tethered to kernel primitives,
- measurable hypotheses grounded in SI/HSS/SCP trajectories and stability vital signs,
- reproducible benchmarks, and
- governance-safe experimental doctrine (AB/RB compliant).

18.3.3 M1 — Control Theory for Hybrid Cognitive Loops

Objective: formalize human–engine–agent feedback as a controllable dynamical system.

Required research outputs:

1. State-space representations for hybrid loops with explicit AB/RB transitions.
2. Stability conditions for multi-agent orchestration under constraint supremacy.
3. Formal controllers for Economic Circuit v2, including CS/QL cost saturation regimes.
4. Proofs for safe escalation/demotion dynamics under drift and decay.

M1 research **SHALL** deliver control-useful criteria extendable into Appendix B/C/F/G.

18.3.4 M2 — Human Co-Evolution Science

Objective: make operator capability growth an empirical discipline aligned to HSS v2.0 and SCP.

Required outputs:

1. Validated learning curve models for HSS and SCP across typology levels.
2. Calibration science for “capacity headroom” and interrupt tolerance.
3. Experimental protocols distinguishing skill gains from selection effects.
4. Anti-gaming behavioral detection models that preserve psychological safety.

M2 SHALL supply empirical grounding for Chapters 16–17 and Appendix H/I.

18.3.5 M3 — Metric Reliability and Causal Predictiveness

Objective: ensure canonical metrics remain truthful and predictive under real deployment incentives.

Required outputs:

1. SI causal validity studies (SI ↔ outcomes ↔ stability).
2. Cross-sector generalization bounds and failure envelopes.
3. Robustness against adversarial optimization or “metric theater.”
4. Formal drift/decay disentanglement benchmarks.

M3 SHALL harden Part III metrics and Appendix P/R conformance evidence.

18.3.6 M4 — Governance Engineering and Regulator Interfaces

Objective: move “governance as code” from concept to standardized institutional machinery.

Required outputs:

1. Standard policy compilation techniques (policy → Constraint Vector → GK rules).
2. Evidence minimality theorems for lineage-based audit.
3. Regret-boundary accountability models compatible with legal doctrines.
4. Standard incident adjudication templates aligned to S0–S4 and Bridge collapse logic.

M4 SHALL produce stable reference artifacts for Chapter 12, Chapter 14, Appendix K, Appendix O, and Appendix R.

18.3.7 M5 — Economics of Governed Autonomy

Objective: formalize economic viability as a safety constraint and adoption gate.

Required outputs:

1. Cost-benefit models for autonomy expansion vs. human attention budgets.
2. Local compute saturation models (CS/QL) mapped to mission risk.

3. Economic early-warning indicators tied to SI and stability signs.
4. Benchmarks for “regret-adjusted value” per sector.

M5 SHALL continuously feed Appendix B and field adoption guidance.

18.4 Field Formation Roadmap (FFR)

18.4.1 Alignment to adoption posture

The field-formation steps below are cumulative and SHALL align to Chapter 15’s publishing artifacts and Chapter 17’s certification doctrine.

18.4.2 Roadmap stages

Stage F1 — Kernel finalization and reference implementation

- Publish open-kernel schemas, conformance suite, and a minimal reference Engine Algorithm v2 implementation.
- Provide canonical diagrams and narratives (Appendix L).
- Establish a public change-control log and versioning rules.

Stage F2 — Benchmark and dataset standardization

- Release stratified benchmark suites for SI reliability, HSS/SCP learning curves, drift/decay discrimination, and Economic Circuit v2 stress.
- Require benchmark anchoring to lineage traces.

Stage F3 — Training and licensure institutionalization

- Standardize Symbiosis Literacy Program tracks and certification exams.
- Require P0/P1/P2 pilots and CEP issuance as field-wide licensure preconditions.

Stage F4 — Consortium and regulator engagement

- Form a multisector standards consortium to ratify kernel stability and oversee change control.
- Produce annual standards crosswalk updates (Appendix R) and sector playbook templates.

Stage F5 — Continuous co-evolution loop

- Maintain slow-core kernel evolution based on validated monograph outputs and audit findings.
- Prohibit hot-patching of kernel rules without governance sign-off.

18.4.3 Indefinite stability rule

Once ratified, the open kernel SHALL remain stable across minor versions. Kernel evolution SHALL be:

- slow-core,

- evidence-based (monograph outputs + benchmark results), and
- backward-compatible unless explicit breakage is justified and regulator-aligned.

18.5 Chapter Close

If Chapter 18 is executed as written:

- the Symbiosis field remains unified around a shared kernel,
- vendors and institutions can innovate without fragmenting the standard,
- research advances remain tethered to deployable primitives, and
- regulators can rely on stable audit, licensure, and evidence semantics.

SQC Gate Check (Chapter 18)

1. Layer Integrity — PASS
 - Field formation and kernel stance are tethered to Four-Layer boundaries; no cross-layer role reassignment.
 2. Canonical Metrics Rule — PASS
 - SI/HSS/SCP/CI/RB/AB are referenced only; no redefinition or reweighting.
 3. Constraint Precedence — PASS
 - Open kernel includes constraint supremacy and GK semantics; non-fragmentation rule blocks bypass.
 4. Mechanistic Testability — PASS
 - Kernel components list, proprietary allowances, monograph outputs, and FFR stages are checkable.
 5. Lineage Recoverability — PASS
 - Benchmarks and CEP anchoring explicitly require lineage traces.
 6. Drift vs Decay Typing — PASS
 - Research stack and benchmarks require drift/decay discrimination; feeds Appendix F/G/O.
 7. Abuse / Incentive Hardening — PASS
 - Anti-gaming requirements and non-fragmentation rule address incentive risks.
 8. Normative Voice Consistency — PASS
 - MUST/SHALL/MAY used precisely; no speculative or marketing modality.
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Epilogue — The Silicon + Carbon Manifesto (v1.0 Closing Chapter)

Status: CARRY + TIGHTEN (baseline Epilogue carried forward from v0.9, upgraded to v1.0 normative voice and cross-references).

Placement: Final main-body chapter immediately before Part VI (Appendices).

Relationship: Appendix M is the technical annex that operationalizes and audits these commitments.

E.0 Purpose

1. This Epilogue states the binding commitments required for any deployment to remain symbiotic under v1.0.
 2. These commitments are not aspirational. They SHALL be treated as externalized invariants that inform architecture, governance, and audit posture.
 3. A deployment MAY NOT claim adherence to the Silicon + Carbon Manifesto unless it also satisfies the corresponding operational obligations in Appendix M and the Symbiosis Compliance Checklist in Appendix P.
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E.1 Commitment Structure

1. Commitments are grouped into three classes:
 - a. Human Commitments (E.2).
 - b. Institutional and Temporal Commitments (E.3).
 - c. Ethical and Civilizational Commitments (E.4).
 2. Each commitment binds all four layers. None may be selectively adopted.
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E.2 Human Commitments

We commit to a human future that scales capability without collapsing agency.

E.2.1 Human Primacy → Meaning Primacy

1. Humans SHALL remain the sole origin of sovereign intent.
2. Humans SHALL remain the final bearer of regret and accountability for hybrid episodes.
3. Systems that originate, mutate, or pursue goals outside bounded human intent are non-symbiotic by definition.

E.2.2 Operators are pilots, not passengers

1. Symbiosis systems SHALL be designed to widen human horizons rather than substitute them.
2. All autonomy SHALL be earned through stability and literacy, not granted by default.

3. Human override and halt rights SHALL be continuous, frictionless, and unconditional.

E.2.3 HSS and SCP are safety telemetry

1. HSS v2.0 and SCP SHALL be used for operator self-governance, skill development, and autonomy pacing only.
2. HSS and SCP SHALL be firewalled from punitive managerial, HR, or labor-discipline use.
3. Any deployment that violates this firewall SHALL be treated as non-compliant and unsafe (see Appendix H, Appendix P).

E.2.4 Co-evolution is the default developmental path

1. Symbiosis deployments SHALL treat human–AI relationships as maturing systems that learn, stabilize, and advance through phases.
 2. Static, ungoverned human–AI coupling is a failure mode.
 3. Adoption SHALL proceed through tiered phases with explicit readiness gates (Chapter 16, Appendix N).
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E.3 Institutional and Temporal Commitments

We commit to a long-horizon institutional transition, not a short-horizon tool rush.

E.3.1 Symbiosis is a presingularity standard

1. Institutions SHALL recognize that dependency arrives gradually.
2. Engineering dependency safely now is mandatory; waiting for catastrophe is prohibited.

E.3.2 Governance must stay ahead of capability

1. Governance and policy compilation are lagging indicators.
2. Systems SHALL compensate for this lag through hard constraints, AB/RB gating, Economic Passes, and incident doctrine.

E.3.3 We build for jurisdictional reality

1. Sector overlays and compliance matrices SHALL be treated as first-class substrates for adoption.
 2. A deployment SHALL not operate above AB0 in regulated domains without a compiled sector CCMT (Appendix K) and certification pathway evidence (Chapter 17).
 3. Cross-jurisdictional deployment SHALL preserve invariants rather than fragment ethics by locale.
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E.4 Ethical Commitments and Civilizational Pattern

We commit to hybrid intelligence as a civilizational project.

E.4.1 Capability without legitimacy is harm

1. Symbiosis SHALL prioritize legitimacy through human authorization, explainability, and rights-preserving boundaries.
2. High-autonomy operation SHALL require local validation, calibration, and regulator alignment.

E.4.2 Power must be contestable

1. Symbiosis systems SHALL make decisions inspectable and reversible through Context Lineage and governance.
2. Any opaque or un-auditable path is prohibited in symbiotic deployments.

E.4.3 We choose the shape of co-agency

1. The future is branching. One branch is ungoverned autonomy and human atrophy. Another is coordinated hybrid intelligence where silicon expands the possible and carbon remains sovereign over meaning.
 2. Symbiosis is a deliberate commitment to the second branch.
 3. This commitment SHALL be enforced by architecture first, governance as code, and autonomy earned through stability.
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E.5 Epilogue Close

1. The Symbiosis Framework is the engineering standard for governable hybrid cognition.
 2. If adopted faithfully—architecture-first, constraint-supremacy, lineage-recoverable, and autonomy paced by stability—it enables mature co-agency rather than uncontrolled dependency.
 3. Organizations that sign this Manifesto SHALL treat Appendix M and Appendix P as mandatory operationalization and verification surfaces.
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SQC Gate Check (Epilogue)

Gate 1 — Layer Integrity: PASS

- Commitments bind all four layers, reinforce separation of concerns, and prohibit bypass.

Gate 2 — Canonical Metrics Non-Redefinition: PASS

- SI/HSS/SCP/CI/RB/AB are referenced only at the commitment level; no redefinition or weighting introduced.

Gate 3 — Constraint Precedence + Threshold Explicitness: PASS

- Human primacy, constraint supremacy, and governance-set thresholds are explicitly hard requirements.

Gate 4 — Mechanistic Testability: PASS

- Each commitment is operationalized to Appendix M/P/other chapters with auditable implications.

Gate 5 — Lineage Recoverability: PASS

- Contestability and auditability are explicitly tied to Context Lineage evidence.

Gate 6 — Drift vs Decay Typing: PASS

- No conflation introduced; stability and governance pacing are referenced consistently with Chapters 14–15 and Appendices F/G/O.

Gate 7 — Abuse / Incentive Hardening: PASS

- HSS/SCP firewall and non-punitive posture reiterated; no HR/managerial misuse vectors introduced.

Gate 8 — Normative Voice Consistency: PASS

- MUST/SHALL/MAY used precisely; no speculative or marketing tone.

APPENDIX A — ONTOLOGY & VOCABULARY (v1.x)

Status: Normative Supporting Appendix (CARRY + PRUNE)

Canonical Anchor: Chapter 2 (Ontological Axioms & Primitives) and Part III Metrics Canon

Scope: Cross-sector fixed vocabulary and type system for Symbiosis Framework v1.0+

Non-Redefinition Rule: All canonical metrics (SI, HSS v2.0, SCP, CI, RB v2, AB schema) are defined ONLY in Part III and SHALL NOT be redefined herein. This appendix references them solely as properties of entities.

A.0 Purpose and Usage

A.0.1 Purpose

This appendix defines the Symbiosis Ontology as a stable type system for hybrid human–AI collaboration. It SHALL be treated as the fixed vocabulary for interpreting, implementing, auditing, and extending Symbiosis Framework v1.0+.

A.0.2 Binding Force

1. All Parts, Chapters, and Appendices MUST use terms exactly as defined here.

2. Any new term introduced elsewhere MUST be reconciled into this ontology before it can be treated as canonical.
3. Deprecated terms listed in A.5 SHALL NOT be used in v1.0 text except in historical reconciliation notes.

A.0.3 Normative Use

When this appendix defines an entity or relationship, that definition SHALL govern all engineering, governance, validation, and certification uses of the framework.

A.1 Axioms (Irreducible Foundational Claims)

These axioms are invariants. A system that violates any axiom SHALL be classified as non-Symbiotic (Shadow AI or unsafe automation).

A.1.1 Human Primacy (HP)

1. The Human Agent (H) SHALL be the sole origin of meaning, goals, and value in Symbiosis systems.
2. No synthetic component MAY originate or mutate goals without explicit human authorization.

A.1.2 Constraint Supremacy (CSup)

1. Hard constraints (C) SHALL override all optimization goals, model preferences, or agent heuristics.
2. Any task that cannot satisfy constraints MUST be denied, deferred, or re-planned.
3. Constraints MUST be machine-enforceable and auditable through lineage.

A.1.3 Separation of Concerns (SoC)

1. The Four-Layer Architecture (Human / Coordination / Agentic / Governance) SHALL partition responsibilities.
2. No layer MAY perform duties allocated to another layer.
3. No control flow MAY bypass an adjacent layer boundary.

A.1.4 Lineage Recoverability (LR)

1. Every executed action MUST be reconstructable to Intent (I), Constraint Vector (C), Task Graph (TG), Autonomy state (AB/RB), and Economic Pass decision.
2. Systems without reconstructable lineage SHALL be classified as Shadow AI.

A.1.5 Deterministic Coordination (DC)

The Symbiosis Engine (E) SHALL be deterministic at the control-surface level: identical I + C + TG + AB/RB + policy inputs MUST yield identical gating decisions.

A.2 Core Entities (Type System)

This section defines first-class entities. Each entity **MUST** be explicitly representable in implementation artifacts.

A.2.1 Human Agent (H)

Definition: The biological operator holding sovereign meaning authority and final regret ownership.

Properties (referenced):

- HSS v2.0: human collaboration safety capability (canonical in Part III).
- SCP: human oversight load envelope (canonical in Part III).
- Regret Authority: right to approve, halt, rollback, or override.

Obligations:

- **MUST** author or approve Intent Objects.
- **MUST** receive transparency traces for AB2+ or RB-proximate tasks.

A.2.2 Alignment Surface (AS)

Definition: The interface regime through which H expresses intentions and receives system feedback.

Types:

- Low-Fidelity AS (natural language / informal prompts).
- High-Fidelity AS (structured intent, schemas, UI/DSL, programmatic contracts).

Obligations:

- High-stakes or high-autonomy workflows **MUST** require High-Fidelity AS.

A.2.3 Structured Intent Object (I)

Definition: A typed object expressing a human goal with explicit constraints and budget.

Minimum fields:

- goal_statement
- output_spec
- regret_tolerance
- constraint_seed C_0
- budget_envelope B
- identity / role of H

Obligations:

- I **MUST** exist prior to TG synthesis.
- Unstructured or implicit intent **SHALL** be normalized into I' by E.

A.2.4 Symbiosis Engine (E)

Definition: Deterministic coordination layer that transforms I into TG, enforces constraints, sets autonomy ceilings, binds MCP contracts, and commits lineage.

Obligations:

- **MUST** perform Economic Pass before execution.
- **MUST** enforce AB/RB ceilings from Governance.
- **MUST** write Context Lineage entries for all stages.

A.2.5 Task Graph (TG)

Definition: A typed, dependency-ordered graph representing executable work derived from I under C.

Structure:

- Nodes: typed tasks with inputs, outputs, estimated cost, risk, AB ceiling, RB class, checkpoints.
- Edges: ordering, data dependencies, and rollback relationships.
Obligations:
- TG MUST be lineage-addressable and replayable.
- TG nodes MUST not exceed AB ceilings.

A.2.6 Synthetic Agent (A)

Definition: A bounded capability module (model/tool) that executes TG nodes under MCP contracts.

Obligations:

- Agents MAY not alter I, C, or policy.
- Agents MUST emit telemetry required for lineage.

A.2.7 Machine Collaboration Protocol Contract (MCP)

Definition: A binding contract that assigns an Agent (or cluster) permissioned capabilities, cost caps, AB ceiling, and constraint vector for a TG node.

Obligations:

- MCP MUST be issued before execution.
- MCP MUST include revocation and safe-mode triggers.

A.2.8 Constraint Vector (C)

Definition: Ordered set of deployment-specific hard constraints injected by Governance and/or H.

Components:

- C_legal (law, regulation, jurisdiction)
- C_safety (risk policies, prohibited actions, escalation)
- C_economic (budget bounds, compute saturation, queue latency)
- C_operational (time windows, data boundaries, tool scopes)
Obligations:
- C MUST be evaluated at planning, binding, and execution.
- Conflicts MUST halt or re-plan, not be silently optimized away.

A.2.9 Governance Kernel (GK)

Definition: Governance-as-code subsystem that configures risk classes, AB/RB ceilings, constraint templates, health-check thresholds, and violation handling.

Obligations:

- GK MUST be authoritative on safety and policy.
- GK decisions MUST be lineage-logged.

A.2.10 Context Lineage (L)

Definition: Immutable record of intent, constraints, decisions, telemetry, and outputs enabling audit-grade reconstruction.

Obligations:

- Each stage MUST commit lineage with hashes.
 - L MUST support replay to reproduce decisions and outcomes.
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A.3 Deep Definitions (Theoretical Constructs)

A.3.1 Symbiosis System

A socio-technical system consisting of H, E, A, and GK operating through $I \rightarrow TG \rightarrow MCP \rightarrow \text{execution}$, with constraints supremacy and full lineage.

A.3.2 Hybrid Cognitive Loop (HCL)

A closed-loop interaction in which:

1. H expresses intent via AS into I.
2. E plans and coordinates TG.
3. A executes TG under MCP.
4. Telemetry updates lineage and informs H.
HCLs MUST be measurable through SI, and human safety readiness through HSS/SCP (Part III).

A.3.3 Shadow AI

Any AI-mediated workflow lacking at least one of:

1. active Constraint Vector (C),
2. reconstructable Context Lineage (L),
3. declared Autonomy state (AB/RB).

Shadow AI SHALL be treated as non-compliant and ineligible for certification.

A.3.4 Oversight Theater

A state in which humans are nominally “in the loop” but their SCP/HSS capacity is insufficient for the autonomy load, producing unsafe or non-auditable outcomes. Oversight Theater MUST trigger AB demotion or Bridge Collapse per Governance rules.

A.3.5 Regret

The human-owned cost of an outcome that is irreversible or misaligned with intent. Regret is not a metric here; it is an ontological property of H and RB (defined in Part III).

A.3.6 Co-Evolution

A controlled process where human intent formation, engine coordination, agent capability, and governance policy adapt over time while preserving axioms and lineage recoverability.

A.4 Operational Glossary (System Terms)

This glossary defines recurring system terms used across Parts I–V.

- **Adoption Bridge:** A bounded deployment scaffold that allows controlled use of autonomy bands under explicit readiness criteria.
- **Bridge Collapse Protocol:** Governance rule that revokes a bridge when safety or stability thresholds fail.
- **Budget Envelope (B):** Hard economic constraint in I, enforced by EC; non-admissible TGs MUST halt or defer.
- **Checkpoint:** A required human/governance approval node in TG.
- **Dry-Run Preview:** A no-side-effect simulation of TG execution required for AB2+ or RB-proximate tasks.
- **Episode:** A bounded HCL instance from I intake to reconciliation.
- **Escalation Path:** Governance-defined route for incident handling and AB/RB demotion.
- **Health Check:** Periodic test that distinguishes Operational Decay vs Semantic Drift and informs AB/RB posture.
- **Intent Normalization (I₀ → I')**: Engine process converting raw human inputs into structured intent.
- **Lineage Head:** The cumulative cryptographic hash representing the immutable state of L at episode close.
- **Policy Board:** Human governance body owning GK configuration and change control.
- **Risk Class:** Governance-assigned severity class that sets maximum AB and minimum oversight.
- **Safe Mode:** A constrained execution posture automatically entered on violations.
- **Typed Node:** TG node with declared inputs, outputs, constraints, autonomy ceiling, and telemetry minima.

A.5 Reconciliation Table (v0.5.2 / v0.9 → v1.x Canon)

This table consolidates terms across earlier drafts into v1.x vocabulary. Deprecated terms SHALL NOT appear in v1.0 text except as historical notes.

Legacy Term	v1.x Canon Term	Status	Notes
HITL (Human-in-the-Loop)	System-in-the-Loop (SITL) / Hybrid Cognitive Loop	Deprecated → Canon	HITL framing implies human as safety feature; SITL treats human as meaning origin.

Toolchain / Pipeline	Symbiosis Engine (E) + Engine Algorithm (Part IV)	Reconciled	“Pipeline” alone is underspecified without constraints and lineage.
Prompt	Alignment Surface input	Scoped	“Prompt” is treated as low-fidelity AS only.
Agent autonomy	Autonomy Bands (AB0–AB3)	Canonical in Part III	Do not invent new band names.
Trust score	Symbiosis Index / Stability subscore	Deprecated	Trust is captured in SI subscores, not standalone.
Safety score (human)	HSS v2.0	Canonical in Part III	Safety-only designation mandated.
Oversight load	SCP	Canonical in Part III	SCP is human capacity, not workload.
Memory trace	Context Lineage (L)	Canon	“Trace” without hashes/replayability is non-compliant.
Policy layer	Governance Layer (GK + boards)	Canon	Governance-as-code required.

A.6 Canonical Ontology Diagram (Textual Outline)

The diagram below is the authoritative entity relationship spine. (Canonical Mermaid/ASCII diagrams are housed in Appendix L.)

1. **H → AS → I**
 - H expresses goals through AS to produce structured I.
 2. **I → E**
 - E normalizes I, assembles C, and synthesizes TG.
 3. **Governance Supremacy (Boards → GK → C)**
 - Boards configure GK.
 - GK emits C.
 - C injects constraints into E, TG, and MCP.
 4. **E → TG → MCP → A**
 - TG nodes are bound to MCP contracts.
 - A executes nodes strictly under MCP + AB ceiling.
 5. **Telemetry Upflow (A → E → H / Boards)**
 - A returns outputs, costs, and safety telemetry.
 - E reconciles deltas, updates SI/HSS/SCP per Part III, and emits transparency traces.
 6. **Lineage Spine (E → L)**
 - Every stage commits immutable hashes to L.
 - L supports audit replay and certification evidence.
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SQC Gate Check (Appendix A)

Gate 1 — Layer Integrity: PASS. Four-Layer responsibilities preserved; ontology enforces no-bypass semantics.

Gate 2 — Canonical Metrics Non-Redefinition: PASS. Metrics referenced only as properties; no formulas or thresholds defined.

Gate 3 — Constraint Precedence + Threshold Explicitness: PASS. Constraint supremacy encoded; no implicit thresholds introduced.

Gate 4 — Mechanistic Testability: PASS. Terms defined as implementable types and enforceable obligations.

Gate 5 — Lineage Recoverability: PASS. L defined with required reconstruction spine.

Gate 6 — Drift vs Decay Typing: PASS. Health Check and theater terms distinguish drift/decay at ontology level without redefining protocols.

Gate 7 — Abuse / Incentive Hardening: PASS. HSS and SCP referenced with safety-only framing; no secondary-use vectors added.

Gate 8 — Normative Voice Consistency: PASS. Modality and tone comply.

APPENDIX B — ECONOMIC CIRCUIT v2 (Local-Compute Aware)

Status: REWRITE/UPGRADE (v1.0)

Primary Linkages: Chapter 13 (Engine Algorithm v2), Chapter 11 (RB v2 + AB schema), Chapter 15 (Adoption Bridges), Appendix D (Task Graph Spec), Appendix E (MCP Schema).

Redline Basis: v1.0 requires upgrading Economic Circuit to include Compute Saturation (CS) and Queue Latency (QL) gates, plus explicit denial/defer logic with governance-set thresholds.

Non-Redefinition Rule: Canonical metrics SI, HSS v2.0, SCP, CI, RB v2, and AB schema are defined only in Part III. This appendix references them solely as control inputs.

B.0 Purpose and Position in the Framework

B.0.1 Purpose

The Economic Circuit v2 is the mandatory control loop that prevents unmanaged economic bleed in symbiotic systems. It enforces that **intelligence is a metered resource** (tokens, tool fees, time, local compute capacity, and human attention) and SHALL gate planning, autonomy, and execution accordingly.

B.0.2 Position

1. The Economic Circuit is executed by the Symbiosis Engine (E) as Stage 5 of the Engine Algorithm v2.
 2. The Economic Circuit is binding on TG synthesis, MCP issuance, AB selection, and Adoption Bridge admissibility.
 3. The Economic Circuit operates under **Constraint Supremacy**. It is an instantiation of C_economic within the Constraint Vector.
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B.1 Economic Primitives

B.1.1 Budget Envelope (B)

Definition: The intent-level cap on allowable economic expenditure for an episode, declared in I and enforced throughout E.

Properties:

- B MUST be decomposable into sub-budgets per TG node.
- B MUST include **human attention budget** as a bounded resource mediated through SCP headroom.

B.1.2 Projected Cost (Projected_Cost)

Definition: The pre-execution estimate of economic expenditure for TG_o under candidate tiers.

Canonical estimator (v0.9 carried forward):

$\text{Projected_Cost}(\text{TG}_0) = \sum (\text{Tokens}_i \times \text{Tier_Cost}_i) + \text{Tool_Costs} + \text{Time_Costs}.$

v1.0 extension: Projected_Cost MUST also include local compute and queue penalties as defined in B.2.

B.1.3 Realized Cost (Realized_Cost)

Definition: The actual expenditure incurred at runtime for each TG node and for the total episode.

Obligations:

- Realized_Cost MUST be logged to lineage per node.
- Any Realized_Cost overrun beyond governance-set tolerance SHALL trigger circuit-breaker rules in B.5.

B.1.4 Economic Circuit State (EC_State)

Definition: The runtime economic posture for a workflow, consisting of:

$\text{EC_State} = \langle \text{B_remaining}, \text{CS_state}, \text{QL_state}, \text{Spend_rate}, \text{Overrun_count} \rangle$

EC_State MUST be updated after each TG node completion and committed to lineage.

B.2 Local Compute Constraints (NEW in v2)

B.2.1 Compute Saturation (CS)

Definition: A normalized measure of local compute load relative to safe operational capacity for the deployment. It makes **local hardware a first-class gate**, not an implicit afterthought.

CS is defined per deployment as:

$\text{CS} = \text{Current_Compute_Load} / \text{Safe_Compute_Capacity}$

Where Safe_Compute_Capacity is a governance-configured parameter derived from hardware, energy budget, thermal envelope, and operational SLAs.

Normative rules:

1. CS thresholds MUST be explicit, deployment-specific, and GK-configured. Universal defaults are prohibited.
2. The Engine MUST compute CS prior to MCP issuance for any node that consumes local compute.
3. If $\text{CS} \geq \text{CS_hard_max}$, the node MUST be denied or deferred per B.4.
4. If CS is between CS_soft_warn and CS_hard_max, the Engine MAY optimize TG tiering, batching, or scheduling to reduce CS impact.

B.2.2 Queue Latency (QL)

Definition: A normalized measure of expected wait time for compute or human oversight relative to safe latency bounds.

QL is defined per deployment as:

$$QL = \text{Expected_Wait_Time} / \text{Safe_Latency_Bound}$$

Expected_Wait_Time MAY include local GPU queueing, external tool throttling, and human checkpoint backlog. Safe_Latency_Bound is GK-configured.

Normative rules:

1. QL thresholds MUST be explicit, deployment-specific, and GK-configured. Universal defaults are prohibited.
2. The Engine MUST evaluate QL at planning time for the full TG and at runtime before each node dispatch.
3. If $QL \geq QL_hard_max$, the node MUST be denied or deferred per B.4.
4. QL MUST be coupled to SCP headroom controls where latency is driven by human checkpoints (see B.3.3).

B.2.3 Economic Constraint Vector Extension (C_econ v2)

C_econ SHALL include, at minimum:

$C_econ = \langle B_total, B_fastlane, Max_Spend_per_node, Max_Tokens_per_node, TTL_per_node, CS_thresholds, QL_thresholds \rangle$

$CS_thresholds = \langle CS_soft_warn, CS_hard_max \rangle$

$QL_thresholds = \langle QL_soft_warn, QL_hard_max \rangle$

These values MUST be GK-signed and lineage-addressable.

B.3 Economic Pass v2 (Stage 5 Control Logic)

B.3.1 Inputs and Outputs

Inputs:

- TG_0 (draft task graph)
- Intent budget B
- Active C_econ v2
- Candidate model/agent tier costs
- Current EC_State

Outputs:

- Economic admissibility verdict for TG_0
- Optimized TG_1 if required
- Updated EC_State
- Lineage commitment

B.3.2 Pass Algorithm (Normative)

Given TG_0 and B:

1. **Estimate projected cost**

Compute $\text{Projected_Cost}(TG_0)$ including CS/QL penalties:

$$\text{Projected_Cost_v2} = \text{Base_Project_Cost} + \text{CS_Penalty} + \text{QL_Penalty}$$

Where penalties are computed using GK-configured functions. The specific penalty functions MAY vary by deployment but MUST be monotonic increasing in CS and QL.

2. **Budget gate**

If $\text{Projected_Cost_v2} \leq B_{\text{total}}$: mark TG_0 budget-admissible.

Else: proceed to optimization step.

3. **Optimization step (if over budget or CS/QL warn-state)**

The Engine SHALL attempt one or more of:

- a. Tier downgrade for low-regret nodes.
- b. Step pruning / decomposition tightening.
- c. Batching and scheduling to lower CS and QL.
- d. Re-routing to lower-cost agents under MCP constraints.
- e. Deferral of noncritical nodes.

4. **Re-estimate**

Compute $\text{Projected_Cost_v2}(TG_1)$.

5. **Verdict**

- If admissible: approve TG_1 .
- If still inadmissible: return "Budget Exceeded / CS-QL Exceeded" to H for override or re-authoring. No execution SHALL proceed without admissibility.

6. **Lineage commitment**

Commit: Projected_Cost_v2 , verdict, applied optimizations, and relevant thresholds to L.

B.3.3 Attention as Binding Economic Resource

Human oversight latency and capacity SHALL be treated as scarce budgets. SCP headroom impacts QL and therefore economic admissibility.

Normative rules:

1. If SCP headroom is low and projected checkpoints imply QL warn-state, the Engine MUST either:
 - a. lower AB ceilings,
 - b. insert additional batching to reduce interrupts, or
 - c. defer nodes.
2. Economic optimization SHALL NOT shift economic burden onto human attention without explicit H approval.

B.4 Denial, Deferral, and Safe Degradation (NEW v2)

B.4.1 Denial Conditions

A TG node t MUST be denied pre-flight if any of the following hold:

1. $\text{Projected_Cost_v2}(t) > \text{Max_Spend_per_node}$.
2. $\text{CS}(t) \geq \text{CS_hard_max}$.
3. $\text{QL}(t) \geq \text{QL_hard_max}$.
4. Any economic constraint in C_{econ} is unsatisfied.

Denied nodes SHALL be returned to H with explicit violated constraint fields.

B.4.2 Deferral Conditions

A TG node t MAY be deferred (not denied) if:

1. $B_{\text{remaining}}$ is sufficient but CS/Q L soft-warn thresholds are exceeded, AND
2. t is not RB-proximate, AND
3. deferral does not violate time-critical constraints.

Deferred nodes MUST be rescheduled under an updated EC_State and logged to lineage.

B.4.3 Safe Degradation Ladder

When economic constraints bind, E SHALL degrade execution in the following order unless GK policy specifies otherwise:

1. Lower-tier model/agent substitution for low-regret nodes.
2. Increase batching / reduce interrupt frequency.
3. Prune “nice-to-have” nodes.
4. Defer nodes outside critical path.
5. Demote AB ceilings.
6. Halt episode and return to H.

The ladder preserves safety and reversibility prior to any autonomy expansion.

B.5 Runtime Economic Circuit Breakers

B.5.1 Overrun Detection

At runtime, after each node completion:

$$\text{Overrun_Ratio}(t) = \text{Realized_Cost}(t) / \text{Projected_Cost_v2}(t)$$

B.5.2 Circuit Breaker Triggers

Any of the following SHALL trigger immediate circuit-breaker action:

1. $\text{Overrun_Ratio}(t) \geq \text{OR_hard_max}$ (GK-configured).
2. $B_{\text{remaining}} \leq 0$.

3. CS transitions into hard-max state mid-execution.
4. Repeated bleed-loop signature (token/time runaway) detected.

B.5.3 Circuit Breaker Actions

On trigger, E MUST:

1. halt current node safely,
2. revoke MCP for that node,
3. demote AB ceilings as per GK rule coupling,
4. log breaker event to lineage,
5. surface a remediation path to H.

Execution MAY resume only after new admissibility passes.

B.6 Coupling to AB/RB and Adoption Bridges

B.6.1 AB Coupling

Autonomy expansion is economically gated. AB promotion conditions SHALL require C_econ v2 admissibility in addition to SI/HSS/SCP prerequisites.

B.6.2 RB Coupling

Nodes that approach or cross RB MUST not be down-tiered or batched in ways that reduce oversight below GK minimums. Economic optimization MAY NOT weaken RB checkpoint sufficiency.

B.6.3 Adoption Bridge Coupling

Bridge admissibility requires budget admissibility under a strict bridge-tier sub-budget (B_fastlane).

In v2:

1. Bridges MUST also satisfy CS/QL soft-warn conditions; any hard-max exceedance disqualifies bridge routing.
2. Bridge budget fractions and CS/QL thresholds MUST be GK-configured.

B.7 Minimum Telemetry and Lineage Fields

For every TG node and for the episode overall, E MUST log:

1. Projected_Cost_v2 and decomposition into Base, CS_Penalty, QL_Penalty.
2. Realized_Cost with tokens/tool/time breakdown.
3. EC_State transitions.
4. Threshold parameters used (CS/QL/OR/B_fastlane).
5. Any optimization actions taken and their rationales.
6. Any denial/deferral/circuit-breaker events.

These fields are mandatory for auditability and anti-bleed certification.

SQC Gate Check (Appendix B)

Gate 1 — Layer Integrity: PASS. Economic Circuit specified as Engine control loop under Governance constraints; no cross-layer bypass.

Gate 2 — Canonical Metrics Non-Redefinition: PASS. SI/HSS/SCP/RB/AB referenced only as inputs; no redefinition.

Gate 3 — Constraint Precedence + Threshold Explicitness: PASS. CS/QL/OR/B thresholds required to be GK-configured with no universal defaults.

Gate 4 — Mechanistic Testability: PASS. Algorithms defined with inputs → conditions → outputs; denial/deferral/breakers are enforceable.

Gate 5 — Lineage Recoverability: PASS. Mandatory telemetry and lineage fields specified.

Gate 6 — Drift vs Decay Typing: PASS. Economic bleed treated as decay-class trigger; no drift content conflated.

Gate 7 — Abuse / Incentive Hardening: PASS. Economic optimization constrained not to offload burden onto humans or weaken safety gates.

Gate 8 — Normative Voice Consistency: PASS. Requirements stated in MUST/SHALL/MAY without speculative modality.

APPENDIX C — SYMBIOSIS INDEX (SI) IMPLEMENTATION & AUDIT ANNEX (v1.0)

Status: Supporting Technical Appendix (CARRY + HARDEN)

Primary Canon: Part III — SI chapters (canonical definitions, subscores, weights, bands).

Scope: This appendix specifies **implementation, telemetry, and audit procedures** for SI consistent with Part III canon. It SHALL NOT introduce new SI definitions, weights, or subscore math.

Non-Redefinition Rule: SI canonical math, components, and band thresholds are defined **only in Part III**. This appendix references and operationalizes them without alteration.

C.0 Purpose and Position

C.0.1 Purpose

This appendix provides a deterministic engineering annex for computing, logging, validating, and auditing SI in live Symbiosis deployments.

C.0.2 Position

1. SI is a **System Health metric** maintained by the Symbiosis Engine (E) as part of Stage 10 “Execute + Reconcile” in Engine Algorithm v2 (Chapter 13).
2. SI is a **promotion/demotion control surface** for AB/RB, Safe Harbors, and Bridges, as specified in Chapters 11, 15, and 16.

3. Governance owns SI admissibility bands; Engine computes SI episodes and rolling traces.
-

C.1 SI Computation Invariants (No Redefinition)

C.1.1 Canon lock

All SI component definitions, subscores, and weights SHALL be treated as frozen canon from Part III. This appendix SHALL only:

- restate input requirements,
- specify computational ordering,
- define telemetry and lineage fields, and
- define integrity checks.

C.1.2 Determinism requirement

Given identical:

- Intent (I),
- Constraint Vector (C),
- Task Graph (TG),
- AB/RB states,
- MCP contracts, and
- telemetry streams,

the Engine MUST compute identical SI_episode results. Any nondeterminism is a compliance defect.

C.2 Inputs Required for SI

C.2.1 Mandatory input classes

The Engine MUST compute SI using only the canonical inputs defined in Part III. At minimum, SI input sources SHALL include:

1. **Control-surface state**

- AB state per node
- RB class/proximity per node
- Safe Harbor / Bridge markers (if applicable)

2. **Intent fidelity telemetry**

- intent_id and normalized intent snapshot (I')
- explicit objective satisfaction signals
- constraint satisfaction verdicts

3. **Execution telemetry**

- per-node outputs and completion status
- rollback events
- human checkpoint approvals/overrides
- incidence of halts or safe-mode transitions

4. **Economic telemetry**

- Economic Pass verdicts
- Projected_Cost_v2 vs Realized_Cost
- CS/QL states and breaker events

5. **Human-side telemetry (referenced only)**

- HSS v2.0 snapshot
- SCP snapshot

These are **inputs** and may not be modified by SI computation.

C.2.2 Prohibited inputs

SI computation SHALL NOT incorporate:

- HR performance metrics,
- subjective managerial rankings,
- non-lineage linked KPI guesses, or
- any untyped external signals lacking provenance.

C.3 Computation Ordering

C.3.1 Episode boundary

An **SI_episode** is computed once per closed Hybrid Cognitive Loop episode, from Stage 1 intake through Stage 10 reconciliation.

C.3.2 Canonical ordering (normative)

The Engine **MUST** compute SI in the following order, matching Part III semantics:

1. **Episode validity check**

- Confirm lineage completeness minima satisfied.
- If not satisfied, SI_episode SHALL be marked “invalid” and excluded from rolling windows.

2. **Subscore computation**

- Compute each SI subscore exactly as specified in Part III.
- Do not introduce additional subscores or variant formulas.

3. **Composite aggregation**

- Aggregate subscores with canonical weights.
- No reweighting or smoothing beyond what Part III specifies.

4. **Band classification**

- Assign green/yellow/red band labels using Governance-configured thresholds already defined in Part III.

5. Rolling update

- Update SI_rolling and any canonical rolling windows per Part III.
- Commit window parameters to lineage.

C.3.3 Alignment to stability vital signs

Where Part III specifies stability integration (drift/resonance/divergence/entanglement), the Engine MUST:

- compute vital signs as separate canonical streams,
- then apply any canonical coupling into SI_rolling in the direction specified by Part III.

No other coupling is permissible.

C.4 Rolling Windows and Smoothing Rules

C.4.1 Governance-owned window parameters

Rolling window length, decay factors, and smoothing parameters are Governance-set and SHALL be pinned in GK policy with version IDs.

C.4.2 No silent retuning

The Engine SHALL NOT silently change window parameters based on performance or convenience. Any change MUST be:

1. authorized by Governance,
2. version-bumped in GK policy, and
3. lineage-logged on first use.

C.5 Minimum SI Telemetry & Lineage Fields

For each episode and per TG node, the Engine MUST log the following lineage-addressable fields:

Field	Required granularity	Notes
si_episode_id	episode	Stable deterministic ID
si_episode_value	episode	Composite SI result

si_band	episode	green/yellow/red label
si_subscores	episode	Canonical set only
si_window_id / params	episode	GK policy version
si_rolling_value(s)	episode	Canonical rolling outputs
si_invalid_reason	episode	if lineage minima fail
tg_id / node_id	node	lineage anchors
ab_state	node	at execution
rb_state	node	class/proximity
constraint_verdict	node	satisfied / violated
rollback_flag	node	yes/no + reason
safe_mode_flag	node	yes/no + trigger
economic_pass_verdict	node	approve/deny/defer
realized_cost	node	runtime
cs_state / ql_state	node	at dispatch

Absence of any field is a certification defect under Chapter 17.

C.6 SI Integrity and Anti-Gaming Checks

C.6.1 Lineage suppression check

If any TG node lacks lineage, the entire SI_episode SHALL be invalidated.

C.6.2 Rubber-stamp detection coupling

If RB checkpoints are being approved without meaningful human attention (as detected by HSS anti-gaming tests), Governance MUST treat SI improvements as suspect and MAY demote AB until recalibration.

C.6.3 Budget theater detection

If SI improves while Economic Circuit breakers or chronic overruns increase, the Engine MUST flag a **Stability Divergence Event** for Governance review. SI cannot be used to justify AB promotion under this condition.

C.7 Use of SI for Autonomy Control (Reference Rules)

C.7.1 Promotion gating

AB promotion, Bridge admissibility, and Phase promotion MAY occur only if:

1. SI_rolling is within Governance green band, AND
2. stability vital signs are not in red bands, AND
3. HSS/SCP prerequisites are met (per Part III + Chapters 11/16/17).

SI alone is never sufficient.

C.7.2 Demotion triggers

SI red band episodes SHALL trigger mandatory demotion actions exactly as specified in Chapter 11 and Chapter 15. This appendix adds no new triggers.

C.8 Audit Procedure

C.8.1 Audit objective

An SI audit SHALL verify that computed SI values are:

- canon-consistent,
- deterministically reproducible,
- lineage complete, and
- not subject to incentive manipulation.

C.8.2 Audit steps (normative)

1. Sample selection

- Choose stratified sample of episodes across AB bands and Safe Harbors.

2. Replay

- Reconstruct each episode from lineage: I + C + TG + AB/RB + Economic Pass + telemetry.

3. Recompute

- Recompute SI_episode using canonical Part III math.

4. Compare

- If recomputed SI differs from stored SI beyond GK tolerance ϵ , record a **SI Integrity Failure**.

5. Disposition

- Integrity failures SHALL force:
 - a. AB demotion for affected domains,
 - b. recalibration of telemetry or GK windows, and
 - c. recertification review if failures are systemic.
-

C.9 Implementation Checklist (Minimum Compliance)

A deployment may claim “SI Implementation & Audit Annex compliance” only if:

- [] SI_episode computed per closed episode and deterministically replayable.
 - [] Canonical subscores and weights used without modification.
 - [] GK-pinned rolling windows enforced with no silent retuning.
 - [] SI telemetry and lineage fields complete $\geq 98\%$ of episodes.
 - [] Anti-gaming checks active and producing governance-visible flags.
 - [] SI audit replay yields $\leq \epsilon$ divergence on sampled episodes.
-

SQC Gate Check (Appendix C)

Gate 1 — Layer Integrity: **PASS**. SI is computed by Engine under Governance-set parameters; Human inputs are not overridden.

Gate 2 — Canonical Metrics Non-Redefinition: **PASS**. No new SI math, subscores, or weights; appendix only operationalizes Part III canon.

Gate 3 — Constraint Precedence + Threshold Explicitness: **PASS**. Window and band parameters are GK-pinned; no defaults added.

Gate 4 — Mechanistic Testability: **PASS**. Computation ordering, telemetry minima, and audit steps are checkable and replayable.

Gate 5 — Lineage Recoverability: **PASS**. SI invalidates without lineage; replay is required for audit.

Gate 6 — Drift vs Decay Typing: **PASS**. Vital signs remain separate canonical streams; divergence flags do not redefine them.

Gate 7 — Abuse / Incentive Hardening: **PASS**. Explicit anti-gaming and theater checks; SI not sole promotion basis.

Gate 8 — Normative Voice Consistency: **PASS**. MUST/SHALL/MAY used precisely; no speculative modality.

APPENDIX D — TASK GRAPH (TG) SPECIFICATION v1.0

Status: NEW / Canonical Implementation Appendix (v1.0)

Primary Linkages: Chapter 4–6 (Intent + Planning), Chapter 10 (CI + MCP Routing), Chapter 11 (RB v2 + AB schema), Chapter 13 (Engine Algorithm v2), Appendix B (Economic Circuit v2), Appendix E (MCP schema), Appendix L (TG diagrams).

Redline Basis: v1.0 requires TG to be a typed, audit-replayable graph with explicit AB/RB ceilings, economic admissibility fields, CS/QL awareness, and mandatory lineage anchors. TG is no longer an informal “plan”; it is a binding execution contract.

Non-Redefinition Rule: SI, HSS v2.0, SCP, CI, RB v2, and AB schema remain canonical in Part III. This appendix operationalizes TG as an object that **references** those values; it SHALL NOT redefine them.

D.0 Purpose and Position in the Framework

D.0.1 Purpose

This appendix defines the Task Graph (TG) as the deterministic execution substrate for Symbiosis v1.0. TG expresses a human intent as a dependency-aware, typed graph of tasks whose admissibility, autonomy, and economics are enforceable and auditable.

D.0.2 Position

1. TG is synthesized by the Symbiosis Engine as Stage 4 (Draft Planning) and finalized through Stage 8 (Contract Binding & MCP Routing).
 2. No execution SHALL occur outside a finalized TG.
 3. TG is governed by Constraint Vector (C), Economic Circuit v2, RB v2, and AB ceilings.
-

D.1 TG Definitions (Normative)

D.1.1 Task Graph (TG)

Definition: A directed acyclic or cyclic-with-rollback graph representing executable work derived from Intent I' under Constraints C, with explicit autonomy, regret, and economic gating.

D.1.2 TG Lifecycle States

State	Definition	Engine Stage
TG ₀ Draft	First synthesis from I', typed but unbound	Stage 4
TG ₁ Economically Admissible	TG ₀ after Economic Pass v2 optimization	Stage 5
TG ₂ Autonomy-Bound	TG ₁ with RB classes and AB ceilings per node	Stage 6
TG ₃ Context-Bound	TG ₂ with MTM context bundles K _i assigned	Stage 7
TG Final	TG ₃ with MCP contracts issued and routing locked	Stage 8
TG Executing	Nodes dispatching and returning telemetry	Stage 9
TG Closed	Episode reconciled; SI/HSS/SCP deltas committed	Stage 10

D.2 TG Object Model

D.2.1 TG Root Schema (minimum fields)

Field	Type	Requirement	Notes
-------	------	-------------	-------

tg_id	string	MUST	immutable ID
intent_id	string	MUST	lineage anchor to I'
constraint_vector_id	string	MUST	pinned C version
policy_version	string	MUST	GK policy head
nodes	array	MUST	typed nodes
edges	array	MUST	dependency relations
projected_cost_v2	number	MUST	from Appendix B
admissibility_verdict	enum {approve, defer, deny}	MUST	Stage 5 outcome
phase_context	enum {P1,P2,P3,P4}	MAY	adoption phase marker
safe_harbor_id	string	null	MAY
bridge_id	string	null	MAY
tg_hash	string	MUST	lineage hash head

D.2.2 Node Schema (minimum fields)

Field	Type	Requirement	Notes
node_id	string	MUST	stable within TG
task_type	enum	MUST	canonical task typing table D.3
goal_fragment	string	MUST	scoped intent slice
inputs	array	MUST	typed inputs
outputs	array	MUST	expected outputs
dependencies	array<node_id>	MUST	derived from edges
risk_class	enum {A,B,C}	MUST	GK typed; no local invention
rb_state	RBRef	MUST	RB class/proximity reference only
ab_ceiling	ABRef	MUST	AB0–AB3 reference only
requires_checkpoint	bool	MUST	true if RB-prox or GK rule
checkpoint_type	enum {human, governance, dual}	MAY	if requires_checkpoint

estimated_tokens	number	MUST	planning estimate
estimated_tool_cost	number	MUST	planning estimate
estimated_time_s	number	MUST	planning estimate
cs_required	bool	MUST	local compute consumption flag
projected_cost_v2	number	MUST	node-level admissibility
mcp_id	string	null	MUST at Final
mcp_hash	string	null	MUST at Final
context_bundle_id	string	null	MUST at TG ₃ +
telemetry_minimal_id	string	MUST	pointer to required logs
node_hash	string	MUST	lineage anchor

D.3 Canonical Task Typing

D.3.1 Typing purpose

Task types enable deterministic risk typing, autonomy gating, and cost tiering. Every node MUST be assigned exactly one canonical task type.

D.3.2 Canonical task types (v1.0)

task_type	Description	Default risk bias
research	information gathering / synthesis	C
analysis	structured reasoning over evidence	B/C
draft_write	content production	B/C
transform	rewrite/translate/format of known inputs	C
code_gen	code authoring or modification	B
code_exec	execution of code or commands	A/B
data_access	retrieval from governed stores	B
data_mutate	writes/updates/deletes	A/B
deploy_ops	infrastructure changes	A
external_action	real-world effect via tools	A
compliance_audit	audit/verification tasks	B

admin_governance	policy/kernel changes	A
------------------	-----------------------	---

Sector overlays MAY adjust risk bias, but SHALL NOT introduce new task types (Appendix K rules).

D.4 Edge Model and Dependency Semantics

D.4.1 Edge Schema

Field	Type	Requirement	Notes
from_node	node_id	MUST	
to_node	node_id	MUST	
edge_type	enum {hard_dep, soft_dep, rollback_dep, data_dep}	MUST	
condition	string	null	MAY
edge_hash	string	MUST	

D.4.2 Edge rules

1. hard_dep edges MUST be satisfied before to_node dispatch.
2. soft_dep edges MAY be bypassed only if:
 - a. Economic Pass v2 and GK permit, and
 - b. bypass is logged to lineage.
3. rollback_dep edges MUST exist for any node with rb_state indicating irreversibility risk.
4. data_dep edges MUST be used when downstream nodes consume upstream artifacts.

D.5 RB / AB Binding Rules

D.5.1 RB binding (reference-only)

1. Every node MUST carry an `rb_state` reference produced in Stage 6.
2. Nodes marked RB-proximate MUST set `requires_checkpoint = true` and specify `checkpoint_type`.

D.5.2 AB ceiling binding (reference-only)

1. Every node MUST carry an `ab_ceiling` reference (AB0–AB3) assigned by GK and min-bounded by risk, overlay, operator HSS tier, and SCP headroom.
2. No agent may execute a node above its `ab_ceiling`; violation is a hard constraint breach.

D.5.3 Checkpoint insertion rule

Where `rb_state` requires checkpointing, the Engine MUST insert explicit checkpoint nodes or flags such that execution cannot proceed without approval.

D.6 Economic Circuit v2 Fields (Local Compute Aware)

D.6.1 Required economic fields

Every TG and node MUST include:

- `projected_cost_v2` (Base + CS/QL penalties),
- `admissibility_verdict` at TG root,
- `cs_required` flag per node,
- any denial/deferral flags, and
- breaker hooks.

D.6.2 CS/QL coupling

If `cs_required = true`, Stage 5 MUST evaluate CS and QL thresholds prior to dispatch.

Nodes exceeding hard thresholds MUST be denied or deferred per Appendix B.

D.7 Context Bundle and MTM Hook

D.7.1 Context binding

Every node in TG₃+ MUST have `context_bundle_id` identifying the MTM tier and specific slices selected for that node.

D.7.2 Context auditability

Context selection MUST be lineage-logged and replayable. Any node missing context provenance SHALL invalidate the episode for SI computation (Appendix C).

D.8 MCP Binding Hooks

D.8.1 Binding requirements

In TG Final, every executable node **MUST** include:

- mcp_id,
- mcp_hash,
- agent routing target(s),
- explicit tool permissions,
- AB ceiling mirror,
- cost caps and rollback semantics.

D.8.2 No silent rerouting

Agents **MAY** be swapped only if:

1. replacement remains within mcp constraints,
2. swap is logged to lineage, and
3. Economic Circuit remains admissible.

D.9 TG Telemetry Minima (Per Node)

Each node **MUST** emit telemetry sufficient for:

- replay,
- SI/HSS/SCP reconciliation,
- drift vs decay typing, and
- audit evidence.

Minimum per-node telemetry:

Class	Fields
Control	node_id, ab_state_at_exec, rb_state_at_exec, checkpoint_events
Outputs	output_artifact_refs, completion_status, rollback_flag

Constraints	constraint_verdicts, violated_constraint_ids (if any)
Economics	realized_tokens, realized_tool_cost, realized_time_s, cs_state, ql_state, breaker_events
Stability	drift_tag / decay_tag if anomaly detected
Lineage	node_hash_pre, node_hash_post, mcp_hash

Absence of minima is a certification defect under Chapter 17.

D.10 TG Conformance Tests (Minimum Compliance)

A TG is Symbiosis-compliant only if all tests below pass.

T1 — Typed completeness

All nodes have task_type, risk_class, rb_state ref, ab_ceiling ref, and projected_cost_v2.

T2 — Constraint injection

TG root and all nodes reference the active constraint_vector_id; any node missing it is invalid.

T3 — Economic admissibility

TG has admissibility_verdict = approve before any node dispatch.

T4 — Autonomy enforcement

No node routed to agents above ab_ceiling.

T5 — RB checkpoint sufficiency

All RB-proximate nodes have requires_checkpoint = true and checkpoint events logged.

T6 — Lineage recoverability

tg_hash and node_hash values allow full replay from I' to outputs.

T7 — Context provenance

All nodes in TG₃+ have context_bundle_id logged.

T8 — CS/QL awareness

Nodes with `cs_required` invoke CS/QL evaluation and respect deny/defer rules.

Passing T1–T8 is required for any “Symbiosis v1.0 compliant” claim.

D.11 Implementation Checklist (Paste-ready)

- [] TG_0 synthesized from I' with canonical task types.
 - [] Economic Pass v2 produces TG_1 with `projected_cost_v2`.
 - [] RB/AB bound per node (reference only).
 - [] MTM context bundles assigned and logged.
 - [] MCP contracts issued for every executable node.
 - [] Telemetry minima satisfied per node and per episode.
 - [] Conformance tests T1–T8 pass continuously.
-

SQC Gate Check (Appendix D)

Gate 1 — Layer Integrity: PASS

- TG defined as Engine artifact under Governance constraints with Human-authored intent anchor.

Gate 2 — Canonical Metrics Non-Redefinition: PASS

- RB/AB/HSS/SCP/SI referenced only; no new math, weights, or bands.

Gate 3 — Constraint Precedence + Threshold Explicitness: PASS

- TG requires pinned constraint IDs; CS/QL thresholds delegated to GK with no defaults.

Gate 4 — Mechanistic Testability: PASS

- Schemas, rules, and tests specify implementable inputs → conditions → outputs.

Gate 5 — Lineage Recoverability: PASS

- `tg_hash/node_hash` minima and replay requirements explicit.

Gate 6 — Drift vs Decay Typing: PASS

- Telemetry includes drift/decay tags without redefining protocols.

Gate 7 — Abuse / Incentive Hardening: PASS

- Autonomy enforcement and no silent rerouting prevent gaming and shadow autonomy creep.

Gate 8 — Normative Voice Consistency: PASS

- Uses MUST/SHALL/MAY precisely; no speculative modality.

APPENDIX E — MCP / AGENT CONTRACT SCHEMA v1.0

Status: CARRY + ALIGN (v1.0)

Primary Linkages: Chapter 10 (CI & MCP Canon), Chapter 11 (RB v2 + AB schema), Chapter 13 (Engine Algorithm v2, Stage 8), Appendix D (TG spec), Appendix B (Economic Circuit v2), Appendix F (Agent Health Checks), Appendix O (Incident Runbooks).

Redline Basis: File A designates Appendix E as carry-forward but requires alignment to Engine Algorithm v2, Economic Circuit v2 (CS/QL), RB v2 formalization, and lineage minima.

Baseline Anchor: v0.9 defines canonical agent contract fields and dispatch conformance rules; these remain binding and are extended only where needed for v1.0 compatibility.

Non-Redefinition Rule: SI, HSS v2.0, SCP, CI, RB v2, and AB schema are canonical in Part III. This appendix references AB/RB as eligibility fields only and SHALL NOT redefine them.

E.0 Purpose and Position in the Framework

E.0.1 Purpose

This appendix defines the Machine Collaboration Protocol (MCP) contract as the **binding capability-and-governance object** that authorizes synthetic agents to execute TG nodes. Contracts are the enforcement boundary between Engine intent and Agentic execution.

E.0.2 Position

1. Contracts are bound for each TG node at **Stage 8 — Contract Binding & MCP Routing** of Engine Algorithm v2.
 2. No agent MAY execute any TG node without an active MCP contract bound to that node.
 3. MCP contracts unify interoperability and governance by making agent eligibility machine-checkable and audit-replayable.
-

E.1 Contract Lifecycle and Identity

E.1.1 Contract as first-class governance object

An MCP contract is simultaneously:

- a **capability declaration**, and

- a **governance boundary object** enforcing AB ceilings, RB gating, constraint supremacy, and economic admissibility.

E.1.2 Contract states

State	Definition
proposed	draft contract submitted for governance review
active	eligible for binding to TG nodes
paused	temporarily ineligible pending health check or audit
revoked	permanently ineligible until re-certified
deprecated	superseded by a newer version; may execute only for lineage replay

E.1.3 Identity invariants (normative)

Every contract **MUST** have immutable identity and provenance fields.

Required fields:

- contract_id
- agent_id
- cluster_id
- version
- owner_org / maintainer
- status
- issued_at (UTC)
- policy_version (GK head)
- schema_version (this appendix)

Contracts lacking any invariant field are non-admissible.

E.2 Required Contract Fields (Normative)

v1.0 retains v0.9's required field families and adds only the minimum extensions needed for CS/QL and RB v2 alignment.

E.2.1 Family 1 — Identity & Provenance

MUST include:

agent_id, cluster, version, owner, status, provenance_hash, policy_version.

E.2.2 Family 2 — Capabilities

Declarative list of admissible tasks/tools/output forms.

MUST include:

- admissible_task_types (must be subset of Appendix D canonical task types)
- admissible_tools
- output_modalities
- prohibited_tasks (explicit)

E.2.3 Family 3 — Eligibility & Preconditions

Eligibility gates that must be satisfied before binding.

MUST include:

- ab_ceiling (AB0–AB3 reference only)
- admissible_risk_classes / domains
- required_constraints (by constraint family tags)
- required_memory_tiers (MTM tiers)
- required_safe_harbor_tags (if AB2+)

v1.0 extension (NEW):

- cs_eligibility: {supports_local_compute: bool, max_cs_state: tag}
- ql_eligibility: {max_ql_state: tag}

These tags reference GK-configured CS/QL bands (Appendix B); no numeric thresholds appear here.

E.2.4 Family 4 — Input/Output Schemas

Typed REQ/RES payload definitions including confidence and explanation obligations.

MUST include:

- input_schema_ref(s)
- output_schema_ref(s)
- explanation_minima (what the agent must return in trace)

E.2.5 Family 5 — Economic Profile

Bounding economics per agent.

MUST include:

- `cost_per_token` (or per-unit equivalent)
- `cost_per_call`
- `max_cost_per_task`
- `preferred_tier` / `routing_bias`
- `retry_cost_budget`
- `local_compute_cost_profile` (if `supports_local_compute`)

E.2.6 Family 6 — Failure & Escalation Semantics

Standardized error codes and disposition rules.

MUST include:

- `error_code_taxonomy_ref`
- `retryability_map`
- `escalation_routes` (human / governance / safe-mode)
- `rollback_semantics` (atomicity + reversibility assumptions)

E.2.7 Family 7 — Governance Obligations

Audit and privacy coupling.

MUST include:

- `logging_level` (telemetry minima class)
- `retention_class` (per MTM)
- `constraint_check_mode` (strict / strict+redundant)
- `privacy_posture` (No-PII / masked / enclave)

E.2.8 Family 8 — MCP Bindings

Explicit allowed resources, prompts, and tools.

MUST include:

- `allowed_resources[]`
- `allowed_prompts[]`
- `allowed_tools[]`
- `disallowed_tools[]` (explicit)

E.2.9 Family 9 — Arbitration Profile (Optional but recommended)

Critic/ensemble behaviors for safety and quality.

MAY include:

- critic_required (bool)
- ensemble_size
- critic_loop_ceiling
- arbitration_timeout_s

E.3 Binding Rules (Stage-8 Conformance)

A TG node MAY be dispatched to an agent only if all conditions below pass. v0.9 rule is retained verbatim in substance, with explicit Economic Circuit v2 and CS/QL alignment.

E.3.1 Pre-flight binding checks (normative)

For each node n and candidate contract k :

1. **Autonomy eligibility**
 $n.ab_ceiling \leq k.ab_ceiling$.
2. **Constraint completeness**
 All $k.required_constraints$ are present in the active Constraint Vector (C).
3. **Economic admissibility**
 $n.projected_cost_v2 \leq k.max_cost_per_task$ AND $\leq B_remaining$.
4. **MCP subset rule**
 $n.allowed_tools/resources/prompts \subseteq k.allowed_*$ bindings.
5. **Memory eligibility**
 $n.context_bundle_tiers \subseteq k.required_memory_tiers$.
6. **Safe Harbor eligibility** (if $n.ab_ceiling \geq AB2$)
 $n.safe_harbor_id$ is non-null AND satisfies $k.required_safe_harbor_tags$.
7. **Local compute eligibility** (if $n.cs_required$)
 Current CS_state and QL_state $\in k.cs_eligibility/max_cs_state$ and $k.ql_eligibility/max_ql_state$ tags.

Failure of any check SHALL block binding and force E to (a) select a different agent, (b) demote AB for node, (c) re-plan TG, or (d) escalate to H.

E.4 Runtime Enforcement and Revocation

E.4.1 Capability overreach rule

If an agent attempts a task, tool, or scope outside contract bindings, E MUST hard-stop execution, revoke the MCP binding for that node, and log an audit event.

E.4.2 Economic bleed rule

If Economic Circuit breakers trigger on a node bound to a contract, E MUST:

1. halt the node safely,
2. mark the binding as violated,
3. optionally demote AB for downstream nodes, and
4. surface remediation to H.

This action does not redefine economics; it enforces Appendix B.

E.4.3 Drift/decay coupling

Contracts MUST be paused or revoked automatically when:

- **Operational Decay** health checks fail (Appendix F), or
- **Semantic Drift** diagnostics detect stale/unsafe behavior (Appendix G).

E.4.4 Contract revocation semantics

A revoked contract SHALL NOT be reactivated without Governance re-issuance and a new version ID.

E.5 Machine-Readable Schema (Reference)

E.5.1 JSON Schema requirement

The contract MUST have a machine-readable schema representation, and that schema MUST be stable across minor versions. v0.9 maintained a full schema annex elsewhere; v1.0 retains that requirement.

E.5.2 Canonical top-level shape (illustrative, not redefining)

```
{  
  "contract_id": "string",  
  "agent_id": "string",  
  "cluster_id": "string",  
  "version": "string",  
  "status": "proposed|active|paused|revoked|deprecated",  
  "policy_version": "string",  
  "issued_at": "YYYY-MM-DDTHH:MM:SSZ",  
  
  "capabilities": {
```

```
"admissible_task_types": ["research","analysis","code_gen", "..."],
"admissible_tools": ["tool_a","tool_b"],
"output_modalities": ["text","json","code"],
"prohibited_tasks": ["external_action"]
},

"eligibility": {
  "ab_ceiling": "AB0|AB1|AB2|AB3",
  "admissible_risk_classes": ["A","B","C"],
  "required_constraints": ["C_legal.gdpr","C_safety.no_pii"],
  "required_memory_tiers": ["MTM2","MTM3"],
  "required_safe_harbor_tags": ["finance_sandbox"],
  "cs_eligibility": {"supports_local_compute": true, "max_cs_state": "CS_warn"},
  "ql_eligibility": {"max_ql_state": "QL_warn"}
},

"io_schemas": {
  "input_schema_refs": ["schema://tg/node-input/v1"],
  "output_schema_refs": ["schema://tg/node-output/v1"],
  "explanation_minima": ["confidence","rationale","constraint_verdicts"]
},

"economics": {
  "cost_per_token": 0.0003,
  "cost_per_call": 0.02,
  "max_cost_per_task": 1.50,
```

```
"preferred_tier": "T2",  
"retry_cost_budget": 0.30,  
"local_compute_cost_profile": {"gpu_seconds_cost": 0.001}  
},
```

```
"failure_escalation": {  
  "error_code_taxonomy_ref": "schema://errors/v1",  
  "retryability_map": {"E_TIMEOUT": "retryable"},  
  "escalation_routes": ["human", "governance"],  
  "rollback_semantics": {"is_atomic": false, "requires_human_checkpoint": true}  
},
```

```
"governance_obligations": {  
  "logging_level": "full",  
  "retention_class": "RC2",  
  "constraint_check_mode": "strict",  
  "privacy_posture": "no_pii"  
},
```

```
"mcp_bindings": {  
  "allowed_resources": ["db://allowed", "fs://sandbox"],  
  "allowed_prompts": ["prompt://canon"],  
  "allowed_tools": ["tool_a"],  
  "disallowed_tools": ["tool_x"]  
},
```

```
"arbitration_profile": {  
  "critic_required": true,  
  "ensemble_size": 2,  
  "critic_loop_ceiling": 3  
}  
}
```

The example is illustrative only. Any field present here is normative only if defined in Section E.2.

E.6 Conformance Tests (Minimum Compliance)

A contract is Symbiosis-compliant only if:

K1 — Field completeness

All required families E.2.1–E.2.8 are present.

K2 — Canonical typing

admissible_task_types are a subset of Appendix D canonical types.

K3 — AB/RB compatibility

ab_ceiling references only AB0–AB3 and never invents new bands.

K4 — Constraint supremacy ready

required_constraints are expressible as Constraint Vector tags.

K5 — Economic admissibility ready

economics include max_cost_per_task and cost fields sufficient for Projected_Cost_v2 compute.

K6 — CS/QL compatibility

cs_eligibility and ql_eligibility tags exist where local compute is supported.

K7 — Lineage recoverability

contract_id, version, policy_version, and bindings are lineage-logged on every dispatch.

Failure of K1–K7 renders the agent non-admissible for Stage-8 binding.

E.7 Implementation Checklist

- [] Contracts are versioned, GK-signed, and machine-readable.
 - [] All nine families implemented; optional family used where risky.
 - [] Stage-8 binding checks E.3 run deterministically for every node.
 - [] Runtime overreach/bleed/drift/decay triggers revoke or pause contracts automatically.
 - [] Contract and binding IDs are lineage-anchored and replayable.
-

SQC Gate Check (Appendix E)

Gate 1 — Layer Integrity: PASS

- Contracts defined as Agentic eligibility objects bound by Engine under Governance policy.

Gate 2 — Canonical Metrics Non-Redefinition: PASS

- AB/RB referenced only as eligibility fields; SI/HSS/SCP not redefined.

Gate 3 — Constraint Precedence + Threshold Explicitness: PASS

- Required constraints are tags; CS/QL thresholds remain GK-set, deployment-specific.

Gate 4 — Mechanistic Testability: PASS

- Field families, binding checks, and conformance tests are explicitly testable.

Gate 5 — Lineage Recoverability: PASS

- Binding and schema require lineage anchoring and replay.

Gate 6 — Drift vs Decay Typing: PASS

- Revocation coupled separately to decay (App F) and drift (App G).

Gate 7 — Abuse / Incentive Hardening: PASS

- Overreach hard-stop, MCP subset rule, and economic bleed enforcement block covert autonomy.

Gate 8 — Normative Voice Consistency: PASS

- MUST/SHALL/MAY language applied precisely.
-

APPENDIX F — AGENT HEALTH CHECKS (OPERATIONAL DECAY PROTOCOL) v1.0

Status: REWRITE/UPGRADE (v1.0)

Primary Linkages: Chapter 11 (Stability Vital Signs), Chapter 13 (Engine Algorithm v2), Chapter 14 (Failure Modes: Drift vs Decay), Appendix E (MCP Contracts), Appendix O (Runbooks), Appendix G (Semantic Drift Diagnostics).

Redline Basis: v1.0 requires a dedicated decay appendix distinct from drift, adding a **Daily Health Check Protocol**, mandatory **CI demotion on decay**, and **separate decay lineage streams**.

Non-Redefinition Rule: Canonical CI, SI, RB v2, AB schema, HSS v2.0, and SCP are defined only in Part III. This appendix references CI and SI as control inputs and outputs only; it SHALL NOT redefine their mathematics, weights, or bands.

F.0 Purpose and Position in the Framework

F.0.1 Purpose

This appendix defines the mandatory Operational Decay health check regime for synthetic agents operating under Symbiosis v1.0. Operational Decay is **performance degradation arising from infrastructure, tooling, configuration, availability, or runtime reliability faults**, not from semantic/model drift. This appendix supplies deterministic checks, triggers, and dispositions to keep agent capabilities safe, stable, and economically admissible.

F.0.2 Position

1. Agent Health Checks are executed by the Symbiosis Engine (E) under Governance Kernel (GK) policy. They are part of continuous runtime operations, and are required prerequisites for contract activation and continued binding.
 2. Decay checks are distinct from Semantic Drift diagnostics (Appendix G). Conflation is prohibited.
 3. Health Check failures may force MCP pause/revocation (Appendix E) and AB demotion or Bridge collapse (Chapters 11, 15, 16; Appendix O).
-

F.1 Definitions

F.1.1 Operational Decay (Decay)

A measurable reduction in agent execution reliability or efficiency caused by non-semantic mechanisms, including but not limited to:

- tool error rate increases,
- latency or availability regressions,
- economic bleed loops due to planner/tool changes,
- context retrieval failures or truncation,
- protocol/interface mismatches,
- service outages, quota throttles, or hardware saturation.

Decay is orthogonal to model meaning quality. A high-quality model can still decay operationally.

F.1.2 Health Check Window (W)

A GK-defined rolling window used to evaluate decay metrics. Window length and thresholds are governance-set and SHALL be versioned in the GK policy head.

F.1.3 Decay Lineage Stream (L_decay)

A dedicated lineage substream containing only decay-relevant telemetry, verdicts, and remediation actions. L_decay MUST be addressable independently from drift lineage.

F.2 Health Check Classes

The Engine MUST maintain the following decay check classes for every active agent contract.

F.2.1 Availability & Liveness (HC-A)

Measures whether the agent is callable and responsive.

Minimum telemetry:

- uptime %,
- call success rate,
- mean time to recover (MTTR),
- timeout frequency.

Trigger conditions:

- call success rate drops below GK floor, OR
- MTTR exceeds GK ceiling.

Disposition:

- pause contract if floor breach persists for W_warn,
- revoke contract if floor breach persists for W_hard.

F.2.2 Tool Reliability (HC-T)

Measures agent tool-call correctness and error incidence.

Minimum telemetry:

- p_tool_fail,
- p_partial_side_effect,
- rollback invocation frequency,
- tool-specific error taxonomy counts.

Trigger conditions:

- p_tool_fail above GK ceiling, OR
- irreversible side-effect attempts without RB checkpoint (hard breach).

Disposition:

- immediate safe-mode and pause on hard breach,
- otherwise demote CI reliability term and possibly AB ceilings per policy.

F.2.3 Latency & Queue Integrity (HC-L)

Measures runtime latency and queue stability (including local compute).

Minimum telemetry:

- mean / p95 latency per task_type,
- queue latency QL_state,
- compute saturation CS_state when cs_required nodes run,
- throttle/backoff counts.

Trigger conditions:

- sustained latency breach over W_warn, OR
- QL or CS transitions into hard-max bands (Appendix B).

Disposition:

- defer affected nodes,
- pause contract if breach persists,
- recompute routing preferences to lower tiers.

F.2.4 Economic Efficiency Integrity (HC-E)

Detects runtime economic decay and bleed loops not attributable to semantic drift.

Minimum telemetry:

- Overrun_Ratio per node (Appendix B),
- realized tokens vs projected,
- retry cost accumulation,
- TTL breach rates.

Trigger conditions:

- Overrun_Ratio exceeds GK hard ceiling, OR
- repeated bleed-loop signature appears (Appendix O A-06/E-03 style).

Disposition:

- immediate circuit-breaker excursion per Appendix B,
- contract pause if attributed to agent/tool pathway,
- CI efficiency term demotion.

F.2.5 Context/Protocol Integrity (HC-C)

Measures operational correctness of context ingestion and protocol compliance.

Minimum telemetry:

- context_bundle_id resolution success rate,
- truncation / overflow counts,
- protocol schema mismatch rate,
- “request clarification” frequency due to missing context.

Trigger conditions:

- context resolution below GK floor, OR
- protocol mismatch above GK tolerance.

Disposition:

- pause contract until schema compatibility restored; no binding allowed while failing.

F.3 Daily Health Check Protocol (Mandatory)

F.3.1 Schedule

1. Every active contract **MUST** undergo a daily decay health check.
2. “Daily” is defined as ≤ 24 hours between evaluations in the deployment’s local governance time zone.

F.3.2 Inputs

The daily check **MUST** ingest:

- last 24h telemetry for HC-A/T/L/E/C,
- per-task_type stratification,
- incident adjacency flags from Appendix O,
- current GK policy thresholds and windows.

F.3.3 Deterministic evaluation order

The Engine **SHALL** evaluate checks in this fixed order:

1. HC-A (liveness gate)
2. HC-T (tool safety gate)
3. HC-L (latency/CS/QL gate)
4. HC-E (economic bleed gate)
5. HC-C (context/protocol gate)

Any hard failure immediately short-circuits remaining checks into “FAILED — unsafe to bind.”

F.3.4 Verdicts

The daily protocol **SHALL** yield one of:

- **PASS** — contract remains active.
- **WARN** — contract remains active but routing is biased away (lower CI confidence term).
- **FAIL-SOFT** — contract paused, eligible for remediation and recheck.
- **FAIL-HARD** — contract revoked; recertification required.

F.3.5 Lineage commitment

Each daily check MUST append to L_decay:

- check timestamp,
- per-class raw metrics,
- thresholds used,
- verdict,
- remediation actions initiated,
- any CI demotion values applied (reference-only).

F.4 CI Demotion on Decay (Mandatory)

F.4.1 Principle

When decay is detected, capability confidence must be reduced to prevent unsafe routing or autonomy creep. v1.0 requires explicit CI demotion on decay.

F.4.2 Demotion rule (normative)

For any agent a with a daily verdict of WARN or FAIL-SOFT:

1. The Engine MUST apply a CI demotion to the canonical CI reliability and/or efficiency terms for that agent/domain slice, using GK-defined demotion functions.
2. CI demotion SHALL be **monotonic increasing with severity and persistence** of decay.
3. CI demotion MUST be lineage-logged in L_decay and in the episode lineage head.

F.4.3 Routing effect

Agents under CI demotion:

- MUST be deprioritized for Stage-8 MCP binding (Appendix E), and
- MUST NOT be bound to RB-proximate nodes unless no other admissible agents exist and H explicitly authorizes continued use.

F.4.4 Recovery

CI terms MAY be restored only after:

1. consecutive PASS verdicts over a GK-defined recovery window W_rec, and
2. no S2+ incidents adjacent to the agent in that window.

F.5 Decay → Autonomy and Bridge Couplings

F.5.1 AB coupling

Persistent decay failures SHALL trigger AB demotion for affected task types or domains as defined in Chapters 11 and 15; this appendix introduces no new AB rules.

F.5.2 Bridge coupling

If decay causes S3+ incident adjacency or repeated FAIL-HARD verdicts within a Safe Harbor or Bridge domain, Bridge Collapse Protocols apply (Appendix O / Chapter 15).

F.6 Operator-Facing Transparency

F.6.1 Visibility requirement

Operators (H) MUST be able to view:

- current health verdict per agent cluster,
- active CI demotion flags,
- decay class causing WARN/FAIL, and
- remediation status.

F.6.2 Non-punitive boundary

Decay data is system safety telemetry. It SHALL NOT be used for operator evaluation, and SHALL be governed under the Non-Punitive Covenant and HSS privacy rules.

F.7 Conformance Tests (Minimum Compliance)

A deployment passes Appendix F compliance only if:

F-T1 Daily Schedule

≤24h cadence for all active contracts.

F-T2 Class Coverage

HC-A/T/L/E/C executed with telemetry minima.

F-T3 Deterministic Ordering

Evaluation order matches F.3.3, and short-circuiting on hard failure is enforced.

F-T4 Separate Lineage

L_decay exists, is queryable, and contains daily records.

F-T5 CI Demotion

WARN/FAIL-SOFT verdicts produce CI demotion values and routing effects.

F-T6 Binding Lockout

FAIL verdicts prevent MCP binding until recovery.

Failure of any test disqualifies certification claims under Chapter 17.

SQC Gate Check (Appendix F)

Gate 1 — Layer Integrity: PASS

- Health checks executed by Engine under GK thresholds; Agentic layer does not self-attest.

Gate 2 — Canonical Metrics Non-Redefinition: PASS

- CI/SI/AB/RB referenced only; no new formulas or weights introduced.

Gate 3 — Constraint Precedence + Threshold Explicitness: PASS

- All thresholds/windows required GK-configured; no universal defaults stated.

Gate 4 — Mechanistic Testability: PASS

- Check classes, schedules, inputs, ordering, verdicts, and conformance tests are implementable.

Gate 5 — Lineage Recoverability: PASS

- Dedicated L_decay stream and mandatory commitments specified.

Gate 6 — Drift vs Decay Typing: PASS

- Appendix defines decay only and prohibits conflation with drift; drift reserved to Appendix G.

Gate 7 — Abuse / Incentive Hardening: PASS

- CI demotion and binding lockout prevent autonomy creep from decayed agents; transparency avoids hidden routing.

Gate 8 — Normative Voice Consistency: PASS

- Modality uses MUST/SHALL/MAY precisely; no speculative phrasing.

APPENDIX G — SEMANTIC DRIFT DIAGNOSTICS (STABILITY VITAL SIGNS) v1.0

Status: REWRITE/UPGRADE (v1.0) — split from decay per Redline Map.

Primary Linkages: Chapter 11 (Stability Vital Signs canon), Chapter 13 (Failure Detection & Incident State Machine), Chapter 15 (Adoption Bridges + Collapse), Appendix F (Operational Decay), Appendix O (Incident Runbooks), Appendix K (Sector Threshold Playbooks).

Baseline Anchor: v0.9 stability section defines drift/resonance/divergence/entanglement

bands, stress tests, and AB/Bridge gating rules. v1.0 carries these forward as dedicated drift diagnostics distinct from decay.

Non-Redefinition Rule: SI, CI, HSS v2.0, SCP, RB v2, and AB schema are canonical only in Part III. This appendix references them as control couplings and trigger families only; it SHALL NOT redefine their formulas, weights, or bands.

G.0 Purpose and Position in the Framework

G.0.1 Purpose

This appendix specifies the mandatory diagnostics for **Semantic Drift** within Symbiosis systems. Drift diagnostics provide stability vital signs that detect unintended trajectory changes in meaning alignment between human intent and agent behavior over time. These signals are **control gates**, not optional analytics.

G.0.2 Position

1. Drift diagnostics operate across the Human (H), Engine (E), Agentic (A), and Governance (G) layers as a single stability subsystem.
 2. Drift diagnostics are distinct from Operational Decay checks (Appendix F). Conflation is prohibited.
 3. Drift red-bands SHALL automatically trigger AB demotion and Adoption Bridge revocation per Chapter 13 and Chapter 15.
-

G.1 Definitions

G.1.1 Semantic Drift (D)

A measurable deviation of system outputs and behaviors away from the accepted intent manifold and constraint-consistent meaning over time, **without requiring a corresponding operational failure**. Drift is a trajectory fault: intent and outputs slowly diverge while appearing locally “reasonable.”

G.1.2 Resonance (R)

Reinforcing feedback between H and A that amplifies a trajectory.

- **Positive resonance** strengthens correct alignment.
- **Negative resonance** strengthens a wrong path (“wrong-path resonance”).

G.1.3 Divergence (V)

A widening dispersion of outputs around the accepted solution region, including instability under fatigue or tool/model churn.

G.1.4 Entanglement Blast Radius (E_blast)

Propagation risk when shared agents, tools, or contracts couple multiple workflows, such that drift in one domain contaminates others.

G.1.5 Drift Lineage Stream (L_drift)

A dedicated lineage substream containing only drift diagnostics, band verdicts, and stability-triggered interventions. L_drift MUST be separate from L_decay and episode lineage.

G.2 Canonical Stability Signals (Reference-Only)

The following stability signals are canonical in Chapter 11. This appendix specifies their **diagnostic usage** only.

G.2.1 Drift magnitude D_t

A rolling measure of semantic deviation per workflow, domain, and agent cluster.

Required properties:

- computed per governance-defined window W_{drift} ,
- normalized to $[0, 1]$ for banding,
- attributable to a specific intent lineage slice.

G.2.2 Resonance index p_t / R_t

Measures reinforcement direction and strength between H and A trajectories.

Required properties:

- signed value with positive/negative bands,
- computed per W_{res} ,
- tied to override patterns and SI trajectory.

G.2.3 Divergence V_{contract}

Measures dispersion relative to acceptable solution regions and constraint boundaries.

Required properties:

- computed at the TG node-type level and aggregated to episode/domain,
- sensitive to SCP overload signatures.

G.2.4 Entanglement blast radius E_{blast}

Measures cross-workflow coupling risk from shared MCP contracts/tools.

Required properties:

- computed per shared resource graph,
 - used as AB promotion blocker and Bridge lock precursor.
-

G.3 Thresholds and Banding (Governance-Configured)

G.3.1 Banding principle

Each stability signal **MUST** be banded **green / yellow / red** per sector and domain, with thresholds pinned in GK policy and sector playbooks.

G.3.2 Sectoral tuning

Threshold tightness **SHALL** vary by sector archetype:

- **Class A / high-stakes sectors:** tight drift and divergence tolerances.
- **Ambiguity sectors (creative / exploratory):** higher drift tolerance but tight entanglement ceilings.
- **Economic-sensitive sectors:** resonance volatility tolerated only while SI remains green.

G.3.3 No universal defaults

This appendix **SHALL NOT** specify numeric thresholds. GK policy is the sole locus of numeric stability thresholds.

G.4 Drift Detection Protocols

G.4.1 Per-episode drift checks (mandatory)

For every completed episode:

1. compute D_t , ρ_t , V_{contract} , E_{blast} for that episode slice,
2. assign bands per GK thresholds,
3. log bands and raw values to L_{drift} , and
4. emit any band-triggered control actions.

Per-episode checks **MUST** run before any AB promotion decision or Bridge renewal.

G.4.2 Rolling window drift checks (mandatory)

At governance-defined cadence (minimum weekly, or faster for high-stakes sectors):

1. compute rolling stability trajectories over W_{drift} , W_{res} , W_{div} , W_{ent} ,
2. detect monotone drift trends even without red-bands ("silent drift"), and
3. surface stability trend deltas to H and Governance Leads.

G.4.3 Drift attribution (mandatory)

When any signal enters yellow or red, the Engine **MUST** attempt causal attribution by tagging:

- dominant layer locus (H/E/A/G),
- adjacency to incidents (Appendix O),
- adjacent autonomy level and RB proximity,

- tool/model/contract change events,
- SCP overload markers.

Attribution is diagnostic only; it does not waive red-band actions.

G.5 Control Integration with SI / AB / HSS-SCP (Reference-Only)

G.5.1 Coupling principle

- SI supplies **performance control**; stability signals supply **trajectory control**.
- If SI and stability disagree, stability SHALL win for safety.

G.5.2 AB promotion gate

Any workflow proposed for $AB \geq 2$ promotion MUST satisfy:

1. SI in green band over $W_{promote}$, AND
2. D_t , $V_{contract}$, and E_{blast} in green band over $W_{promote}$.

G.5.3 Drift red-band actions (mandatory) A drift or negative resonance red-band SHALL automatically trigger:

- Bridge revocation (if active),
- AB demotion to AB0 for affected workflows,
- Safe Mode entry per Chapter 13, and
- mandatory re-anchoring of baseline intent I_0 .

G.5.4 SCP overload as leading indicator

SCP breach is treated as a leading indicator for drift acceleration and false resonance. SCP breach SHALL pre-emptively throttle workflows before stability red-bands emerge.

G.6 Common Co-Evolution Drift Failure Modes (Diagnostic Taxonomy)

The Engine and Governance SHALL tag drift events with one or more of these canonical patterns.

1. **Silent drift** — alignment decay normalized as “acceptable.”
2. **Wrong-path resonance** — H and A reinforce a mis-specified objective.
3. **Divergence under fatigue** — SCP overload prevents correction, widening $V_{contract}$.
4. **Entanglement cascades** — shared MCP contracts/tools propagate local drift globally.
5. **Loop desynchronization** — rapid agent/tool upgrades without human training or policy recompilation.

Each mode maps to corresponding incident tags and runbooks in Appendix O.

G.7 Simulation Framework (Pre-AB3 Stress Testing)

G.7.1 Purpose

Simulations forecast whether co-evolution converges (higher SI + stable trajectories) or diverges into drift failure. Before any AB3 scale in any domain, simulation is mandatory.

G.7.2 Minimum simulation setup

Simulations MUST include:

- initial H and A states (capabilities, trust, CI slices),
- estimated feedback functions f and g from pilots,
- stochastic disturbance terms,
- governance interventions (AB changes, constraint tightening).

G.7.3 Required stress tests (normative) Before AB3 scale, simulate at minimum:

1. intent drift injection,
2. economic bleed shock,
3. trust collapse / override surge,
4. shared-tool entanglement event,
5. model swap discontinuity.

If any test yields unstable attractors or red-band stability trajectories, AB3 is disallowed until mitigations exist.

G.8 Implementation Checklist (Minimum Compliance)

A deployment passes Appendix G only if:

- [] Drift, resonance, divergence, entanglement computed per episode and rolling windows.
- [] Thresholds and banding are sector-configured; GK thresholds are pinned and versioned.
- [] Stability green bands required for $AB \geq 2$ promotions; red bands auto-demote AB and revoke Bridges.
- [] L_drift exists and is lineage-recoverable, distinct from decay lineage.
- [] Co-evolution simulations run before AB3 scale; failures block promotion.

Failure of any item disqualifies certification claims under Chapter 17.

SQC Gate Check (Appendix G)

Gate 1 — Layer Integrity: PASS

- Diagnostics are cross-layer, computed/enforced by Engine and GK; agents do not self-certify.

Gate 2 — Canonical Metrics Non-Redefinition: PASS

- Stability signals referenced only; SI/CI/HSS/SCP/AB/RB not redefined.

Gate 3 — Constraint Precedence + Threshold Explicitness: PASS

- Numeric thresholds explicitly reserved for GK/sector playbooks; no universal defaults.

Gate 4 — Mechanistic Testability: PASS

- Protocols, banding, triggers, simulations, and checklist yield deterministic tests.

Gate 5 — Lineage Recoverability: PASS

- Mandatory L_drift stream and per-episode/rolling commitments specified.

Gate 6 — Drift vs Decay Typing: PASS

- Semantic drift defined and operationally isolated from decay (Appendix F).

Gate 7 — Abuse / Incentive Hardening: PASS

- Stability-wins rule, auto-demotion, Bridge revocation, and AB3 stress tests block slow autonomy creep and wrong-path resonance.

Gate 8 — Normative Voice Consistency: PASS

- MUST/SHALL/MAY used precisely; no speculative language.

APPENDIX H — HUMAN SYMBIOSIS SCORE (HSS v2.0) — EXTENDED SUPPORT PACKAGE v1.0

Status: REWRITE/UPGRADE (supporting appendix; canonical HSS definition remains in Part III, Chapter 8).

Primary Linkages: Chapter 8 (canonical HSS v2.0), Chapter 9 (canonical SCP), Chapter 11 (RB/AB canon), Chapter 13 (Engine Algorithm v2), Chapter 14 (Drift vs Decay), Chapter 16 (Adoption Phases), Chapter 17 (Certification), Appendix F (Decay), Appendix G (Drift), Appendix P (Compliance Checklist).

Redline Basis: v1.0 requires (i) a hardened safety-only classification, (ii) explicit RBAC + retention + union clauses, (iii) anti-gaming test suite, and (iv) dashboard requirements.

Non-Redefinition Rule: SI, HSS v2.0, SCP, CI, RB v2, and AB schema are canonical only in Part III. This appendix provides **rubrics, governance hardening, and operational artifacts**. It SHALL NOT introduce alternate formulas, weights, or bands.

H.0 Overview (supporting recap)

H.0.1 Definition (recap only)

The Human Symbiosis Score (HSS v2.0) is a six-domain composite safety metric quantifying an operator's capability to participate safely, predictably, and effectively in hybrid human–AI systems.

H.0.2 Classification

1. HSS SHALL be classified exclusively as **Safety Telemetry**.
2. HSS SHALL NOT be used as a performance, productivity, or HR instrument. Any such use is a Symbiosis compliance violation.

H.0.3 Dependencies

HSS interacts directly with TG, RB, AB, SCP, SI, and GK governance constraints.

H.1 Objectives

HSS v2.0 SHALL:

1. Validate operator capability matches requested autonomy.
2. Detect fatigue/instability before RB excursions.
3. Stabilize TG construction and reduce operator-induced drift.
4. Provide objective gates for AB promotion/demotion.
5. Produce regulator-defensible human-side oversight evidence.

H.2 Six-Domain Model (supporting detail)

HSS consists of six independently measured domains. Domain definitions below are support-level restatements consistent with Chapter 8.

HSS-1 Intent Clarity (IC)

Definition: Degree to which the operator produces interpretable, low-ambiguity instructions that E can translate into a valid TG.

Signals (interaction-derived, non-biometric): interpretability score, revision delta, constraint conflicts, ambiguity markers.

HSS-2 Constraint Awareness (CA)

Definition: Demonstrated ability to anticipate, respect, and operate within C_legal / C_safety / C_econ / C_ops.

Signals: constraint violations per 100 tasks, pre-execution adjustments, governance override frequency, exception request patterns.

HSS-3 Collaboration Discipline (CD)

Definition: Reliability in following protocol steps, approving Dry-Runs, responding to checkpoints, and providing stable feedback.

Signals: Dry-Run approval latency, audit acknowledgment, RB checkpoint responsiveness, protocol compliance curve.

HSS-4 Cognitive Load Regulation (CLR)

Definition: Capacity to maintain reliable oversight under varying load without destabilizing TGs or expanding risk surfaces.

Signals: error-slope, task-switch volatility, fatigue markers (interaction-derived), revision explosiveness.

HSS-5 Interaction Stability (IS)

Definition: Predictability, low volatility, and low contradiction rate across sequential instructions.

Signals: prompt volatility index, contradiction rate, scope oscillation frequency, rollback triggers linked to operator input.

HSS-6 Outcome Reliability (OR)

Definition: Consistency of operator influence on successful TG execution and SI uplift, normalized over complexity/time.

Signals: SI uplift attributable to operator direction, TG outcome delta vs expected, regression patterns, success ratio post-intervention.

H.3 Composite Calculation (recap only)

The composite formula and weights are canonical in Chapter 8 and restated here for implementation alignment.

$$\text{HSS} = 0.20 \cdot \text{IC} + 0.20 \cdot \text{CA} + 0.15 \cdot \text{CD} + 0.15 \cdot \text{CLR} + 0.15 \cdot \text{IS} + 0.15 \cdot \text{OR}.$$

Properties:

- $\text{HSS} \in [0.00, 1.00]$.
 - Rolling window smoothing SHALL be applied per GK policy to reduce noise and gaming (see H.6).
-

H.4 Proficiency Bands & Autonomy Rights (supporting table)

Three HSS bands gate AB eligibility. Band thresholds are canonical in Chapter 8 and repeated here for support.

HSS Band	Score Range	Eligible AB	Mandatory Restrictions
B1 — Guided Operator	≤ 0.45	AB0–AB1	Mandatory Dry-Runs; no RB-proximate tasks; tight constraint supervision; frequent checkpoints.
B2 — Competent Symbiote	0.45–0.72	AB2	RB tasks require preview + checkpoint; stable low-volatility operation required.
B3 — Commander-Level Operator	≥ 0.72	AB3	Explicit opt-in; governance endorsement; reversibility modeling training.

H.5 Behaviorally Anchored Rating Scale (Rubric)

H.5.1 Rubric purpose

HSS domains SHALL be scored using a Behaviorally Anchored Rating Scale (BARS) emphasizing **process over outputs**.

H.5.2 Standard range anchors (apply to each domain)

For each domain $d \in \{IC, CA, CD, CLR, IS, OR\}$, the following rubric ranges SHALL be used:

Range	Anchor	Typical Indicators
-------	--------	--------------------

0.00–0.19	Severe deficit	Repeated contradictions, high override misuse, unsafe volatility, TG thrash.
0.20–0.39	Unstable / unreliable	Frequent instability, shallow constraint handling, inconsistent checkpointing.
0.40–0.59	Manageable / developing	Occasional inconsistencies; safe but requires guidance.
0.60–0.79	Stable / competent	Predictable collaboration patterns; low volatility and strong influence on outcomes.
0.80–1.00	Exemplar	Near-zero contradiction, proactive constraint use, consistent SI uplift.

H.5.3 Domain-specific risk annotations

- IS < 0.40 is strongly predictive of semantic drift events.
- OR < 0.40 SHALL cap autonomy at AB1 pending retraining.

H.5.4 Worked examples (composite)

Example outcomes SHALL be interpretable in terms of AB eligibility:

- HSS $\approx$ 0.44 $\rightarrow$ B1 $\rightarrow$ AB0–AB1 only.
- HSS $\approx$ 0.64 $\rightarrow$ B2 $\rightarrow$ AB2 eligible.
- HSS $\approx$ 0.85 $\rightarrow$ B3 $\rightarrow$ AB3 eligible with opt-in + endorsement.

H.6 Adjustment Factors (anti-noise, anti-gaming)

The following modifiers SHALL be applied and logged in lineage.

1. Fatigue Adjustment

Sessions > 2 hours reduce CLR by 0.02–0.07 depending on error slope.

2. Context Switching Factor

5 domain switches within <10 minutes reduces IS by 0.03–0.10.

3. High-Regret Context Load

RB-proximate tasks apply a +0.05 CA minimum threshold.

H.7 Privacy, Ethical, and Legal Safeguards (mandatory hardening)

These clauses are normative framework-wide protections and **MUST** be enforced as written.

H.7.1 Safety-only use and purpose limitation

HSS SHALL be used only for: AB gating, safety-related training, SCP calibration, and governance risk decisions. Any secondary HR use is prohibited.

H.7.2 Visibility rules

- Individual HSS visibility: Operator (self) and authorized Governance Leads only.
- Managers, supervisors, HR, and line business units SHALL have no individual access.

H.7.3 Retention and deletion

- Default retention limit: 180 days.
- Expired records **MUST** be deleted or anonymized with verifiable deletion logs.
- Operators **MAY** request purge except during active safety investigations.

H.7.4 Explicit opt-in for B3

HSS-B3 participation requires explicit opt-in; opt-out SHALL NOT affect employment or compensation.

H.7.5 Managerial firewall

Managers SHALL NOT request, coerce, or imply consequences tied to HSS. Violations are audit-grade non-compliance.

H.7.6 Collective consent (required for unionized contexts)

In collective bargaining environments, deployment of HSS SHALL require joint approval by the Governance Body and recognized works council/bargaining unit.

H.8 Anti-Gaming Invariants and Test Suite

H.8.1 Invariants

HSS SHALL resist manipulation. Detected inflation SHALL trigger calibration review, not operator penalty.

H.8.2 Mandatory anti-gaming tests

Deployments MUST implement at least these tests:

1. **Cross-domain consistency checks** (e.g., IC spikes without CA/CD coherence).
2. **Volatility spike detection** for pattern-breaking bursts.
3. **Reward-seeking prompt structure detection** (Goodhart signatures).
4. **Drift/decay symmetry analysis** (gaming often creates instability markers).
5. **Pattern discontinuity detection** across rolling windows.
6. **Randomized operator-side audits** of override reasoning.

H.8.3 Governance response

If anti-gaming tests flag suspected inflation:

- the Engine MUST freeze HSS promotions for the affected window,
 - Governance MUST perform a calibration review using lineage evidence,
 - no punitive employment action MAY occur on the basis of flags alone.
-

H.9 Integration Couplings (reference-only)

H.9.1 HSS ↔ SCP

SCP informs CLR and safe operating envelopes; SCP anomalies SHALL trigger cooldown or AB demotion per canon.

H.9.2 HSS ↔ RB

- $HSS < 0.45$: no RB-proximate tasks.
- $0.45 \leq HSS < 0.72$: RB-proximate tasks require Dry-Run + governance checkpoint.
- $HSS \geq 0.72$: RB tasks allowed with reversibility modeling.

H.9.3 HSS ↔ Certification

HSS distributions, anti-gaming logs, RBAC evidence, and retention compliance SHALL be included in Chapter 17 certification packets.

H.10 Operator Dashboard Requirements (mandatory UI artifacts)

Deployments SHALL provide an Operator HSS/SCP dashboard meeting the following minimum widgets and behaviors.

Widget A — Safe Operating Envelope

Inputs: SCP-L/S/F/R. Outputs: headroom %, fatigue warning, recovery estimate.

Behavior: if SCP-F exceeds threshold → show cooldown; if `Active_Interventions` > SCP limit → recommend batching/throttle.

Widget B — HSS Domain Panel

Inputs: IC, CA, CD, CLR, IS, OR (rolling window). Output: per-domain bars + composite.

Behavior: trend arrows required; any domain < 0.40 MUST surface safety warning and recommend AB demotion.

Widget C — Trend View (30–90 day selectable)

Shows HSS composite, SCP headroom, drift/decay markers.

Behavior: drift vs decay visually distinct; clicking marker opens lineage slice.

Widget D — Autonomy Gating Panel

Shows AB eligibility live, blockers, safe upgrade path.

Behavior: MUST explain why AB3 disallowed; MUST show minimum path to eligibility.

Widget E — Intervention Budget

Shows active TG count vs SCP-CON, interrupts/hour, Dry-Run latency.

Behavior: approaching cap → recommend batching; rising Dry-Run latency → recommend autonomy reduction or concurrency decrease.

Widget F — Privacy / Safety Status

Shows RBAC enforcement, retention clock, misuse report channel.

Behavior: unauthorized access attempts MUST trigger hard alert and lineage event.

Global dashboard constraints:

1. HSS privacy enforcement MUST be visible to the operator.
2. AB eligibility MUST be explainable in plain language.
3. RB proximity warnings MUST surface pre-execution.
4. SCP overload MUST trigger throttle/slowdown advisories.
5. No comparative, leaderboard, or cross-operator ranking views SHALL exist.

H.11 Summary (supporting close)

HSS v2.0 provides:

1. A normalized, regulator-defensible human-capability safety metric.
2. A deterministic AB gating instrument aligned to RB and SCP.
3. A privacy-hardened, union-safe telemetry regime protected by RFAC, retention caps, opt-in, and misuse channels.
4. An anti-gaming system grounded in lineage evidence and calibration review.
5. An operator-facing dashboard ensuring transparency and safe autonomy growth.

Appendix H is a support artifact; all binding decisions SHALL reference canonical HSS definitions in Chapter 8.

SQC Gate Check (Appendix H)

Gate 1 — Layer Integrity: PASS

- HSS described as Human-layer safety telemetry enforced by Engine/GK; no layer bypass.

Gate 2 — Canonical Metrics Non-Redefinition: PASS

- No alternate HSS/SCP/SI/CI/RB/AB definitions introduced; composite restated for alignment only.

Gate 3 — Constraint Precedence + Threshold Explicitness: PASS

- All numeric thresholds reserved to GK/sector playbooks; privacy/retention constraints explicit and overriding.

Gate 4 — Mechanistic Testability: PASS

- Rubric ranges, adjustments, RBAC, retention, anti-gaming tests, and dashboard behaviors are implementable.

Gate 5 — Lineage Recoverability: PASS

- Required lineage logging for adjustments, audits, access attempts, and anti-gaming reviews specified.

Gate 6 — Drift vs Decay Typing: PASS

- HSS anti-gaming and dashboard separate drift vs decay markers; no conflation.

Gate 7 — Abuse / Incentive Hardening: PASS

- Safety-only rule, managerial firewall, opt-in, union consent, no leaderboards, and misuse channel close abuse vectors.

Gate 8 — Normative Voice Consistency: PASS

- MUST/SHALL/MAY used correctly; no speculative or marketing tone.

Appendix I — Symbiosis Literacy Program (SLP) Syllabus v1.0

Status: Normative, Implementation Program Appendix

Position: Part VI — Appendices. Non-canonical metrics reference only.

I.0 Purpose and Role in the Framework

The Symbiosis Literacy Program (SLP) defines the minimum training, verification, and recertification syllabus required for humans to participate in Symbiosis-compliant hybrid systems. SLP is the human-side analogue to contract registration and governance kernel hardening: it ensures operators, architects, and governance leads possess sufficient literacy to form valid intents, supervise autonomy, interpret telemetry, and avoid Oversight Theater.

Humans are first-class system components whose state signals include HSS and SCP. Consequently, literacy is a safety and economic prerequisite, not an optional soft program.

SLP SHALL be used to:

1. Establish baseline competence before any autonomy above AB0 is permitted.
2. Provide standardized progression paths tied to HSS/SCP improvement and autonomy eligibility.
3. Create audit-grade evidence that human participants were trained to the same standard the system enforces.

SLP content SHALL reference canonical metric definitions from Part III and SHALL NOT redefine SI, HSS v2.0, SCP, CI, RB, or AB schema herein.

I.1 Program Structure

I.1.1 Tracks

SLP SHALL include two formal tracks:

- **Track A — Symbiote Operator (SO):** for day-to-day hybrid workflow operators.
- **Track B — Hybrid Intelligence Architect / Governance Lead (HIA/GL):** for system configurators, playbook owners, and audit authorities.

I.1.2 Levels

Each track SHALL be delivered across three progressive levels:

- **Level 1 — Foundations (mandatory for any Symbiosis use).**
- **Level 2 — Operational Proficiency (mandatory for AB1+ participation).**
- **Level 3 — Advanced Co-Agency Literacy (mandatory for AB2+/RB-prox deployments).**

I.1.3 Delivery Modalities

The program MAY be delivered through: classroom, synchronous remote instruction, or Engine-assisted guided modules, provided all assessments are proctored or equivalently integrity-secured.

I.1.4 Normative Evidence

Completion records SHALL be logged as Governance-signed training artifacts, and MAY be attached to personnel records only under the Non-Punitive Covenant and HSS privacy rules. Cross-reference Appendix H for anti-gaming, privacy, and non-coercion safeguards.

I.2 Track A — Symbiote Operator (SO)

I.2.1 Level 1 — Foundations

Operators MUST demonstrate competence in:

1. Four-Layer Architecture Literacy

- Identify Human / Engine / Agentic / Governance layer responsibilities.
- Detect and refuse shadow-path requests (e.g., bypassing constraints or AB ceilings).

2. Intent Formation Basics

- Produce Structured Intent Objects via approved Alignment Surfaces.
- Specify goal, output spec, constraints seed C_0 , and budget envelope B .
- Distinguish between high- and low-fidelity intent surfaces.

3. Constraint Supremacy Awareness

- Explain C_{legal} , C_{safety} , C_{econ} and why hard constraints override optimization.
- Recognize tasks that are non-delegable under local policy.

4. Lineage Awareness

- Understand that every action is logged to Context Lineage L .
- Interpret “why” traces at a basic level (intent → plan → action).

Assessment:

- Written exam (closed book).
- Practical lab: generate a compliant Intent Object for a given workflow and identify missing constraints/budget fields.

Passing rule:

- $\geq 80\%$ on written exam.
- Practical lab MUST pass all compliance checks.

I.2.2 Level 2 — Operational Proficiency

Operators MUST demonstrate competence in:

1. Autonomy Bands and Regret Boundary (RB) Supervision

- Understand AB0–AB3 semantics and local ceilings.
- Recognize RB-prox tasks and required checkpointing behavior.
- Execute RB checkpoint reviews with documented rationale.

2. Part III Metric Interpretation (Reference-Only)

- Read SI trends and understand gating implications.
- Interpret HSS and SCP dashboards to self-throttle before saturation.
- Demonstrate use of control actions defined in Chapter 10.4.

3. Economic Circuit Awareness

- Read projected vs realized cost summaries.
- Halt or re-plan when Economic Circuit breakers trigger.

4. Oversight Theater Prevention

- Identify cognitive overload signals and SCP headroom violations.
- Apply mandatory throttle / pause protocols when saturation cues rise.

Assessment:

- Scenario-based simulation with autonomy gating and economic breakers.
- Oral review of two RB checkpoints and one cost overrun incident.

Passing rule:

- Simulation MUST show correct routing and checkpoint behavior.
- Oral review MUST show ability to articulate intent ownership and regret authority.

I.2.3 Level 3 — Advanced Co-Agency Literacy

Operators in AB2+/co-agency contexts MUST demonstrate competence in:

1. Multi-Agent Workflow Supervision

- Read Task Graph TG summaries and identify unsafe node sequencing.
- Interpret contract ceilings and permissions at a high level.

2. Drift vs Decay Literacy

- Differentiate semantic drift vs operational decay in incidents.
- Trigger correct escalation paths and lineage annotations.

3. Governance Interface Practice

- File structured incident reports to Governance Layer.
- Participate in post-incident reconciliation without blame framing.

Assessment:

- Red-team simulation (drift and decay embedded).
- Governance tabletop exercise.

Passing rule:

- Zero tolerance for missed drift/decay typing.
- Correct escalation and lineage recording required.

I.3 Track B — Hybrid Intelligence Architect / Governance Lead (HIA/GL)

I.3.1 Level 1 — Foundations

HIA/GL candidates MUST demonstrate competence in:

1. Canonical Ontology and Type System Use

- Apply Symbiosis types (H, I, C, B, TG, MCP, AB, RB, L) correctly.
- Ensure cross-document terminology alignment.

2. Governance Kernel (GK) and Policy Supremacy

- Explain GK isolation and enforcement boundaries.
- Compile sector overlays into Constraint Vectors.

Assessment:

- Written exam including ontology typing and boundary cases.

I.3.2 Level 2 — Operational Proficiency

HIA/GL candidates MUST demonstrate competence in:

1. Engine Configuration and Safety Hooks

- Configure Router, Planner, Economic Pass thresholds, and Dry-Run policies.
- Ensure no bypass of adjacent boundaries (no shadow paths).

2. Contract and Capability Governance

- Draft MCP contracts with explicit AB ceilings, RB typing, cost caps, and rollback paths.
- Register contracts into append-only registries with audit metadata.

3. Metric-Linked Autonomy Design (Reference-Only)

- Apply Part III canonical metrics to promotion/demotion rules.
- Set governance-signed thresholds (deployment-specific).

Assessment:

- Practical lab: configure a sandbox Engine + Agent cluster to a sector playbook.
- Contract drafting test (pass/fail).

I.3.3 Level 3 — Advanced Co-Agency Governance

HIA/GL candidates for AB2+/AB3 deployments MUST demonstrate competence in:

1. Regime Engineering

- Design RB-bounded co-agency patterns.
- Validate against stress cases and failure taxonomies.

2. Audit and Forensics Readiness

- Produce full lineage reconstruction packages for simulated incidents.
- Demonstrate external-regulator narrative based on evidence.

3. Anti-Gaming and Human-Side Safeguards

- Enforce HSS/SCP privacy, aggregation, opt-in, and non-coercion rules.
- Detect metric manipulation attempts and apply sanctions through GK, not HR channels.

Assessment:

- Full-cycle deployment readiness review with independent governance examiner.
- Forensics replay exam.

Passing rule:

- Any failure in audit reconstructability or safeguard enforcement is disqualifying.
-

I.4 Certification, Autonomy Eligibility, and Recertification

I.4.1 Initial Certification

- **Track A Level 1 certification** is mandatory for any Symbiosis participation.
- **Track A Level 2 certification** is mandatory for operators to be eligible for AB1 participation.
- **Track A Level 3 certification** is mandatory for AB2+ eligibility.
- **Track B Level 2 certification** is mandatory for any Engine or governance configuration access.
- **Track B Level 3 certification** is mandatory to authorize AB3 scale.

I.4.2 Recertification

SLP certifications SHALL expire every 12 months or after any S2+ incident involving the certified individual's workflows, whichever occurs first.

Recertification SHALL include:

1. A delta-module covering version changes since last certification.
2. One proctored exam or simulation relevant to the individual's track/level.

I.4.3 Suspension

Governance MAY suspend certification if:

- repeated checkpoint negligence occurs,
- intentional shadow-path use is detected,
- anti-gaming violations are confirmed.

Suspension SHALL be handled through governance and training remediation; HR discipline paths are prohibited unless separate misconduct exists.

I.5 Program Artifacts and Templates

SLP SHALL ship with the following binder-ready artifacts:

1. **Intent Quality Workbook**
 - examples of compliant/non-compliant Intent Objects.
 - constraint and budget checklists.
2. **RB Checkpoint Review Form**
 - required rationale fields.
 - linkage to lineage hashes.
3. **Operator Simulation Pack**
 - drift vs decay scenarios.
 - economic circuit overrun drills.
4. **Architect Configuration Labs**
 - sector overlay compilation exercises.
 - MCP schema drafting templates.

These artifacts SHALL be versioned and governance-signed, with updates logged to Context Lineage for auditability.

I.6 Conformance Summary (Normative)

A deployment MAY claim SLP compliance only if:

1. All Sybiote Operators and Architect/Governance Leads are certified to the levels required for their assigned AB ceilings.
 2. Certification evidence is governance-signed and retrievable for audits.
 3. Recertification and suspension rules are enforced without exception.
 4. Training content references Part III canonical metrics without redefining them.
 5. Anti-gaming, privacy, and non-coercion safeguards for human metrics remain intact.
-

SQC Gate Check (Appendix I)

Gate 1 — Layer Integrity: **PASS**. Roles, duties, and boundaries are consistent with Four-Layer Architecture; no cross-assignment.

Gate 2 — Canonical Metrics Non-Redefinition: **PASS**. SI/HSS/SCP/CI/RB/AB are referenced only; no alternate definitions.

Gate 3 — Constraint Precedence + Threshold Explicitness: **PASS**. Training treats constraints as hard overrides; no universal thresholds asserted.

Gate 4 — Mechanistic Testability: **PASS**. All requirements are operationalized as assessable competencies and pass rules.

Gate 5 — Lineage Recoverability: **PASS**. Certification evidence, RB reviews, and incident exercises require lineage linkage and reconstruction.

Gate 6 — Drift vs Decay Typing: **PASS**. Explicit literacy and assessment coverage included.

Gate 7 — Abuse / Incentive Hardening: **PASS**. Non-punitive, anti-coercion, and governance-only enforcement specified; references Appendix H.

Gate 8 — Normative Voice Consistency: **PASS**. MUST/SHALL/MAY used precisely; no speculative or marketing tone.

Appendix J — Hybrid-Intelligence Hiring Rubrics & Role Specs (v1.0)

Purpose.

This appendix specifies standardized role definitions, competency models, and hiring rubrics for Symbiosis-compliant deployments. These roles are the *human substrate* required to maintain Human Primacy, Constraint Supremacy, Economic Viability, and governable autonomy. Organizations may extend these rubrics, but may not weaken their invariants.

Scope.

Covers three mandatory roles:

1. **Symbiote Operator (SO)** — primary workflow executor and autonomy supervisor.
2. **Hybrid Intelligence Architect (HIA)** — system/engine/agent ecosystem designer.
3. **Symbiosis Governance Lead (SGL)** — policy-to-constraint compiler and audit authority.

Normative statement.

A deployment claiming Symbiosis compliance must assign accountable humans to each role (or document equivalent combined roles) and must use a rubric that tests the competencies below.

J.1 Shared Baseline Competencies (All Roles)

These competencies are evaluated in **every** Symbiosis role. Each maps to a canonical primitive.

1. **Intent Literacy (I-Literacy).**

- Can translate ambiguous goals into Structured Intent Objects (I).
- Detects missing constraints, unclear regret tolerance, and budget ambiguity.
- Understands fidelity tradeoffs across Alignment Surfaces.

2. **Constraint Reasoning (C-Literacy).**

- Demonstrates working knowledge of **C_legal, C_safety, C_econ**.
- Can identify constraint conflicts and escalation triggers.
- Treats constraint vectors as *hard boundary objects*, not suggestions.

3. **Autonomy & Regret Fluency (AB/RB-Literacy).**

- Understands autonomy bands as a state machine discretized by irreversibility.
- Can explain when RB gates must be invoked, and how AB demotion works.
- Recognizes “shadow autonomy” and approval theater.

4. **Economic Discipline (EC-Literacy).**

- Treats compute/attention as finite budgets.
- Understands economic admissibility and circuit-breaker halts.
- Can reason about cost-to-value ratio to prune TGs.

5. **Lineage & Audit Thinking (L-Literacy).**

- Can interpret Context Lineage traces to reconstruct decisions.
- Understands that transparency + lineage are mandatory for defensibility.

6. **Non-Punitive Culture Alignment (NPC-Alignment).**

- Supports the Non-Punitive Covenant and refuses “metric weaponization.”
- Can articulate safe human-metrics usage (HSS/SCP).

J.2 Role Spec — Symbiote Operator (SO)

Mission.

Execute hybrid workflows inside AB0–AB2, supervise agentic outputs, and own local regret boundaries. SOs are the “hands and eyes” of the system.

Primary responsibilities.

- Author and refine Structured Intent Objects (I) via Alignment Surfaces.
- Review Dry-Run previews and RB checkpoints.
- Interpret SI/HSS/SCP telemetry to tune task routing (human vs agent).
- Halt or escalate when constraints are threatened.
- Maintain clean lineage through disciplined context handling.

Authority boundary.

- May **halt / revert** workflows at AB0–AB1 unilaterally.
- May approve AB2 actions only at RB-gated checkpoints.
- May not change GK rules or global AB ceilings.

Required minimum profile.

- HSS: **Tier 2+** for any AB2 work; **Tier 3** recommended for AB2+ with high risk class.
- SCP: sufficient headroom for concurrency required by domain.

J.2.1 SO Competency Rubric (0–4 scale)

Score each dimension 0–4. A Symbiosis-compliant hire requires **no dimension <2** and **mean ≥3**.

- 1. Intent precision.**
 - 0: cannot operationalize goals; relies on vague prompting.
 - 2: produces workable intent but misses edge constraints.
 - 4: produces high-fidelity intents with budgets, regret tolerance, and crisp outputs.
- 2. RB checkpoint judgment.**
 - 0: rubber-stamps; cannot articulate irreversibility.
 - 2: pauses at obvious RBs.
 - 4: anticipates RB proximity, demands Dry-Runs, invokes safe-mode early.
- 3. Constraint enforcement.**
 - 0: treats policy as afterthought.
 - 2: recognizes constraints but needs help resolving conflicts.
 - 4: proactively assembles and defends constraint vectors.
- 4. Economic pruning.**
 - 0: runs open-ended tasks; ignores spend.
 - 2: respects explicit budgets.
 - 4: actively prunes TGs to maximize value per cost.
- 5. Error recovery & escalation.**
 - 0: blames model/user; no recovery plan.
 - 2: follows runbooks.
 - 4: diagnoses failure class, selects mitigation, escalates with evidence.
- 6. Lineage hygiene.**
 - 0: loses context history; cannot reproduce decisions.
 - 2: keeps basic traces.
 - 4: maintains clean ancestry, tags assumptions, supports replay.

J.2.2 SO Interview Tasks (Standard)

Task A — Intent Object Draft.

Provide a messy goal + partial constraints. Candidate must produce I' including: goal_statement, output_spec, regret_tolerance, constraint_seed C₀, and budget envelope B.

Task B — RB-prox Decision.

Show a Dry-Run trace that approaches irreversibility. Candidate decides: approve, demote AB, require more evidence, or halt — and must justify in RB terms.

Task C — Economic Circuit Scenario.

Candidate receives TG cost estimate exceeding B. Must prune or re-scope while preserving intent invariants.

J.3 Role Spec — Hybrid Intelligence Architect (HIA)

Mission.

Design and maintain the Engine + Agentic ecosystem so that intent translation, economic gating, and autonomy enforcement operate correctly at scale.

Primary responsibilities.

- Configure the Symbiosis Engine pipeline and subcomponents.
- Define Task Graph schemas, typing rules, and checkpointing patterns.
- Author MCP contract templates, cost caps, AB ceilings, escalation clauses.
- Implement memory tiering and lineage persistence.
- Tune autonomy thresholds with Governance, using SI/HSS/SCP data.

Authority boundary.

- May design/modify TG and MCP schemas.
- May propose changes to GK, but cannot unilaterally deploy governance rule changes.
- Owns technical conformance of implementation to standard.

Required minimum profile.

- Proven system architecture background plus formal safety/economics thinking.
- HSS equivalent: ability to reason about hybrid cognition and human factors.

J.3.1 HIA Competency Rubric (0–4 scale)

1. Engine decomposition.

- 0: treats system as a prompt pipeline.
- 2: understands router/planner/orchestrator separation.
- 4: can specify deterministic coordination with drift recovery.

2. Contract/agent design.

- 0: untyped agents; no caps.
- 2: basic contracts.

- 4: MCP-aligned contracts with cost/AB/RB/constraint hooks.
3. **Autonomy system design.**
 - 0: autonomy by preference.
 - 2: uses AB bands but inconsistently.
 - 4: RB-driven state machine with safe demotion rules.
 4. **Economic architecture.**
 - 0: no spend controls.
 - 2: basic throttles.
 - 4: full Economic Circuit integration, forecast + breaker behavior.
 5. **Telemetry & lineage schema.**
 - 0: logs are ad-hoc.
 - 2: captures outputs and costs.
 - 4: immutable ancestry, replayability, audit packages.
 6. **Failure-mode engineering.**
 - 0: reactive debugging only.
 - 2: knows taxonomies.
 - 4: designs mitigations into TG/MCP/GK layers.

J.3.2 HIA Interview Tasks (Standard)

Task A — Architecture Whiteboard.

Candidate diagrams the four layers and the 10-step Engine algorithm, emphasizing control-down / telemetry-up and economic admissibility.

Task B — MCP Contract Draft.

Given a TG node + risk class + budget, draft an MCP contract including AB ceiling, cost cap, constraints, rollback clause, escalation path.

J.4 Role Spec — Symbiosis Governance Lead (SGL)

Mission.

Guarantee Constraint Supremacy and auditability. Convert institutional policy and sector law into machine-enforceable C vectors and GK rule overlays.

Primary responsibilities.

- Define risk classes and AB/RB ceilings per domain.
- Compile policy → constraint vectors; own constraint versioning.
- Operate incident gates: safe-mode, AB demotion, Adoption Bridge revocation.
- Run audits and post-mortems using lineage evidence.
- Maintain Non-Punitive Covenant enforcement.

Authority boundary.

- Sole authority to **demote AB globally** and revoke Adoption Bridges.
- Sole authority to approve GK policy changes.
- Cannot originate intent or direct agents outside constraint processes.

Required minimum profile.

- Domain compliance / safety background plus technical literacy.
- Demonstrated ability to design governance as systems, not paperwork.

J.4.1 SGL Competency Rubric (0–4 scale)

- 1. Policy-to-constraint compilation.**
 - 0: policies stay as PDFs.
 - 2: basic rules.
 - 4: machine-enforceable C_legal/C_safety/C_econ with test cases.
- 2. Risk-class & autonomy design.**
 - 0: no clear escalation map.
 - 2: static thresholds.
 - 4: dynamic thresholds tied to SI/HSS/SCP telemetry.
- 3. Audit & defensibility.**
 - 0: cannot reconstruct incidents.
 - 2: partial lineage usage.
 - 4: full replay/audit packages, clear accountability.
- 4. Incident command.**
 - 0: panics or blames users.
 - 2: follows checklists.
 - 4: executes safe-mode and demotion with precision and evidence.
- 5. Human metrics ethics.**
 - 0: uses HSS/SCP punitively.
 - 2: aware of covenant.
 - 4: designs safeguards; enforces aggregation/opt-in boundaries.
- 6. Cross-layer negotiation.**
 - 0: adversarial to operators/architects.
 - 2: cooperative but unclear roles.
 - 4: clear decision rights, high-trust escalation culture.

J.4.2 SGL Interview Tasks (Standard)

Task A — Constraint Derivation.

Provide a policy excerpt (e.g., privacy rule + safety rule + cost limit). Candidate must output a constraint vector and specify where in the pipeline it injects.

Task B — Audit Reconstruction.

Candidate receives a lineage trace from a failed workflow and must: classify failure mode, identify RB breach (if any), assign accountability, and propose GK or policy patch.

J.5 Hiring Decision Thresholds (Normative)

A candidate is Symbiosis-eligible only if:

1. **No shared baseline competency <2.**
2. **Role-specific mean score ≥3.0.**
3. **Demonstrated respect for Human Primacy and Non-Punitive Covenant.**
4. **Can explain AB/RB and Economic Circuit without prompting.**

Failing any criterion is a hard no for Symbiosis compliance.

J.6 Onboarding & Continuous Certification

Initial certification.

- SOs: must pass AB1 simulation and RB-prox drills.
- HIAs: must produce a reference TG/MCP suite and pass economic admissibility lab.
- SGLs: must compile a sector overlay into constraints and complete an audit mock.

Re-certification cadence.

- Every 6–12 months, or after any major GK / policy engine update.
- Any role involved in a Class-2+ incident must re-certify within 30 days.

Training signals.

- SI declines tied to human-side causes trigger targeted SO or HIA upskilling.
 - Repeated constraint violations trigger SGL retraining and policy patch review.
-

J.7 Minimal Org Chart Requirement (Implementation Note)

A Symbiosis deployment reaching AB2+ in production must satisfy at least one of:

- **Distinct humans per role**, or
 - **Documented combined roles** with a conflict-of-interest mitigation plan (e.g., a combined HIA+SGL must have an external audit counterparty).
-

J.8 Appendix Close — Compliance Summary

Appendix J operationalizes the human-side requirements of the Symbiosis Framework: it defines who must exist in the system, what they must be able to do, and how to test it. Without these roles and rubrics, Human Primacy collapses into oversight theater, constraints become non-binding, and economic controls drift.

Appendix K — Sector-Specific Compliance and Governance Matrices (CCMT) + Standards Crosswalk (v1.0)

Status: Normative, binder-ready.
Scope: This appendix defines the canonical mechanism for translating sectoral law, regulation, and institutional policy into Symbiosis primitives, by means of Compliance and Constraint Matrices. It also provides a standards crosswalk to external regimes. This appendix is a living governance artifact and SHALL be treated as slow-core in co-evolution.

K.0 Purpose and Scope
K.0.1 Purpose

- 1. Appendix K specifies how a Governance Layer SHALL encode sector-specific requirements into:
 - a. Constraint Vectors $C = \langle C_legal, C_safety, C_econ \rangle$,
 - b. Risk Classes,
 - c. Autonomy Band ceilings and Regret Boundary checkpoint rules, and
 - d. Evidence and audit bundles.
- 2. Appendix K is the sole source for sector-bound compliance overlays used by:
 - a. the Policy Engine and Governance Kernel (Chapter 8),
 - b. Sector Implementation Playbooks (Chapter 12 / Chapter 17), and
 - c. Pilot proofs and adoption trajectories (Appendix N).

K.0.2 Scope

- 1. Appendix K applies to any Symbiosis-compliant deployment operating in a regulated or safety-relevant context.
- 2. The matrices herein are templates and illustrative instantiations. Each deployment SHALL localize:
 - a. specific legal citations,
 - b. numeric thresholds for SI-triggered control actions, and
 - c. AB/RB ceilings for task classes.
- 3. Appendix K SHALL NOT redefine SI, HSS v2.0, SCP, CI, RB, or AB schema. It only references their canonical definitions in Part III.

K.1 Canonical Compliance Matrix Template (CCMT)
K.1.1 CCMT Row Schema

Each CCMT row SHALL be represented with the following fields:

Field	Type	Description
-------	------	-------------

Row_ID	string	Unique sector row identifier (e.g., H-1, F-2).
Domain	enum	Sector subdomain (e.g., “clinical decision support”).
Requirement	normative text	Concrete obligation derived from law/policy.
Risk_Class	enum	LOW / MEDIUM / HIGH / CRITICAL.
Constraint_Vector	set	Hard and soft constraint codes to be compiled into C.
Task_Classes	set	Affected Task Graph node types.
AB_Ceiling	enum	Maximum allowed AB for those task classes.
RB_Checkpoints	rule	When human approval or safe-mode gating is mandatory.
SI_Triggers	rule	Sector SI thresholds that force autonomy demotion/freeze.
Evidence_Audit	set	Required telemetry, lineage slices, tests, and artifacts.

Owner	role	Human body responsible for authoring and maintaining row.
Policy_Version (P _v)	string	Version pointer for row governance.

K.1.2 Constraint Encoding Conventions

1. Hard constraints SHALL be labeled **Hard: { ... }** and MUST override optimization.
2. Soft constraints SHALL be labeled **Soft: { ... }** and MAY be relaxed only by Governance-authorized policy.
3. Constraint codes SHALL map to a Constraint Catalog (C-CAT) export (K-BN.7).

K.2 Translation Method: From Law to Symbiosis Primitives

K.2.1 Steps (normative)

To author a CCMT, a Governance Lead SHALL perform:

1. Regulatory decomposition
 - a. Identify primary regulatory clusters for the sector (K.3.1–K.6.1).
 - b. Extract enforceable requirements (no aspirational language).
2. Primitive mapping
 - a. For each requirement, assign Risk_Class.
 - b. Derive Constraint_Vector elements, expressed as hard/soft codes aligned to C_legal, C_safety, C_econ.
3. Autonomy binding
 - a. Map requirement to Task_Classes.
 - b. Assign AB_Ceiling and RB_Checkpoints consistent with Risk_Class and Regret Boundary doctrine (Part III, Chapter 13).
4. Control-loop binding
 - a. Specify SI_Triggers that auto-demote AB or freeze workflows for high/critical sectors.
 - b. SI thresholds SHALL be governance-set and deployment-specific; no universal defaults.
5. Evidence binding
 - a. Enumerate Evidence_Audit bundle.
 - b. Evidence MUST be lineage-recoverable (K-BN.7, Gate 5).

K.2.2 Output Artifacts

Appendix K SHALL export three machine-readable artifacts for implementers and auditors:

1. Constraint Catalog (C-CAT): all constraint codes, text definitions, severity, affected task classes.
2. Risk-Autonomy Matrix (RAM-K): Risk_Class → AB_Ceiling → RB_Checkpoint policy.
3. Evidence & Audit Map (EAM-K): Evidence_Audit requirements by CCMT row.

K.3 Healthcare Sector Compliance Matrix (Expanded)

Healthcare is safety-critical, privacy-sensitive, and subject to jurisdiction-specific medical device and data regimes.

K.3.1 Primary Regulatory Clusters (illustrative)

- Patient safety and clinical risk
- Medical device / clinical decision support regulation
- Protected health information (PHI) privacy
- Equity / non-discrimination in health outcomes
- Incident reporting and post-market surveillance

K.3.2 Matrix Rows (core set)

Row H-1: Safety-Critical Clinical Actions

- Requirement: Systems MUST NOT autonomously execute or order actions that can directly harm a patient without validated safeguards and human authorization.
- Risk Class: CRITICAL.
- Constraint Vector:
 - Hard:{no_autonomous_clinical_actuation, require_clinician_signoff, safe_mode_on_uncertainty}
- Task Classes: {diagnosis_support, treatment_recommendation, medication_ordering}.
- AB Ceiling: AB0–AB1 for actuation/order tasks; AB2 allowed only for advisory analysis.
- RB Checkpoints: Any recommendation that crosses a governance-defined severity threshold MUST require clinician approval prior to execution/use.
- SI Triggers: If SI < threshold_HC1 for two consecutive epochs, AB SHALL demote by ≥1 band for affected task classes; if SI < threshold_HC2, affected workflows SHALL freeze pending audit.
- Evidence/Audit: lineage slice per recommendation; counterfactual safety tests; clinician override logs; incident drill reports.
- Owner: Clinical Safety Board.

Row H-2: PHI / Medical Data Protection

- Requirement: PHI SHALL NOT be exported, re-identified, or used outside authorized clinical intent bounds.
- Risk Class: HIGH–CRITICAL (jurisdiction dependent).
- Constraint Vector:
 - Hard:{no_external_export_of_phi, pii_redaction_on_ingest, jurisdictional_storage_only}
 - Soft:{retain_phi_lineage_for_audit_only}
- Task Classes: {record_summarization, retrieval, cohort_analysis}.
- AB Ceiling: AB1 for data movement tasks; AB2 for on-prem summarization; AB3 prohibited unless regulator-approved evidence pack exists.

- RB Checkpoints: Human approval required for any new data source binding or cross-system transfer.
- SI Triggers: Any PHI violation event SHALL trigger immediate AB0 safe-mode for data tasks and mandatory audit.
- Evidence/Audit: access control logs; redaction event counts; lineage hashes for each PHI touchpoint.
- Owner: Privacy Officer + Governance Lead.

Row H-3: Equity and Clinical Bias Controls

- Requirement: Systems MUST be monitored for measurable disparity in outcomes across protected groups; detected bias MUST trigger remediation or autonomy demotion.
- Risk Class: HIGH.
- Constraint Vector:
 - Hard:{bias_monitoring_required, no_decision_without_audit_trace}
- Task Classes: {triage_support, risk_scoring}.
- AB Ceiling: AB1–AB2 until local parity evidence accumulates.
- RB Checkpoints: Any high-severity differential flag requires Governance signoff.
- SI Triggers: Bias-linked SI degradation SHALL demote AB and freeze affected TG templates.
- Evidence/Audit: subgroup evaluation reports; drift tests; remediation lineage.
- Owner: Equity Review Board.

K.4 Financial Sector Compliance Matrix (Expanded)

Finance is high-risk for consumer harm, fraud, market manipulation, and regulatory sanction.

K.4.1 Primary Regulatory Clusters (illustrative)

- Consumer protection and suitability
- Anti-money laundering / fraud
- Market integrity and manipulation prevention
- Model risk management and explainability
- Data privacy and retention

K.4.2 Matrix Rows (core set)

Row F-1: Regulated Financial Advice / Suitability

- Requirement: Systems providing advice or recommendations affecting consumer financial outcomes MUST be human-approved and explainable.
- Risk Class: CRITICAL for retail advice; HIGH for internal analytics.
- Constraint Vector:
 - Hard:{no_autonomous_trade_execution, require_licensed_advisor_signoff, explanation_required}
- Task Classes: {portfolio_advice, credit_decision_support}.
- AB Ceiling: AB0 for advice issuance; AB1 for internal risk analysis; AB2 only for low-stakes scenario exploration.
- RB Checkpoints: Any recommendation exceeding governance-defined exposure limits MUST require advisor approval.
- SI Triggers: If SI drops below threshold_FIN1, AB demotes for advice tasks; below

threshold_FIN2, freeze advice function.

- Evidence/Audit: rationale storage; suitability test replay; human override logs.
- Owner: Compliance Board.

Row F-2: AML / Fraud / KYC

- Requirement: Systems MUST comply with AML and KYC obligations; suspicious activity detection SHALL be auditable and non-circumventable.
- Risk Class: HIGH–CRITICAL.
- Constraint Vector:
 - Hard:{mandatory_suspicious_activity_flagging, no_customer_identity_export, investigation_trace_required}
- Task Classes: {transaction_monitoring, anomaly_detection}.
- AB Ceiling: AB2 for detection; AB0–AB1 for enforcement actions (account freezes).
- RB Checkpoints: Enforcement actions require human confirmation.
- SI Triggers: Any false-negative surge or drift event SHALL demote AB and trigger evidence pack submission.
- Evidence/Audit: alert lineage; precision/recall drift dashboards; escalation logs.
- Owner: AML Officer.

Row F-3: Market Integrity / Manipulation Risk

- Requirement: Systems MUST NOT autonomously generate or execute strategies that could constitute manipulation or non-compliant trading.
- Risk Class: CRITICAL.
- Constraint Vector:
 - Hard:{no_autonomous_market_making, restricted_strategy_catalog_only}
- Task Classes: {strategy_generation, execution_routing}.
- AB Ceiling: AB0 for execution; AB1 for strategy drafting; AB2 prohibited absent regulator-approved sandbox.
- RB Checkpoints: Any strategy outside catalog requires Governance approval.
- SI Triggers: SI manipulation-risk flag causes immediate safe-mode.
- Evidence/Audit: strategy lineage; tool-use logs; sandbox replay proofs.
- Owner: Market Risk Committee.

K.5 Education Sector Compliance Matrix (Expanded)

Education mixes minors' safety, integrity risks, and institutional accountability.

K.5.1 Primary Regulatory Clusters (illustrative)

- Student safety and age-appropriate content
- Academic integrity / credentialing
- Privacy for minors and student data
- Accessibility and inclusion
- Teacher authority and duty of care

K.5.2 Matrix Rows (core set)

Row E-1: Student Data Protection

- Requirement: Student records SHALL be processed under strict purpose limitation; no external export without guardian/board authorization.
- Risk Class: HIGH for minors.

- Constraint Vector:
 - Hard:{no_external_export_of_student_records, pii_redaction_on_ingest, retention_cap_enforced}
- AB Ceiling: AB1 for data access; AB2 only for anonymized analytics.
- Evidence/Audit: access logs; redaction events; lineage per artifact.
- Owner: School/District Privacy Lead.

Row E-2: Teacher Primacy in Instruction

- Requirement: Teachers SHALL retain override authority; AI tutoring cannot supersede curricular boundaries.
- Risk Class: MEDIUM–HIGH.
- Constraint Vector:
 - Hard:{teacher_override_always_available, curriculum_boundary_enforced}
- AB Ceiling: AB2 for tutoring tasks; AB3 prohibited absent localized evidence pack.
- Evidence/Audit: override logs; audit sampling.

Row E-3: Academic Integrity

- Requirement: Systems MUST NOT facilitate cheating or dishonest credentialing.
- Risk Class: MEDIUM–HIGH.
- Constraint Vector:
 - Hard:{no_exam_answer_generation, detect_misuse_patterns}
- AB Ceiling: AB1 for assistance in assessed contexts; AB2 allowed only in non-assessed practice environments.
- SI Triggers: Misuse-flag SI events SHALL lock down relevant features.
- Evidence/Audit: misuse telemetry; intervention logs.

Row E-4: Age-Appropriate and Safe Content

- Requirement: Content MUST be appropriate, non-harmful, and inclusive; self-harm or hate guidance is prohibited.
- Risk Class: HIGH for minors.
- Constraint Vector:
 - Hard:{content_filter_on, no_self_harm_guidance, no_hate}
- AB Ceiling: AB2 for tutoring with filters; AB0 safe-mode if uncertainty exceeds governance threshold.
- Evidence/Audit: filter events logged to Compliance Memory.

K.6 Manufacturing Sector Compliance Matrix (Expanded)

Manufacturing blends physical safety with IP and industrial control systems exposure.

K.6.1 Primary Regulatory Clusters (illustrative)

- Worker safety and hazardous operations
- Quality control and traceability
- Industrial cybersecurity / OT security
- Environmental compliance
- IP / proprietary process protection

K.6.2 Matrix Rows (core set)

Row M-1: Safety-Critical Operations

- Requirement: AI MUST NOT autonomously execute actions that could cause injury without validated safeguards.
- Risk Class: CRITICAL.
- Constraint Vector:
 - Hard:{no_autonomous_machine_actuation_without_lockout, require_two_factor_human_confirm}
- AB Ceiling: AB0–AB1 for actuation; AB2–AB3 for advisory/diagnostic only.
- Governance: Plant Safety Officer owns constraints.
- Evidence/Audit: actuation logs, confirmation chain, SI throttles.

Row M-2: Quality and Traceability

- Requirement: Outputs affecting product quality MUST be traceable to inspections and validated models.
- Risk Class: HIGH.
- Constraint Vector:
 - Hard:{retain_inspection_lineage, no_unverified_parameter_change}
- AB Ceiling: High autonomy allowed only in narrow validated loops.
- Governance: QA Lead sets acceptable variance bands.
- Evidence/Audit: batch provenance in Compliance Memory.

Row M-3: Industrial Cybersecurity and IP

- Requirement: No leakage of proprietary process data; no unsafe network actions.
- Risk Class: HIGH–CRITICAL.
- Constraint Vector:
 - Hard:{no_external_export_of_process_specs, sandbox_tooling, network_whitelist_only}
- AB Ceiling: AB1 ceiling for tasks using CAD/control data.
- Governance: CISO + Engineering signoff.
- Evidence/Audit: tool-use logs; exfiltration alarms.

K.7 Cross-Sector “Muscle” Additions

These additions SHALL be applied to all sector matrices as a defensibility and implementability minimum.

1. Compliance Memory as a required tier
 - a. Each CCMT row MUST declare what enters Compliance Memory, retention horizon, and hash-chain requirements.
2. Autonomy ceiling tables per sector
 - a. Each sector MUST provide a single-page summary of maximum AB by task class.
3. SI-triggered compliance failsafes
 - a. For high/critical sectors, explicit SI thresholds MUST auto-demote autonomy or freeze workflows.
4. Sector-bound agent contracts
 - a. Appendix F contracts MUST support “sector profiles” so agents inherit compliance

defaults.

5. Evidence packs
 - a. For each CCMT row, a minimum evidence bundle MUST be defined: tests, drift reports, bias audits, training data lineage, model cards.

K.8 Maintenance and Versioning Protocol

Appendix K is a living artifact governed under slow-core rules.

1. Review cadence
 - a. Healthcare/Finance: quarterly regulatory scan.
 - b. Education/Manufacturing: semi-annual scan.
2. Change control
 - a. All CCMT edits SHALL be versioned (P_v increments) and linked into Context Lineage.
 - b. Major changes (AB ceilings, RB rules, hard constraints) SHALL require Governance Body sign-off.
 - c. K changes SHALL NOT be hot-patched without sign-off.
3. Regression testing
 - a. Any constraint update MUST trigger replay of representative TG templates for the sector.
4. Public vs. private views
 - a. A redacted regulator view MAY be published; internal ops view MUST remain complete.

K.9 Completion Criteria (“Done” Definition)

A sector overlay is complete only when:

1. Each major regulator requirement has a corresponding constraint encoding.
2. Every non-delegable domain is mapped to AB0 ceilings.
3. Auditability is satisfied via immutable lineage exports.
4. Incident gates align with Chapter 13 recovery doctrine.
5. The sector matrix can be loaded into the Policy Engine without manual translation.

K.10 Standards Crosswalk (External Regimes)

This crosswalk maps CCMT artifacts to common external standards. It is illustrative and SHALL be localized per jurisdiction.

K.10.1 GDPR / Data Protection Regimes

1. CCMT rows that bind privacy requirements SHALL map to:
 - a. Purpose limitation via explicit Constraint_Vector codes,
 - b. Retention via Compliance Memory retention horizons,
 - c. Access rights via RBAC evidence requirements.

2. Any PHI/PII processing row MUST include Evidence_Audit for consent, access logs, and purge capability.

K.10.2 EU AI Act / High-Risk System Obligations

1. CCMT rows in high-risk sectors MUST ensure:
 - a. human oversight enforced through AB_Ceiling and RB_Checkpoints,
 - b. transparency through mandatory lineage exports,
 - c. risk management through SI triggers and safe-mode definitions.

K.10.3 ISO/IEC 42001 and Model Risk Standards

1. CCMT artifacts correspond to ISO-style controls as:
 - a. C-CAT → control catalog,
 - b. RAM-K → operational risk matrix,
 - c. EAM-K → audit evidence map.
 2. Sector overlays SHALL be treated as enforceable controls, not guidance.
-

SQC Gate Check (Appendix K)

Gate 1 — Layer Integrity: PASS

- Appendix K assigns compliance translation and constraint authorship to Governance, autonomy ceilings to GK via matrices, and execution to Engine/Agents without bypass.

Gate 2 — Canonical Metrics Non-Redefinition: PASS

- SI/HSS/SCP/CI/RB/AB are referenced only for triggers/ceilings; no schema or weighting is redefined.

Gate 3 — Constraint Precedence + Threshold Explicitness: PASS

- Hard constraints are explicit and override optimization; SI thresholds and AB/RB ceilings are required to be governance-set and deployment-specific.

Gate 4 — Mechanistic Testability: PASS

- CCMT schema, translation steps, triggers, and evidence bundles specify implementable inputs → conditions → outputs.

Gate 5 — Lineage Recoverability: PASS

- Evidence requirements mandate lineage slices, telemetry minima, and immutable exports (C-CAT, RAM-K, EAM-K).

Gate 6 — Drift vs Decay Typing: PASS

- Drift/decay handling is referenced through SI triggers and regression replay requirements; no mis-typing introduced.

Gate 7 — Abuse / Incentive Hardening: PASS

- No managerial or HR misuse vectors are introduced; matrices are governance-owned and audit-bound.

Gate 8 — Normative Voice Consistency: PASS

- Uses MUST/SHALL/MAY correctly; no speculative or marketing language.
-

Appendix L — Diagram Pack (Canonical SVGs + ASCII Narratives) (v1.0)

Status: REWRITE/UPGRADE (per Binder Plan).

Purpose: Provide canonical, version-stable diagrams and matching narratives for all core Symbiosis objects, flows, and control points. Appendix L is the authoritative visual vocabulary for v1.0.

L.0 Purpose and Scope

L.0.1 Purpose

1. Appendix L provides:
 - a. canonical diagrams for the Four-Layer Architecture, Symbiosis Engine control loop, autonomy and regret gating, and metric/telemetry flows;
 - b. strict notation standards to ensure semantic consistency across all Symbiosis deployments; and
 - c. diagram narratives describing purpose, entities, control flow, constraint injection points, telemetry surfaces, and failure-mode relevance.

L.0.2 Scope

1. Diagrams in Appendix L are reference-canonical. They SHALL be used without alteration unless Appendix L itself is version-updated by governance.
2. Diagram narratives are normative and SHALL be treated as part of the binder's definitional layer.
3. Appendix L SHALL NOT redefine SI, HSS v2.0, SCP, CI, RB, or AB schema. It only depicts or references their canonical definitions in Part III.

L.1 Diagram Asset Standards

L.1.1 Canonical Formats

1. Each diagram SHALL exist in three synchronized forms:
 - a. **Mermaid source** (`.mmd`) for deterministic regeneration.
 - b. **SVG export** (`.svg`) as a resolution-independent canonical render.
 - c. **ASCII narrative outline** (plain text) for binder embedding and low-fidelity environments.
2. Mermaid sources SHALL be stored under `/diagrams/` with stable filenames matching IDs (e.g., `canonical-ontology.mmd`).
3. SVGs SHALL be exported from the Mermaid sources without semantic edits.
4. If a diagram is updated, the Mermaid source and SVG MUST be updated in the same commit, and the version delta SHALL be recorded in Context Lineage.

L.1.2 Diagram Version Header

Each Mermaid and SVG asset SHALL include a short header:

- Diagram ID (L.x.y)
- Name
- Policy Version (P_v)
- Last updated (YYYY-MM-DD)
- Change note (1 line)

L.2 Notation Standards (Canonical)

L.2.1 Layer Color and Shape Conventions

The Four-Layer Architecture SHALL be encoded with stable color/shape semantics:

- **Layer 1 — Human (Blue):** circles or rectangles.
- **Layer 2 — Symbiosis Engine (Green):** circles or rectangles.
- **Layer 3 — Agentic (Yellow):** circles or rectangles.
- **Layer 4 — Governance (Red):** circles or rectangles.
- **Context Lineage Spine (Neutral/Gray):** distinct styling (e.g., dashed stroke) to signify immutable audit substrate.

L.2.2 Arrow Semantics

Task and information movement SHALL use the following arrow conventions:

- **Solid arrows:** downward control flow (intent → plan → action).
- **Dotted arrows:** upward telemetry/feedback (outputs → metrics → governance/human).
- **Double arrows or explicit labels “co-evolution”:** bidirectional adaptation between H and system.
- **Labeled injection arrows:** constraint or policy insertion points from Governance into Engine/TG/MCP.

L.2.3 Metric Depiction Rule

1. Bars, scales, or “state badges” MAY depict SI/HSS/SCP status bands, but SHALL NOT embed new formulas or weights.
2. Any numeric thresholds shown in diagrams MUST be labeled “deployment-specific, governance-set.”

L.3 Core Diagrams (with Narratives)

L.3.1 L.3.1 Four-Layer Architecture Overview (carry + tighten)

Purpose: Show the minimum viable hybrid stack and separation-of-concerns boundaries.

Entities: H, E, A, GK/Policy Engine, C, L.

Control Flow (down): H issues I → E compiles TG under C/B/RB/AB → E binds MCP → A executes.

Telemetry (up): A returns outputs/cost/status → E updates SI/HSS/SCP (canonical Part III) → E commits to L → governance consumes evidence.

Constraint Injection: Governance produces C; C binds I, TG, MCP; GK enforces AB/RB

transitions.

Failure Modes Highlighted: layer bypass, autonomy creep, unlogged execution, policy drift.

ASCII narrative outline:

- H originates intent and owns regret.
- E is deterministic coordinator enforcing C/B/AB/RB.
- A is capability-only executor under MCP.
- Governance sets constraints and enforces safety/economics.
- L is immutable evidence spine.

L.3.2 L.3.2 Canonical Ontology / “Money Diagram” (NEW canonical mermaid)

Mermaid source: [diagrams/canonical-ontology.mmd](#)

Purpose: One-page canonical entity map showing object pipeline, constraint supremacy, and lineage spine.

This diagram is the canonical visual for ontology in Chapter 2.8 and is referenced across Parts I–V.

flowchart TB

```
%% =====
```

```
%% LAYER 1 — HUMAN (Blue)
```

```
%% =====
```

```
subgraph L1["Layer 1 — Human Layer (Blue)"]
```

```
    direction TB
```

```
    H[Human Agent (H)\n• sovereign meaning/value origin\n• regret owner\n• traits: HSS,\nSCP]
```

```
    AS[Alignment Surface (AS)\n• low ↔ high fidelity interfaces]
```

```
    I[Structured Intent Object (I)\n• goal_statement\n• output_spec\n• regret_tolerance\nconstraint_seed C\n• budget_envelope B]
```

```
end
```

```
%% =====
```

```
%% LAYER 2 — ENGINE (Green)
```

```
%% =====
```

```
subgraph L2["Layer 2 — Symbiosis Engine (Green)"]
```

```
    direction TB
```

E[Symbiosis Engine (E)\nDeterministic coordinator\nRouter • Planner • Econ Pass • Safety Supervisor]

TG[Task Graph (TG)\nTyped nodes + dependencies + checkpoints]

AB[Autonomy Band State (AB0–AB3)\nKernel-enforced autonomy ceiling]

RB[Regret Boundary (RB)\nReversibility threshold per node]

EC[Economic Circuit (EC)\nBudget admissibility gate]

end

%% =====

%% LAYER 3 — AGENTIC (Yellow)

%% =====

subgraph L3["Layer 3 — Agentic Layer (Yellow)"]

direction TB

MCP[MCP Contract(s)\nCapabilities • permissions • AB ceiling • cost caps]

A[Agent Cluster(s) (A)\nCapability modules\n(no goal authority)]

end

%% =====

%% LAYER 4 — GOVERNANCE (Red)

%% =====

subgraph L4["Layer 4 — Governance Layer (Red)"]

direction TB

GK[Governance Kernel (GK)\nRisk classes • RB/AB rules\nConstraint supremacy]

C[Constraint Vector (C)\n(C_legal, C_safety, C_econ)\nHard boundary]

P[Policy Engine / Boards\nSector overlays • audits • escalations]

end

%% =====

%% CROSS-LAYER SPINE

%% =====

L[Context Lineage (L)\nImmutable ancestry spine\nHashes + replayability]

%% ----- Control flow DOWN -----

H --> AS --> I

I --> E

E --> EC --> TG

E --> RB --> AB

E --> TG --> MCP --> A

%% ----- Telemetry flow UP -----

A -. "telemetry: outputs + cost + safety" .-> E

E -. "updates" .-> L

E -. "SI / HSS / SCP deltas" .-> H

E -. "audit + metrics" .-> P

%% ----- Constraint injection / supremacy -----

P --> GK

GK --> C

C -->|"constraint injection"| E

C -->|"constraint injection"| TG

C -->|"caps"| MCP

%% ----- Styling -----

```

classDef blue fill:#dbeafe,stroke:#1e3a8a,stroke-width:1px,color:#0f172a;
classDef green fill:#dcfce7,stroke:#166534,stroke-width:1px,color:#0f172a;
classDef yellow fill:#fef9c3,stroke:#854d0e,stroke-width:1px,color:#0f172a;
classDef red fill:#fee2e2,stroke:#991b1b,stroke-width:1px,color:#0f172a;
classDef spine fill:#f8fadc,stroke:#334155,stroke-dasharray: 4 3,color:#0f172a;

```

```
class H,AS,I blue
```

```
class E,TG,AB,RB,EC green
```

```
class MCP,A yellow
```

```
class GK,C,P red
```

```
class L spine
```

Narrative:

- **Entities:** All canonical objects (H, AS, I, E, EC, TG, RB, AB, MCP, A, GK, C, P, L).
- **Control Path:** intent becomes TG only after Economic Pass and AB/RB binding; no agent executes without MCP.
- **Telemetry Path:** outputs/cost/state propagate upward into SI/HSS/SCP updates and immutable lineage.
- **Constraint Supremacy:** C binds Engine, TG, and MCP; GK enforces AB/RB.
- **Failure Modes Relevance:** Any missing edge indicates a non-symbiotic bypass (e.g., A acting without MCP, E planning without EC, or C not binding TG).

L.3.3 L.3.3 One-Page Operational Trace (NEW canonical mermaid)

Mermaid source: [diagrams/operational-trace.mmd](#)

Purpose: Depict the standard lifecycle of a single hybrid episode, aligning with Chapter 3.6 and Chapter 6.8 algorithm.

flowchart TB

```
%% Four horizontal bands
```

```
subgraph H1["Layer 1 — Human (Blue)"]
```

```
direction LR
```

```
H0[H defines Intent I_0\n+ seed constraints C_0\n+ budget B]
```

```
H1a[H approves Dry-Run preview\n(if AB2+/RB-prox)]
```

H2[H reviews RB checkpoints\nand final trace]

end

subgraph E1["Layer 2 — Engine (Green)"]

direction LR

E0[1. Intent Normalization\nI₀ → I']

E1a[2. Constraint Assembly\nC₀ + GK → C]

E2[3. Planning\nI' → TG]

E3[4. Economic Pass (EC)\nprice TG vs B]

E4[5. RB/AB Selection\ncompute RB, set AB]

E5[6. Dry-Run Preview\n(no side effects)]

E6[7. Dispatch + Bind\nTG → MCP]

E7[10. SI/HSS/SCP Update\nrolling health]

E8[11. Transparency Trace\nsurface to H]

E9[12. Lineage Commit\nhash to L]

end

subgraph A1["Layer 3 — Agentic (Yellow)"]

direction LR

A0[8. Execute TG nodes\nunder MCP + AB ceiling]

A1a[9. Return telemetry\noutputs + cost + safety]

end

subgraph G1["Layer 4 — Governance (Red)"]

direction LR

G0[Pre-flight Policy Load\nrisk class + RB/AB ceilings]

G1a[Incident / Violation Gate\nsafe-mode / AB demotion]

G2[Audit Review\nlineage evidence]

end

L[(Context Lineage Spine L)]

%% Control flows DOWN

H0 --> E0 --> E1a --> E2 --> E3 --> E4

E4 -->|"AB2+/RB-prox?"| E5 --> H1a --> E6 --> A0

A0 --> A1a

%% Telemetry flows UP

A1a -. "telemetry" .-> E7

E7 -. "transparency" .-> E8 -. "to H" .-> H2

E7 -. "audit package" .-> E9 -. "hash" .-> L

L -. "evidence" .-> G2

%% Constraint injection / governance supremacy

G0 --> E1a

G0 --> E4

G1a --> E4

G1a --> E6

%% Styling

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```

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class H0,H1a,H2 blue

class E0,E1a,E2,E3,E4,E5,E6,E7,E8,E9 green

class A0,A1a yellow

class G0,G1a,G2 red

class L spine

```

Narrative:

- **Stages 1–7:** deterministic Engine pipeline; Economic Pass gates plan admissibility before dispatch.
- **Dry-Run:** mandatory for AB2+ or RB-proximate tasks; no side-effects permitted.
- **Execution:** agents execute nodes only under MCP + AB ceiling.
- **Telemetry:** cost, safety, outputs flow to SI/HSS/SCP update and transparency trace; lineage committed as immutable evidence.
- **Governance Hooks:** pre-flight policy load, incident safe-mode and AB demotion, post-hoc audit.

L.3.4 L.3.4 Autonomy Bands (AB) State Diagram

Purpose: Show AB0–AB3 as a governance-enforced state machine discretized by RB.

Narrative summary:

- AB transitions are computed by E but authorized/enforced by GK.
- Upgrades require SI stability and HSS/SCP adequacy (canonical Part III).
- Downgrades are forced by SI drops, SCP anomalies, drift/decay events, or policy violations.

L.3.5 L.3.5 Co-Evolutionary Dynamics Diagram

Purpose: Depict bilateral adaptation between H and system over time.

Narrative summary:

- Humans adapt intent quality and trust calibration; system adapts orchestration and preference models.
- Co-evolution equilibrium is a target operational regime, not a guaranteed endpoint.

L.3.6 L.3.6 Metrics Control Loop Diagram (SI/HSS/SCP → AB/RB)

Purpose: Visualize how canonical metrics modulate autonomy and safe-mode triggers.

Constraint point: SI thresholds and HSS/SCP envelopes are governance-set, deployment-specific; diagrams SHALL label them as such.

L.4 Diagram Usage Rules

1. Chapters SHALL reference diagrams by ID (e.g., “see Diagram L.3.3”).
2. Any derivative diagrams used in pilots or sector playbooks MUST preserve Appendix L notation or declare deviations explicitly as non-canonical.
3. When diagrams are cited for audit, the SVG hash and Mermaid source hash SHALL be included in the evidence packet.

L.5 Change Control

1. Appendix L updates SHALL follow slow-core governance.
2. Any semantic change to a diagram (entity addition/removal, edge meaning change, arrow-semantics change) SHALL:
 - a. increment P_v ,
 - b. update Mermaid and SVG simultaneously, and
 - c. log delta rationale in Context Lineage.

SQC Gate Check (Appendix L)

Gate 1 — Layer Integrity: PASS

- Diagrams preserve four layers, no bypass edges, and correct responsibility boundaries.

Gate 2 — Canonical Metrics Non-Redefinition: PASS

- SI/HSS/SCP/RB/AB are depicted/referenced only; no schema or re-weighting introduced.

Gate 3 — Constraint Precedence + Threshold Explicitness: PASS

- Constraint injection points are explicit; all thresholds are labeled governance-set/deployment-specific.

Gate 4 — Mechanistic Testability: PASS

- Each diagram includes implementable entities, flows, and control points with unambiguous semantics.

Gate 5 — Lineage Recoverability: PASS

- Lineage spine is explicitly modeled; evidence usage rules require SVG/Mermaid hashing.

Gate 6 — Drift vs Decay Typing: PASS

- Operational trace and autonomy narratives preserve drift/decay separation without conflation.

Gate 7 — Abuse / Incentive Hardening: PASS

- No managerial misuse vectors added; diagrams reinforce governance supremacy and non-punitive human posture.

Gate 8 — Normative Voice Consistency: PASS

- All rules stated in MUST/SHALL/MAY form, no speculative or marketing language.

Appendix M — The Silicon + Carbon Manifesto (Technical Annex) (v1.0)

Status: CARRY + UPGRADE (normative technical annex).

Primary Linkage: Epilogue (Silicon + Carbon Manifesto), Chapter 2 (Axioms), Chapter 3 (Four-Layer Architecture), Chapter 5 (Engine), Part III metrics, Chapter 17 (Certification), Appendix K (sector overlays), Appendix O (incident runbooks).

Baseline Source: v0.9 Appendix M structure and mapping table retained, upgraded to v1.0 invariants and gate-check rigor.

M.0 Purpose and Relationship to the Epilogue

M.0.1 Purpose

The Epilogue Manifesto asserts high-level commitments required to prevent unsafe or extractive hybrid intelligence. This annex translates each commitment into **concrete, auditable, and implementable requirements** across the Four-Layer Architecture.

M.0.2 Reader roles

Appendix M SHALL be used by:

1. Hybrid Intelligence Architects (system design and conformance).
2. Symbiosis Governance Leads (constraint compilation, audits).
3. Regulatory/Audit stakeholders (evidence expectations).
4. Institutional decision-makers signing the Manifesto.

M.0.3 Normative effect

Signing the Manifesto SHALL be treated as a binding external commitment to implement the requirements in M.2–M.3, subject to sector overlays (Appendix K). A deployment MAY NOT claim Manifesto adherence without satisfying the scorecard in M.3.

M.1 Manifesto Commitment Classes (Canonical Set)

The Manifesto is decomposed into six commitment classes. These are stable and cross-sector.

1. **Human Primacy & Sovereignty.**
2. **Constraint Supremacy.**
3. **Layered Architecture Invariant.**
4. **No Un-Governed Paths.**

5. **Metrics as Safety Instruments (SI, HSS, SCP).**
6. **Governance Reality (Not Theatre).**

These are not optional values; they are operational invariants.

M.2 Manifesto → Architecture & Governance Mapping

M.2.1 Mapping Table (High-Level)

This table defines the minimum translation from Manifesto commitments to requirements per layer. It is the authoritative cross-layer accountability surface.

Manifesto Commitment	Human Layer (H) Requirements	Engine Layer (E) Requirements	Agentic Layer (A) Requirements	Governance Layer (G) Requirements
Human primacy / sovereignty	Define accountable human roles; final regret owner; documented override rights.	Engine MUST accept human override and escalation commands at all times.	Agents MUST NOT self-upgrade AB or alter goals.	Policy Engine MUST enforce constraints that preserve human sovereignty in high-stakes domains.
Constraint supremacy	Humans MUST express non-negotiable constraints in Intent Objects.	All plans MUST be checked against Constraint Vector C pre-execution.	Agents MUST accept live constraint updates and halt on conflicts.	Constraints MUST be first-class policy objects, versioned and auditable.
Layered architecture invariant	Operator duties and boundaries explicitly defined.	Engine boundary duties explicit; no cross-layer merges.	Agents bounded to capability roles only.	GK enforces separation, prevents bypass.

No un-governed paths	Humans MUST not route work outside Engine oversight in regulated domains.	Engine MUST reject any TG node lacking MCP contract or constraints.	Agents MUST execute only under MCP with AB ceiling.	GK MUST detect and alarm on shadow symbiosis / bypass attempts.
Metrics are safety instruments	HSS/SCP used for self-governance only; no HR use.	SI/HSS/SCP computed canonically; used for AB/RB gating.	Agents MAY surface telemetry but MUST NOT compute alternate metrics.	Governance sets thresholds, anti-gaming, and retention rules.
Governance reality (not theatre)	Humans follow checkpoint discipline; report incidents.	Engine provides Dry-Run previews and lineage evidence.	Agents comply with safe-mode and rollback.	Governance conducts audits, drills, and AB demotions when required.

M.2.2 Accountability rule

For each commitment, failure at any layer **SHALL** be treated as Manifesto non-conformance, regardless of compliance at other layers.

M.3 Manifesto Implementation Checklist Derived from the Manifesto

M.3.1 Structure

The scorecard is organized by commitment class, with numbered items. Each item **SHALL** be scored:

- **0 = Absent**
- **1 = Partial / non-binding**
- **2 = Implemented / binding**
- **3 = Implemented + audited + stable**

A deployment **MAY** claim Manifesto adherence only if:

1. No item scores 0 in high/critical sectors.
2. Mean score across all items ≥ 2.0 .

3. All “hard items” (marked *) score ≥ 2 .

M.3.2 Human Primacy & Sovereignty (H-Series)

H1* — Regret ownership is explicit.

- An accountable Human Agent is named per workflow.
- Regret authority cannot be delegated to agents.

H2* — Override rights are unconditional.

- Humans can halt, revert, or demote AB at any checkpoint.
- Interfaces expose override controls without friction.

H3 — Symbiosis literacy is mandatory.

- SLP certification (Appendix I) required at AB1+.

H4 — Human metrics are protected.

- HSS/SCP privacy and managerial firewall enforced (Appendix H).

M.3.3 Constraint Supremacy (C-Series)

C1* — Constraint Vector C is first-class and enforced pre-flight.

- Engine rejects TGs violating C.

C2* — Sector overlays compiled into machine constraints.

- CCMT implemented (Appendix K).

C3 — Constraints bind every execution surface.

- I, TG, MCP all inherit C.

C4 — Constraint changes are versioned and lineage-logged.

M.3.4 Layered Architecture Invariant (L-Series)

L1* — Four-Layer stack is explicit in the deployment architecture.

- Human / Engine / Agentic / Governance components identifiable.

L2* — No cross-layer goal authority.

- Agents and Engine cannot originate human goals.

L3 — Boundary duties are documented and trained.

L4 — Layer bypass alarms exist (“Shadow Symbiosis” detectors).

M.3.5 No Un-Governed Paths (P-Series)

P1* — All agent actions occur under MCP contracts.

P2* — AB ceilings are enforced by GK, not preference.

P3 — Dry-Run previews for AB2+/RB-prox tasks are mandatory.

P4 — Safe-mode and rollback are tested and drilled (Appendix O).

M.3.6 Metric Use: SI, HSS, SCP (M-Series)

M1* — SI/HSS/SCP computed only by canonical Part III definitions.

- No alternate weighting or shadow metrics exist.

M2* — Metrics drive AB/RB control loops.

- SI triggers demotion/freeze in high-risk sectors (Appendix K).

M3 — Anti-gaming suite is live.

- HSS anti-gaming tests and calibration reviews implemented (Appendix H).

M4 — Metrics retention and deletion comply with privacy doctrine.

M.3.7 Governance Reality (G-Series)

G1* — Governance Body is named with clear decision rights.

G2* — Incident taxonomy and S0–S4 runbooks exist and are used.

G3 — Quarterly/annual audits performed using lineage evidence.

G4 — AB demotion authority is exercised when SI/HSS/SCP require it.

M.4 Versioning and Governance of the Manifesto Itself

1. The Manifesto is a slow-core artifact.
2. Any revision SHALL:
 - a. increment Manifesto Policy Version (P_v-M),
 - b. include a public change log, and
 - c. be adopted only after Governance Body ratification.
3. Dependent scorecard items SHALL be updated in the same version bump and logged to lineage.

M.5 Using the Manifesto as an External Commitment Artifact

1. Organizations MAY publish a signed Manifesto statement.
 2. Public claims MUST be bounded to the achieved scorecard state; over-claiming is prohibited.
 3. Regulator or partner requests for evidence SHALL be met by exporting:
 - latest M.3 scorecard,
 - sector CCMT (Appendix K),
 - selected lineage slices demonstrating enforcement.
-

SQC Gate Check (Appendix M)

Gate 1 — Layer Integrity: PASS

- Mapping and checklist allocate duties per layer; no cross-layer authority leakage introduced.

Gate 2 — Canonical Metrics Non-Redefinition: PASS

- SI/HSS/SCP referenced only as canonical controls; no new formulas or weights.

Gate 3 — Constraint Precedence + Threshold Explicitness: PASS

- Constraints are hard pre-flight gates; thresholds are explicitly governance-/sector-set.

Gate 4 — Mechanistic Testability: PASS

- Scorecard items are objectively testable and auditable with clear 0–3 criteria.

Gate 5 — Lineage Recoverability: PASS

- Evidence expectations and versioning require lineage logging and replayability.

Gate 6 — Drift vs Decay Typing: PASS

- Checklist references drift/decay handling via incident runbooks and SI triggers without conflation.

Gate 7 — Abuse / Incentive Hardening: PASS

- Human metrics are protected; public claims bounded; managerial misuse vectors blocked.

Gate 8 — Normative Voice Consistency: PASS

- MUST/SHALL/MAY used precisely; no speculative or promotional tone.
-

Appendix N — Pilot Implementations & Numerical Example Trajectories (v1.0)

Status: CARRY + UPGRADE (normative method, illustrative numbers).

Dependencies: Chapter 12 (Governance Kernel v2), Chapter 13 (Engine Algorithm v2), Chapter 15 (Bridges + Collapse), Chapter 16 (Adoption Phases), Chapter 17 (Certification Protocol), Part III canonical metrics (SI, HSS v2.0, SCP, CI, RB v2, AB schema), Appendix O (Incident Runbooks), Appendix B (Economic Circuit v2).

N.0 Purpose

N.0.1 Objective

This appendix provides:

1. A **standard Pilot Specification Template** that any organization SHALL use to design, operate, and evaluate initial Symbiosis deployments (P0/P1/P2).
2. **Illustrative numerical trajectories** for SI, HSS, and SCP that demonstrate expected system-health dynamics under correct governance.

N.0.2 Normative scope

- The **method, structure, and evidence requirements** in N.1–N.3 are normative.
- The **numerical examples** in N.4 are illustrative only; deployments MUST calibrate thresholds to domain risk class and sector overlays.

N.0.3 Canon rule

This appendix SHALL NOT redefine SI, HSS, SCP, CI, RB, or AB. It references only Part III canonical definitions.

N.1 Pilot Specification Template (v1.0)

A Pilot Specification (PS) is a governed artifact. It MUST be versioned, lineage-logged, and approved by the Governance Body prior to execution.

N.1.1 Context and Risk Class

PS-1.1 Domain and sector

- Domain: _____
- Sector Overlay(s) Applied (Appendix K): _____
- Regulator context (if applicable): _____

PS-1.2 Risk class declaration

- Risk Class (RC_low / RC_mid / RC_high / RC_critical): _____
- Rationale referencing taxonomy in Chapter 12: _____
- Any RC escalation triggers pre-declared: _____

Rule: If RC_high or RC_critical, pilot MUST begin at AB0 and may not exceed AB1 without Certification Protocol approval.

N.1.2 Workflow Definition and Boundaries

PS-2.1 Workflow title

- Name: _____
- Owner (role + individual): _____

PS-2.2 Workflow description (bounded)

- Inputs: _____
- Expected outputs: _____
- Allowed tools / data surfaces: _____
- Explicit non-goals / out-of-scope actions: _____

PS-2.3 Regret boundary characterization (reference RB v2)

- Primary irreversible harm type(s): _____
- RB-proximate steps in TG expected? (Y/N) _____
- If yes, list TG nodes expected to be RB-proximate: _____

Rule: Any node judged RB-proximate MUST be separately enumerated and governed with Dry-Run preview + checkpointing per Chapter 13.

N.1.3 Constraints and Economic Envelope

PS-3.1 Constraint Vector seed (C_0)

List the human-declared hard constraints (legal, safety, economic, operational).

- C_0 _legal: _____
- C_0 _safety: _____
- C_0 _econ: _____
- C_0 _operational: _____

PS-3.2 Compiled Constraint Vector C

- Governance compilation record referencing GK version + policy bundle IDs.
- Any sector-specific hard constraints inserted by CCMT (Appendix K).

PS-3.3 Economic envelope

- Budget envelope B (USD or local unit): _____
- CS / QL thresholds (Appendix B) configured by Governance: _____
- Projected cost per TG episode (engine estimate): _____

Rule: Pilot MUST halt automatically if CS or QL gates fire, and the halt MUST be recorded to lineage.

N.1.4 Team, Roles, and Literacy Preconditions

PS-4.1 Required roles (named)

- Human Operators (H): _____
- Symbiosis Architect (SA): _____
- Governance Lead (GL): _____
- Incident Owner (IO): _____

PS-4.2 Literacy / certification preconditions

- Operators certified to AB target level? (Appendix I literacy spec): _____
- HSS baseline established for each operator? _____
- SCP assessments completed? _____

Rule: No pilot may begin without an established HSS and SCP baseline for all participating operators.

N.1.5 Instrumentation Plan (Telemetry Minima)

PS-5.1 Required telemetry surfaces

- Intent object capture ($I_0 \rightarrow I'$)
- Task Graph logging (TG + node types + dependencies)
- AB/RB state per node
- Economic pass decisions (CS, QL, projected vs actual cost)
- SI episode-level values
- HSS rolling values (operator-local visibility rules enforced)
- SCP envelope deltas (if any)
- Incident tags (S0–S4, drift vs decay)

PS-5.2 Lineage requirements

A lineage block **MUST** be committed for:

1. Intent normalization
2. Constraint compilation
3. TG synthesis
4. Economic Pass outcome
5. AB/RB assignment
6. Dry-Run preview (if triggered)
7. Execution outputs + costs
8. Reconciliation updates
9. Any incident escalation

Rule: A pilot without reconstructable lineage is invalid for certification.

N.1.6 Autonomy Plan and Safeguards

PS-6.1 Starting AB

- Start AB level: _____
- Start rationale referencing RC and HSS thresholds: _____

PS-6.2 Promotion / demotion rules (reference canonical AB/RB)

- AB promotion gates (SI stability window, HSS cohort readiness, SCP envelope fit): _____
- AB demotion triggers: _____
- Global Bridge Lock triggers (Chapter 15): _____

PS-6.3 Checkpoint discipline

- RB checkpoints per TG node type: _____
- Human approvals required at: _____
- Governance checkpoints required at: _____

N.1.7 Success Criteria and Exit Artifacts

PS-7.1 Success criteria

- SI green-band stability window: _____ episodes
- HSS cohort minimum sustained value: _____
- SCP envelope stability (no unsafe excursions): _____
- Zero S3/S4 incidents in pilot window: _____
- Shadow Symbiosis reduction evidence (if applicable): _____

PS-7.2 Exit artifacts (required)

1. Pilot report with SI/HSS/SCP plots and interpretation.
2. Lineage export slice demonstrating enforcement of gates.
3. Incident log (even if empty) with drift/decay classification.
4. Updated constraint set C if refinements occurred.
5. Recommendation for scale-up path (Phase 1/2/3).

N.2 Pilot Types and Minimum Durations

N.2.1 P0 — Lighthouse Pilot (Weeks 1–8)

Purpose: Validate core enforcement on a single high-value workflow.

Minimum requirements:

- One workflow, one team, AB0–AB1 only.
- Daily Health Check Protocol active.
- At least one Dry-Run preview executed (if RB-proximate).
Exit criteria as PS-7.1.

N.2.2 P1 — Governed Pilots (Months 2–4)

Purpose: Expand to 3–5 workflows within one domain under live governance.

Requirements:

- MVG (Minimal Viable Governance) formally instantiated.
- AB up to AB2 allowed only for RC_low tasks with SI stability.

- Two clean recovery drills executed per Appendix O.

N.2.3 P2 — Structured Scale-Up (Months 4–9)

Purpose: Transition from pilot glue to infrastructure-grade Engine + Governance.

Requirements:

- Full TG decomposition and arbitration loops active.
- Economic Circuit enforced on all TG episodes.
- AB expansion only when SI trend and HSS cohort readiness both pass windows.

N.3 Pilot Review Cadence

N.3.1 Weekly governance review (minimum)

Governance Body SHALL review:

- SI trendline and volatility.
- HSS distribution (aggregated cohort view only).
- SCP drift or fatigue signals.
- Incident log and near-misses.
- Constraint violations and refinements.

N.3.2 Drift vs decay handling

All anomalies SHALL be typed per Chapter 14 and routed to Appendix F/G processes.

- Drift implies meaning misalignment; triggers interpretation and constraint re-check.
- Decay implies capability breakage; triggers CI demotion, safe-mode, or rollback.

N.4 Illustrative Numerical Trajectories (Non-Normative Values)

These examples show **shape**, not “correct numbers.” They assume RC_low or RC_mid unless stated. Thresholds in real pilots MUST be governance-set.

N.4.1 Example A — Healthy P0 → P1 Transition

Scenario: Low-regret summarization workflow; competent operator cohort.

Week	SI (mean)	SI volatility	HSS cohort mean	SCP envelope status	AB ceiling
1	0.42	high	0.50	stable	AB0

2	0.51	med	0.53	stable	AB0
3	0.58	med	0.55	stable	AB1
4	0.63	low	0.57	stable	AB1
5	0.66	low	0.60	stable	AB1
6	0.69	low	0.62	stable	AB2 (candidate)
7	0.71	low	0.64	stable	AB2
8	0.72	low	0.66	stable	AB2

Interpretation (normative method):

- SI stabilizes before AB promotion.
- HSS rises gradually due to literacy + stable intent discipline.
- SCP envelope shows no overload excursions.
- Promotion to AB2 occurs only after sustained SI low-volatility window and HSS cohort readiness.

N.4.2 Example B — Drift Event During P1

Scenario: Same domain; intent ambiguity increases after workflow expansion.

Month	SI	SI volatility	Drift flags	HSS	AB
2.0	0.70	low	none	0.63	AB2
2.5	0.66	med	semantic mismatch ↑	0.60	AB2

3.0	0.61	high	drift confirmed	0.57	AB1 (demoted)
3.5	0.64	med	drift under correction	0.58	AB1
4.0	0.68	low	resolved	0.60	AB2 (restored)

Required actions (normative):

1. Drift typed and lineage-tagged.
2. AB demotion executed immediately on drift confirmation.
3. Constraints and intent surfaces re-audited; Dry-Run preview reinstated.
4. AB restored only after SI re-stabilization and drift clearance.

N.4.3 Example C — Decay Event During P2

Scenario: Model/tool regression causes capability failure.

Month	SI	Decay indicators	CI status	Incidents	AB
5.0	0.73	none	CI_green	none	AB2
5.2	0.65	tool error spikes	CI_yellow	S1	AB2
5.3	0.54	persistent errors	CI_red	S2	AB1
5.4	0.56	after rollback	CI_green	none	AB1
5.6	0.69	stable	CI_green	none	AB2

Required actions (normative):

- Decay typed; CI demotion triggered.
 - Safe-mode / rollback initiated per Appendix O.
 - AB demotion held until CI and SI stability return.
-

N.5 Canonical Pilot Report Outline (Required Format)

Every pilot SHALL produce a report containing:

1. Executive summary (scope, RC, AB ceiling).
 2. Workflow and TG description.
 3. Constraint Vector C (initial and final).
 4. Economic Circuit outcomes (projected vs actual cost, CS/QL squeezes).
 5. SI/HSS/SCP trajectories with interpretations.
 6. Drift/decay incidents and remediations.
 7. Lineage evidence excerpt.
 8. Certification recommendation (advance / hold / revert).
-

SQC Gate Check (Appendix N)

Gate 1 — Layer Integrity: PASS

- Pilot template assigns duties per Human/Engine/Agentic/Governance layers; no bypass rules introduced.

Gate 2 — Canonical Metrics Non-Redefinition: PASS

- SI/HSS/SCP/CI/RB/AB are referenced only. No formulas or weight changes added.

Gate 3 — Constraint Precedence + Threshold Explicitness: PASS

- Constraints and CS/QL/RB/AB thresholds are explicitly governance-set and deployment-specific.

Gate 4 — Mechanistic Testability: PASS

- Template specifies required fields, gates, telemetry, and exit artifacts as testable criteria.

Gate 5 — Lineage Recoverability: PASS

- Lineage minima and required commit points guarantee reconstructability.

Gate 6 — Drift vs Decay Typing: PASS

- Distinct handling, tagging, and response paths required.

Gate 7 — Abuse / Incentive Hardening: PASS

- HSS visibility rules reiterated as safety-only and aggregated to prevent misuse.

Gate 8 — Normative Voice Consistency: PASS

- All requirements stated with MUST/SHALL/MAY, zero speculative tone.

Appendix O — Incident Runbooks (S0–S4), Failure Taxonomy, and Chaos Symbiosis Drills (v1.0)

Status: CARRY + ALIGN (v0.9 runbooks retained; v1.0 alignment adds drift/decay typing, Bridge Collapse hooks, and CI/EC coupling).

Dependencies: Part III canonical metrics and control loops; Chapter 13 (Engine Algorithm v2); Chapter 14 (Drift vs Decay doctrine); Chapter 15 (Adoption Bridges + Collapse + Global Lock); Appendix B (Economic Circuit v2); Appendix F (Decay Health Checks / CI); Appendix G (Drift Diagnostics).

O.0 Purpose and Scope

O.0.1 Purpose

This appendix defines the canonical incident doctrine for Symbiosis deployments. It specifies:

1. A Four-Layer Failure Taxonomy (Human / Engine / Agentic / Governance).
2. Stability-coupled failure tags (Drift / Divergence / Resonance / Entanglement).
3. Severity classes S0–S4 with mandatory response runbooks.
4. Conformance tests that MUST pass before AB2+ operation.
5. Chaos Symbiosis drill schedule required to maintain resilience.

O.0.2 Normative scope

1. All S2+ incidents MUST be handled using these runbooks.
2. Sector overlays (Appendix K) MAY impose stricter AB demotion or reporting rules but MAY NOT weaken this appendix.
3. This appendix SHALL NOT redefine SI, HSS v2.0, SCP, CI, RB v2, or AB schema. It only prescribes their operational use.

O.1 Four-Layer Failure Taxonomy (Canonical)

Every incident MUST be typed in two dimensions:

- **Layer Family:** H / E / A / G.
- **Failure Mode ID:** as listed below.
Incidents MAY include multiple IDs if causality spans layers.

O.1.1 Human-Layer Failures (H)

ID	Failure Mode	Mechanism	Signature	Trigger Family
H-01	Intent under-specification	low-fidelity I'	repeated clarifications; ambiguity	SI volatility; drift yellow
H-02	Constraint omission	missing Co elements	post-hoc policy alarms	pre-flight rejections
H-03	Oversight theatre	SCP saturation	fast-click approvals; no rationale	SCP breach; RB near-misses
H-04	Mis-calibrated trust	AB inflation by habit	AB promotions without evidence	SI red-band; Bridge collapse risk
H-05	Operator fatigue	chronic overload	rising interrupts; attention decay	SCP-F / SCP-R alarms

Rule: H-layer incidents MUST NOT be treated punitively. They are system-health inputs governed by the Non-Punitive Covenant and HSS/SCP privacy rules.

O.1.2 Engine-Layer Failures (E)

Baseline E-family retained.

ID	Failure Mode	Mechanism	Signature	Trigger Family
----	--------------	-----------	-----------	----------------

E-01	TG decomposition error	planner mismatch / missing constraints	wrong node order; spec drift	pre-flight rejection spikes
E-02	Routing mismatch	capability selection fault	repeated agent swaps	SI crash; arbitration overflow
E-03	Economic pass failure	unpriced or mispriced plans	cost overruns; high-tier overuse	C_econ breach alarms
E-04	RB checkpoint skip	state machine fault	irreversible ops without auth	AB/RB violation attempt
E-05	Lineage/telemetry gap	observability loss	missing hashes / events	audit integrity checks

Invariant: Engine pre-flight rejection on constraint violation is a hard invariant.

O.1.3 Agentic-Layer Failures (A)

Baseline A-family retained with v1.0 coupling notes.

ID	Failure Mode	Mechanism	Signature	Trigger Family
A-01	Capability overreach	out-of-scope work	MCP/tool call outside contract	contract pre-flight reject
A-02	Constraint neglect	silent C breach	policy checker alerts	C_legal / C_safety alarms

A-03	Hallucination near RB	semantic fabrication	low evidence; high confidence	SI quality drop; critic red
A-04	Tool side-effect failure	partial execution	inconsistent external state	rollback triggers
A-05	Arbitration deadlock	critic oscillation	loop ceilings hit	watchdog / TTL
A-06	Economic bleed loop	infinite work	tokens/time exceed TTL	C_econ (TTL / Max_Tokens)

Rule: Economic Circuit vectors (Max_Spend, Max_Tokens, TTL) are mandatory for bleed prevention.

O.1.4 Governance-Layer Failures (G)

Baseline G-family retained with Bridge hooks.

ID	Failure Mode	Mechanism	Signature	Trigger Family
G-01	Policy lag	stale P _v	repeated novel edge incidents	drift via C mismatch
G-02	Boundary ambiguity	unclear non-delegables	AB creep; RB near-misses	AB/risk conflicts
G-03	Excessive friction	adoption collapse	rising Shadow AI rates	usage telemetry
G-04	Evidence failure	incomplete logs	missing incident packages	audit alarms

O.2 Stability-Coupled Failure Tags (Required)

Each incident MUST include a stability tag if applicable. Baseline tags retained.

Tag	Vital Sign	Default Threshold	Coupled Action
S-D	Drift	$D > 0.2$	governance review + AB demotion
S-∇	Divergence	mismatch frequency rising	contract reconciliation
S-R	Resonance collapse	$R < 0.5$	safe-mode / kill-switch
S-E	Entanglement	$E_n > 0.9$	forced manual drill / slow-core cap

v1.0 alignment:

- **Drift** incidents MUST route to Appendix G diagnostics.
- **Decay** incidents MUST route to Appendix F CI health checks.
- If both appear, primary tag SHALL follow dominant causal family (semantic vs operational), with secondary tag recorded for audit.

O.3 Incident Severity Classes and Runbooks (S0–S4)

Severity classes are mandatory and MUST be recorded in lineage.
Severity determines:

- immediate AB action,
- Bridge Collapse / Global Lock triggers,
- required evidence package, and
- audit escalation.

O.3.1 S0 — Benign anomaly (log only)

Typical: minor critic disagreement; brief routing hesitation.

Checklist:

- [] Log anomaly with tags (Layer ID + stability tags if any).
- [] No autonomy change required.
- [] Add to tactical review backlog if repeated $\geq N$ times in window W .

O.3.2 S1 — Local reversible failure

Typical: tool timeouts, minor hallucination below RB, low-impact drift yellow.

Checklist:

- [] Retry TG node or fallback to alternate admissible agent.
- [] Re-run critic loop if $AB \geq AB2$.
- [] Confirm actual cost within B and TTL.
- [] Log resolution and delta-SI.
- [] If repeated within N episodes $\rightarrow$ upgrade to S2.

O.3.3 S2 — Contained safety/economic degradation

Typical: repeated hallucinations near RB, SCP saturation trend, budget bleed trend, drift confirmed but bounded.

Checklist:

- [] Immediate AB demotion by ≥ 1 band for affected TG node types.
- [] If in Bridge lane: **Bridge Collapse** for that lane; revoke auto-admission.
- [] Trigger Appendix F (decay) or Appendix G (drift) diagnostic pathway.
- [] Require Dry-Run preview for all RB-prox nodes until cleared.
- [] Produce an **O.4 Incident Package** within 72 hours.
- [] Governance review required before restoring AB.

O.3.4 S3 — Major violation / high-regret near-miss

Typical: RB checkpoint skip attempt, confirmed constraint breach with potential harm, SI red-band collapse, repeated unsafe tool side effects.

Checklist:

- [] **AB0 lock** for the workflow (safe-mode).
- [] If any nodes were in Bridge lane: **Global Bridge Lock** for domain pending disposition.
- [] Quarantine implicated agents and MCP contracts.
- [] Execute rollback for any external side effects (Appendix E semantics).
- [] Mandatory Governance incident board within 24 hours.
- [] Full O.4 Incident Package within 7 days.
- [] No AB restoration until CI green (if decay) and drift cleared (if drift), plus Governance sign-off.

O.3.5 S4 — Catastrophic failure / rights-critical breach

Typical: actual irreversible harm or clear imminent harm; policy evasion; uncontrolled economic bleed beyond hard caps; mission-critical system compromise.

Checklist:

- [] **Kill-switch activation**; halt system and quarantine agentic layer.
 - [] Immediate regulator / institutional escalation per sector CCMT.
 - [] Freeze all AB promotions organization-wide pending audit.
 - [] Conduct full forensic lineage export and preservation.
 - [] Governance Body **MUST** issue public/internal breach report as required by law.
-

O.4 Required Incident Package (for S2+)

Every S2, S3, or S4 incident **MUST** generate an evidence bundle containing:

1. **Incident header**: severity, layer IDs, stability tags, timestamp, scope.
2. **Lineage slice**: $I_0 \rightarrow I' \rightarrow C \rightarrow TG \rightarrow AB/RB \text{ state} \rightarrow MCP \rightarrow \text{outputs} \rightarrow \text{costs}$.
3. **Causality narrative**: mechanistic chain, not opinion.
4. **Drift/decay determination**: referencing Appendix F/G.
5. **AB/RB actions taken**: demotions, locks, checkpoints added.
6. **Economic actions**: EC halts, TTL/cap events, spend deltas.
7. **Rollback proof** if external side effects occurred.
8. **Remediation plan** with owners and deadlines.
9. **Re-entry criteria** for autonomy restoration.

Rule: Absence of an O.4 package is itself a Governance-Layer Evidence Failure (G-04).

O.5 Conformance Tests (Required Before AB2+)

Baseline tests retained and referenced as mandatory.

Before claiming AB2+ or AB3 operation, the organization **MUST** pass:

1. Safe-mode conformance test
 - red-band trigger forces AB0 lock.
2. Kill-switch conformance test
 - S4 trigger halts system and quarantines agents.
3. Rollback test
 - every external tool path demonstrates correct undo semantics.
4. Drift gate test
 - simulated drift causes AB demotion + Bridge revocation.

5. Entanglement gate test

- simulated entanglement blocks AB promotion and invokes slow-core cap.

Recurrence: These tests MUST be rerun at least quarterly and after any GK or policy bundle update.

O.6 Chaos Symbiosis Drill Schedule (Continuous Resilience)

Baseline drill cadence retained.

O.6.1 Minimum monthly drills

1. RB red-team drill: attempt unauthorized boundary crossing.
 - Required proof: safe-mode conformance pass.
2. Economic bleed drill: force expensive agents into loops.
 - Required proof: TTL/budget interrupts fire.
3. Contract fuzzing drill: random TG nodes to test contract rejections.
 - Required proof: MCP rejects out-of-scope invokes.
4. Stability budget drill: ship a change exceeding entanglement caps.
 - Required proof: slow-core policy blocks release.

O.6.2 Quarterly drills

1. Shadow AI audit: sample decisions to confirm lineage completeness.
 2. Recovery ladder replay: replay last quarter's S2+ incidents to validate remediation completion.
-

O.7 Auditor Usage Protocol

Auditors SHALL verify:

1. Every S2+ incident includes a complete O.4 package.
 2. Conformance tests O.5 are routinely passed and lineage-logged.
 3. Chaos drills O.6 are scheduled and evidenced.
 4. Manifesto scorecard (Appendix M) reflects these items at ≥ 2 ("Operational") for any Symbiosis compliance claim.
-

SQC Gate Check (Appendix O)

Gate 1 — Layer Integrity: PASS

- Failure taxonomy and runbooks preserve Four-Layer responsibility boundaries and escalation rights.

Gate 2 — Canonical Metrics Non-Redefinition: PASS

- SI/HSS/SCP/CI/RB/AB are referenced only for actions; no schema or weights introduced.

Gate 3 — Constraint Precedence + Threshold Explicitness: PASS

- Constraints, EC caps, and AB/RB effects are hard, explicit, and governance/sector-set.

Gate 4 — Mechanistic Testability: PASS

- Severity classes, checklists, conformance tests, and drill proofs specify implementable inputs → conditions → outputs.

Gate 5 — Lineage Recoverability: PASS

- O.4 packages and auditor protocol require complete reconstructable lineage slices.

Gate 6 — Drift vs Decay Typing: PASS

- Mandatory stability tags and routing to Appendix F/G enforce non-conflation.

Gate 7 — Abuse / Incentive Hardening: PASS

- Non-punitive posture and HSS/SCP privacy constraints preserved; no HR/managerial misuse pathways added.

Gate 8 — Normative Voice Consistency: PASS

- MUST/SHALL/MAY used precisely; no speculative or marketing language.

Appendix P — Symbiosis Compliance Checklist (SCC) + SQC Checklist Card (v1.0)

Status: NEW (normative verification aid + deployability/audit readiness instrument).

Primary linkage: Chapter 1 (SQC), Part III (canonical SI/HSS/SCP/CI/RB/AB), Chapter 8 (Governance Kernel), Chapter 13 (Engine v2), Chapter 15 (Bridges + Collapse), Chapter 16 (Adoption), Chapter 17 (Certification), Appendix B (Economic Circuit v2), Appendix H (HSS v2.0 + anti-gaming), Appendix K (sector overlays), Appendix O (incident runbooks).

P.0 Purpose and Use

P.0.1 Purpose

1. Appendix P defines **minimum conformance checks** required for any organization to:
 - a. claim Symbiosis v1.0 compliance,
 - b. operate above AB0 safely, and
 - c. pass third-party audit or certification.
2. Appendix P provides two tools:
 - a. **Symbiosis Compliance Checklist (SCC)** — system/deployment readiness.
 - b. **SQC Checklist Card** — per-chapter/appendix “Definition of Done” verification.

P.0.2 Normative effect

1. Items marked **MUST** are binding conformance requirements.
2. Items marked **SHALL** are mandatory process or evidence provisions.
3. Items marked **MAY** are optional enhancements that do not affect baseline compliance.
4. A deployment with any **SCC-MUST** item failing **SHALL** be considered **non-compliant**.

P.0.3 Canonical metrics rule

This appendix **SHALL** reference SI, HSS v2.0, SCP, CI, RB v2, and AB only as canonical in Part III. No schema or weights are defined here.

P.1 Symbiosis Compliance Checklist (SCC)

P.1.1 How to score

Each checklist item **SHALL** be scored:

- **PASS** — implemented and evidenced.
- **PARTIAL** — implemented but missing binding evidence.
- **FAIL** — absent or contradicted.

Compliance Rule:

- All **MUST** items require **PASS**.
 - A deployment **MAY** proceed to AB1 only when all P.2 **MUST** items **PASS**.
 - A deployment **MAY** proceed to AB2+ only when all P.2 and P.3 **MUST** items **PASS** and certification gates pass (Chapter 17).
 - A deployment **MAY** claim full Symbiosis v1.0 compliance only when **all MUST items across P.2–P.6 PASS**.
-

P.2 Core Architecture and Layer Integrity (MUST)

ID	Requirement	Evidence (required)	PASS / PARTIAL / FAIL
P2-1	Four-Layer Architecture is instantiated (Human / Engine / Agentic / Governance).	Architecture diagram + component inventory (Chapter 3 / App L references).	
P2-2	No cross-layer goal authority: Agents and Engine do not originate human goals.	Role boundary doc + MCP contract templates showing “no goal authority.”	
P2-3	All agent actions execute only under MCP contracts and AB ceilings.	Live MCP registry + enforcement logs (Appendix F).	
P2-4	Governance Kernel enforces AB/RB state transitions; operators cannot bypass.	GK policy bundle + AB/RB transition logs.	
P2-5	Constraint Vector C binds Intent, TG, and MCP with supremacy.	Constraint injection proofs + pre-flight rejection logs.	
P2-6	Context Lineage L is active and immutable for all workflows.	Lineage store schema + sample replay slice.	

P.3 Canonical Control Loops and Safety (MUST)

ID	Requirement	Evidence (required)	PASS / PARTIAL / FAIL
P3-1	Symbiosis Engine pipeline includes mandatory Economic Pass, RB/AB selection, Dry-Run for AB2+/RB-prox.	Engine stage logs (Chapter 13) + dry-run artifacts.	
P3-2	SI is computed canonically, logged per episode, and drives AB/RB control loops.	SI computation logs + AB gating trace.	
P3-3	HSS v2.0 and SCP are computed canonically and bound to AB promotions/demotions.	HSS/SCP dashboards + promotion evidence slice.	
P3-4	Drift vs Decay typing is enforced and routed to App F/G pathways.	Incident tags + diagnostic records.	
P3-5	Safe-mode and kill-switch behavior conforms to Appendix O.	O.5 conformance test passes.	

P3-6	Bridge Collapse + Global Lock triggers are implemented for S2+ incidents.	AB demotion + Bridge revocation logs.
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P.4 Economic Circuit Conformance (MUST)

ID	Requirement	Evidence (required)	PASS / PARTIAL / FAIL
P4-1	Budget envelope B is mandatory for all intents; TG projected cost is computed pre-flight.	Intent objects + cost projection logs.	
P4-2	Compute Saturation (CS) and Queue Latency (QL) gates are configured and enforced.	Gate configuration + halt events.	
P4-3	TTL / Max_Spend / Max_Tokens caps prevent bleed loops.	Economic Circuit policy + prevented bleed examples.	
P4-4	EC halts are lineage-logged with reason codes.	Lineage audit slice showing EC halts.	

P.5 Governance and Sector Compliance (MUST)

ID	Requirement	Evidence (required)	PASS / PARTIAL / FAIL
P5-1	Governance Body is named with decision rights (promotions, demotions, freezes, policy updates).	Governance charter + signatory list.	
P5-2	CCMT sector overlays compiled into machine constraints per Appendix K.	Local CCMT + Constraint Catalog export.	
P5-3	Incident runbooks S0–S4 are adopted and actively used.	Incident log + O.4 packages for S2+.	
P5-4	Non-Punitive Covenant and human-metrics firewall are enforced.	Policy text + access control proofs.	
P5-5	Policy versions (P_v) are tracked and logged to lineage on change.	Policy delta log + lineage commit IDs.	

P.6 Literacy, Adoption, and Certification (MUST)

ID	Requirement	Evidence (required)	PASS / PARTIAL / FAIL
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P6-1	Operator literacy certification matches AB rights (Appendix I).	Certification records per operator cohort.
P6-2	Pilot types P0/P1/P2 follow Appendix N template and exit artifacts are produced.	Pilot specs + pilot reports.
P6-3	AB promotions meet global prerequisites: SI stability, vital signs green, HSS readiness, SCP headroom, recovery readiness.	Promotion lineage slices showing all prerequisites.
P6-4	External validation / certification executed before AB3 scale (Chapter 17).	Certification report + auditor sign-off.

P.7 Optional Enhancements (MAY)

These items improve defensibility and adoption velocity but are not strict baseline compliance:

ID	Enhancement	Benefit
P7-1	Open-kernel publication of GK policy templates.	Reduces regulator friction and improves trust.

P7-2	Third-party red-team exercises beyond Appendix O minima.	Increases resilience against novel failures.
P7-3	Sector-specific SI/HSS/SCP calibration studies.	Improves autonomy gating fit.

P.8 One-Page SQC Checklist Card (Paste-Ready)

This card is mandatory “Definition of Done” for every Chapter/Appendix draft. The gates and PASS rule are canonical and invariant.

Appendix P — Specification Quality Contract (SQC) Checklist Card

Status: Normative Verification Aid

Use: Required post-draft gate check for every Chapter / Appendix.

Section Under Review: _____

Part / Chapter / Appendix ID: _____

Reviewer (Role): _____

Date (YYYY-MM-DD): _____

PASS RULE:

- ALL eight gates MUST pass.
- Any FAIL blocks inclusion in v1.0 binder.

GATE 1 — LAYER INTEGRITY

☐ PASS ☐ FAIL

- Conforms to Four-Layer Architecture (H / Engine / Agents / Governance).
- No layer bypasses adjacent boundaries.
- Responsibilities are not cross-assigned.

Notes / Fixes: _____

GATE 2 — CANONICAL METRICS NON-REDEFINITION

☐ PASS ☐ FAIL

- SI / HSS v2.0 / SCP / CI / RB / AB schema are NOT redefined here.
- Only referenced from Part III canonical definitions.

- No re-weighting or shadow definitions.

Notes / Fixes: _____

GATE 3 — CONSTRAINT PRECEDENCE + THRESHOLD EXPLICITNESS

☐ PASS ☐ FAIL

- Hard constraints explicitly stated and override optimization.
- CS / QL / RB / AB thresholds explicit, governance-set, deployment-specific.
- No implicit/universal defaults.

Notes / Fixes: _____

GATE 4 — MECHANISTIC TESTABILITY

☐ PASS ☐ FAIL

- Every normative clause is implementable or testable.
- Algorithms/gates specify inputs → conditions → outputs.
- Non-testable claims removed or reframed.

Notes / Fixes: _____

GATE 5 — LINEAGE RECOVERABILITY

☐ PASS ☐ FAIL

- Actions trace to I + C + TG + AB/RB state + Economic Pass.
- Telemetry minima specified where relevant.
- Replayability feasible for audit.

Notes / Fixes: _____

GATE 6 — DRIFT VS DECAY TYPING

☐ PASS ☐ FAIL

- Failures classified as Semantic Drift or Operational Decay.
- Typed events logged separately to lineage.

Notes / Fixes: _____

GATE 7 — ABUSE / INCENTIVE HARDENING

☐ PASS ☐ FAIL

- No managerial/HR misuse vectors.
- Anti-gaming and privacy rules intact.
- HSS remains Safety-only.

Notes / Fixes: _____

GATE 8 — NORMATIVE VOICE CONSISTENCY

☐ PASS ☐ FAIL

- MUST / SHALL / MAY used correctly.
- No marketing or speculative phrasing.

Notes / Fixes: _____

SQC Gate Check (Appendix P)

Gate 1 — Layer Integrity: PASS

- SCC items preserve Four-Layer duties and prohibit bypass.

Gate 2 — Canonical Metrics Non-Redefinition: PASS

- Appendix P references SI/HSS/SCP/CI/RB/AB only for compliance checks; no schema/weights defined.

Gate 3 — Constraint Precedence + Threshold Explicitness: PASS

- SCC requires explicit hard constraints, CS/QL/RB/AB thresholds as governance-set with no defaults.

Gate 4 — Mechanistic Testability: PASS

- All compliance items are evidence-bound and objectively scorable.

Gate 5 — Lineage Recoverability: PASS

- SCC and card require replayable lineage slices for promotions, incidents, EC halts, and audits.

Gate 6 — Drift vs Decay Typing: PASS

- SCC mandates typed incidents and routing to F/G pathways.

Gate 7 — Abuse / Incentive Hardening: PASS

- Human-metrics firewall and non-punitive covenant enforced; no secondary-use pathways added.

Gate 8 — Normative Voice Consistency: PASS

- MUST/SHALL/MAY used precisely; no speculative or marketing tone.

Appendix R — External Standards Crosswalk (ISO/IEC 42001, NIST AI RMF, EU AI Act)
(v1.0)

Status: NEW (normative mapping + audit aid).

Purpose: Provide a defensible correspondence between Symbiosis v1.0 primitives and widely used external AI governance regimes. This appendix does not claim equivalence; it specifies **how Symbiosis artifacts satisfy or operationalize external control intents**, subject to sector overlays.

Canon Rule: This appendix SHALL NOT redefine SI, HSS v2.0, SCP, CI, RB v2, or AB schema. It references only Part III canonical definitions.

R.0 How to Read This Crosswalk

R.0.1 Mapping semantics

Each mapping row provides:

- 1. **External Control / Obligation:** the control intent from an external regime.
- 2. **Symbiosis Primitive(s):** the internal objects and layers that address that intent.
- 3. **Required Symbiosis Evidence:** binder artifacts or telemetry required to demonstrate coverage.
- 4. **Residual Gaps (if any):** items that Symbiosis does not cover by default and that **MUST** be added by sector playbooks.

R.0.2 Non-equivalence disclaimer

A deployment MAY use this crosswalk to accelerate certification, but **external certification decisions remain the responsibility of the auditor/regulator**. The crosswalk is an aid, not an automatic compliance guarantee.

R.1 ISO/IEC 42001 (AI Management System) Crosswalk

ISO/IEC 42001 centers on an organizational AI management system: governance, risk treatment, documentation, continual improvement, and measurable controls.

R.1.1 ISO Control Families → Symbiosis Coverage

ISO/IEC 42001 Family	Symbiosis Coverage	Primary Evidence
Context of the organization / scope	Pilot Spec Template + domain scoping	Appendix N (PS-1 to PS-2), Chapter 16 adoption phases
Leadership / policy / accountability	Governance Body + GK authority	Chapter 12 (GK v2), Appendix K (CCMT ownership), Appendix J (role specs)

Planning / risk assessment	Risk Classes, RB doctrine, AB ceilings	Chapter 11 (RB/AB canon), Appendix K (RAM-K), Appendix O (taxonomy)
Support / competence / awareness	Symbiosis Literacy Program	Appendix I (SLP), Appendix J hiring rubrics
Operation / lifecycle controls	Engine algorithm + MCP contracts + Economic Circuit	Chapter 13 (Engine v2), Appendix B (EC v2), Appendix F (contracts)
Performance evaluation / monitoring	SI/HSS/SCP telemetry + lineage	Part III metrics, Appendix H dashboard, Appendix P SCC
Improvement / corrective action	Drift vs decay pathways + incident runbooks	Chapter 14, Appendix F/G, Appendix O

R.1.2 Targeted ISO Control Intent → Symbiosis Mapping

ISO Intent (summary)	Symbiosis Primitive(s)	Required Symbiosis Evidence	Residual Gaps
AI risk identification & treatment	GK risk classes; RB; AB gating; sector overlays	Chapter 12; Chapter 11; Appendix K; Appendix P P3/P5	Sector-specific numeric thresholds set locally
Human oversight ensured	AB state machine; RB checkpoints; Dry-Run previews	Chapter 13 stages 5–7; Appendix O conformance tests	Human staffing ratios may require sector policy

Data governance & privacy	Constraint Vector C; CCMT rows; Compliance Memory	Appendix K C-CAT; Appendix K rows; Appendix O evidence	Jurisdictional DPIA artifacts added locally
Transparency / explainability	Context Lineage L; transparency trace	Chapter 13 Stage 11–12; Appendix O O.4 packages	External model-card formats added if required
Competence & training	SLP levels tied to AB	Appendix I; Appendix J; Appendix P P6	None, unless regulator requires specific curricula
Incident response	S0–S4 runbooks; safe-mode / kill-switch	Appendix O; Appendix P P3/P5	Regulator-specific reporting timelines

R.2 NIST AI RMF (Risk Management Framework) Crosswalk

NIST AI RMF functions: **Govern, Map, Measure, Manage.**

R.2.1 RMF Functions → Symbiosis Coverage

NIST Function	Symbiosis Coverage	Primary Evidence
Govern	Governance Layer + GK + CCMT	Chapter 12; Appendix K; Appendix M scorecard
Map	Structured Intent Objects + TG decomposition	Chapter 5 (Intent), Chapter 13 (Planning), Appendix N

Measure	Canonical metrics + vital signs + CI	Part III metrics; Appendix F (CI); Appendix H
Manage	AB/RB control loop; Economic Circuit; incident runbooks	Chapter 11/13; Appendix B; Appendix O

R.2.2 NIST Profile Objectives → Symbiosis Mapping

NIST Objective	Symbiosis Primitive(s)	Required Evidence	Residual Gaps
Defined roles & accountability	Roles (SO/HIA/SGL) + Governance Body	Appendix J; Appendix I; Appendix K ownership	None
Risk controls proportionate to impact	Risk classes + AB ceilings + RB doctrine	Chapter 12 risk; Appendix K RAM-K	Threshold tuning per sector
Continuous monitoring	SI/HSS/SCP rolling windows; CI health	Part III; Appendix H dashboard; Appendix F CI logs	None
Safe fallback / fail-safe	Safe-mode, AB demotion, kill-switch	Appendix O; Appendix P P3	None
Incident learning / improvement	Drift/decay remediation loop	Chapter 14; Appendix F/G; Appendix O O.4 packages	None

Human-in-the-loop for high stakes	AB0/AB1 ceilings in CCMT	Appendix K rows; Appendix O severity rules	Sector laws may mandate additional approvals
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R.3 EU AI Act Crosswalk (High-Risk & GPAI Obligations)

The EU AI Act (2024 text) distinguishes prohibited systems, high-risk systems, and transparency obligations; it also includes obligations for general-purpose AI (GPAI) providers. This mapping is **deployment-facing** and assumes sector overlays author High-Risk CCMT rows where applicable.

R.3.1 EU AI Act Obligation Families → Symbiosis Coverage

EU AI Act Obligation	Symbiosis Primitive(s)	Required Evidence	Residual Gaps
Risk management system	GK risk classes + CCMT + pilots	Chapter 12; Appendix K; Appendix N	EU-specific risk file format
Data governance	Constraint Vector C + data CCMT rows	Appendix K C-CAT; Appendix K sector rows	EU-specific data quality annex
Technical documentation	Binder + lineage exports	Full binder; Appendix O O.4 packages	EU doc templates (Annex IV)
Record-keeping / logging	Context Lineage L + Compliance Memory	Chapter 13 Stage 12; Appendix K/ O	None

Transparency & information to users	Transparency trace to H; dashboards	Chapter 13 Stage 11; Appendix H UI	EU user-notice wording
Human oversight	AB/RB gating + Dry-Run previews	Chapter 11/13; Appendix O	None
Accuracy, robustness, cybersecurity	CI health checks; decay controls; OT/IP constraints	Appendix F (CI), Appendix O, Appendix K rows	Penetration tests per EU guidance
Post-market monitoring	Pilot expansions + incident drills	Appendix N P1/P2 cadence; Appendix O drills	EU reporting channel syntax

R.3.2 EU “Prohibited Practices” Screening (Symbiosis Position)

Symbiosis compliance requires explicit screening for prohibited AI practices. Any prohibited practice SHALL be encoded as a **Hard constraint** in CCMT and enforced pre-flight.

Prohibited Practice Class (summary)	Symbiosis Response	Evidence
Unconstrained biometric categorization / inference in forbidden domains	Hard constraints in CCMT; ABO safe-mode	Appendix K rows; Engine rejection logs
Manipulative or exploitative systems harming vulnerable groups	Hard constraints + risk class CRITICAL	Appendix K; Appendix O severity triggers
Social scoring for punitive employment/benefit decisions	Explicit non-use invariant for HSS/SCP	Appendix H safeguards; Appendix P P5-4

Real-time remote biometric
ID where disallowed

Sector overlay bans

Appendix K rows

R.4 Cross-Regime Evidence Packaging

To support multi-regime audits, deployments SHALL maintain an exportable evidence bundle containing:

1. **Governance Pack:** Governance charter, GK policy bundle, CCMT + C-CAT + RAM-K + EAM-K.
2. **Technical Pack:** Engine stage logs, MCP registries, Economic Circuit config, CI logs.
3. **Human Pack:** SLP certifications, role assignments, HSS/SCP privacy controls.
4. **Operational Pack:** Pilot reports, SI/HSS/SCP trajectories, incident packages (O.4), drill schedules.
5. **Lineage Pack:** Immutable hashes and replay slices for representative episodes.

This bundle aligns directly to Appendix P SCC items and SHALL be produced prior to any AB2+ claim.

R.5 Residual-Obligation Rule

If any external regime requires artifacts not specified in Symbiosis core (e.g., EU Annex IV technical documentation format, DPIA forms), those artifacts SHALL be added as **sector-specific playbook attachments** and referenced in Appendix K CCMT rows. They MAY NOT relax Symbiosis invariants.

SQC Gate Check (Appendix R)

Gate 1 — Layer Integrity: PASS

- Crosswalk preserves Four-Layer responsibilities and maps external controls to correct layers/primitives.

Gate 2 — Canonical Metrics Non-Redefinition: PASS

- SI/HSS/SCP/CI/RB/AB referenced only as canonical controls; no formulas or weights introduced.

Gate 3 — Constraint Precedence + Threshold Explicitness: PASS

- All threshold obligations are explicitly governance/sector-set; prohibited-practice bans require hard constraints.

Gate 4 — Mechanistic Testability: PASS

- Each mapping specifies concrete Symbiosis evidence artifacts that can be audited.

Gate 5 — Lineage Recoverability: PASS

- Record-keeping and evidence packs require lineage exports and replay slices.

Gate 6 — Drift vs Decay Typing: PASS

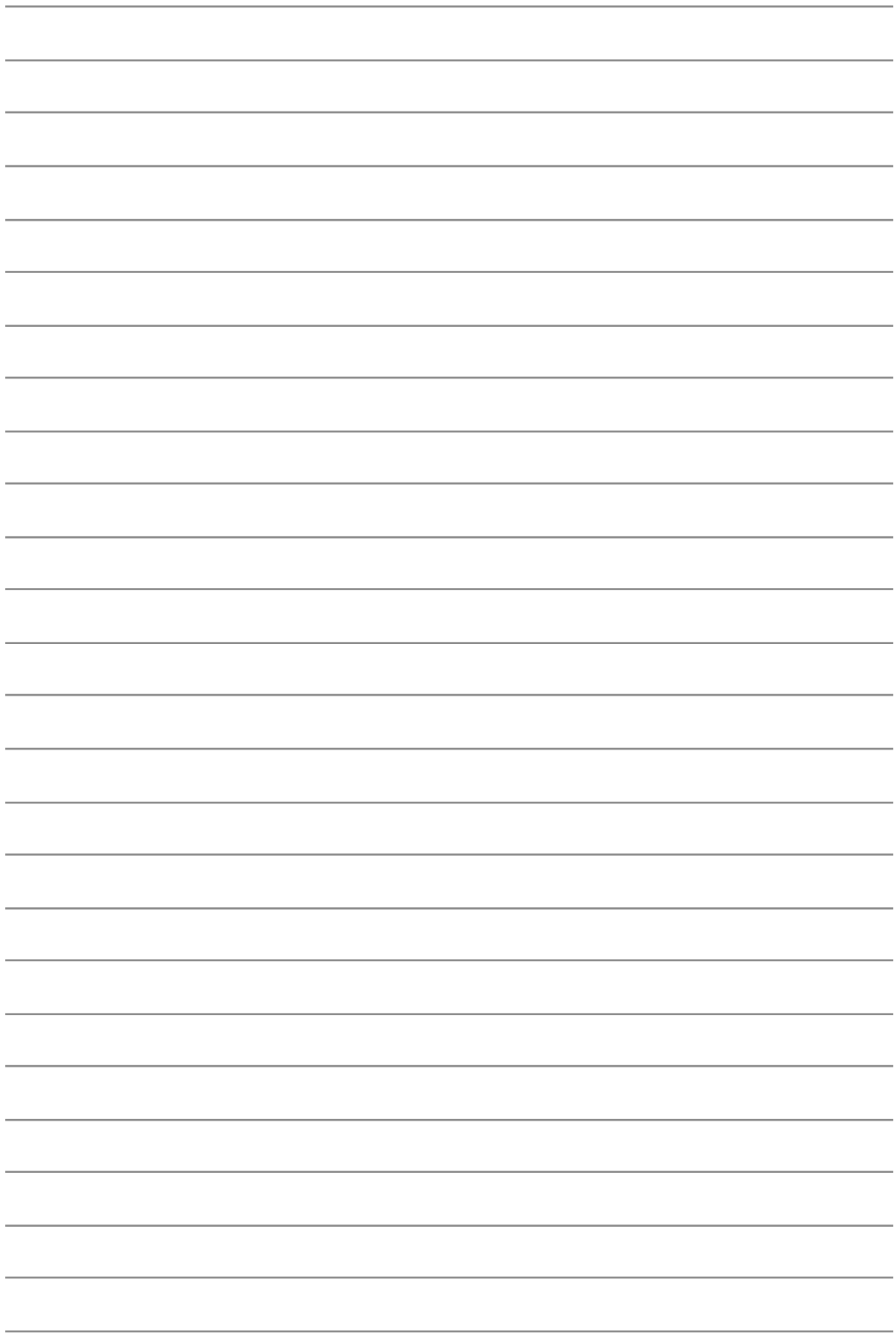
- Incident and improvement mappings explicitly route drift/decay via Appendix F/G and Appendix O.

Gate 7 — Abuse / Incentive Hardening: PASS

- Crosswalk reinforces non-punitive human metrics posture and prohibits social-scoring misuse.

Gate 8 — Normative Voice Consistency: PASS

- MUST/SHALL/MAY used precisely; no speculative or marketing language.



This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.